Archival Research Edition · Scalar–Tensor & Disformal Action Mapping · Steps 1–25 Audit Package

Time Attraction Law — Scalar–Tensor Yukawa Fit (V26: V25 & Steps 23–25)

Derivation Steps 1–22, Extended Observational Tests (Steps 23–25: GW170817, Lensing, Bullet Cluster, Cosmology & CMB), and Reproducibility Manifest

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V25 Deposit & Status: **V25** (10.5281/zenodo.23699685): 4-component curved exterior BVP, scalar_m_sign = -
and V25 diagnostics (stability closing statement conditional on independent variational/BC
Earlier V24 Deposit: 10.5281/zenodo.23680715 (refers strictly to the prior V24 release; V25 changes are not par
Extended Modules (V26): step23_en (GW170817 & Lensing), step24_en (Bullet Cluster), step25_en (Cosmology &
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Included Archival Files in This English Edition (27 Text Files)

This document compiles the 27 English text files of the Steps 1–25 audit package from scalar-tensor-yukawa-fit/:

- **Package Overview & Numerical Summary:** README_en.md
- **Derivation & Observational Chapters (Steps 1–25):** step1_dilaton_form_en.md through step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md, plus the extended observational modules step23_gw170817_and_lensing_en.md, step24_bullet_cluster_en.md, and step25_cmb_and_cosmology_en.md
- **Reproducibility & SHA-256 Checksum Manifest:** MANIFEST_en.txt

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1 Time Attraction Law — Scalar-Tensor Yukawa Fit, V26 — Scalar–Tensor and Yukawa/Disformal Action Mapping (V26; V23 ↔ Jordan–Brans–Dicke / Dilaton Framework) — Steps 1–25 Derivation and Audit Reproduction Package

Source file: `README_en.md`

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
V26 current package — Time Attraction Law — Scalar-Tensor Yukawa Fit, V26 (25 steps): The current 25-step package is designated **V26**. Historical delineation: **V24** (DOI: 10.5281/zenodo.23680715) refers strictly to the prior V24 release; **V25** (DOI: 10.5281/zenodo.23699685) covers Steps 1–22. V26 adds the extended observational modules `step23`–`step25` with the following locked numbers: $M_{\mathrm{ADM,max}} = 2.033214\ M_\odot$ (SLy), synthetic 1D Bullet Cluster metric $M_{\mathrm{synth}} = 32.089$, GW170817 scalar suppression $\log_{10} A = -5.360 \times 10^{19}$ (at 40 Mpc), and a dedicated 1.0 $M_\odot$ Solar–Cassini table (`step23`, Table 3; `source_body = Sun_1p00Msun`), separate from the conditional diagnostic `NS_far_field_b_*` rows on the Step 22 1.50999814 $M_\odot$ `Branch II-Fluid` neutron-star background (`step23`, Table 2) and the self-consistent 1.56203491 $M_\odot$ `Branch II-Conf` trace-sourced ($\tilde{\varepsilon} - 3\tilde{P}$) BVP solution (`step23`, Table 2B). Locked convention: $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$, $g^{(J)}_{\mu\nu} = \exp(-2\alpha_Y \phi_L) g_{\mu\nu}$, $\alpha_{\mathrm{DEF}} \equiv d \ln A / d\phi_{\mathrm{canon}} = -1/\sqrt{4\pi} \approx -0.2820947918$, $\beta_{\mathrm{DEF}} \equiv d^2 \ln A / d\phi^2_{\mathrm{canon}} = 0$, $\beta_{\mathrm{Yuk}} \equiv \alpha^2_{\mathrm{DEF}} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) \approx 0.0795774715$, $\omega_{\mathrm{BD}} = \frac{1}{2\beta_{\mathrm{Yuk}}} - \frac{3}{2} = \frac{2\pi}{\alpha_Y^3} - \frac{3}{2} \approx 4.7831853072$ (vs. legacy `step5` 4π-convention $\frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.066371$). $c_T = c$ is proven on the purely conformal minimal branch ($G_{4,X} = 0$, $G_5 = 0$), not automatically on a disformal/refractive branch ($D(\phi_L, X) \neq 0$).

Publication status, version delineation (V24 vs. V25), and extended observational modules (`step23`–`step25`): The earlier Zenodo publication (**V24**, DOI: 10.5281/zenodo.23680715) refers strictly to the prior V24 release. The sign fixes in `curved_yukawa_adm_and_stability.py` (the exterior metric coupling term `a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0` at line 593 of `linearized_exterior_response`, and setting the exact left-hand side sign `scalar_m_sign = -1.0`, i.e., `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`, as the default in the $L_u u = R_h\, \delta h$ operator of `solve_yukawa_coupled_omega2_gl2`) and the three new diagnostics in `curved_yukawa_audit_summary.json` (`full_jacobian_multi_eos_and_points`, `baryonic_mass_and_first_law_audit`, `eigenmode_xi_nodes_and_static_tangent_audit`) belong to **V25** (Zenodo DOI: 10.5281/zenodo.23699685), with the coupled radial stability closing statement formulated conditionally until independent verification of the variational derivation and boundary-condition matching is concluded. Building directly on the V25 curved-exterior solution at fixed $L = 10$ km and $\alpha_Y = 1$, modules `step23`, `step24`, and `step25` provide the extended observational tests for gravitational-wave propagation (`GW170817` $c_T = c$) and gravitational lensing (`step23`), the `Bullet Cluster` (`1E 0657-558`, `step24`), and `CMB` / cosmological background dynamics (`step25`).

V25 diagnostics summary (exact data from `curved_yukawa_audit_summary.json` and `curved_yukawa_coupled_stabilit`

- Derivative verification (SLy, $\log_{10} \rho_{0c} = 15.0$, $L = 10$ km): `exterior_dphi_prime_dm_R` $= -1.683651 \times 10^{-4}$ km^{-2} (rel. diff 3.98×10^{-10}); `interior_r6km_a_m_sc` $(\partial \phi_L'/\partial m) = -1.728334 \times 10^{-4}$ km^{-3} (rel. diff 6.08×10^{-9}).
- Full 4×4 Jacobian: 4 configurations (SLy and APR4), 108 checks, max relative error 2.74×10^{-5}.
- Baryonic mass / First Law: `delta_dex_ADM_vs_baryon_turning` between M_{ADM} and the adaptive Gauss–Kronrod baryonic turning point (M_B^{quad}) is -4.58×10^{-6} to $+8.43 \times 10^{-6}$ dex; non-conformal excess $8.23 \times 10^{-5}\ M_\odot$/dex.
- Main stability zero crossing and eigenmode audit (with exact LHS sign `scalar_m_sign = -1.0`): the main dynamical $\omega_0^2 = 0$ zero crossing in `curved_yukawa_coupled_stability.csv` for SLy $L = 10$ km, $\alpha_Y = 1$ is `15.316410` ($+5.80 \times 10^{-6}$ dex above the ADM peak `15.316404` and $+1.33 \times 10^{-6}$ dex above the M_B^{quad} peak `15.316409`; across all 5 sequences $\leq 6.9 \times 10^{-6}$ dex offset, nodeless $n = 0$ fundamental mode, and (0.6–5.0) $\times 10^{-4}$ relative L^2 agreement with the static equilibrium tangent; the diagnostic `scalar_m_sign = +1.0` root is `15.319129`, $+2.725 \times 10^{-3}$ dex — details: `step22`, Sections 4 and 7).

Date: 2026-10-27
Working Directory: `scalar-tensor-yukawa-fit`
Self-Contained Audit Archive: `scalar-tensor-yukawa-fit_audit_package.zip`
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Official Archival DOIs:

- **V14:** 10.5281/zenodo.23217189
- **V17 (SPARC 155 / 123 blind sample):** 10.5281/zenodo.19666220

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- **V19 (3D spherical top-hat SPARC, 175 galaxies):** 10.5281/zenodo.23497128
- **V20 (Two-clock granite chamber and 8-coil magnetic test):** 10.5281/zenodo.23540741
- **V21 (Compact stars, TOV, and Sirius B audit):** 10.5281/zenodo.23602501
- **V22 Book Edition (Trilingual monograph, HU/EN/DE):** 10.5281/zenodo.23604423
- **V23 (Covariant action closure and coupled radial stability, 50aa2db0):** 10.5281/zenodo.23639084
- **V24 (Earlier 22-step scalar–tensor Yukawa derivation archive):** 10.5281/zenodo.23680715
- **V25 (4-component curved exterior BVP, GL2 scalar_m_sign = -1.0 sign fixes & diagnostics):** 10.5281/zenodo.23699685

1.1 1. Research Question, Theoretical Framework & Branch Taxonomy

This audit package contains the exact mathematical correspondence and numerical verification mapping the covariant action candidates developed in the V19–V23 manuscripts and monograph edition of the **Time Attraction Law (TAL)** (10.5281/zenodo.23604423, 10.5281/zenodo.23639084) — **Formulation I** (nonlocal, positive-definite bilateral top-hat kernel action with auxiliary fields) and **Formulation II** (local, screened Helmholtz–Yukawa scalar-field action principle) — onto general **scalar–tensor theories of gravity** (Jordan–Brans–Dicke family, low-energy string-theory **dilaton** coupling, and Bekenstein-type **disformal** clock metrics).

1.1.1 1.1. Core Equations & Canonical Normalization

1. **Exponential Time Factor and Scalar/Dilaton Field (step1_dilaton_form.md / step1_dilaton_form_en.md):**

$$f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\,\bar\rho_L(\mathbf{x})\big) \;\equiv\; e^{-\phi_L(\mathbf{x})}, \qquad \phi_L(\mathbf{x}) \;=\; \gamma_L\,\bar\rho_L(\mathbf{x}) \;\geq\; 0,$$

where the canonical coupling coefficient originates from the gravitational volumetric capacity ($C = \frac{4\pi G}{c^2} = 9.33306 \times 10^{-27}$ m/kg):

$$\gamma_L \;=\; \frac{GL^2}{c^2} \;=\; \frac{C}{4\pi}\,L^2, \qquad m_\phi \;=\; \frac{\hbar}{Lc}.$$

2. **Conformal Jordan Metric vs. Medium-Anchored Disformal Clock Metric (step2, step5, step8, step14):**

$$\tilde g^{\rm conf}_{\mu\nu} \;=\; A^2(\phi_L)\,g_{\mu\nu} \;=\; f_L^2\,g_{\mu\nu} \qquad \text{vs.} \qquad \tilde g^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} + \frac{1-f_L^2}{c^2}\,u_\mu u_\nu \;\Longrightarrow\; \tilde g^{\rm disf}_{00} \;=\; f_L^2\,g_{00} \text{ (in the comoving frame).}$$

In a comoving ($u^\mu = (c/\sqrt{-g_{00}}, \mathbf{0})$, $u_\mu = (-\sqrt{-g_{00}}\,c, \mathbf{0}) \;\Longrightarrow\; u_0u_0/c^2 = -g_{00}$) baryonic medium and in atomic clocks, both metrics yield the same proper-time factor ($d\tilde\tau = f_L\,d\tau_g$). For ultrarelativistic light rays ($g_{\mu\nu}k^\mu k^\nu = 0$) and PPN tests, however, the two branches separate sharply:

- In the universal **conformal Jordan branch (Branch II-Conf, step23–step25)**, $\tilde g^{\rm conf}_{\mu\nu} = A^2(\phi_L)g_{\mu\nu} = e^{-2\alpha_Y\phi_L}g_{\mu\nu}$, $\sqrt{-\tilde g^{\rm conf}} = e^{-4\alpha_Y\phi_L}\sqrt{-g}$, the Maxwell action is conformally invariant, and for the canonically normalized scalar $\phi_{\rm canon} = \phi_L/\sqrt{2\tilde\alpha_L} = \sqrt{4\pi/\alpha_Y}\,\phi_L$ the unambiguous Damour–Esposito-Farèse / Yukawa / Brans–Dicke dictionary holds:

$$\alpha_{\rm DEF} \;\equiv\; \frac{d\ln A}{d\phi_{\rm canon}} \;=\; -\alpha_Y\sqrt{\frac{\alpha_Y}{4\pi}} \;=\; -\frac{1}{\sqrt{4\pi}}, \qquad \beta_{\rm DEF} \;\equiv\; \frac{d^2\ln A}{d\phi^2_{\rm canon}} \;=\; 0, \qquad \beta_{\rm Yuk} \;\equiv\; \alpha^2_{\rm DEF} \;=\; 2\tilde\alpha_L\alpha_Y^2 \;=\; \frac{\alpha_Y^3}{4\pi} \;=\; \frac{}{4}$$

$$\omega_{\rm BD} \;=\; \frac{1}{2\beta_{\rm Yuk}} - \frac{3}{2} \;=\; \frac{2\pi}{\alpha_Y^3} - \frac{3}{2} \;\approx\; 4.7831853072\ (\alpha_Y = 1), \qquad \gamma^{\rm conf}_{\rm PPN}(r) \;=\; \frac{1-\beta_{\rm Yuk}(1+r/L)e^{-r/L}}{1+\beta_{\rm Yuk}(1+r/L)e^{-r/L}},$$

whereas the earlier step5 chapter used the 4π-convention $\alpha_{\rm std} = 2\tilde\alpha_L = \frac{\alpha_Y}{4\pi}$ and $\omega^{(\rm step5)}_{\rm BD} = \frac{1-3\alpha_{\rm std}}{2\alpha_{\rm std}} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.066371$ (at the baseline $\alpha_Y = 1$, both conventions share $\beta_{\rm Yuk} = \alpha_{\rm std} = 1/(4\pi) \approx 0.0795774715$).

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- In the medium-anchored **disformal branch (`Branch II-Disf`)**, $\sqrt{-\tilde{g}^{\rm disf}} = f_L\sqrt{-g}$, magnetic energy ($\rho_B = |\mathbf{B}|^2/(2\mu_0 c^2)$) directly sources ϕ_L (`step15`), and light is deflected by the achromatic refractive index $n_t(s) = f_L(s)^{-1} = e^{+\phi_L(s)}$ (`step9`, `step14`), so the conformal Brans–Dicke $\gamma_{\rm PPN}^{\rm conf}$ formula does not apply to this branch.

3. **V23 Local Helmholtz–Yukawa Action Principle, Exact Normalization, and Curved Exterior Boundary-Value Problem (`step3`, `step5`, `step7`, `step22`):** The exact SI-unit coefficient of the scalar kinetic and mass terms (with $\alpha_Y \equiv 8\pi\tilde{\alpha}_L = 4\pi\alpha_{\rm std}$) is:

$$S_{\rm II}[g_{\mu\nu},\phi_L,\Psi] \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\left(R \;-\; \frac{1}{\tilde{\alpha}_L}\Big(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\Big)\right) \;+\; e^{-\phi_L}\,\mathcal{L}_{\rm matter}\right] \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}R \;-\; \right.$$

which in geometrized ($G = c = 1$) units yields the V23 code form $\mathcal{L}_\phi = -\frac{1}{2\alpha_Y}\big((\nabla\phi_L)^2 + \phi_L^2/L^2\big)$ and the exact curved-spacetime Euler–Lagrange equation (`yukawa_action_el`, $B(r) = 1 - 2m(r)/r$):

$$B\,\phi_L'' \;+\; \left[\frac{2B}{r} + B\nu' + \frac{B'}{2}\right]\phi_L' \;-\; \frac{\phi_L}{L^2} \;=\; -\frac{\alpha_Y G}{c^2}\,e^{-\phi_L}\,\varepsilon(r), \qquad \rho_\phi(r) \;=\; \frac{c^2}{2G\,\alpha_Y}\left[B\big(\phi_L'\big)^2 + \frac{\phi_L^2}{L^2}\right].$$

In the curved exterior vacuum outside the stellar surface ($r \geq R_*$, where $\varepsilon = P = 0$, $B\nu' + B'/2 = \frac{2m}{r^2} - \frac{4\pi r}{\alpha_Y L^2}\phi_L^2$), the coupled exterior Einstein–scalar system (`step22`, `curved_yukawa_adm_and_stability.py`) is:

$$\left(1 - \frac{2m(r)}{r}\right)\phi_L'' \;+\; \left[\frac{2}{r}\left(1 - \frac{m(r)}{r}\right) - \frac{4\pi r}{\alpha_Y L^2}\phi_L^2\right]\phi_L' \;-\; \frac{\phi_L}{L^2} \;=\; 0, \qquad \frac{dm(r)}{dr} \;=\; 4\pi r^2\rho_\phi(r),$$

which uniquely determines the genuine curved Dirichlet-to-Neumann slope $\phi'_{L,\rm curved}(R_*) = -k_{\rm curved}\phi_L(R_*)$ (for SLy $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$, $k_{\rm curved} = 0.233542\ \mathrm{km}^{-1}$, which is +24.74% steeper than the flat-space Robin approximation $k_{\rm flat} = 1/R_* + 1/L = 0.187216\ \mathrm{km}^{-1}$) and the asymptotic **ADM mass**:

$$M_{\rm ADM} \;\equiv\; \lim_{r\to\infty} m(r) \;=\; M_{\rm surf} \;+\; \Delta M_{\phi,\rm ext}^{\rm curved}, \qquad M_{\rm surf} \;\equiv\; m(R_*).$$

(Note on `Branch II-Fluid` vs. `Branch II-Conf`, `step23` Section 1.1 and Table 2B: outside the star [$r > R_*$, vacuum], both branches obey the exact same 4-component curved exterior Einstein–scalar ODE system above and coincide in the weak-field dust limit [$\tilde{P} \ll \tilde{\varepsilon}$] at $\alpha_Y = 1$; inside a relativistic star [$0 \leq r \leq R_*$], however, `Branch II-Fluid` [$\mathcal{L}_m = -e^{-\phi_L}\varepsilon(n)$] has scalar source $-(\alpha_Y G/c^2)e^{-\phi_L}\varepsilon$ and yields $M_{ADM}^{(II\text{-}Fluid)} = 1.50999814\ M_\odot$, whereas `Branch II-Conf` [$S_m[e^{-2\alpha_Y\phi_L}g_{\mu\nu},\Psi_m]$] has trace source $-(\alpha_Y^2 G/c^2)e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$ [$3\tilde{P}_c/\tilde{\varepsilon}_c = 0.566065$] and yields $M_{ADM}^{(II\text{-}Conf)} = 1.56203491\ M_\odot$ for SLy $\log_{10}\rho_{0c} = 15.0$.)

1.1.2 1.2. Explicit Branch Taxonomy Table

The assumptions, kernel moments, and domain of validity of the six theoretical branches examined in this package are separated in the table below (`step2`, `step5`, `step10`, `step11`, `step21`, `step22`):

Branch	Action Principle / Metric Definition	Kernel Moments (Expansion of $\bar{\rho}_L(\mathbf{x})$)	Interior Hydrostatic Balance	Exterior ($r > R_*$) Vacuum and Mass	Domain of Validity & Steps
Branch I-A (V21 nonlocal 3D top-hat: `tophat_mass`, `tophat_gradient`)	Phenomenological 3D spherical top-hat average + algebraic Bianchi X-sector	$\left(1 + \frac{L^2}{10}\nabla^2 + \mathcal{O}(L^4)\right)\rho$	`mass`: $h' = -\nu'$; `gradient`: $h' = -\nu' = -f_L A_{\rm GR}$	Compact support $[R_*, R_* + L]$; $m(r) = M_{\rm surf}$ constant for $r \geq R_*$	V21 769 models, V23 Eqs. (8)–(9) static tangent BVP (`step7`, `step16`–`step18`, `step21`)
Branch I-B (V23 Formulation I: `bilateral_lapse_weighted`)	Auxiliary-field (ϕ_L, λ_L) bilateral kernel action: $S_{\rm int} = -\int\sqrt{-g}\,e^{-\phi_L}\varepsilon$	$\left(1 + \frac{L^2}{10}\nabla^2 + \mathcal{O}(L^4)\right)\rho$	$h' = -\nu' - (\ln f_{\rm recip})'$ (reciprocal lapse-weighted force)	Compact support $[R_*, R_* + L]$	V23 reciprocal bilateral potential $V_{{\rm eff},N}(r)$ and Synge boundary variation (`step10`, `step13`)

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Branch	Action Principle / Metric Definition	Kernel Moments (Expansion of $\bar{\rho}_L(\mathbf{x})$)	Interior Hydrostatic Balance	Exterior ($r > R_*$) Vacuum and Mass	Domain of Validity & Steps
Branch II-Fluid (V23 Formulation II: `yukawa_action_el` / `yukawa_action_el_curved_adm`)	Local Helmholtz–Yukawa action with fluid-density coupling: $\mathcal{L}_m = -e^{-\phi_L}\varepsilon(n)$	$(1 - L^2\nabla^2)^{-1}\rho = (1 + L^2\nabla^2 + \mathcal{O}(L^4))\rho$	$h'(r) = -\nu'(r) + \phi_L'(r)$ (`sym_yukawa_el_hydrostat` = 0)	Curved Einstein–scalar vacuum BVP in (R_*, ∞); $M_{\rm ADM} = M_{\rm surf} + \Delta M^{\rm curved}_{\phi,\rm ext}$	Curved exterior ADM mass, mass–radius curves, and coupled $(\delta h, \delta\phi_L, \delta m)$ stability (`step3`, `step4`, `step7`, `step12`, `step22`, `step23`)
Branch II-Conf (Universal conformal Jordan branch)	$\tilde{g}^{\rm conf}_{\mu\nu} = e^{-2\alpha_Y\phi_L} g_{\mu\nu}$, $\sqrt{-\tilde{g}^{\rm conf}} = e^{-4\alpha_Y\phi_L}\sqrt{-g}$, $S_m[\tilde{g}^{\rm conf}, \Psi]$, $\rho_m^{(E)} = e^{-4\alpha_Y\phi_L}\tilde{\varepsilon}$, $P_m^{(E)} = e^{-4\alpha_Y\phi_L}\tilde{P}$	Interior trace source $-\alpha_Y^2 e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$; dust limit $(1 - L^2\nabla^2)^{-1}\rho_m$	$h' = -\nu' + \alpha_Y\phi_L'$ (trace-free EM does not source ϕ_L)	Identical 4-component curved Einstein–scalar vacuum BVP ($r > R_*$); conformally invariant null geodesics	Canonical DEF / PPN dictionary ($\alpha_{\rm DEF} = -1/\sqrt{4\pi}$, $\beta_{\rm DEF} = 0$, $\beta_{\rm Yuk} = \frac{\alpha_Y^3}{4\pi} = \frac{1}{4\pi}$, $\omega_{\rm BD} = \frac{2\pi}{\alpha_Y^3} - \frac{3}{2} \approx 4.7832$ [vs. `step5` $\frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.07$]; `step1`, `step2`, `step5`, `step14`, `step23`–`step25`)
Branch II-Disf (Medium-anchored purely temporal disformal branch)	$\tilde{g}^{\rm disf}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$, $\sqrt{-\tilde{g}^{\rm disf}} = f_L\sqrt{-g}$	Top-hat or Yukawa kernel on comoving energy density	Comoving clocks $d\tilde{\tau} = f_L d\tau_g$; u_μ variation yields extra constraint	Magnetic energy ($\rho_B = B^2/(2\mu_0 c^2)$) sources ϕ_L; refractive index $n_t = e^{+\phi_L}$	V20 8-coil magnetic test, V20 granite chamber H1, Sirius B H1, VHE gamma dispersion (`step2`, `step8`, `step9`, `step14`, `step15`, `step19`, `step20`, `step23`–`step25`)
Branch II-Cham (Nonlinear chameleon/symmetron extension)	Hypothetical density-dependent $\beta_{\rm eff}(\rho)$ and self-interaction potential $V_0(\phi_L)$	Density-dependent $L_{\rm eff}(\rho) = [\partial^2 V_{\rm eff}/\partial\phi_L^2]^{-1/2}$	Screened (thin shell) at high density, amplified at low density	Environment-dependent Yukawa range	Separate theoretical extension hypothesis beyond the constant-α_Y V23 action (`step6`, `step11`, `step24`, `step25`)

(Note on the top-hat ↔ Yukawa correspondence: the second-order Taylor moment of the 3D spherical top-hat kernel is $\frac{1}{10}L^2\nabla^2$, whereas that of the Yukawa Green's function is $L^2\nabla^2$; hence choosing $\alpha_Y = 1$ matches the zeroth-order homogeneous [$\nabla^2\rho \to 0$] limit, while matching the gradient-sensitive second moment requires $L_{Yukawa} = L_{top\text{-}hat}/\sqrt{10}$ or rescaling α_Y — see `step10`.)

1.2 2. Step-by-Step Summary of Hungarian Files (`step1`–`step25`, `README.md`, `MANIFEST.txt`) and English Files (`step1_*_en.md`–`step25_*_en.md`, `README_en.md`, `MANIFEST_en.txt`)

1.2.1 2.1. Hungarian Derivation and Observational Audit Chapters (`step1_*.md`–`step25_*.md`)

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
1	`step1_dilaton_form.md`	11 205	`ecd894436341cbdc`	Derives the exact equivalence between the Cauchy-derived exponential factor $f_L = \exp(-\gamma_L \bar{\rho}_L) = e^{-\phi_L}$ and the string-theory dilaton matter coupling $A(\phi) = e^{\beta\phi}$.
2	`step2_conformal_disformal.md`	18 402	`7a3761a4adcca6fd`	Derives the Jacobian, inverse metric, comoving-clock equivalence, and explicit branch taxonomy of the conformal and disformal metric frames.
3	`step3_yukawa_helmholtz_eom.md`	12 647	`77fa5ad7646bad0b`	Helmholtz–Yukawa-egyenletet (`yukawa_action_el`), a Yukawa Green-függvényt és a lapos téri Robin-közelítést. / Variationally derives the curved-spacetime massive Klein–Gordon / Helmholtz–Yukawa field equation, Green function, and flat-space Robin approximation at $r = R_*$.
4	`step4_covariant_conservation_hydrostatic.md`	12 356	`52b1cab653da99d6`	Proves exact covariant stress-energy conservation $\nabla_\mu(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_{(\phi)}) = 0$ and the modified hydrostatic Euler–TOV balance $h' = -\nu' + \phi'_L$.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
5	step5_coupling_dictionary_ppn_screening.md	14 949	d9c9d69ff93e6625	Establishes the exact $-\frac{c^4}{2G\alpha_Y}$ normalized coupling dictionary ($\alpha_{\rm std} = 1/(4\pi)$, $\omega_{\rm BD} \approx 11.07$) and separates conformal vs. disformal PPN behavior.
6	step6_sparc_galactic_ngc3198_consistency.md	14 615	8fbdd2157d0a4aa9	4,311 igényét az izolált konstans csatolású modell $C_{\rm pred}^{\rm isol} \leq 1{,}0796$ ($V_{\rm pred}^{\rm isol} =$ 88,53–91,99 km/s) előrejelzésével. / Verifies 0 negative-density points on 175 SPARC galaxies and contrasts NGC 3198's required $C_{\rm req} = 2.720$ against the isolated constant-coupling prediction $C_{\rm pred}^{\rm isol} \leq 1.0796$ as a negative control.
7	step7_neutron_star_tov_stability.md	21 215	d656ac9249a495ae	Derives the curved-spacetime TOV equations, compares flat-Robin surface mass vs. curved-exterior ADM mass, and separates Branch I-A vs. Branch II-Fluid stability.
8	step8_microscopic_clock_coupling_sirius_b.md	16 306	f4d7f20f47b5ab8d4	Derives H0 vs. conditional H1 clock coupling from the Dirac–Maxwell action and quantifies Sirius B redshift exclusions, the 8558 kg granite chamber ($\sim 10^{-24}$), and the 8-coil solenoid ($\sim 10^{-48}$).

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
9	`step9_vhe_photon_propagation_hess_dispersion.md`	12 164	`ad5b8d0ab66ff02c`	Derives the achromatic refractive index $n_t(s) = f_L(s)^{-1}$ in the disformal metric and the multi-source H.E.S.S./MAGIC VHE gamma-ray timing audit.
10	`step10_bilateral_kernel_reciprocity_sym.md`	13 420	`3cfe4aaa90e66f3f`	Derives Formulation I's auxiliary-field bilateral kernel action, reciprocal potential $V_{\mathrm{eff},N}(r)$, and zeroth- vs. second-moment correspondence to Formulation II.
11	`step11_chameleon_screening_multischeft.md`	13 297	`69a908a620c75626`	Formulates environment-dependent chameleon/symmetron screening (`Branch II-Cham`) as an explicit nonlinear extension hypothesis beyond the constant-α_Y V23 action.
12	`step12_second_variation_coupled_pulsation.md`	19 550	`35e161b76cd419e0a`	Explicitly derives $\mathcal{Q}_{\xi\xi}, \mathcal{Q}_{\phi\phi}, \mathcal{C}_{\xi\phi}, \mathcal{C}_{\phi\xi}$ after metric constraint elimination, moving-surface matching, and numerical stability on the curved `yukawa_action_el` sequence.
13	`step13_curved_window_metric_variation_synge.md`	15 595	`261d6ad0f89cf4ad`	Derives the exact metric variation $\delta[\sqrt{h}K_L]/\delta g^{\mu\nu}$ of the curved-spacetime top-hat window via Synge's world function and the $\delta_D(L - s_{\mathrm{geo}})$ surface stress.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
14	`step14_gravitational_lensing_null_geodesics.md`	16 324	d37321ab8dff	Derives conformal vs. disformal light deflection, FLRW cancellation in $\nabla \ln f_L$, and exact horizon scales for both $\Omega_b = 0.0493$ (57.98 Gpc) and $\Omega_m = 0.315$ (22.94 Gpc).
15	`step15_electromagnetic_trace_anomaly_disformal.md`	13 340	75382	Proves $T^{\mu}_{\mu,\mathrm{EM}} = 0$ while $T^{\mu}_{\mu,\mathrm{EM}} = 0$ suppresses conformal trace coupling, disformal coupling $u_\mu u_\nu T^{\mu\nu}_{\mathrm{EM}}$ sources ϕ_L via $\rho_B = \lvert\mathbf{B}\rvert^2/(2\mu_0 c^2)$ with exact polarity-even cross-term cancellation.
16	`step16_anisotropic_x_sector_bianchi.md`	14 176	cbafe291bab7	Derives the V21/V23 algebraic X-sector anisotropic stress reconstruction from the Bianchi identity ($\max \lvert p_{tX}/P_c \rvert = 0.140$) and compares it with the dynamical Yukawa stress tensor.
17	`step17_chandrasekhar_factored_difference_identity.md`	14 383	fe81adf9f49c	Analytically derives the V23 Eq. (8) factorized Chandrasekhar–hydrostatic difference identity and proves the non-positivity of the `gradient`-mode `metric` inertia weight for $L \geq 367$ m.
18	`step18_nonlocal_tangent_bvp_shell_kernel.md`	21 179	44dee1493f22	Derives the V23 Eq. (9) nonlocal static tangent BVP on the V21 background, the 3D spherical-shell kernel K_{ij}, and the exact inertia-scaling invariance $\omega^2_{4W}/\omega^2_W = 0.2500$.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
19	`step19_in_out_differential_clock_quadrature_geometry_and_vs_sirius_3D_common_pro`	18 882	bf6eea32fe1a730d0…	Dual vs. 3D IN–OUT differential quadrature of the 8558 kg granite chamber across three geometric L-regimes and the Sirius B–Sirius A `common_background` vs. `fixedM` redshift proxy.
20	`step20_eight_coil_elliptic_biot_savart_and_compactified_quadrature.md`	17 366	56eb853ee017a0d3…	Eight-coil elliptic Biot–Savart field, peak signal $\Delta y(L_p) = -1.2017 \times 10^{-48}$, $r = L/t$ compactified exterior quadrature (7.44×10^{-16} agreement), and the $L = 0.95$ m switching identity.
21	`step21_enthalpy_tov_even_parity_s_r2_regularized_halo_lapse.md`	12 431	29eb46950de70ca57…	Dual vs. 7-segment piecewise-polytrope log-enthalpy TOV transformation, $s = r^2$ even-parity PCHIP regularization, exterior vacuum-halo lapse $\Delta\nu_{\text{ext}}$, and contrasts V21 $h' = -\nu'$ against Yukawa $h' = -\nu' + \phi'_L$.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
22	`step22_curved_exterior_yukawa_adm_mass_mr_fd79d42b81e7b7201bi...`	35 778	f679d42b81e7b720	V25. Solves the 4-component curved exterior Einstein–scalar BVP ($\Delta k_{\rm curved}/k_{\rm flat} = +24.74\%$), recomputes the $M_{\rm ADM}$–R_* sequences ($M^{\rm SLy}_{\rm ADM,max} = 2.03321\,M_\odot$, $R^{\rm ADM}_{1.4} = 11.5359$ km), and documents the coupled $(\delta h, \delta\phi_L, \delta m)$ GL2 stability diagnostics and conditional closing statement.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
23	`step23_gw170817_and_lensing.md`	54 254	`c40714093b83206e`	Extended observational test ($L = 10$ km, $\alpha_Y = 1$): Proves via Horndeski/DHOST mapping that $G_4 = M_{\rm Pl}^2/2$, $G_5 = 0 \implies c_T = c$ identically (GW170817 / GRB 170817A), the scalar mode is evanescent below $f_{\rm cut} = 4771.35$ Hz ($\lambda_{\rm ev} =$ 10.00–10.08 km, $A_{\rm scalar}(40\ {\rm Mpc}) \sim 10^{-5.36\times10^{19}}$), and evaluates curved-spacetime light deflection and the Cassini PPN slip ($|1 - \gamma_{\rm PPN}(1.6\,R_\odot)| = 10^{-48343}$); $c_T = c$ is proven on the purely conformal, minimal Einstein–Hilbert + canonical scalar + universal Maxwell coupling branch ($G_4 = M_{\rm Pl}^2/2$, $G_{4,X} = 0$, $G_5 = 0$, $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$, $\beta_{\rm Yuk} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795775$); this does not by itself prove $c_T = c$ on a disformal/refractive branch ($D(\phi_L, X) \neq 0$); both the Step 22 1.50999814 $M_\odot$ `Branch II-Fluid` diagnostic SLy neutron-star background (`NS_far_field_b_*`, Table 2, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) =$ 1.290852 rad) and the self-consistent 1.56203491 $M_\odot$ `Branch II-Conf` trace-sourced ($\varepsilon = 3P$) BVP solution (Table 2B, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) =$

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
24	`step24_bullet_cluster.md`	23 334	`571036611966f5e1`	Extended observational test ($L = 10$ km, $\alpha_Y = 1$): Demonstrates on the 4-component baryonic model of the Bullet Cluster (`1E 0657-558`, $z_L = 0.296$) that at $R_c/L \sim (2.0\text{–}4.6) \times 10^{17}$ the local Yukawa filter yields $\kappa_{\rm local} = \kappa_{\rm GR,bary}$ ($\lvert\Delta\kappa_{\rm disf}/\kappa_{\rm bary}\rvert \leq 2.17 \times 10^{-36}$), leaving the peaks on the X-ray gas (-175.44 kpc, $+137.01$ kpc vs. galaxies at -350 kpc, $+300$ kpc), and derives the collisionless clock-weighting threshold ($\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$model-dependent threshold of the chosen 1D King geometry and $L_{\rm macro} = 300$ kpc; $\eta_{\rm gas} = 0$ and the two separately fitted galactic weights ($\eta_{\rm gal,main}, \eta_{\rm gal,sub}$) are phenomenological assumptions not derived from the covariant action); labeled as a synthetic 1D illustrative test against the self-constructed 1D King-profile target curve `kappa_synth_target_1D` (not an observational χ^2 fit to Clowe/Bradač data, which would require a published 2D convergence map, covariance matrix, pre-registered selection, and 5 degree-of-freedom subtraction).

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
25	`step25_cmb_and_cosmology.md`	37 301	`8a58496edc72d408`	Extended observational test ($L = 10$ km, $\alpha_Y = 1$): Derives the ultra-stiff quasi-static FLRW attractor ($m_\phi c/H_0 = 1.3725 \times 10^{22}$), the conditional separate-branch hypothesis for the V25 negative screening density $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 ($\rho_{\rm clock,cosmo} = +1.192 \times 10^{-26}$ kg/m^3), CMB $G_{\rm eff}(a,k)$ and ISW suppression ($(G_{\rm eff} - G)/G = 2.50 \times 10^{-36}$ at $\ell_1 = 220$), the external Planck input ratio $\Omega_c h^2/\Omega_b h^2 = 5.3643$ (conditional separate branch; not a TAL result), and the 6-scale grand synthesis table; Only the FLRW background quantities ($H(z)$, q_0, $a_{\max}$, $r_s(z_*)$, $D_M(z_*)$, $D_A(z_*)$, $100\,\theta_*$, ℓ_A) are numerically integrated; ℓ_1, ℓ_2, ℓ_3 are fixed analytic phase-shift template estimates, and the peak-height ratios ($H_2/H_1 = 0.458$, $H_3/H_2 = 0.985$ for $\Omega_c > 0$ vs. 0.312, 0.415 for $\Omega_c = 0$) are fixed external literature reference inputs (no Einstein–Boltzmann perturbation solver was run). The ratio $5.364327 = 0.1200/0.02237$ is the input Planck 2018 CDM/baryon density ratio, not a prediction computed from the TAL clock sector;

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1.2.2 2.2. English Translation Chapters (`step1_*_en.md`–`step25_*_en.md`, `README_en.md`, `MANIFEST_en.txt`)

Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
1 (EN)	`step1_dilaton_form_en.md`	10 890	`83a2e8e888083739`	Derives the exact equivalence between the Cauchy-derived exponential factor $f_L = \exp(-\gamma_L \bar{\rho}_L) = e^{-\phi_L}$ and the string-theory dilaton matter coupling $A(\phi) = e^{\beta\phi}$.
2 (EN)	`step2_conformal_disformal_en.md`	18 014	`abd9db8d104e3bcf`	Derives the Jacobian, inverse metric, comoving-clock equivalence, and explicit branch taxonomy of the conformal and disformal metric frames.
3 (EN)	`step3_yukawa_helmholtz_eom_en.md`	12 258	`8ca1db6f110e2617`	Helmholtz–Yukawa-egyenletet (`yukawa_action_el`), a Yukawa Green-függvényt és a lapos téri Robin-közelítést. / Variationally derives the curved-spacetime massive Klein–Gordon / Helmholtz–Yukawa field equation, Green function, and flat-space Robin approximation at $r = R_*$.
4 (EN)	`step4_covariant_conservation_hydrostatic_en.md`	11 986	`4d1aba07182e7cfa`	Proves exact covariant stress-energy conservation $\nabla_\mu(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_{(\phi)}) = 0$ and the modified hydrostatic Euler–TOV balance $h' = -\nu' + \phi'_L$.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
5 (EN)	`step5_coupling_dictionary_ppn_screening_en.md`	14 701	e57d506cdd3ba02b	Establishes the exact $-\frac{c^4}{2G\alpha_Y}$ normalized coupling dictionary ($\alpha_{\rm std} = 1/(4\pi)$, $\omega_{\rm BD} \approx 11.07$) and separates conformal vs. disformal PPN behavior.
6 (EN)	`step6_sparc_galactic_ngc3198_consistency_en.md`	13 870	6f75d1f0ccba91de	4,311 igényét az izolált konstans csatolású modell $C_{\rm pred}^{\rm isol} \leq 1{,}0796$ ($V_{\rm pred}^{\rm isol} =$ 88,53–91,99 km/s) előrejelzésével. / Verifies 0 negative-density points on 175 SPARC galaxies and contrasts NGC 3198's required $C_{\rm req} = 2.720$ against the isolated constant-coupling prediction $C_{\rm pred}^{\rm isol} \leq 1.0796$ as a negative control.
7 (EN)	`step7_neutron_star_tov_stability_en.md`	20 516	e219a8cfaf4c81a2	Derives the curved-spacetime TOV equations, compares flat-Robin surface mass vs. curved-exterior ADM mass, and separates Branch I-A vs. Branch II-Fluid stability.
8 (EN)	`step8_microscopic_clock_coupling_sirius_en.md`	15 629	1dcaf38be27692ea	Derives H0 vs. conditional H1 clock coupling from the Dirac–Maxwell action and quantifies Sirius B redshift exclusions, the 8558 kg granite chamber ($\sim 10^{-24}$), and the 8-coil solenoid ($\sim 10^{-48}$).

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
9 (EN)	`step9_vhe_photon_propagation_hess_dispersion_def17584b`	11 454	dab6f01def17584b	Derives the achromatic refractive index $n_t(s) = f_L(s)^{-1}$ in the disformal metric and the multi-source H.E.S.S./MAGIC VHE gamma-ray timing audit.
10 (EN)	`step10_bilateral_kernel_reciprocity_symmetry_2a41d1e1`	12 890	74a6e20c625a41d1e1	Derives Formulation I's auxiliary-field bilateral kernel action, reciprocal potential $V_{\mathrm{eff},N}(r)$, and zeroth- vs. second-moment correspondence to Formulation II.
11 (EN)	`step11_chameleon_screening_multischeme_3de84ad7`	12 508	375040d0e83de84ad7	Formulates environment-dependent chameleon/symmetron screening (`Branch II-Cham`) as an explicit nonlinear extension hypothesis beyond the constant-α_Y V23 action.
12 (EN)	`step12_second_variation_coupled_pulsation_a32075f3a7`	19 410	798f21ac32075f3a7	Explicitly derives $\mathcal{Q}_{\xi\xi}, \mathcal{Q}_{\phi\phi}, \mathcal{C}_{\xi\phi}, \mathcal{C}_{\phi\xi}$ after metric constraint elimination, moving-surface matching, and numerical stability on the curved `yukawa_action_el` sequence.
13 (EN)	`step13_curved_window_metric_variation_emd`	23	4bb76e505	Derives the exact metric variation $\delta[\sqrt{h}K_L]/\delta g^{\mu\nu}$ of the curved-spacetime top-hat window via Synge's world function and the $\delta_D(L - s_{\mathrm{geo}})$ surface stress.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
14 (EN)	`step14_gravitational_lensing_null_geodesics_en.md`	15 515	48051ef7f1852a025ad	Derives conformal vs. disformal light deflection, FLRW cancellation in $\nabla \ln f_L$, and exact horizon scales for both $\Omega_b = 0.0493$ (57.98 Gpc) and $\Omega_m = 0.315$ (22.94 Gpc).
15 (EN)	`step15_electromagnetic_trace_anomaly_disformal_en.md`	12 947	8b537673ab66d4	Proves while $T^{\mu}_{\mu,\mathrm{EM}} = 0$ suppresses conformal trace coupling, disformal coupling $u_\mu u_\nu T^{\mu\nu}_{\mathrm{EM}}$ sources ϕ_L via $\rho_B = \lvert\mathbf{B}\rvert^2/(2\mu_0 c^2)$ with exact polarity-even cross-term cancellation.
16 (EN)	`step16_anisotropic_x_sector_bianchi_en.md`	13 580	f3886101a66459	Derives the V21/V23 algebraic X-sector anisotropic stress reconstruction from the Bianchi identity ($\max \lvert p_{tX}/P_c \rvert = 0.140$) and compares it with the dynamical Yukawa stress tensor.
17 (EN)	`step17_chandrasekhar_factored_difference_en.md`	31 765	672da1dfe28d46	Analytically derives the V23 Eq. (8) factorized Chandrasekhar–hydrostatic difference identity and proves the non-positivity of the `gradient`-mode `metric` inertia weight for $L \geq 367$ m.
18 (EN)	`step18_nonlocal_tangent_bvp_shell_kernel_en.md`	20 832	7267ec0072e983	Derives the V23 Eq. (9) nonlocal static tangent BVP on the V21 background, the 3D spherical-shell kernel K_{ij}, and the exact inertia-scaling invariance $\omega^2_{4W}/\omega^2_W = 0.2500$.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
19 (EN)	`step19_in_out_differential_clock_quadrature_earth_chamber_and_sirius_3D_common_pro`	18 067	9fab3449db6858bb	Dual-vessel IN–OUT differential quadrature of the 8558 kg granite chamber across three geometric L-regimes and the Sirius B–Sirius A `common_background` vs. `fixedM` redshift proxy.
20 (EN)	`step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md`	16 820	804ff01b69664079	Eight-coil elliptic Biot–Savart field, peak signal $\Delta y(L_p) = -1.2017 \times 10^{-48}$, $r = L/t$ compactified exterior quadrature (7.44×10^{-16} agreement), and the $L = 0.95$ m switching identity.
21 (EN)	`step21_enthalpy_tov_even_parity_s_r2_regularization_halo_lapse_en.md`	17 542	1a06b9ea97d195f4	Dual-vessel 7-segment piecewise-polytrope log-enthalpy TOV transformation, $s = r^2$ even-parity PCHIP regularization, exterior vacuum-halo lapse $\Delta\nu_{\text{ext}}$, and contrasts V21 $h' = -\nu'$ against Yukawa $h' = -\nu' + \phi'_L$.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
22 (EN)	`step22_curved_exterior_yukawa_adm_mass_mr_bond_stability_en.md`	36 896	2d1c8044e2fcba	V25: Solves the 4-component curved exterior Einstein–scalar BVP ($\Delta k_{\rm curved}/k_{\rm flat} = +24.74\%$), recomputes the $M_{\rm ADM}$–R_* sequences ($M^{\rm SLy}_{\rm ADM,max} = 2.03321\,M_\odot$, $R^{\rm ADM}_{1.4} = 11.5359$ km), and documents the coupled $(\delta h, \delta\phi_L, \delta m)$ GL2 stability diagnostics and conditional closing statement.

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
23 (EN)	`step23_gw170817_and_lensing_en.md`	52 257	`3f723eb9f5c06d01`	Extended observational test ($L = 10$ km, $\alpha_Y = 1$): Proves via Horndeski/DHOST mapping that $G_4 = M_{\rm Pl}^2/2$, $G_5 = 0 \implies c_T = c$ identically (GW170817 / GRB 170817A), the scalar mode is evanescent below $f_{\rm cut} = 4771.35$ Hz ($\lambda_{\rm ev} =$ 10.00–10.08 km, $A_{\rm scalar}(40\ {\rm Mpc}) \sim 10^{-5.36\times10^{19}}$), and evaluates curved-spacetime light deflection and the Cassini PPN slip ($|1 - \gamma_{\rm PPN}(1.6\,R_\odot)| = 10^{-48343}$); $c_T = c$ is proven on the purely conformal, minimal Einstein–Hilbert + canonical scalar + universal Maxwell coupling branch ($G_4 = M_{\rm Pl}^2/2$, $G_{4,X} = 0$, $G_5 = 0$, $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$, $\beta_{\rm Yuk} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795775$); this does not by itself prove $c_T = c$ on a disformal/refractive branch ($D(\phi_L, X) \neq 0$); both the Step 22 1.50999814 $M_\odot$ `Branch II-Fluid` diagnostic SLy neutron-star background (`NS_far_field_b_*`, Table 2, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) =$ 1.290852 rad) and the self-consistent 1.56203491 $M_\odot$ `Branch II-Conf` trace-sourced ($\varepsilon = 3P$) BVP solution (Table 2B, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) =$

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
24 (EN)	`step24_bullet_cluster_en.md`	21 841	`0c57b832dd1be029`	Extended observational test ($L = 10$ km, $\alpha_Y = 1$): Demonstrates on the 4-component baryonic model of the Bullet Cluster (`1E 0657-558`, $z_L = 0.296$) that at $R_c/L \sim (2.0\text{–}4.6) \times 10^{17}$ the local Yukawa filter yields $\kappa_{\rm local} = \kappa_{\rm GR,bary}$ ($\lvert\Delta\kappa_{\rm disf}/\kappa_{\rm bary}\rvert \leq 2.17 \times 10^{-36}$), leaving the peaks on the X-ray gas (-175.44 kpc, $+137.01$ kpc vs. galaxies at -350 kpc, $+300$ kpc), and derives the collisionless clock-weighting threshold ($\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$model-dependent threshold of the chosen 1D King geometry and $L_{\rm macro} = 300$ kpc; $\eta_{\rm gas} = 0$ and the two separately fitted galactic weights ($\eta_{\rm gal,main}, \eta_{\rm gal,sub}$) are phenomenological assumptions not derived from the covariant action); labeled as a synthetic 1D illustrative test against the self-constructed 1D King-profile target curve `kappa_synth_target_1D` (not an observational χ^2 fit to Clowe/Bradač data, which would require a published 2D convergence map, covariance matrix, pre-registered selection, and degree-of-freedom subtraction).

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
25 (EN)	`step25_cmb_and_cosmology_en.md`	36 001	`eb4e4cfcfa03890d`	Extended observational test ($L = 10$ km, $\alpha_Y = 1$): Derives the ultra-stiff quasi-static FLRW attractor ($m_\phi c/H_0 = 1.3725 \times 10^{22}$), the conditional separate-branch hypothesis for the V25 negative screening density $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 ($\rho_{\rm clock,cosmo} = +1.192 \times 10^{-26}$ kg/m^3), CMB $G_{\rm eff}(a,k)$ and ISW suppression ($(G_{\rm eff} - G)/G = 2.50 \times 10^{-36}$ at $\ell_1 = 220$), the external Planck input ratio $\Omega_c h^2/\Omega_b h^2 = 5.3643$ (conditional separate branch; not a TAL result), and the 6-scale grand synthesis table; Only the FLRW background quantities ($H(z)$, q_0, $a_{\max}$, $r_s(z_*)$, $D_M(z_*)$, $D_A(z_*)$, $100\,\theta_*$, ℓ_A) are numerically integrated; ℓ_1, ℓ_2, ℓ_3 are fixed analytic phase-shift template estimates, and the peak-height ratios ($H_2/H_1 = 0.458$, $H_3/H_2 = 0.985$ for $\Omega_c > 0$ vs. 0.312, 0.415 for $\Omega_c = 0$) are fixed external literature reference inputs (no Einstein–Boltzmann perturbation solver was run). The ratio $5.364327 = 0.1200/0.02237$ is the input Planck 2018 CDM/baryon density ratio, not a prediction computed from the TAL clock sector;

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Step	File Name	Size (B)	SHA-256 (prefix)	One-Line Gist
README (EN)	`README_en.md`	97 492	—	Full English overview, equations, branch taxonomy, V25 & Steps 23–25 numerical summary, and reproduction instructions.
MANIFEST (EN)	`MANIFEST_en.txt`	45 077	—	Full English manifest listing all tracked files (including the 25 Hungarian and 25 English chapters) with exact SHA-256 checksums and byte sizes.

1.3 3. Summary of Key Numerical Results

Physical Domain	Quantity	Exact Numerical Value	Source Step & Script
Scalar–Tensor / Yukawa Dictionary (`Branch II-Conf`)	Canonical DEF derivatives $(\alpha_{\rm DEF}, \beta_{\rm DEF})$, Yukawa strength $(\beta_{\rm Yuk})$, and Brans–Dicke $\omega_{\rm BD}$	$\alpha_{\rm DEF} \equiv d\ln A/d\phi_{\rm canon} = -1/\sqrt{4\pi} \approx -0.2820948$, $\beta_{\rm DEF} \equiv d^2\ln A/d\phi^2_{\rm canon} = 0$; $\beta_{\rm Yuk} \equiv \alpha^2_{\rm DEF} = 2\tilde{\alpha}_L\alpha_Y^2 = \frac{\alpha_Y^3}{4\pi} = \frac{1}{4\pi} \approx 0.0795775$; V26 canonical $\omega_{\rm BD} = \frac{1}{2\beta_{\rm Yuk}} - \frac{3}{2} = \frac{2\pi}{\alpha_Y^3} - \frac{3}{2} \approx 4.783185$ $(\alpha_Y = 1)$, whereas in the `step5` 4π-convention $(\alpha_{\rm std} = \frac{\alpha_Y}{4\pi})$ $\omega_{\rm BD}^{(\rm step5)} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.066371$ $(\alpha_Y = 1)$ or 36.1991 $(\alpha_Y = 1/3)$	`step5`, `step23`, `covariant_closure_50aa2db0.py`, `gw170817_and_lensing_audit.py`
SPARC 175 Galaxies ($\rho < 0$ positivity test)	Number of negative-density points (across 3391 radial points)	**0 points** (positivity check of photometric + 3D top-hat + 3D Yukawa kernels) vs. **239 points** (140 non-holdout galaxies, $\Upsilon_d = 0.5$) / **301–303 points** (175 galaxies, spherical derivative formula)	`step6`, `SPARC_mode.py`, `sparc_tophat.py`

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Physical Domain	Quantity	Exact Numerical Value	Source Step & Script
NGC 3198 Negative Control ($R = 30.08$ kpc)	Observed $C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2$ vs. isolated constant-coupling $C_{\rm pred}^{\rm isol}$	$V_{\rm obs} = 146 \pm 2$ km/s; $\Upsilon_d = 1.0$: $V_{\rm bar} = 88.53$ km/s, $C_{\mathbf{req}} = 2.720$, whereas for the isolated constant-coupling model $C_{\mathbf{pred}}^{\mathbf{isol}} \leq 1 + \alpha_{\mathbf{std}} = 1.0796$ ($V_{\rm pred}^{\rm isol} = 88.53$–$91.99$ km/s; $\Upsilon_d = 0.5$: $V_{\rm bar} = 70.32$ km/s, $C_{\rm req} = 4.311$, $V_{\rm pred}^{\rm isol} = 70.32$–$73.06$ km/s)	`step6`, `SPARC_mode.py`, `data/NGC3198_table.csv`
Cosmological Horizon Scales (Ω_b vs. Ω_m)	$L_{\rm hor} = \frac{c}{\sqrt{G\rho_{\rm bg}}} = \sqrt{\frac{8\pi}{3\Omega}}\,\frac{c}{H_0}$ ($H_0 = 67.4$ km/s/Mpc)	**Baryonic background ($\Omega_b = 0.0493$, $\rho_{b,0} = 4.2067 \times 10^{-28}$ kg/m^3):** $\sqrt{8\pi/(3\Omega_b)} = 13.031 \implies L_{\rm hor}^{(b)} = 1.7892 \times 10^{27}$ m $\approx$ **57.98** Gpc;**Matter background ($\Omega_m = 0.315$, $\rho_{m,0} = 2.686 \times 10^{-27}$ kg/m^3):** $\sqrt{8\pi/(3\Omega_m)} = 5.160 \implies L_{\rm hor}^{(m)} = 7.081 \times 10^{26}$ m $\approx$ **22.94** Gpc	`step14`, `sparc_tophat.py`
Bianchi / Noether Conservation	$\max_r \lvert r \cdot \nabla_\mu (T^\mu_{r,\rm matter} + T^\mu_{r,X}) \rvert / \varepsilon_c$ ($s = r^2$ PCHIP)	Symbolic: `0.0`; continuum: 1.82×10^{-11} (`tophat_mass`), 1.59×10^{-11} (`tophat_gradient`), 3.65×10^{-10} (`yukawa_action_el`)	`step4`, `step16`, `step21`, `covariant_closure_50aa2db0.py`
Curved Exterior Yukawa BVP and ADM Mass (`Branch II-Fluid`)	$R_{\rm max}$ convergence ($[R_*, R_* + N_{\rm mult}L]$, $N_{\rm mult} \in \{10, 12, 14, 16\}$), $k_{\rm curved}$ vs. $k_{\rm flat}$, and $M_{\rm ADM,max}$	$R_{\rm max}$ spread: $\Delta k_{\rm curved}/k_{\rm curved} = 6.17 \times 10^{-12}$, $\Delta M_{\rm ADM}/M_{\rm ADM} = 5.54 \times 10^{-13}$; SLy 15.0, $L = 10$ km, $\alpha_Y = 1$: $k_{\rm curved} = 0.233542$ km^{-1} vs. $k_{\rm flat} = 0.187216$ km^{-1} (+24.74%); $M_{\rm ADM,max}^{\rm SLy} = \mathbf{2.03321}\,\mathbf{M}_\odot$ ($M_{\rm surf,max} = 2.02911\,M_\odot$, $M_{B,\rm max}^{\rm quad} = \mathbf{2.42339}\,\mathbf{M}_\odot$, $\Delta M_{\phi,\rm ext}^{\rm curved} = 0.00411\,M_\odot$), $R_{1.4}^{\rm ADM} = \mathbf{11.5359}$ km ($M_{\rm ADM,max}^{\rm APR4} = 2.18068\,M_\odot$, $M_{\rm ADM,max}^{\rm H4} = 1.97809\,M_\odot$)	`step7`, `step22`, `curved_yukawa_adm_and_stability`

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Physical Domain	Quantity	Exact Numerical Value	Source Step & Script
Coupled $(\delta h, \delta\phi_L, \delta m)$ **Stability on the Curved** `yukawa_action_el` **Branch**	$\omega_0^2 = 0$ zero crossing vs. $\frac{dM_{\rm ADM}}{d\rho_c} = 0$ and $\frac{dM_B}{d\rho_c} = 0$ (`scalar_m_sign = -1.0`)	SLy $L = 10$ km, $\alpha_Y = 1$: $\log_{10}\rho_{0c}(M_{\rm ADM,max}) = 15.316404$, $\log_{10}\rho_{0c}(M_{B,\rm max}^{\rm quad}) = 15.316409$, coupled $\omega_0^2 = 0$: 15.316410 ($+\mathbf{5.80\times10^{-6}}$ dex, 2097× reduction over frozen-ϕ_L shift $+1.22\times10^{-2}$ dex); $\alpha_Y = 1/3$: 15.307170 ($+4.20\times10^{-6}$ **dex**, 995× reduction); $L = 1$ km: 15.303080 (-6.86×10^{-6} **dex**, 165× reduction); APR4 ($L = 10$ km): 15.293025 ($+3.57\times10^{-6}$ **dex**, 2392× reduction); H4 ($L = 10$ km): 15.227783 (-1.34×10^{-6} **dex**, 17593× reduction)	`step12`, `step22`, `curved_yukawa_adm_and_stability`
V21 Nonlocal 3D Top-Hat Tangent BVP (`Branch I-A`)	SLy $L = 10$ km `mass` / `gradient` $\omega_0^2 = 0$ vs. $\partial M/\partial\rho_c = 0$ and $W \to 4W$ scaling	In `mass` mode: 15.29931 vs. 15.29924 ($+6.97\times10^{-5}$ dex, 58.3× improvement); in `gradient` mode: 15.30113 vs. 15.30106 ($+7.39\times10^{-5}$ dex, 59.3× improvement); inertia scaling: $\omega_{4W}^2/\omega_W^2 = 0.2500$	`step7`, `step17`, `step18`, `covariant_closure_50aa2db0.py`
Sirius B H0 vs. H1 Redshift	$v_{\rm obs} = 80.65 \pm 0.77$ km/s ($M = 1.018\,M_\odot$, $R = 5846.6$ km)	H0: $\lvert\Delta v\rvert \le 1.27$ m/s (unconstrained); conditional H1 2σ ($\Delta\chi^2 = 4$) excludes: $L \in [4.08\times10^6, 3.37\times10^7]$ m ($n = 1.5$, max $\Delta\chi^2 = 52.3$ / 72.0)	`step8`, `step19`, `covariant_closure_50aa2db0.py`
Granite Chamber (8558 kg) IN–OUT Test	Conditional H1 differential clock signal $\Delta y_{\rm H1}$(IN − OUT)	At $L = 1.00$ m: Top-hat -1.436×10^{-24} ($\equiv 0$ for $L \ge 2.944$ m); Yukawa -3.781×10^{-25} (-4.818×10^{-25} at $L = 10$ m)	`step8`, `step19`, `covariant_closure_50aa2db0.py`
8-Coil Solenoid ($N = 2000$, 14 mA)	Magnetic peak signal ($L_p = 0.630957$ m), $r = L/t$ quadrature, and $L = 0.95$ m	$\Delta W(L_p) = 1.96114452$ J/A^2, $k_g = -6.13117767\times10^{-45}$ A^{-2}, $\Delta y(L_p) = -1.2017108\times10^{-48} \sim -10^{-48}$; Path A vs. B ($L = 3$ m): 7.44×10^{-16}; $L = 0.95$ m identity: 8.88×10^{-16} J/A^2	`step15`, `step20`, `make_revision3_checks.py`

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Physical Domain	Quantity	Exact Numerical Value	Source Step & Script
GW170817 Tensor Speed c_T and Curved-Spacetime Lensing (`step23`)	Horndeski/DHOST c_T/c, scalar cutoff $f_{\rm cut}$, Cassini $\gamma_{\rm PPN}$, and NS grazing deflection $\hat{\alpha}(b)$	$G_4 = M_{\rm Pl}^2/2$, $G_5 = 0 \implies c_T = c$ identically ($\lvert c_T/c - 1\rvert = 0 \le 3 \times 10^{-15}$; $c_T = c$ is proven on the purely conformal, minimal Einstein–Hilbert + canonical scalar + universal Maxwell coupling branch ($G_4 = M_{\rm Pl}^2/2$, $G_{4,X} = 0$, $G_5 = 0$, $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$, $\beta_{\rm Yuk} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795775$); this does not by itself prove $c_T = c$ on a disformal/refractive branch ($D(\phi_L, X) \neq 0$)); $f_{\rm cut} = c/(2\pi L) = 4771.35$ Hz (scalar mode evanescent in LIGO band $[24, 600]$ Hz with $\lambda_{\rm ev} = 10.00$–10.08 km, $A_{\rm scalar}(40\ {\rm Mpc}) \sim 10^{-5.36 \times 10^{19}}$); Cassini: both the Step 22 1.50999814 $M_\odot$ `Branch II-Fluid` diagnostic SLy neutron-star background (`NS_far_field_b_*`, Table 2, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) = 1.290852$ rad) and the self-consistent 1.56203491 $M_\odot$ `Branch II-Conf` trace-sourced ($\tilde{\varepsilon} - 3\tilde{P}$) BVP solution (Table 2B, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) = 1.334024$ rad) are evaluated and kept separate from the dedicated 1.000000 $M_\odot$ Solar–Cassini calculation ($b = R_\odot$: $\hat{\alpha}_\odot = 1.751201''$; $b_{\rm Cassini} = 1.6\, R_\odot = 1.11312 \times 10^6$ km: $\hat{\alpha}_\odot = 1.094498''$, round-trip Earth–Cassini Shapiro delay 262.4234 μs / 242.7215 μs with geometric -1, $\log_{10}\lvert\gamma_{{\rm PPN},\odot} - 1\rvert = -48\,342.99 \ll 2.3 \times 10^{-5}$); NS grazing ray on the Step 22 `Branch II-Fluid` diagnostic background ($M_{\rm ADM} = 1.50999814\ M_\odot$, $b = 14.749$ km): $\hat{\alpha}_{\rm Yuk,conf} = 1.290852$ rad ($+0.3652\%$ vs. $M_{\rm surf}$, -0.0287% vs. $M_{\rm ADM}$; asymptotic $+0.1550\%$ for $b \ge 200$ km), and on the self-consistent `Branch II-Conf` trace-sourced ($\tilde{\varepsilon} - 3\tilde{P}$) BVP ($R_* = 11.6627$ km, $M_{\rm surf} = 1.560636\ M_\odot$, $M_{\rm ADM} = 1.562035\ M_\odot$,	`step23`, `gw170817_and_lensing_audit.py`

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Physical Domain	Quantity	Exact Numerical Value	Source Step & Script
Bullet Cluster (1E 0657-558, $z_L = 0.296$) Lensing Convergence κ (step24)	Local $L = 10$ km, $\alpha_Y = 1$ convergence $\kappa_{\rm local}(x)$ as a synthetic 1D illustrative test against the self-constructed 1D King-profile target curve kappa_synth_target_1D (not an observational χ^2 fit to Clowe/Bradač data, which would require a published 2D convergence map, covariance matrix, pre-registered selection, and degree-of-freedom subtraction), and the model-dependent clock-weighting threshold	At $R_c/L \sim (2.0\text{–}4.6) \times 10^{17}$, $\Delta\kappa^{\rm conf}_{\rm local} = 0$ and $\lvert\Delta\kappa^{\rm disf}_{\rm local}/\kappa_{\rm GR,bary}\rvert \leq 2.171 \times 10^{-36}$, so $\kappa_{\rm local}(x) = \kappa_{\rm GR,bary}(x)$ with peaks on the X-ray gas ($x_{\rm peak} = -175.44$ kpc, $+137.01$ kpc; $\kappa_{\rm peak} = 0.04503, 0.03697$; $M_{\rm synth} = 32.089$) rather than the galaxies (-350 kpc, $+300$ kpc); shifting the peaks onto the galaxies requires collisionless compact-well clock weighting $\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$ ($M_{\rm synth} = 1.637$) (model-dependent threshold of the chosen 1D King geometry and $L_{\rm macro} = 300$ kpc; $\eta_{\rm gas} = 0$ and the two separately fitted galactic weights $(\eta_{\rm gal,main}, \eta_{\rm gal,sub})$ are phenomenological assumptions not derived from the covariant action)	step24, bullet_cluster_lensing_audit.py

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Physical Domain	Quantity	Exact Numerical Value	Source Step & Script
Cosmological Background ($\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3) and CMB Acoustic Peaks (step25)	Stiff attractor $m_\phi c/H_0$, negative ρ_Λ turnaround vs. compensated background, CMB $G_{\rm eff}(a,k)$, ISW, and $\Omega_{\rm eff,clock}/\Omega_b$	$m_\phi c/H_0 = 1.3725 \times 10^{22}$ (5.92×10^{17} at $z_* = 1089.80$); uncompensated $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 ($\Omega_\Lambda = -0.685$) causes flat-space turnaround at $a_{\max} = 0.7718$ ($z_{\rm turn} = 0.2957$) or open curvature $\Omega_K = +1.3702$ ($q_0 = +0.8426$, $\ell_1 = 553.96$), whereas the conditional separate-branch $\rho_{\rm clock,cosmo} = +1.192 \times 10^{-26}$ kg/m^3 restores $\Omega_{\rm DE,eff} = +0.6847$, $q_0 = -0.5276$, $100\,\theta_* = 1.04109$; at z_*, $(G_{\rm eff} - G)/G = 2.5015 \times 10^{-36}$ at $\ell_1 = 220$ (4.6516×10^{-34} at $\ell = 3000$), satisfying Planck 2018 ($\lvert G_{\rm eff}/G - 1\rvert \le 0.02$) by > 31 orders of magnitude; $\Omega_{\rm eff,clock}(z_*)/\Omega_b(z_*) = 5.3643$ is the input Planck ratio. Only the FLRW background quantities ($H(z)$, q_0, $a_{\max}$, $r_s(z_*)$, $D_M(z_*)$, $D_A(z_*)$, $100\,\theta_*$, ℓ_A) are numerically integrated; ℓ_1, ℓ_2, ℓ_3 are fixed analytic phase-shift template estimates, and the peak-height ratios ($H_2/H_1 = 0.458$, $H_3/H_2 = 0.985$ for $\Omega_c > 0$ vs. 0.312, 0.415 for $\Omega_c = 0$) are fixed external literature reference inputs (no Einstein–Boltzmann perturbation solver was run). The ratio $5.364327 = 0.1200/0.02237$ is the input Planck 2018 CDM/baryon density ratio, not a prediction computed from the TAL clock sector; "Planck compatibility" refers strictly to the negligibility of the local $L = 10$ km Yukawa fifth-force modification ($\Delta G_{\rm eff}/G = 2.50 \times 10^{-36}$ at $\ell_1 = 220$, 4.65×10^{-34} at $\ell = 3000$). In the minimal canonical action of step23 ($V(\phi_L) = +\frac{1}{2} m_\phi^2 \phi_L^2 \ge 0$) a homogeneous static scalar has positive potential energy density in $T^{(\phi)}_{\mu\nu}$; $\rho_\Lambda = -V/c^2 = -5.96 \times 10^{-27}$ kg/m^3 and	step25, cmb_and_cosmology_audit.py

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1.3.1 3.1. V25 Fixes and Diagnostics: 4-Component Exterior BVP, Exact Exterior Linearization, Dynamical Scalar Wave, and GL2 N-Convergence

(Source: `curved_yukawa_adm_and_stability.py`, `curved_yukawa_audit_summary.json`, `curved_yukawa_coupled_stability`, `curved_yukawa_adm_sequences.csv`.)

- **Exterior Lapse (Issue 5):** the exterior ν is co-integrated as the fourth ODE component (ϕ_L, ψ_L, m, ν) with `solve_bvp tol=1e-12` and outer boundary condition $\nu(R_{\max}) = \frac{1}{2}\ln(1 - 2m(R_{\max})/R_{\max})$. Spreads for SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $N_{\rm mult} \in \{10, 12, 14, 16\}$: $\delta k/k = 6.17 \times 10^{-12}$, $\delta M_{\rm ADM}/M_{\rm ADM} = 5.54 \times 10^{-13}$, $\delta\nu_R = 1.63 \times 10^{-14}$.

$N_{\rm mult}$	$R_{\max}$ (km)	$k_{\rm curved}$ (km^{-1})	$M_{\rm ADM}$ ($M_\odot$)	ν_R
10	111.4657	0.233542329280	1.509998138706	-0.246306687228
12	131.4657	0.233542329278	1.509998138707	-0.246306687228
14	151.4657	0.233542329278	1.509998138707	-0.246306687228
16	171.4657	0.233542329278	1.509998138707	-0.246306687228

- **Exterior Linearization and Finite-Difference Derivative Tests (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$):** `linearized_exterior_response` (line 593: `a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0`) and the left-hand side ($L_u u = R_h\,\delta h$) interior scalar operator coefficient `a_m_sc` (`scalar_m_sign = -1.0`: `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`) verified against central finite differences:
  - Exterior response $\partial\phi_L'(R_*)/\partial m(R_*)$: linearized BVP `-1.68365059e-04` km^{-2} vs. central finite difference of the full 4-component exterior BVP `-1.68365059e-04` km^{-2} (relative difference: 3.98×10^{-10}).
  - Interior $r = 6.0$ km scalar–mass derivative ($\partial\phi_L''/\partial m$): analytical `-1.72833429e-04` km^{-3} vs. central finite difference of the interior scalar RHS `-1.72833428e-04` km^{-3} (relative difference: 6.08×10^{-9}).
- **Dynamical Scalar Block:** the scalar block includes the $-\omega^2 e^{-2\nu}\delta\phi_L$ term and the exterior response is supplied by the ω-dependent BVP (`dynamical_scalar_wave=True`); solver: GL2 piecewise-segment Volterra discretization, $N = 480$, with exact LHS sign `scalar_m_sign = -1.0`. The quasi-static and dynamical roots agree to all 6 reported decimal places.
- **Turning Points and Zero Crossings:** `minimize_scalar` + 7-point quadratic fit over ± 0.002 dex; zero crossings from a 3-point quadratic fit over ± 0.0005 dex around the root.

Sequence	$\log_{10}\rho_{0c}(dM_{\rm ADM}=0)$	$\log_{10}\rho_{0c}(dM_B=0)$	Dynamical $\omega_0^2 = 0$	$\Delta_{\rm ADM}$ Dynamical (dex)	Δ_{M_B} Dynamical (dex)	Quasi-Static $\omega_0^2 = 0$	Frozen $\delta\phi_L = 0$	$\Delta_{\rm ADM}$ Frozen (dex)
SLy, $L = 10$ **km,** $\alpha_Y = 1$	15.316404	15.316409	**15.316410**	**+5.80e-06**	+1.33e-06	15.316410	15.328565	+1.22e-02
SLy, $L = 10$ **km,** $\alpha_Y = 1/3$	15.307166	15.307169	**15.307170**	**+4.20e-06**	+6.41e-07	15.307170	15.311340	+4.17e-03
SLy, $L = 1$ **km,** $\alpha_Y = 1$	15.303086	15.303082	**15.303080**	**-6.86e-06**	-2.61e-06	15.303080	15.304220	+1.13e-03
APR4, $L = 10$ **km,** $\alpha_Y = 1$	15.293022	15.293025	**15.293025**	**+3.57e-06**	-1.40e-07	15.293025	15.301564	+8.54e-03
H4, $L = 10$ **km,** $\alpha_Y = 1$	15.227785	15.227787	**15.227783**	**-1.34e-06**	-4.05e-06	15.227783	15.251384	+2.36e-02

GL2 Convergence on the $\omega_0^2 = 0$ Root (across N and $R_{\max}$) and at $\log_{10}\rho_{0c} = 15.311404$ (SLy, $L = 10$ km, $\alpha_Y = 1$): on the root, $p_{\rm eff,root} = 4.248$, Richardson root $\log_{10}\rho_{0c,\rm Rich}(\omega_0^2 = 0) =$ '15.31640981' ($N = 480$ discretization

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error 6.72×10^{-8} dex; spread across $R_{\max} \in [R_* + 10L, R_* + 20L]$ is 2.42×10^{-13} dex); at $\log_{10} \rho_{0c} = 15.311404$, $p_{\rm eff} = 5.223$, Richardson value $\omega_0^2 = 6369417.611\ \mathrm{s}^{-2}$ ($N = 480$ relative error 1.29×10^{-5}).

N	$\log_{10} \rho_{0c}(\omega_0^2 = 0)$	$\lvert\Delta \log_{10} \rho_{0c}\rvert$ vs. Richardson Root (dex)	$\Delta_{\rm ADM}$ (dex)	Δ_{M_B} (dex)	$\omega_0^2(15.311404)$ (s^{-2})	Rel. Diff. (ω_0^2)
80	15.31645808	4.83e-05	+5.40e-05	+4.95e-05	6432300.503336	9.87e-03
120	15.31642223	1.24e-05	+1.82e-05	+1.37e-05	6385729.603235	2.56e-03
160	15.31641426	4.46e-06	+1.02e-05	+5.72e-06	6375276.486537	9.20e-04
240	15.31641079	9.80e-07	+6.71e-06	+2.24e-06	6370703.105233	2.02e-04
360	15.31641001	2.06e-07	+5.94e-06	+1.47e-06	6369680.451299	4.13e-05
480	15.31640987	6.72e-08	+5.80e-06	+1.33e-06	6369499.539632	1.29e-05
600	15.31640983	2.56e-08	+5.76e-06	+1.29e-06	6369446.532399	4.54e-06
800	15.31640982	7.55e-09	+5.74e-06	+1.27e-06	6369424.048161	1.01e-06
1200	15.31640981	1.35e-09	+5.73e-06	+1.26e-06	6369418.385568	1.22e-07

Interpretive Note (Boundary of Theoretical Closure and Conditional Stability Closing Statement): With the exact left-hand side sign (`scalar_m_sign = -1.0`), the root of the GL2 operator converges with respect to N and $R_{\max}$ to the reported precision ($< 10^{-7}$ dex, `15.31640981`), and lies within $\leq 6.9 \times 10^{-6}$ dex of the ADM mass maximum across all 5 sequences (for SLy $L = 10$ km, $\alpha_Y = 1$: 15.316410 vs. 15.316404, $+5.80 \times 10^{-6}$ dex; the diagnostic comparison with `scalar_m_sign = +1.0` is documented in Section 7.4 of `step22`). For SLy $L = 10$ km, $\alpha_Y = 1$, the relative L^2 differences between the eigenfunction and the static tangent are $\delta h = 5.04 \times 10^{-4}$, $\delta m = 1.77 \times 10^{-4}$, $\delta\phi_L = 2.27 \times 10^{-4}$, $\delta\phi'_L = 3.09 \times 10^{-4}$, while the relative difference of the free-surface displacements is 3.54% (`rel_diff_xi_R_exact_minus_vs_static_tangent = 0.035406`). While the minus-sign root and grid convergence represent a major step forward, because of the 3.54% free-surface difference and the ODE vs. quadrature baryonic-mass turning-point spread they do not by themselves constitute an independent proof of the full radial fundamental spectrum and free-surface variation; therefore, the stability closing statement is formulated conditionally until the independent verification of the coupled variational derivation and boundary-condition matching is concluded.

1.3.2 3.2. Extended Observational Tests at Fixed $L = 10$ km, $\alpha_Y = 1$ (`step23`–`step25`) and Grand Synthesis Across All 6 Physical Scales

(Source: `step23_gw170817_and_lensing_en.md`, `step24_bullet_cluster_en.md`, `step25_cmb_and_cosmology_en.md`, `gw170817_and_lensing_audit.py`, `bullet_cluster_lensing_audit.py`, `cmb_and_cosmology_audit.py`, and their 6 CSV + 3 JSON outputs.)

1. **Module 1 (`step23` — GW170817 $c_T = c$, Scalar Cutoff $f_{\rm cut} = 4771.35$ Hz, and Curved-Spacetime Lensing):**

   - **Exact Luminal Tensor Speed ($c_T = c$):** In the Horndeski / DHOST classification of the local Helmholtz–Yukawa action $S_{\rm II}[g_{\mu\nu}, \phi_L, \Psi]$, the Einstein-frame tensor kinetic function is $G_4(\phi_L, X) = M_{\rm Pl}^2/2 = \text{const}$ ($\partial_X G_4 = 0$) and $G_5 = 0$, while in the Jordan frame ($\tilde{g}_{\mu\nu}^{\rm conf} = e^{-2\phi_L} g_{\mu\nu}$) null cones are invariant ($d\tilde{s}^2 = 0 \iff ds^2 = 0$) (the medium-anchored disformal branch `Branch II-Disf`, with $u^\mu h_{\mu\nu}^{TT} = 0$, is not covered by this proof). Hence, on this branch, $c_T = c$ ($|c_T/c - 1| = 0 \leq 3 \times 10^{-15}$, Abbott et al. 2017): $c_T = c$ is proven on the purely conformal, minimal Einstein–Hilbert + canonical scalar + universal Maxwell coupling branch ($G_4 = M_{\rm Pl}^2/2$, $G_{4,X} = 0$, $G_5 = 0$, $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$, $\beta_{\rm Yuk} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795775$); this does not by itself prove $c_T = c$ on a disformal/refractive branch ($D(\phi_L, X) \neq 0$).
   - **Massive Yukawa Scalar Dispersion ($f_{\rm cut} = 4771.35$ Hz):** With $L = 10$ km, the scalar Compton mass is $m_\phi = \hbar/(Lc) = 1.97327 \times 10^{-11}$ eV/c^2 and the cutoff frequency is $f_{\rm cut} = c/(2\pi L) = 4771.3452$ Hz. Across binary-pulsar orbital frequencies ($f \sim 10^{-4}$ Hz) and the entire LIGO/Virgo inspiral band of GW170817 ($f \in [24, 600]$ Hz), $f < f_{\rm cut}$, so the scalar breathing mode is strictly evanescent with skin depth $\lambda_{\rm ev}(f) = L/\sqrt{1 - (f/f_{\rm cut})^2} \in [10.0001, 10.0800]$ km and amplitude suppression $A_{\rm scalar}(40\ \mathrm{Mpc}) = \exp(-\kappa_{\rm ev} D) \sim 10^{-5.36 \times 10^{19}} = 0.0$.
   - **Weyl Lensing Potential, Cassini PPN Slip, and Curved-Exterior Deflection:** In the conformal Jordan frame, the linear scalar shift cancels out of the Weyl lensing sum ($\Phi_{\rm lens} = (\tilde{\Psi} + \tilde{\Phi})/2 = -GM_{\rm ADM}/r$), while the PPN slip is $\gamma_{\rm PPN}(b) = (1 - \beta_{\rm Yuk} e^{-b/L})/(1 + \beta_{\rm Yuk} e^{-b/L})$. At the Cassini impact parameter $b = 1.6\, R_\odot = 1.11312 \times 10^6$ km ($b/L = 1.11312 \times 10^5$), $|1 - \gamma_{\rm PPN}| = 10^{-48343} = 0.0 \ll 2.3 \times 10^{-5}$. On the V25

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Step 22 `Branch II-Fluid` 4-component curved-exterior diagnostic background (SLy reference neutron star, $\log_{10}\rho_{0c} = 15.0$, $R_* = 11.4657$ km, $M_{\rm surf} = 1.507661\,M_\odot$, $M_{\rm ADM} = 1.509998\,M_\odot$, Table 2), a surface-grazing ray ($b = 14.7490$ km) experiences $\hat\alpha_{\rm Yuk,conf} = 1.290852$ rad (+0.3652% above vacuum Schwarzschild with $M_{\rm surf}$, and −0.0287% below vacuum Schwarzschild with $M_{\rm ADM}$ because the ray traverses inside the exterior scalar cloud), whereas on the self-consistent `Branch II-Conf` trace-sourced ($\tilde\varepsilon - 3\tilde P$) interior + exterior BVP solution ($R_* = 11.6627$ km, $M_{\rm surf} = 1.560636\,M_\odot$, $M_{\rm ADM} = 1.562035\,M_\odot$, $\Delta M^{\rm curved}_{\phi,\rm ext} = +0.001399\,M_\odot$, Table 2B) the grazing ray ($b = 15.0649$ km) experiences $\hat\alpha_{\rm Yuk,conf} = 1.334024$ rad (+0.2184% above Schwarzschild with $M_{\rm surf}$, and −0.0167% below Schwarzschild with $M_{\rm ADM}$; at the shared impact parameter $b = 15.064914$ km, `Branch II-Fluid` gives 1.224744 rad vs. `Branch II-Conf` 1.334024 rad; for the `Branch II-Conf` BVP, the converged interior scalar ODE residual is $\max|{\rm res}_{\rm ODE}| = 1.480\times10^{-14}$ km^{-2} banded / 6.958×10^{-14} km^{-2} iterated, the enforced Robin boundary derivative matches to $|\phi'_{L,\rm int,Robin}(R_*) - \phi'_{L,\rm ext}(R_*)| = 1.465\times10^{-9}$ km^{-1} under 4-step and 3.937×10^{-15} km^{-1} under every-step 4-component Robin updates (`tol_iter = 1e-11`), whereas the independent one-sided FD2 derivative differs by 4.723×10^{-10} km^{-1} at $N = 4000$ and converges in second order (empirical scaling `4.0332` consistent with expected `4.0020`), and the `DOP853` (`max_step = 0.02 km`, `tol_iter = 1e-10`) $N = 2000$ vs. $N = 4000$ mesh refinement gives $|\Delta R_*| = 4.213\times10^{-8}$ km, $|\Delta M_{\rm ADM}| = 1.097\times10^{-8}\ M_\odot$, $|\Delta\phi_L(R_*)| = 1.920\times10^{-9}$, Section 2B.1); for $b \geq 200$ km, both branches converge exponentially ($\leq 8.03\times10^{-13}$ and $\leq 1.79\times10^{-13}$) to their respective $\hat\alpha_{\rm Schw}(M_{\rm ADM}, b)$.

2. **Module 2 (`step24` — Bullet Cluster `1E 0657-558` Weak-Lensing Convergence κ and Peak-Offset Analysis):**

   - **Local $L = 10$ km Yukawa No-Go on Cluster Scales ($R_c/L \sim (2.0\text{–}4.6)\times10^{17}$):** Modeling the main cluster and subcluster X-ray gas ($M_{\rm gas,tot} = 3.25\times10^{13}\,M_\odot$ centered at $x = -170$ kpc and +140 kpc) and collisionless galaxies ($M_{\rm gal,tot} = 7.30\times10^{12}\,M_\odot$ centered at $x = -350$ kpc and +300 kpc), the local $L = 10$ km Yukawa operator reduces to an algebraic contact term ($\phi_L^{\rm local} \simeq -\alpha_{\rm std}L^2(4\pi G/c^2)\rho_b \sim -10^{-42}$). On `Branch II-Conf`, $\Delta\kappa^{\rm conf}_{\rm local} = 0$ identically; on `Branch II-Disf`, $|\Delta\kappa^{\rm disf}_{\rm local}/\kappa_{\rm GR,bary}| \leq 2.171\times10^{-36}$. Thus $\kappa_{{\rm local}\,L=10\ {\rm km}}(x) = \kappa_{\rm GR,bary}(x)$, leaving the convergence peaks on the X-ray gas ($x_{\rm peak,main} = -175.44$ kpc, $\kappa = 0.04503$; $x_{\rm peak,sub} = +137.01$ kpc, $\kappa = 0.03697$; $M_{\rm synth} = 32.089$), 7.11× below the observed amplitude ($\kappa_{\rm obs} = 0.320$) and displaced by 163–175 kpc from the galaxies (synthetic 1D illustrative test against the self-constructed 1D King-profile target curve `kappa_synth_target_1D` (not an observational χ^2 fit to Clowe/Bradač data, which would require a published 2D convergence map, covariance matrix, pre-registered selection, and degree-of-freedom subtraction)).
   - **Condition for Shifting Lensing Peaks to the Galaxies:** Even an unscreened macroscopic Yukawa/top-hat kernel ($L_{\rm macro} = 300$ kpc) coupled uniformly to bulk baryon density (`Macro_unscreened_bulk_baryon`) keeps the peaks on the dominant X-ray gas ($x_{\rm peak,main} = -176.02$ kpc, $M_{\rm synth} = 20.723$) because $M_{\rm gas}/M_{\rm gal} = 4.452$. In this synthetic 1D illustrative test against the self-constructed 1D King-profile target curve `kappa_synth_target_1D` (not an observational χ^2 fit to Clowe/Bradač data, which would require a published 2D convergence map, covariance matrix, pre-registered selection, and degree-of-freedom subtraction), reproducing galaxy-centered peaks in the synthetic target ($x_{\rm peak,main} = -347.86$ kpc, $x_{\rm peak,sub} = +299.45$ kpc, $M_{\rm synth} = 1.637$) requires a collisionless or compact-potential-well clock weighting with $\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$ (model-dependent threshold of the chosen 1D King geometry and $L_{\rm macro} = 300$ kpc; $\eta_{\rm gas} = 0$ and the two separately fitted galactic weights ($\eta_{\rm gal,main}, \eta_{\rm gal,sub}$) are phenomenological assumptions not derived from the covariant action).

3. **Module 3 (`step25` — Cosmological Background, Conditional Separate-Branch Hypothesis for $\rho_\Lambda = -5.96\times10^{-27}$ kg/m^3, and CMB Acoustic Peaks):**

   - **Ultra-Stiff Quasi-Static Attractor and the Negative Screening Energy Density as a Conditional Separate-Branch Hypothesis:** Because $m_\phi c/H_0 = 1.3725\times10^{22}$ today and $m_\phi c/H(z_*) = 5.921\times10^{17}$ at recombination ($z_* = 1089.80$), the homogeneous background field is locked to $\phi_{\rm attr}(z) = +\alpha_Y^2 GL^2\bar\rho_m(z)/c^2 = 1.995\times10^{-46}\,(1+z)^3$ (SI ODE $\ddot{\bar\phi}_L + 3H\dot{\bar\phi}_L + (c^2/L^2)\bar\phi_L = \alpha_Y^2 G\bar\rho_m$ [s^{-2}]; V25 std: $1.587\times10^{-47}\,(1+z)^3$), contributing a negligible energy density fraction $\Omega_{\phi,\rm can}(0) = 1.582\times10^{-50}$ (std) / 3.140×10^{-47} (unit), $\rho_{\phi,\rm can}(0) = 1.350\times10^{-76}$ / 2.679×10^{-73} kg/m^3. If the V23/V25 negative vacuum screening density $\rho_\Lambda = -5.96\times10^{-27}$ kg/m^3 ($\Omega_{\Lambda,\rm bare} = -0.6847$) is inserted uncompensated into the Friedmann equation, a flat universe ($\Omega_K = 0$) suffers premature turnaround at $a_{\rm max} = 0.7718$ ($z_{\rm turn} = 0.2957$, $E^2(z=0) = -0.3693 < 0$), while an open universe ($\Omega_K = +1.3693$) decelerates today ($q_0 = +0.8426$) and shifts the first CMB peak to $\ell_1 = 553.96$. Embedding ρ_Λ in a macroscopic cosmic clock background $\rho_{\rm clock,cosmo} = +1.192\times10^{-26}$ kg/m^3 restores $\rho_{\rm DE,eff} = +5.96\times10^{-27}$ kg/m^3 ($\Omega_{\rm DE,eff} = +0.6847$), $q_0 = -0.5276$, and $100\,\theta_* = 1.04109$ ($\ell_1 = 220.0$). In the minimal canonical action of `step23` ($V(\phi_L) = +\frac{1}{2}m_\phi^2\phi_L^2 \geq 0$) a homogeneous static scalar has positive potential energy density in $T^{(\phi)}_{\mu\nu}$; $\rho_\Lambda = -V/c^2 = -5.96\times10^{-27}$ kg/m^3 and its compensating $\rho_{\rm clock,cosmo} = +2|\rho_\Lambda| = +1.192\times10^{-26}$ kg/m^3 are therefore treated strictly as a conditional separate-branch hypothesis that recovers the input ΛCDM $\rho_{\rm DE,eff} = +5.96\times10^{-27}$ kg/m^3 by algebraic construction, not as an independent cosmological prediction.

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- **CMB Perturbations ($G_{\mathrm{eff}}(a,k)$, ISW, and Acoustic Peak Ratios $\ell_1 : \ell_2 : \ell_3$):** On CMB and LSS scales, $(G_{\mathrm{eff}}(a,k) - G)/G = \beta_{\mathrm{Yuk}}(k_{\mathrm{phys}}L)^2/(1 + (k_{\mathrm{phys}}L)^2) = 2.5015 \times 10^{-36}$ at the first acoustic peak $\ell_1 = 220$ ($k = 0.01586$ Mpc^{-1} at $z_* = 1089.80$) and 4.6516×10^{-34} at the Silk damping tail $\ell = 3000$, satisfying the Planck 2018 recombination bound ($|G_{\mathrm{eff}}/G - 1| \leq 0.02$) by > 31 orders of magnitude, with ISW Weyl-potential slip $< 4.13 \times 10^{-38}$. A purely baryonic universe ($\Omega_c h^2 = 0$) with only the local $L = 10$ km Yukawa force expands the sound horizon to $r_s(z_*) \in [195.11, 209.14]$ Mpc (shifting ℓ_1 to 303.45–315.03 flat or 1309.70 open) and collapses the third-to-second peak ratio from $(H_3/H_2)_{\mathrm{Planck}} = 0.985$ to 0.415; matching the Planck 2018 acoustic peaks requires a pre-recombination Thomson-decoupled clock-metric driving ratio $\Omega_{\mathrm{eff,clock}}(z_*)/\Omega_b(z_*) = 0.1200/0.02237 = 5.364327$ (input Planck 2018 ratio). Only the FLRW background quantities ($H(z)$, q_0, $a_{\max}$, $r_s(z_*)$, $D_M(z_*)$, $D_A(z_*)$, $100\,\theta_*$, ℓ_A) are numerically integrated; ℓ_1, ℓ_2, ℓ_3 are fixed analytic phase-shift template estimates, and the peak-height ratios ($H_2/H_1 = 0.458$, $H_3/H_2 = 0.985$ for $\Omega_c > 0$ vs. 0.312, 0.415 for $\Omega_c = 0$) are fixed external literature reference inputs (no Einstein–Boltzmann perturbation solver was run). The ratio $5.364327 = 0.1200/0.02237$ is the input Planck 2018 CDM/baryon density ratio, not a prediction computed from the TAL clock sector; "Planck compatibility" refers strictly to the negligibility of the local $L = 10$ km Yukawa fifth-force modification ($\Delta G_{\mathrm{eff}}/G = 2.50 \times 10^{-36}$ at $\ell_1 = 220$, 4.65×10^{-34} at $\ell = 3000$).

#	Observational Regime & Steps	Characteristic Length / Wavenumber Scale	Scale Ratio (r/L or $k_{\mathrm{phys}}L$)	Yukawa Screening / Suppression Factor	Exact Prediction at Fixed $L = 10$ km, $\alpha_Y = 1$	Observational Benchmark	Multi-Scale Audit Verdict at Fixed $L = 10$ km, $\alpha_Y = 1$
1	**Neutron Star Interior & Surface** (`step7`, `step17`, `step21`, `step22`)	$r \sim 10$ km ($R_* = 11.17$ km)	$r/L \sim 1.0$	$e^{-r/L} \sim 0.368$ ($\mathcal{O}(1)$ active fifth force)	$M^{\mathrm{SLy}}_{\mathrm{ADM,max}} = 2.033214\ M_\odot$ ($R(M_{\mathrm{ADM,max}}) = 9.8853$ km, $M_{\mathrm{surf}} = 2.029106\ M_\odot$, $M^{\mathrm{quad}}_{B,\max} = 2.423385\ M_\odot$, $R^{\mathrm{ADM}}_{1.4} = 11.5359$ km; APR4: $2.18068\ M_\odot$); $\omega_0^2 = 0$ at $\log_{10}\rho_{0c} = 15.316410$ ($\Delta \log_{10}\rho_{0c} = +5.80 \times 10^{-6}$ dex)	PSR J0740+6620 $\neq$ J0348+0432 ($M_{\max} > 2.0\ M_\odot$) & radial stability alignment	**SATISFIED** (curved-exterior BVP & coupled Sturm–Liouville stability)
2	**GW170817 / GRB 170817A Propagation** (`step23`)	$D = 40$ Mpc, $f \in [24, 600]$ Hz ($\lambda_{\mathrm{GW}} \sim$ 500–12500 km)	$f/f_{\mathrm{cut}} \in [0.0050, 0.1257]$ ($f_{\mathrm{cut}} = 4771.35$ Hz)	Scalar mode evanescent ($\lambda_{\mathrm{ev}} = 10.00$–$10.08$ km, $A_{\mathrm{scalar}} \sim 10^{-5.36\times10^{19}}$)	$c_T = c$ on the purely conformal minimal branch ($G_{4,X} = 0$, $G_5 = 0$; $\|c_T/c - 1\| = 0$; not automatic for $D(\phi_L, X) \neq 0$); zero scalar breathing flux at detector	$\|c_T/c - 1\| \leq 3 \times 10^{-15}$ (Abbott et al. 2017, GW170817)	**SATISFIED IDENTICALLY** (by > 15 orders of magnitude; evanescent scalar)

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#	Observational Regime & Steps	Characteristic Length / Wavenumber Scale	Scale Ratio (r/L or $k_{\rm phys}L$)	Yukawa Screening / Suppression Factor	Exact Prediction at Fixed $L = 10$ km, $\alpha_Y = 1$	Observational Benchmark	Multi-Scale Audit Verdict at Fixed $L = 10$ km, $\alpha_Y = 1$
3	**Solar System / Cassini Shapiro Delay** (`step5`, `step14`, `step23`)	$b = 1.6\,R_\odot = 1.11312 \times 10^6$ km	$b/L = 1.11312 \times 10^5$	$e^{-b/L} = 10^{-48\,342.19} = 0.0$ (`float64`)	$\gamma_{\rm PPN} = 1.0$, $\lVert 1 - \gamma_{\rm PPN} \rVert = 10^{-48\,343.00}$	$\lVert \gamma_{\rm PPN} - 1 \rVert \leq 2.3 \times 10^{-5}$ (Bertotti et al. 2003, Cassini)	**SATISFIED** (by $> 48\,337$ orders of magnitude)
4	**Galactic Rotation Curves & Lensing** (`step6`, `step11`, `step14`, `step23`)	$r \sim$ 10–30.08 kpc (SPARC / NGC 3198)	$r/L = 3.09 \times 10^{16}$–$9.28 \times 10^{16}$	$e^{-r/L} = 10^{-1.34\times10^{16}}$ to $10^{-4.03\times10^{16}} = 0.0$	Local $L = 10$ km Yukawa gives pure baryonic GR; flat rotation curves require macroscopic clock branch $L_{\rm eff}(\rho)$	SPARC 175 galaxies & galaxy-galaxy weak lensing ($\tilde\Phi + \tilde\Psi$)	**REQUIRES MACROSCOPIC CLOCK BRANCH** (local $L = 10$ km Yukawa screened to zero)
5	**Bullet Cluster `1E 0657-558` Lensing** (`step24`)	$r \sim$ 150–350 kpc ($z_L = 0.296$)	$r/L = 4.63 \times 10^{17}$–$1.08 \times 10^{18}$	$e^{-r/L} = 10^{-2.01\times10^{17}}$ to $10^{-6.70\times10^{17}} = 0.0$	Local $L = 10$ km Yukawa gives $\kappa = \kappa_{\rm GR,bary}$ (peaks on X-ray gas at -175.44 kpc, $+137.01$ kpc, synthetic 1D test $M_{\rm synth} = 32.089$)	Clowe et al. (2006) lensing peaks on galaxies (-350 kpc, $+300$ kpc; 8σ offset)	**REQUIRES COLLISIONLESS / COMPACT-WELL CLOCK WEIGHTING** ($\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$, model-dependent 1D King / $L_{\rm macro} = 300$ kpc threshold; phenomenological η assumptions)

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#	Observational Regime & Steps	Characteristic Length / Wavenumber Scale	Scale Ratio (r/L or $k_{\rm phys}L$)	Yukawa Screening / Suppression Factor	Exact Prediction at Fixed $L = 10$ km, $\alpha_Y = 1$	Observational Benchmark	Multi-Scale Audit Verdict at Fixed $L = 10$ km, $\alpha_Y = 1$
6	**Cosmology,** $\rho_\Lambda = -5.96 \times 10^{-27}$ **kg/m^3 & CMB** (`step25`)	$z_* = 1089.80$, $k \in [10^{-4}, 10^{1}]$ Mpc^{-1} ($\ell \in [2, 3000]$)	$k_{\rm phys}(z_*)\, L = 5.61 \times 10^{-18}$ (at $\ell_1 = 220$); $m_\phi c/H_0 = 1.37 \times 10^{22}$	$(G_{\rm eff} - G)/G = 2.50 \times 10^{-36}$ at $\ell_1 = 220$ (4.65×10^{-34} at $\ell = 3000$); ISW slip $\leq 4.13 \times 10^{-38}$	Zero perturbative fifth-force slip on CMB; uncompensated $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 causes turnaround ($a_{\max} = 0.7718$ flat) & shifts ℓ_1	Planck 2018: $100\,\theta_* = 1.04109 \pm 0.00030$, $\ell_1 = 220.0$, $H_3/H_2 = 0.985$, $\|G_{\rm eff}/G - 1\| \leq 0.02$	$G_{\rm eff}$ **SATISFIED BY** > 31 **ORDERS OF MAGNITUDE**; background & peaks require $\rho_{\rm clock,cosmo} = +1.192 \times 10^{-26}$ kg/m^3 and $\Omega_{\rm eff,clock}/\Omega_b = 5.3643$

1.4 4. Local Archival Manuscripts Used (`manuscripts/`)

All referenced V19–V23 LaTeX (`.tex`) source files are included in the `manuscripts/` subdirectory:

1. **`manuscripts/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex`** and **`manuscripts/tal_covariant_closure_50aa`**
   - *Version & DOI:* V23 (`50aa2db0`), Zenodo DOI `10.5281/zenodo.23639084`.
   - *Role in the Derivation:* Primary source for the two covariant action candidates (Formulation I bilateral kernel action and Formulation II local Helmholtz–Yukawa scalar–tensor action), Bianchi residuals, coupled radial stability BVP (`eq:factored_diff`, `eq:coupled_bvp`), and H0/H1 clock coupling (Sirius B and 8558 kg granite chamber).
2. **`manuscripts/hu_p1.tex`**, **`manuscripts/hu_p2.tex`**, **`manuscripts/hu_p3.tex`**, **`manuscripts/en_part1.tex`**, **`manuscripts/en_part2.tex`**
   - *Version & DOI:* V22 Book Edition chapter sources, Zenodo DOI `10.5281/zenodo.23604423`.
   - *Role in the Derivation:* Source of verbatim citations for the Cauchy derivation (Chapter 2), 3D top-hat and SPARC 175-galaxy analysis (Chapters 4–5), V17 H.E.S.S./MAGIC gamma-ray timing audit (Chapter 6), 7-segment piecewise-polytrope TOV and $s = r^2$ regularization (Chapter 7), coupled radial pulsation (Chapter 8), Sirius B redshift (Chapter 9), 8558 kg granite chamber and 8-coil solenoid experiment (Chapter 10), and the numerical Appendix.
3. **`manuscripts/tal_book_HU_2026-10-24.tex`**, **`manuscripts/tal_book_EN_2026-10-24.tex`**, **`manuscripts/tal_book_DE_`**
   - *Version & DOI:* V22 Book Edition unified Hungarian, English, and German LaTeX monograph files, Zenodo DOI `10.5281/zenodo.23604423`.
4. **`manuscripts/sparc_tophat_record_2026-10-22_v5.tex`**
   - *Version & DOI:* V19 SPARC 3D spherical top-hat manuscript, Zenodo DOI `10.5281/zenodo.23497128`.
5. **`manuscripts/tal_v20_fop.tex`**
   - *Version & DOI:* V20 laboratory two-clock and 8-coil magnetic manuscript (*Foundations of Physics* format), Zenodo DOI `10.5281/zenodo.23540741`.
6. **`manuscripts/tal_v21_fop.tex`**
   - *Version & DOI:* V21 compact stellar TOV and Sirius B manuscript, Zenodo DOI `10.5281/zenodo.23602501`.

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1.5 5. Python Programs & SPARC Data Files

1.5.1 5.1. Python Programs (in the Root Directory and code/ Subdirectory)

1. **curved_yukawa_adm_and_stability.py** — Module solving the coupled interior + curved exterior ($r \in [R_*, R_* + 12L]$) Einstein–scalar boundary-value problem of yukawa_action_el (**Branch II-Fluid**), verifying $R_{\max}$-independence, curved vs. flat Robin slope, recomputed mass–radius sequences in $M_{\rm ADM}$ and M_B (curved_yukawa_adm_sequences.csv), and coupled ($\delta h, \delta\phi_L, \delta m$) dynamical radial pulsation eigenvalues (curved_yukawa_coupled_stability.csv, curved_yukawa_audit_summary.json).
2. **covariant_closure_50aa2db0.py** — Main computational and symbolic verification module of V23 (10.5281/zenodo.23639084: SymPy Bianchi identities, Formulation I and II TOV comparison — including yukawa_action_el_curved_adm rows —, coupled coupled_bvp / sym_galerkin radial stability, $W \to 4W$ scaling invariance, Sirius B and 8558 kg granite chamber H0/H1 clock coupling).
3. **radial_modes.py** — Finite-element / banded-matrix eigenvalue solver for the Chandrasekhar Sturm–Liouville radial pulsation operator (std, modA, modB, and fluid vs. metric inertia weight).
4. **tal_profiles.py** — Profile module implementing $s = r^2$ even-parity central PCHIP interpolation (Audit Fix 01), the 3D top-hat shell integral, and the gradient-mode exterior vacuum-halo lapse correction $\Delta\nu_{\rm ext}$ on $r \in [R_*, R_* + L]$.
5. **stellar_tal_v21.py** (and code/stellar_tal.py, code/diagnostics.py) — V21/V23 module containing the Read et al. (2009) 7-segment piecewise-polytrope equations of state (SLy, APR4, H4), the log-enthalpy (h) TOV integrator (769 models, 33 causal endpoints), and the white-dwarf (Sirius B) integrator.
6. **make_revision3_checks.py** (along with code/energy_integrator.py, code/fields.py, code/tophat_finite_source.p code/v20_fixed_case_runner.py, code/build_pdf.py) — Scripts running the V20 (10.5281/zenodo.23540741) 8-coil solenoid elliptic Biot–Savart check, $L_p = 0.630957$ m peak recalculation, $r = L/t$ compactified exterior integral (Paths A and B), and $L = 0.95$ m interior/exterior switching identity.
7. **sparkto_b_raw.py** and **sparc_tophat.py** — V19 program computing the 3D spherical top-hat mean density ($\bar\rho_L(\mathbf{x})$) and $f_L = \exp(-\gamma_L \bar\rho_L)$ factor of the 3D axisymmetric baryonic disk+gas+bulge model for all 175 SPARC galaxies.
8. **SPARC_mode.py**, **ta_pipeline.py**, **ta_foundation.py**, **sparc_full_runner.py**, **tal_factor_7th_model.py**, **ta_gammaL.py**, **ta_run.py**, **ta_run_l0.py**, **ta_run_l0_v2.py** — Programs running the SPARC 175 / 155 / 123 galaxy photometric vs. derivative-based (foundation) density comparison (0 vs. 239 / 301–303 negative-density points), the NGC 3198 ($R = 30.08$ kpc) control point, and V17/V19 7th-model rotation-curve fits.
9. **gw170817_and_lensing_audit.py** (and code/gw170817_and_lensing_audit.py) — Module 1 (step23) solver verifying Horndeski/DHOST luminal gravitational-wave speed ($c_T = c$), massive Yukawa scalar dispersion ($f_{\rm cut} = 4771.35$ Hz), scale-dependent PPN slip $\gamma_{\rm PPN}(b)$, and exact 128-point Gauss–Legendre curved-spacetime light deflection $\hat\alpha(b)$ and Shapiro delay on both the V25 Step 22 Branch II-Fluid diagnostic background ($M_{\rm ADM} = 1.50999814\ M_\odot$) and the self-consistent Branch II-Conf trace-sourced ($\tilde\varepsilon - 3\tilde P$) BVP solution ($M_{\rm ADM} = 1.56203491\ M_\odot$), plus the dedicated 1.000000 $M_\odot$ Solar–Cassini table (Sun_1p00Msun) and the single conformal convention $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\alpha_{\rm DEF} = -1/\sqrt{4\pi}$, $\beta_{\rm DEF} = 0$, $\beta_{\rm Yuk} = \alpha_{\rm DEF}^2 = 2\tilde\alpha_L \alpha_Y^2 = \alpha_Y^3/(4\pi)$ (gw170817_dispersion_and_cT.csv, lensing_deflection_and_ppn_slip.csv, gw170817_and_lensing_summary.json).
10. **bullet_cluster_lensing_audit.py** (and code/bullet_cluster_lensing_audit.py) — Module 2 (step24) solver evaluating the 4-component baryonic + dark-matter / clock-metric convergence profiles $\kappa(x)$ of the Bullet Cluster (1E 0657-558, $z_L = 0.296$), proving the local $L = 10$ km Yukawa algebraic contact reduction ($|\Delta\kappa_{\rm disf}/\kappa_{\rm bary}| \le 2.171 \times 10^{-36}$), and computing the collisionless compact-well clock-weighting threshold $\eta_{\rm gal}/\eta_{\rm gas} \ge 25.6650$ (model-dependent), reported as the synthetic 1D target metric $M_{\rm synth}$ (synth_mse_norm_vs_target) (bullet_cluster_convergence_profile.csv, bullet_cluster_peak_offsets_summary.csv, bullet_cluster_audit_s
11. **cmb_and_cosmology_audit.py** (and code/cmb_and_cosmology_audit.py) — Module 3 (step25) solver integrating the FLRW cosmological background models (including the V23/V25 negative vacuum screening density $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 and its conditional separate-branch compensation $\rho_{\rm clock,cosmo} = +1.192 \times 10^{-26}$ kg/m^3), scale-dependent $G_{\rm eff}(a,k)$ and Weyl slip across $\ell \in [2, 3000]$, and CMB acoustic peak template estimates; peak heights are fixed literature inputs (boltzmann_spectrum_computed = False) (cosmological_background_models.csv, cmb_geff_and_acoustic_scales.csv, cmb_and_cosmology_summary.json).

1.5.2 5.2. SPARC and V19–V23 Data Files (data/, csv/, run_output/)

- **data/Rotmod_LTG.zip** (SHA-256: 0a80cc90714828cc28b7dd57923576714d209f2490328c087c4a4ad607faf588) and **data/Rotmod_LTG/*_rotmod.dat** — Original Lelli, McGaugh & Schombert (2016) rotation-curve data files for all 175 SPARC galaxies (including data/Rotmod_LTG/NGC3198_rotmod.dat).

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- **data/SPARC_Lelli2016c.mrt** (SHA-256: 5aa0501f6b0d881fa579030e315e7b5b6ef561a5bd3a07472f9929c7e5728243) — SPARC Table 1 (distances, inclinations, $L_{3.6}$, $R_{\rm disk}$, $M_{\rm HI}$, $R_{\rm HI}$, quality flags).
- **data/blind_123.txt** (and v17_package/github/data/blind_123.txt, SHA-256: 3637d2f5bda6b2b97e645c5fba5cf39d3 — Hash-locked 123-galaxy blind sample list of V17/V19.
- **data/NGC3198_table.csv** and **data/NGC3198_table.md** — Detailed table of all 43 radial points of NGC 3198 (highlighting the $R = 30.08$ kpc control point).
- **data/sparc_tophat_per_galaxy_max.csv** and **data/sparc_tophat_summary.json** — V19 3D spherical top-hat peak values and Monte Carlo cross-checks across all 175 SPARC galaxies.
- **data/v19_7th_model/** and **run_output/** — Reference outputs of the V17/V19 baryonic baselines (points_vbar.csv, summary_vbar.csv) and the 7th model (points_tal_fL_FITTED_forward.csv, fit_scan_L.csv, comparison.json, run_meta.json).
- **csv/** and **data/v23/** — Reference files of the V20 laboratory granite chamber and 8-coil solenoid, together with the V23 (50aa2db0) and Step 22 curved exterior ADM mass and coupled stability output files (curved_yukawa_adm_sequences.csv, curved_yukawa_coupled_stability.csv, curved_yukawa_audit_summary.json, conservation_Residuals.json, closure_tov_comparison.csv, coupled_stability_results.csv, clock_coupling_HO plus the 6 CSV and 3 JSON outputs of the extended observational modules step23–step25: gw170817_dispersion_and_cT.c lensing_deflection_and_ppn_slip.csv, gw170817_and_lensing_summary.json, bullet_cluster_convergence_profi bullet_cluster_peak_offsets_summary.csv, bullet_cluster_audit_summary.json, cosmological_background_mode cmb_geff_and_acoustic_scales.csv, and cmb_and_cosmology_summary.json).
- **data/v21/** and **observational_inputs.csv** — V21/V23 769 stellar models (mr_sequences.csv, stellar_summary.csv, causal_endpoint_convergence.csv, profiles_L10km.csv, radius_shifts.csv, sirius_redshift.csv, numerical_diagnostics.json, wd_summary.json) and NICER/LIGO/pulsar observational inputs.

1.6 6. Exact Reproduction Instructions

1.6.1 6.1. Software Dependencies

- **Operating System:** Linux / macOS / POSIX-compatible environment
- **Python Version:** Python >= 3.10 (tested under Python 3.11–3.13; on the local workstation, /workspace/scratch/venv_o has all scientific packages pre-installed)
- **Required Python Packages:** numpy, scipy, pandas, sympy, matplotlib
  Installation (if needed):

```
pip install numpy scipy pandas sympy matplotlib
```

1.6.2 6.2. Running Each Program & Expected Outputs

Every command below can be executed directly from the unpacked scalar-tensor-yukawa-fit working directory without network access:

1. **Curved Exterior yukawa_action_el Einstein–Scalar BVP, ADM Mass ($M_{\rm ADM}$), Mass–Radius Curves, and Coupled ($\delta h, \delta\phi_L, \delta m$) Radial Stability (curved_yukawa_adm_and_stability.py):**

```
python3 curved_yukawa_adm_and_stability.py
```

   - *Expected Output (~2–3 minutes):*
     - Generates curved_yukawa_adm_sequences.csv, curved_yukawa_coupled_stability.csv, and curved_yukawa_audit_summary.json (and updates data/v23/).
     - Verifies $R_{\rm max}$-independence ($\Delta k_{\rm curved}/k_{\rm curved} = 6.17 \times 10^{-12}$, $\Delta M_{\rm ADM}/M_{\rm ADM} = 5.54 \times 10^{-13}$), curved vs. flat Robin slope (+24.74% for SLy $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$), ADM mass peaks ($M^{\rm SLy,10km}_{\rm ADM,max} = 2.03321\,M_\odot$, $R^{\rm ADM}_{1.4} = 11.5359$ km), finite-difference derivative tests ($\partial\phi'_L(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ km^{-2}, interior $r = 6$ km a_m_sc $= -1.72833 \times 10^{-4}$ km^{-3}), and coupled stability zero crossings (scalar_m_sign = -1.0: $\log_{10}\rho_{0c}(\omega_0^2 = 0) = 15.316410$, $\Delta_{\rm ADM} = +5.80 \times 10^{-6}$ dex, and -6.86×10^{-6} dex at $L = 1$ km).

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2. **SPARC 175 Galaxies ($\rho < 0$ Audit), NGC 3198 ($R = 30.08$ kpc) Control Point, and V19 3D Top-Hat Peaks (`SPARC_mode.py`):**

```
python3 SPARC_mode.py --mode summary
```

   - *Expected Output (JSON on stdout, ~0.2 s):*
     - `"photometric_and_3D_tophat_yukawa_rho_le_0_points": 0` vs. `"derivative_formula_rho_lt_0_points_vi` 239 (and 301 across all 175 galaxies).
     - `"NGC3198_control_point": "R_kpc": 30.08, "V_obs_km_s": 146.0, "V_bar_Ydisk_1p0_km_s":` 88.53, `"C_req_Ydisk_1p0": 2.72`, `"V_bar_Ydisk_0p5_km_s": 70.32`, `"C_req_Ydisk_0p5": 4.311`.

3. **SPARC 3D Spherical Top-Hat Mean Density and f_L Factor Calculation (`sparkto_b_raw.py` / `sparc_tophat.py`):**

```
python3 sparkto_b_raw.py --galaxies NGC3198,UGC02487,UGC03546 --skip_checks --out ./output_tophat_check
```

4. **V20 Eight-Coil Magnetic-Field Experiment, Elliptic Biot–Savart, $r = L/t$ Exterior Quadrature, and $L = 0.95$ m Identity (`make_revision3_checks.py`):**

```
python3 make_revision3_checks.py . ./csv/revision3_checks_reproduced.json
```

5. **V23 Covariant Action Closure, Bianchi Residuals, Coupled TOV Stability BVP, Sirius B, and 8558 kg Granite Chamber (`covariant_closure_50aa2db0.py`):**

```
python3 covariant_closure_50aa2db0.py
```

6. **Extended Observational Tests at Fixed $L = 10$ km, $\alpha_Y = 1$: GW170817 & Lensing (`step23`), Bullet Cluster (`step24`), and Cosmology & CMB (`step25`):**

```
python3 gw170817_and_lensing_audit.py
python3 bullet_cluster_lensing_audit.py
python3 cmb_and_cosmology_audit.py
```

   - *Expected Output (~5–10 s total):*
     - Generates and synchronizes `gw170817_dispersion_and_cT.csv`, `lensing_deflection_and_ppn_slip.csv`, `gw170817_and_lensing_summary.json`, `bullet_cluster_convergence_profile.csv`, `bullet_cluster_peak_c` `bullet_cluster_audit_summary.json`, `cosmological_background_models.csv`, `cmb_geff_and_acoustic_sca` and `cmb_and_cosmology_summary.json` across the root directory, `code/`, `csv/`, and `data/v23/`.

7. **SHA-256 Integrity Verification of Steps 1–25 and All Package Files:**

```
sha256sum -c SHA256SUMS.txt
```

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2 Step 1: Mathematical Equivalence of the Form of the TAL Factor $f_L = \exp(-\gamma_L \bar{\rho}_L)$ and the String-Theory Dilaton Coupling

Source file: `step1_dilaton_form_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step1_dilaton_form_en.md` (English translation of `step1_dilaton_form.md`)
License: CC BY-NC 4.0
Local Archival Sources:

- V23 (`50aa2db0`): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50` & `tal_covariant_closure_50aa2db0_HU_2026-10-24.tex` (DOI: `10.5281/zenodo.23639084`)
- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex` & `tal_book_HU_2026-10-24.tex` (DOI: `10.5281/zenodo.23604423`)
- V19 (model 7): 3D spherical top-hat law (DOI: `10.5281/zenodo.23497128`)
- Local working note: `docs/C_derivation_status.md` (line 15: `Time_Attraction.pdf (v1-v7)`)

2.1 1. Verbatim definitions of the Time Attraction Law (TAL) from the local manuscripts

In the local V19–V23 manuscripts, the positive-definite exponential time-flow / damping factor $f_L(\mathbf{x})$, the 3D spherical top-hat mean density $\bar{\rho}_L(\mathbf{x})$, and the scale-dependent coupling coefficient γ_L (together with the coupling constant C) appear verbatim in the following form:

2.1.1 1.1. The exponential law and the definition of the constants (V22 `hu_p1.tex`, lines 141–144 and 245–253)

- **Exponential time-flow factor:**

$$f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\,\bar{\rho}_L(\mathbf{x})\big), \qquad 0 < f_L(\mathbf{x}) \le 1$$

- **3D spherical top-hat mean density (with smoothing radius L and volume $V_L = \frac{4}{3}\pi L^3$):**

$$\bar{\rho}_L(\mathbf{x}) \;=\; \frac{3}{4\pi L^3}\int_{|\mathbf{y}-\mathbf{x}|\le L} \rho(\mathbf{y})\, d^3y \;\ge\; 0$$

- **Fundamental gravitational–temporal constants (in SI units, CODATA $G = 6{,}67430\times10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $c = 2{,}99792458\times10^{8}\ \mathrm{m\,s^{-1}}$):**

$$C \;\equiv\; \frac{4\pi G}{c^2} \;=\; 9{,}33306\times10^{-27}\ \mathrm{m\,kg^{-1}}, \qquad \frac{G}{c^2} \;=\; \frac{C}{4\pi} \;=\; 7{,}425915\times10^{-28}\ \mathrm{m\,kg^{-1}}$$

$$\gamma_L \;=\; \frac{G\,L^2}{c^2} \;=\; \frac{C\,L^2}{4\pi} \;=\; 7{,}425915\times10^{-28}\,L^2\ \mathrm{m^3\,kg^{-1}}$$

2.1.2 1.2. The covariant auxiliary scalar and the matter-Lagrangian coupling in V23 (`tal_covariant_closure_50aa2` lines 32–46 and 54; `C_derivation_status.md`, line 15)

- The early `Time_Attraction.pdf` (v1–v7, cited in line 15 of `docs/C_derivation_status.md`) and the V23 `50aa2db0` manuscript (Section 1, Eq. 1–3 and line 54) weight the matter action by the exponential factor f_L:

$$S \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_X(g_{\mu\nu},\phi_L,\lambda_L) + \mathcal{L}_{\mathrm{matter}}\right], \qquad \mathcal{L}_{\mathrm{fluid}}(n,\phi_L) \;=\; -f_L(\phi_L)\,\varepsilon(n)$$

or, in the formulation of versions 1–7:

$$S \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + f_L\,\mathcal{L}_m\right]$$

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- In Formulation I of V23 (Eq. 2–3), the dimensionless scalar $\phi_L(x)$ is tied by the Lagrange multiplier $\lambda_L(x)$ to the 3D spherical top-hat average taken on the spatial hypersurface Σ_u ($K_L(x,y) = \delta(\tau_u(x) - \tau_u(y))\,\Theta(L - d_\Sigma(x,y))/V_L(x)$):

$$\frac{\delta S}{\delta \lambda_L(x)} = 0 \quad \Longrightarrow \quad \phi_L(x) \;=\; \gamma_L\,\bar{\rho}_L(x), \qquad f_L(x) \;=\; e^{-\phi_L(x)} \;=\; \exp\!\big(-\gamma_L\,\bar{\rho}_L(x)\big).$$

2.2 2. The low-energy effective action of string theory and the conventions of the dilaton coupling

In the low-energy effective field theory of string theory (bosonic and superstring), reduced to 4 dimensions, the gravitational sector necessarily contains a scalar field, the **dilaton (ϕ)** (see e.g. Polchinski: *String Theory*, Vol. I, Sec. 3.7; Gasperini & Veneziano, *Phys. Rept.* **373**, 1, 2003; Damour & Polyakov, *Nucl. Phys. B* **423**, 532, 1994).

1. **String frame:** At tree level (weighted by the square of the string coupling constant $g_s = e^{\phi}$, i.e. by $g_s^{-2} = e^{-2\phi}$), the effective action of the gravitational and closed/open string sector is:

$$S_{\text{string}} \;=\; \int d^4x\sqrt{-g_{(s)}}\,e^{-2\phi}\left[\frac{c^4}{16\pi G_0}\Big(R_{(s)} + 4\,g^{\mu\nu}_{(s)}\partial_\mu\phi\,\partial_\nu\phi\Big) - \dots\right] + \int d^4x\sqrt{-g_{(s)}}\,e^{-\eta\phi}\,\mathcal{L}_m\big(g_{(s)\mu\nu},\Psi\big)\,,$$

   where:

   - in the closed-string / tree-level sector $\eta = 2$, i.e. the matter Lagrangian is multiplied directly by the exponential factor $e^{-2\phi}\,\mathcal{L}_m$;
   - in the open-string / D-brane (Born–Infeld) sector $\eta = 1$, i.e. the coupling factor is $e^{-\phi}\,\mathcal{L}_m$.

2. **Einstein frame — exponential matter coupling alongside a canonical R term:** In order for the factor in front of the curvature scalar R (as in TAL V23 Eq. (1): $\frac{c^4}{16\pi G}R$) to take a canonical, ϕ-independent form, we perform the conformal rescaling $g_{(s)\mu\nu} = e^{2\phi}g_{\mu\nu}$ (Damour & Polyakov 1994; Fujii & Maeda: *The Scalar-Tensor Theory of Gravitation*, Cambridge Univ. Press, 2003). The curvature term then becomes purely Einstein–Hilbert in form, while an exponential dilaton multiplier appears in the matter sector:

$$S_{\text{Einstein}} \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}\Big(R - 2\,g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi\Big) \;+\; e^{-2\beta_{\text{dil}}\phi}\,\mathcal{L}_m(\Psi, g_{\mu\nu})\right],$$

   (where, depending on the convention and the matter sector, the usual notation of the exponential multiplier is $e^{-2\phi}$, $e^{-\phi}$, or in general scalar–tensor notation $e^{-2\beta_{\text{dil}}\phi}$).

2.3 3. The formal mathematical correspondence (Step 1 Mapping)

Comparing the TAL matter-coupling term ($f_L(\mathbf{x})\,\mathcal{L}_m$, where $f_L(\mathbf{x}) = e^{-\phi_L(\mathbf{x})} = \exp(-\gamma_L\bar{\rho}_L(\mathbf{x}))$) with the exponential matter coupling of the string-theory dilaton ($e^{-2\phi}\,\mathcal{L}_m$ in the 2ϕ convention, or $e^{-\phi}\,\mathcal{L}_m$ in the 1ϕ convention), the **mathematical functional form of the two expressions is exactly the same**:

$$\boxed{f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\,\bar{\rho}_L(\mathbf{x})\big) \;=\; e^{-\phi_L(\mathbf{x})} \quad\Longleftrightarrow\quad e^{-2\phi(\mathbf{x})} \;\;(\text{or } e^{-\phi(\mathbf{x})})}$$

i.e. the direct correspondence of the dimensionless exponents:

$$\boxed{\phi_L(\mathbf{x}) \;\equiv\; \gamma_L\,\bar{\rho}_L(\mathbf{x}) \;=\; \frac{GL^2}{c^2}\,\bar{\rho}_L(\mathbf{x}) \quad\Longleftrightarrow\quad 2\phi(\mathbf{x}) \;\;(\text{in the } e^{-2\phi} \text{ convention}),\ \text{or } \phi(\mathbf{x}) \;\;(\text{in the } e^{-\phi} \text{ convention}).}$$

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2.3.1 Open question recorded for later steps (not resolved in this step)

- **The similarity:** In both theories, a negative exponential function of a dimensionless scalar quantity ($e^{-\phi_L}$ and $e^{-2\phi}$, respectively) multiplies the matter Lagrangian.
- **The difference (open question for Step 2 and beyond):** In the basic definition of the Time Attraction Law (and in Formulation I of V23), the exponent $\phi_L(\mathbf{x}) = \gamma_L \bar{\rho}_L(\mathbf{x})$ is determined algebraically / via a nonlocal integral directly by the mass-energy density $\bar{\rho}_L$ averaged over the scale L, whereas in string theory the dilaton ϕ is an independent dynamical field with its own kinetic term ($\propto (\partial\phi)^2$). The connection between this dynamical kinetic term and the local Helmholtz–Yukawa operator of Formulation II of V23, $(-L^2\Box_g + 1)\phi_L = \alpha_Y \gamma_L \rho_{\rm src}$, is examined in later steps; we do not touch it here.

2.4 4. Checks

2.4.1 4.1. Dimensional analysis (inverse-density dimension of γ_L)

We check whether the exponent $\gamma_L \bar{\rho}_L$ is dimensionless in SI base units $(\mathrm{m}, \mathrm{kg}, \mathrm{s})$:

- Dimension of Newton's gravitational constant:
$$[G] \;=\; \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$$
- Dimension of the square of the speed of light:
$$[c^2] \;=\; \mathrm{m}^2\,\mathrm{s}^{-2}$$
- Dimension of the mass–length ratio (G/c^2, or $C = 4\pi G/c^2$):
$$\left[\frac{G}{c^2}\right] \;=\; \frac{\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}}{\mathrm{m}^2\,\mathrm{s}^{-2}} \;=\; \mathrm{m}\,\mathrm{kg}^{-1}$$
- Dimension of the square of the smoothing scale:
$$[L^2] \;=\; \mathrm{m}^2$$
- Consequently, the dimension of the coupling coefficient $\gamma_L = \frac{GL^2}{c^2}$:
$$[\gamma_L] \;=\; \left[\frac{G}{c^2}\right]\cdot[L^2] \;=\; \left(\mathrm{m}\,\mathrm{kg}^{-1}\right)\cdot \mathrm{m}^2 \;=\; \mathrm{m}^3\,\mathrm{kg}^{-1} \;=\; \frac{1}{\mathrm{kg}\,\mathrm{m}^{-3}} \;=\; \frac{1}{[\text{density}]} \quad \checkmark$$
- Since the dimension of the 3D volumetric mass density is $[\bar{\rho}_L] = \mathrm{kg}\,\mathrm{m}^{-3}$, the product is:
$$[\gamma_L\,\bar{\rho}_L] \;=\; \left(\mathrm{m}^3\,\mathrm{kg}^{-1}\right)\cdot\left(\mathrm{kg}\,\mathrm{m}^{-3}\right) \;=\; 1 \quad (\text{dimensionless}) \quad \checkmark$$

(Note for the written-out check: the ratio $[G/c^2]$ by itself is $m\,kg^{-1}$ [the reciprocal of a linear mass density], and multiplication by the factor L^2 [m^2] makes the dimension of $\gamma_L = GL^2/c^2$ equal to $m^3\,kg^{-1}$, which is exactly the reciprocal of the 3D volumetric density [$kg\,m^{-3}$], consistent with Eq. (3) and Eq. (5) of V22 `hu_p1.tex`*.)*

2.4.2 4.2. Checks of limits and signs

1. **Vacuum limit ($\bar{\rho}_L \to 0$):** When the averaged mass-energy density in the neighbourhood of radius L vanishes ($\bar{\rho}_L \to 0$), the exponent $\phi_L = \gamma_L \bar{\rho}_L \to 0$, so:
$$\lim_{\bar{\rho}_L \to 0} f_L(\bar{\rho}_L) \;=\; \exp(0) \;=\; 1 \quad \checkmark$$
This exactly recovers the unmodified standard matter coupling ($\mathcal{L}_{\rm matter} \to 1 \cdot \mathcal{L}_m$), which in the dilaton picture corresponds to the asymptotic vacuum $\phi \to 0$ ($e^{-2\phi} \to 1$).
2. **Positive-density domain ($\bar{\rho}_L > 0$):** Since $G > 0$, $c > 0$, $L > 0 \implies \gamma_L > 0$, and the 3D spherical top-hat average is strictly positive ($\bar{\rho}_L > 0$) under the non-negative energy condition ($\rho(\mathbf{y}) \geq 0$), we have $\gamma_L \bar{\rho}_L > 0$, therefore:
$$0 \;<\; f_L(\bar{\rho}_L) \;=\; \exp(-\gamma_L \bar{\rho}_L) \;<\; 1 \qquad (\forall\, \bar{\rho}_L > 0) \quad \checkmark$$
and the function is strictly monotonically decreasing:
$$\frac{df_L}{d\bar{\rho}_L} \;=\; -\gamma_L \exp(-\gamma_L \bar{\rho}_L) \;<\; 0 \quad \checkmark$$

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3 Step 2: Conformal, Disformal, and Effective Fluid-Coupled Jordan Metrics ($\tilde{g}_{\mu\nu}$) and the Taxonomy of V19–V23 Model Branches

Source file: step2_conformal_disformal_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step2_conformal_disformal_en.md
Previous Step: step1_dilaton_form_en.md (step1_dilaton_form.md)
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Local Archival Sources:

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_co (Sections 1, 1.1, 1.2, and 3, Eqs. 1–12; DOI: 10.5281/zenodo.23639084)
- V23 (50aa2db0 English manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_cova
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex and hu_p2.tex (DOI: 10.5281/zenodo.23604423)
- V23 SymPy verification script: covariant_closure_50aa2db0.py (lines 105–138)

3.1 1. Verbatim Citations from the V23 Manuscript (tal_covariant_closure_50aa2db0_EN_2026- / HU)

3.1.1 1.1. Formulation of the Covariant Action (V23 EN/HU Section 1, Eq. (1), lines 33–35)

Section 1 of the V23 manuscript (*"Covariant Action Principle: Two Explicit Formulations"*) establishes the following covariant action using the $(-+++)$ metric signature convention:

$$S\big[g_{\mu\nu},\,\phi_L,\,\Psi_{\rm matter}\big] \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\,R(g) \;+\; \mathcal{L}_{\rm matter}\big(\Psi_{\rm matter},\,\tilde{g}_{\mu\nu}(g,\phi_L)\big) \;+\; \mathcal{L}_{\phi}\big(g_{\mu\nu},\,\phi_L\big)\right]. \quad \text{(V23 Eq. 1)}$$

3.1.2 1.2. Verbatim Definition of the Microscopic H_0 vs. H_1 Coupling and the Two Effective Jordan Metrics (V23 Section 3, Eqs. (11)–(12), lines 95–101)

Section 3 of the V23 manuscript (*"Microscopic Clock and Spectral-Line Coupling from the Action (H_0 vs. H_1)"*) defines the two coupling branches and the two Jordan metrics verbatim as follows: > *"In the covariant action (1), the microscopic coupling of atomic clocks and electromagnetic spectral lines is uniquely determined by which metric contracts the indices of the Dirac + Maxwell Lagrangian $\mathcal{L}_{atom}(\psi_e, A_\mu)$: >* - **Minimal physical-metric coupling (H_0 — default covariant closure):** *$\mathcal{L}_{atom}(\psi_e, A_\mu; g_{\mu\nu})$ couples directly to the physical spacetime metric $g_{\mu\nu}$, while the macroscopic baryonic energy–momentum source is weighted by f_L through Formulation I or II. Then the proper time of a stationary atom is $d\tau_{H_0} = \sqrt{-g_{00}}\,dt = e^{\nu(r)}dt$, the local transition frequencies $\omega_{atom} = m_e c^2\alpha_{EM}^2/(2\hbar)$ are spacetime constants, and atomic clocks measure precisely the metric lapse function arising from the f_L-modified Einstein equations. The macroscopic clock anomaly appears exclusively through the gravitational potential change $\Delta\Phi_N$ originating from the modified mass: >*

$$\left.\frac{\Delta\nu}{\nu}\right|_{H_0} \;=\; \frac{\Delta\Phi_{\rm metric}}{c^2}, \qquad \Delta z_{H_0}(\text{Sirius B}) \;=\; z_{\rm GR}\left(\frac{\Delta M_B}{M_B}\right)_{\rm TAL}. \quad \text{(V23 Eq. 11)}$$

> - **Disformal / conformal proper-time coupling (H_1 — direct clock-coupling hypothesis):** *$\mathcal{L}_{atom}(\psi_e, A_\mu; \tilde{g}_{\mu\nu})$ couples to the effective metric >*

$$\tilde{g}^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} \;+\; \frac{1-f_L(\phi_L)^2}{c^2}\,u_\mu u_\nu \qquad \text{or} \qquad \tilde{g}^{\rm conf}_{\mu\nu} \;=\; f_L(\phi_L)^2\,g_{\mu\nu} \quad \text{(V23 Eq. 12)}$$

> *(with fixed m_e), so that the proper-time element is $d\tilde{\tau}_{H_1} = \sqrt{-\tilde{g}_{00}}\,dt = f_L(\mathbf{x})\,e^{\nu(\mathbf{x})}dt$. This induces a direct non-metric frequency shift: >*

$$\left.\frac{\Delta\nu}{\nu}\right|_{H_1}(\mathbf{x}) \;=\; \frac{\Delta\Phi_{\rm metric}(\mathbf{x})}{c^2} \;+\; \big(f_L(\mathbf{x})-1\big) \;\simeq\; \frac{\Delta\Phi_{\rm metric}(\mathbf{x})}{c^2} \;-\; \phi_L(\mathbf{x})."$$

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3.2 2. Derivation of the Conformal ($\tilde{g}^{\rm conf}_{\mu\nu}$) Transformation and the Brans–Dicke Relation

In Step 1 (`step1_dilaton_form.md`), we proved that $f_L(\phi_L) = e^{-\phi_L}$. Substituting this into the second equation of V23 Eq. (12):

$$\boxed{\tilde{g}^{\rm conf}_{\mu\nu}(x) \;=\; f_L(\phi_L(x))^2\, g_{\mu\nu}(x) \;=\; e^{-2\phi_L(x)}\, g_{\mu\nu}(x), \qquad \Omega(\phi_L) \;\equiv\; f_L(\phi_L) \;=\; e^{-\phi_L}.}$$

3.2.1 2.1. Inverse Metric, Determinant, and Proper-Time Element

1. **Inverse conformal metric:** Since $\tilde{g}^{\rm conf}_{\mu\alpha}\tilde{g}^{\alpha\nu}_{\rm conf} = \delta^\nu_\mu$,
$$\tilde{g}^{\mu\nu}_{\rm conf} \;=\; f_L(\phi_L)^{-2}\, g^{\mu\nu} \;=\; e^{+2\phi_L}\, g^{\mu\nu}.$$
2. **Four-volume element (determinant):** In $D = 4$ spacetime dimensions, $\det(e^{-2\phi_L} g_{\mu\nu}) = (e^{-2\phi_L})^4 \det(g_{\mu\nu})$, therefore:
$$\sqrt{-\tilde{g}_{\rm conf}} \;=\; f_L(\phi_L)^4 \sqrt{-g} \;=\; e^{-4\phi_L}\sqrt{-g}.$$
3. **Proper-time element of a static observer ($dx^i = 0$):** In a static spherically symmetric physical metric ($g_{00} = -e^{2\nu(r)}$):
$$d\tilde{\tau}^2_{\rm conf} \;=\; -\frac{1}{c^2}\tilde{g}^{\rm conf}_{00}\, dt^2 \;=\; -f_L^2\,\frac{g_{00}}{c^2}\, dt^2 \;=\; f_L(r)^2\, e^{2\nu(r)}\, dt^2 \quad\Longrightarrow\quad \boxed{d\tilde{\tau}_{\rm conf} \;=\; f_L(r)\, e^{\nu(r)}\, dt \;=\; e^{\nu(r)-\phi_L(r)}\, dt.}$$

3.2.2 2.2. Conformal Transformation of the Ricci Scalar (Jordan Frame ↔ Einstein Frame)

From standard $D = 4$ conformal geometry (e.g., Wald 1984, Appendix D; Fujii & Maeda 2003), if $g_{\mu\nu} = \Omega^{-2}\tilde{g}^{\rm conf}_{\mu\nu} = e^{+2\phi_L}\tilde{g}^{\rm conf}_{\mu\nu}$ (that is, $\tilde{g}^{\rm conf}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ with $\Omega = e^{-\phi_L}$), the relation between the two Ricci scalars is:

$$R(g) \;=\; \Omega^2\left[\tilde{R}(\tilde{g}_{\rm conf}) \;+\; 6\,\tilde{\Box}\ln\Omega \;-\; 6\,\tilde{g}^{\mu\nu}_{\rm conf}\,\partial_\mu \ln\Omega\,\partial_\nu\ln\Omega\right].$$

Accordingly, rewriting the Einstein–Hilbert term in the Jordan frame ($\sqrt{-g} = \Omega^{-4}\sqrt{-\tilde{g}_{\rm conf}}$) yields:

$$\int d^4x\,\sqrt{-g}\,\frac{c^4}{16\pi G}\,R(g) \;=\; \int d^4x\,\sqrt{-\tilde{g}_{\rm conf}}\,\frac{c^4}{16\pi G}\left[\Omega^{-2}\tilde{R}(\tilde{g}_{\rm conf}) \;-\; 6\,\Omega^{-4}\tilde{g}^{\mu\nu}_{\rm conf}\partial_\mu\Omega\,\partial_\nu\Omega\right],$$

where the boundary divergence term $\nabla_\mu(\sqrt{-g}\,6\Omega\nabla^\mu\Omega)$ has been dropped via integration by parts. Introducing the **Brans–Dicke scalar field**:

$$\boxed{\Phi_{\rm BD}(x) \;\equiv\; \Omega(x)^{-2} \;=\; f_L(x)^{-2} \;=\; e^{+2\phi_L(x)},}$$

for which $\frac{\partial_\mu\Phi_{\rm BD}}{\Phi_{\rm BD}} = +2\partial_\mu\phi_L$, the local scalar kinetic term of V23 Formulation II ($\mathcal{L}_\phi = -\frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\frac{1}{2}g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right]$, V23 Eq. 8) in the Jordan frame becomes:

$$-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L \;\longrightarrow\; -\frac{\omega_{\rm BD}}{\Phi_{\rm BD}}\,\tilde{g}^{\mu\nu}_{\rm conf}\partial_\mu\Phi_{\rm BD}\,\partial_\nu\Phi_{\rm BD}.$$

Matching the coefficients yields the standard scalar–tensor identity:

$$2\omega_{\rm BD}+3 \;=\; \frac{1}{\tilde{\alpha}_L} \;=\; \frac{1}{2\beta^2} \quad\Longleftrightarrow\quad \boxed{\omega_{\rm BD} \;=\; \frac{1}{2\tilde{\alpha}_L} - \frac{3}{2} \;=\; \frac{1}{4\beta^2} - \frac{3}{2}.}$$

(The exact numerical values of $\tilde{\alpha}_L$, $\alpha_Y = 8\pi\tilde{\alpha}_L$, $\alpha_{std} = 2\tilde{\alpha}_L = \frac{\alpha_Y}{4\pi}$, and ω_{BD}, together with the Cassini PPN bound, are analyzed in detail in Step 5.)

3.3 3. The Disformal ($\tilde{g}^{\rm disf}_{\mu\nu}$) Coupling and Algebraic Projector Decomposition

The first equation of V23 Eq. (12) defines a purely timelike **Bekenstein-type disformal metric** (Bekenstein 1993, *Phys. Rev. D* 48, 3641) using the four-velocity u^μ of the medium ($g_{\mu\nu}u^\mu u^\nu = -c^2$, $u_\mu = g_{\mu\nu}u^\nu$):

$$\boxed{\tilde{g}^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} \;+\; \frac{1-f_L(\phi_L)^2}{c^2}\,u_\mu u_\nu \;=\; g_{\mu\nu} \;+\; \frac{1-e^{-2\phi_L}}{c^2}\,u_\mu u_\nu.}$$

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3.3.1 3.1. Exact Projector Decomposition (SymPy-Verified Identity, `covariant_closure_50aa2db0.py` lines 108–125)

Let us introduce the spatial projector of the spacetime metric orthogonal to u^μ:

$$h_{\mu\nu} \;\equiv\; g_{\mu\nu} + \frac{1}{c^2} u_\mu u_\nu \qquad \big(h_{\mu\nu}u^\nu = 0, \quad h^\mu{}_\alpha h^\alpha{}_\nu = h^\mu{}_\nu, \quad h^\mu{}_\mu = 3\big)\,.$$

Then $g_{\mu\nu} = h_{\mu\nu} - \frac{1}{c^2}u_\mu u_\nu$, so the separated projector form of the disformal metric is:

$$\boxed{\tilde{g}^{\rm disf}_{\mu\nu} \;=\; h_{\mu\nu} \;-\; f_L(\phi_L)^2\,\frac{u_\mu u_\nu}{c^2} \;=\; \Big(g_{\mu\nu} + \frac{u_\mu u_\nu}{c^2}\Big) \;-\; e^{-2\phi_L}\,\frac{u_\mu u_\nu}{c^2}.}$$

From this, the physical interpretation follows immediately:

1. **The disformal coupling leaves the spatial geometry untouched:** for every spatial displacement δx^μ orthogonal to u^μ ($u_\mu \delta x^\mu = 0$), $\tilde{g}^{\rm disf}_{\mu\nu}\delta x^\mu \delta x^\nu = g_{\mu\nu}\delta x^\mu \delta x^\nu$.
2. **It scales the timelike (lapse) component by precisely the factor $f_L(\phi_L) = e^{-\phi_L}$:** the proper time of a clock comoving with the medium ($\delta x^\mu = \frac{u^\mu}{c} d\tau$) is:

$$d\tilde{\tau}^2_{\rm disf} \;=\; -\frac{1}{c^2}\tilde{g}^{\rm disf}_{\mu\nu}\,dx^\mu dx^\nu \;=\; f_L(\phi_L)^2\left(-\frac{1}{c^2}g_{\mu\nu}dx^\mu dx^\nu\right) \;=\; f_L(\phi_L)^2\,d\tau^2.$$

3. **Inverse disformal metric and determinant:** Since $h^{\mu\alpha}u_\alpha = 0$ and $\big(-\frac{u_\mu u_\alpha}{c^2}\big)\big(-\frac{u^\alpha u^\nu}{c^2}\big) = -\frac{u_\mu u^\nu}{c^2}$, the inverse metric and volume element are exactly:

$$\boxed{\tilde{g}^{\mu\nu}_{\rm disf} \;=\; h^{\mu\nu} \;-\; f_L(\phi_L)^{-2}\,\frac{u^\mu u^\nu}{c^2} \;=\; g^{\mu\nu} \;-\; \frac{f_L(\phi_L)^{-2}-1}{c^2}\,u^\mu u^\nu, \qquad \sqrt{-\tilde{g}_{\rm disf}} \;=\; f_L(\phi_L)\,\sqrt{-g} \;=\; e^{-\phi_L}\sqrt{-g}.}$$

3.3.2 3.2. General Static Spherically Symmetric Background and `SymPy` Verification

In a static spherically symmetric physical background metric:

$$ds^2 \;=\; g_{\mu\nu}dx^\mu dx^\nu \;=\; -e^{2\nu(r)}c^2dt^2 + e^{2\lambda(r)}dr^2 + r^2 d\Omega^2, \qquad g_{00} = -e^{2\nu(r)},$$

the four-velocity of a stationary (comoving) observer is $u^\mu = \big(c/\sqrt{-g_{00}},\,0,\,0,\,0\big) = \big(c\,e^{-\nu(r)},\,0,\,0,\,0\big)$, and its covariant form is $u_\mu = g_{\mu\nu}u^\nu = \big(-\sqrt{-g_{00}}\,c,\,0,\,0,\,0\big) = \big(-c\,e^{\nu(r)},\,0,\,0,\,0\big)$. Thus:

$$\frac{u_0 u_0}{c^2} \;=\; -g_{00} \;=\; +e^{2\nu(r)},$$

and substituting this yields the $(0,0)$-component of the disformal metric:

$$\tilde{g}^{\rm disf}_{00} \;=\; g_{00} \;+\; \big(1 - f_L(r)^2\big)\frac{u_0 u_0}{c^2} \;=\; g_{00} \;+\; \big(1-f_L(r)^2\big)\big(-g_{00}\big) \;=\; f_L(r)^2\,g_{00} \;=\; -f_L(r)^2\,e^{2\nu(r)}.$$

In `covariant_closure_50aa2db0.py` (lines 108–125), within the local orthonormal tetrad ($g_{\hat{a}\hat{b}} = \eta_{\hat{a}\hat{b}} = \mathrm{diag}(-1,1,1,1)$, $u^{\hat{a}} = (c,0,0,0)$, $u_{\hat{a}} = (-c,0,0,0)$), the same identity is verified: $\tilde{g}^{\rm disf}_{\hat{0}\hat{0}} = -1 + (1 - f_L^2) = -f_L^2$, so the general coordinate expression is $\tilde{g}^{\rm disf}_{00} = e^{\hat{0}}{}_0 e^{\hat{0}}{}_0 \tilde{g}^{\rm disf}_{\hat{0}\hat{0}} = e^{2\nu}(-f_L^2) = f_L^2 g_{00}$.

3.4 4. Explicit Taxonomy of V19–V23 Model Branches (Branch Taxonomy Table)

For peer-review transparency, the table below unambiguously separates the model branches appearing across the V19–V23 manuscripts and the present derivation chain, including their action principle/definition, volume element, hydrostatic equation, clock/light coupling, and domain of validity:

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Branch Code	Name and Manuscript Source	Scalar / Kernel Definition and 2nd Moment	Matter Coupling / Volume Element	Internal Hydrostatic Equilibrium	Electromagnetic / Clock / PPN Behavior
Branch I-A (`tophat_mass`, `tophat_gradient`)	V21 phenomenological 3D spherical top-hat TOV (`stellar_tal_v21.py`, V23 Eqs. 4–5)	Nonlocal 3D spherical window: $\phi_L = \gamma_L(K_L * \varepsilon)$, $K_L = \frac{3}{4\pi L^3}\Theta(L-r)$, $1 + \frac{L^2}{10}\nabla^2 + \dots$	Phenomenological f_L multiplier in m' (`mass`) or ν' (`gradient`), with algebraic $T^\mu_\nu(X)$ closure	$h'(r) = -\nu'(r)$ (`mass`: $\nu' = A_{\rm geo}$; `gradient`: $\nu' = f_L A_{\rm geo}$)	H_0: minimal physical metric; H_1: direct f_L clock shift (excluded by Sirius B for $L \geq 1$ km)
Branch I-B (`bilateral_eom`, `bilateral_lapse_weighted`)	V23 Formulation I: bilateral weighted kernel action (V23 Eqs. 2–3, 6–7)	Symmetric two-point kernel $K_L(x,y) = K_L(y,x)$, $1 + \frac{L^2}{10}\nabla^2 + \dots$	$S_{{\rm I},m} = -\int d^4x\sqrt{-g}\, e^{-\phi_L}\varepsilon$ (in fixed-metric Euclidean kernel approximation)	$V_{\rm eff}(r) = f_L(r) - \gamma_L \int K_L f_L \varepsilon$ with reciprocity force	Macroscopic hydrostatic test branch; metric variation inside the kernel is not closed
Branch II-Fluid (`yukawa_action_el`)	V23 Formulation II: local Yukawa-screened Einstein–scalar–fluid action (V23 Eqs. 8–10)	Local Helmholtz–Yukawa PDE: $(-\Box_g + L^{-2})\phi_L = \frac{\alpha_Y G}{c^2} e^{-\phi_L}\varepsilon$, $1 + L^2\nabla^2 + \dots$	$S_m = -\int d^4x\sqrt{-g}\, e^{-\phi_L}\varepsilon(n)$, $P_{\rm matter} = e^{-\phi_L}P$, $\rho_\phi = \frac{B(\phi_L')^2 + \phi_L^2/L^2}{2\alpha_Y}$	$h'(r) = \nu'(r) + \phi_L'(r)$ (fifth force + curved exterior scalar tail $M_{\rm ADM}$)	Under H_0, atomic clocks couple to physical $g_{\mu\nu}$; turning points of $M_{\rm ADM}$ and M_B coincide
Branch II-Conf ($\tilde{g}^{\rm conf}_{\mu\nu} = e^{-2\phi_L} g_{\mu\nu}$)	Universal conformal Jordan–Brans–Dicke / dilaton metric (V23 Eq. 12b, `step1`, `step5`)	Same local Yukawa PDE; $\Phi_{\rm BD} = e^{+2\phi_L}$	$\sqrt{-\tilde{g}_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$; conformally invariant Maxwell action ($\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu}\sqrt{-\tilde{g}} = F_{\mu\nu}F^{\mu\nu}\sqrt{-g}$)	In the non-relativistic limit ($\varepsilon \simeq \rho_0 c^2 \gg 3P$), coincides with the $\beta = \sqrt{\alpha_{\rm std}/2}$ fifth force	Standard conformal Brans–Dicke / PPN dictionary applies to this branch: $\alpha_{\rm std} = \frac{\alpha_Y}{4\pi}$, $\omega_{\rm BD} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.07$ ($\alpha_Y = 1$), light deflection is conformally invariant
Branch II-Disf ($\tilde{g}^{\rm disf}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$)	Comoving timelike disformal clock and EM metric (V23 Eq. 12a, V20/V22 solenoid, `step14`–`step15`)	Same local Yukawa PDE, augmented with magnetic source $\rho_B = \frac{\lvert\mathbf{B}\rvert^2}{2\mu_0 c^2}$	$\sqrt{-\tilde{g}_{\rm disf}} = f_L\sqrt{-g} = e^{-\phi_L}\sqrt{-g}$ (coincides with the $e^{-\phi_L}$ weight of Branch II-Fluid!)	If u^μ is fixed by a comoving khronon/aether field, energy weight is $e^{-\phi_L}$; free u^μ variation yields an extra constraint	Refractive index $n_{\rm refr} = e^{+\phi_L}$ (refractive light deflection), magnetic solenoid source (ρ_B); **not** described by the conformal Brans–Dicke PPN formula
Branch II-Cham ($\beta_{\rm eff}(\rho)$ extension)	Density-dependent chameleon / symmetron nonlinear EFT hypothesis (`step11`)	Density-dependent effective mass $m^2_{\rm eff}(\rho)$ and coupling $\beta_{\rm eff}(\rho)$	Nonlinear $V(\phi) + A(\phi)\rho$ extension beyond the quadratic constant-α_Y V23 action	Environment-dependent fifth force (strong screening in laboratory/NS, enhancement in dilute galactic medium)	**Standalone extension hypothesis**, not a consequence of the constant-α_Y V23 `yukawa_action_el` equations

3.5 5. Checks

1. **Limit Check ($\phi_L \to 0$, i.e., $f_L \to 1$):**

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- When $\phi_L = 0$ ($f_L = 1$), $\tilde{g}^{\rm conf}_{\mu\nu} = g_{\mu\nu}$ and $\tilde{g}^{\rm disf}_{\mu\nu} = g_{\mu\nu} + 0 = g_{\mu\nu}$, so both Jordan metrics reduce to the physical Einstein metric $g_{\mu\nu}$, and $d\tilde{\tau}_{H_1} = d\tau_{H_0}$ ✓.

2. **Proper-Time Agreement for Stationary Clocks (along u^μ):**

   - For a clock comoving with the baryonic medium ($dx^\mu \parallel u^\mu$), the conformal and disformal metrics yield the **identical** proper-time element and frequency shift:

$$d\tilde{\tau}_{\rm conf} \;=\; d\tilde{\tau}_{\rm disf} \;=\; f_L(x)\, d\tau \;=\; e^{-\phi_L(x)}\, \sqrt{-g_{00}}\, dt,$$

   which is also verified symbolically by `covariant_closure_50aa2db0.py` (`sym_res["proper_time_ratio_H1_over_H0" = "f_L"`, `sym_res["weak_field_H1_clock_shift"] = "Phi_N/c^2 - phi_L"`) ✓.

3. **Difference in the Spatial Sector, Volume Element, and Light Propagation:**

   - Whereas $\sqrt{-\tilde{g}_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$ and the conformal metric leaves null geodesics (light rays) invariant ($\tilde{g}^{\rm conf}_{\mu\nu} k^\mu k^\nu = 0 \iff g_{\mu\nu} k^\mu k^\nu = 0$), the disformal metric satisfies $\sqrt{-\tilde{g}_{\rm disf}} = e^{-\phi_L}\sqrt{-g}$ and induces an effective refractive index ($n = f_L^{-1} = e^{+\phi_L}$) for light while leaving spatial distances unscaled ✓.

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4 Step 3: Mapping the V23 Local Screened Helmholtz–Yukawa Action and Field Equation to the Massive Scalar-Tensor / Dilaton Klein–Gordon Equation

Source file: step3_yukawa_helmholtz_eom_en.md

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Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step3_yukawa_helmholtz_eom_en.md
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md), step2_conformal_disformal_en.md (step2_conformal_disformal.md)
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Local Archival Sources:

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_co (Subsection 1.2, lines 56–65, Eqs. 6–7; DOI: 10.5281/zenodo.23639084)
- V23 (50aa2db0 English manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_cova (Sec. 1.2, lines 56–65, Eqs. 6–7)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (lines 200–206 and 245–254; DOI: 10.5281/zenodo.23604423)
- V23 SymPy verification script: covariant_closure_50aa2db0.py (lines 126–149) and conservation_Residuals.json

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4.1 1. Verbatim Citations from the V23 (50aa2db0) and V22 Manuscripts

In Step 1 (step1_dilaton_form.md), we recorded as an open question how the nonlocal exponent $\phi_L(\mathbf{x}) = \gamma_L \bar{\rho}_L(\mathbf{x})$ becomes a local dynamical scalar field endowed with its own kinetic term. **Subsection 1.2** of the V23 manuscript (*Formulation II: Local Screened Scalar-Tensor (Helmholtz–Yukawa) Action*, lines 56–65, **Eqs. 6–7**) defines this verbatim as follows:

- **Lagrangian and Field Equation of the Scalar Field (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 57–60, Eq. 6):** > *"In Formulation II, I replace the nonlocal integral kernel with a local canonical screened scalar action:"*

$$\mathcal{L}_\phi = -\frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right], \quad \left(-\Box_g + \frac{1}{L^2}\right)\phi_L = \frac{4\pi G}{c^2}\alpha_{\rm std}\,\rho_{\rm src} \iff (-L^2\nabla^2+1)\phi_L = \alpha_Y\gamma_L\rho_{\rm src}. \quad \text{(V23}$$

- **Static Spherically Symmetric Green's Function Solution and Scalar Stress–Energy Tensor (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 61–65, Eq. 7):** > *"In a static spherical system, the exact positive-definite Green's function solution of the radial Helmholtz operator* $-\frac{1}{r^2}\frac{d}{dr}(r^2\phi_L') + L^{-2}\phi_L = \frac{4\pi G}{c^2}\alpha_{std}\rho_{src}$ *(*$\alpha_Y = 4\pi\alpha_{std}$*) is"*

$$\phi_L^{\rm Yuk}(r) = \alpha_Y\,\frac{G}{c^2}\int_0^{R_*} dr'\,\frac{r'L}{r}\,\sinh\frac{r_<}{L}\,e^{-r_>/L}\rho_{\rm src}(r'), \qquad \phi_L^{\rm Yuk}(0) = \alpha_Y\,\frac{G}{c^2}\int_0^{R_*} r' e^{-r'/L}\rho_{\rm src}(r')\,dr', \quad \text{(V23 Eq. 7)}$$

> *"and the variation* $\delta S_\phi/\delta g^{\mu\nu}$ *provides the local covariant stress–energy tensor* $T_{\mu\nu}^{(\phi)} = \frac{c^4}{8\pi G\tilde{\alpha}_L}\big[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\big((\nabla\phi_L)^2 + L^{-2}\phi_L^2\big)\big]$*. (**SymPy:** sym_yukawa_center_limit_residual $= 0$*, no extra L-factor). The unit-normalized kernel corresponds to* $\alpha_Y = 1$*, i.e.,* $\alpha_{std} = 1/(4\pi) \approx 0.0796$*, which recovers the homogeneous top-hat limit (*$\phi_L = \gamma_L\rho_{src}$*);* $\alpha_Y = 1/3$ *(*$\alpha_{std} = 1/(12\pi)$*) is the volume-matched variant."*

- **Action-Consistent (yukawa_action_el) Euler–Lagrange Branch (tal_covariant_closure_50aa2db0_EN_2026-10-2 / HU, line 65):** > *"The* yukawa_action_el *rows follow from the action* $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ *(*$G = c = 1$*; SymPy:* sym_yukawa_el_source_factor *and* _hydrostatic *residual* $= 0$*): *$B\phi'' + (2B/r + B\nu' + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L\varepsilon$*,* $B = 1 - 2m/r$*,* $P_{matter} = f_L P$*,* $h' = -\nu' + \phi'$*,* $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$*, exterior matching* $\phi'(R) = -(1/R + 1/L)\phi(R)$*."*

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4.2 2. Derivation of the Euler–Lagrange Field Equation and Mapping to a Massive Dilaton / Scalar-Tensor Field

4.2.1 2.1. Variation with Respect to ϕ_L (with Matter-Sector Coupling $f_L(\phi_L) = e^{-\phi_L}$)

Consider the local action of V23 Formulation II, where the matter energy density couples with the factor $f_L(\phi_L) = e^{-\phi_L}$ derived in Steps 1 and 2 ($\mathcal{L}_m = -f_L(\phi_L)\,\rho c^2 = -e^{-\phi_L}\varepsilon$):

$$S_{\rm II}[g_{\mu\nu}, \phi_L, \Psi] \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R \;-\; \frac{c^4}{8\pi G\tilde{\alpha}_L}\left(\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right) \;-\; e^{-\phi_L}\,\rho c^2\right].$$

Varying the action with respect to ϕ_L ($\delta S_{\rm II}/\delta\phi_L = 0$):

$$\frac{1}{\sqrt{-g}}\frac{\delta S_{\rm II}}{\delta\phi_L} \;=\; \frac{c^4}{8\pi G\tilde{\alpha}_L}\left(\Box_g\phi_L - \frac{1}{L^2}\phi_L\right) \;+\; e^{-\phi_L}\,\rho c^2 \;=\; 0,$$

where $\Box_g\phi_L = \frac{1}{\sqrt{-g}}\partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi_L\big)$. Rearranging and multiplying by $-\frac{8\pi G\tilde{\alpha}_L}{c^4}$ yields:

$$\left(-\Box_g + \frac{1}{L^2}\right)\phi_L \;=\; \frac{8\pi G}{c^2}\,\tilde{\alpha}_L\,\rho_{\rm src}, \qquad \rho_{\rm src} \;=\; \begin{cases}\rho & \text{(weak-field / linear source, } f_L \simeq 1\text{)},\\ f_L(\phi_L)\,\rho = e^{-\phi_L}\rho & \text{(exact } \texttt{yukawa_action_el} \text{ source)}.\end{cases}$$

4.2.2 2.2. Identification of the Constants ($\tilde{\alpha}_L$, $\alpha_{\rm std}$, α_Y, γ_L, and C)

Comparing term by term with V23 Eq. (6):

1. **Relation between the Poisson and Yukawa normalizations:**
$$\frac{8\pi G}{c^2}\,\tilde{\alpha}_L \;=\; \frac{4\pi G}{c^2}\,\alpha_{\rm std} \;=\; C\,\alpha_{\rm std} \quad\Longrightarrow\quad \tilde{\alpha}_L \;=\; \frac{\alpha_{\rm std}}{2},$$
where $C = \frac{4\pi G}{c^2}$ is the fundamental coupling constant C of the V22 monograph (`hu_p1.tex`, Eq. 4).
2. **Factoring out $\gamma_L = \frac{GL^2}{c^2}$ ($-L^2\Box_g + 1$ form):** Multiplying both sides of the field equation by L^2 (in the quasi-static limit $\Box_g \to \nabla^2$):
$$(-L^2\nabla^2 + 1)\phi_L \;=\; 4\pi\alpha_{\rm std}\left(\frac{GL^2}{c^2}\right)\rho_{\rm src} \;=\; \alpha_Y\,\gamma_L\,\rho_{\rm src}, \qquad \boxed{\alpha_Y \;\equiv\; 4\pi\,\alpha_{\rm std} \;=\; 8\pi\,\tilde{\alpha}_L.}$$
Or in an equivalent notation (as written in Section 1.1 of `README.md`):
$$(\nabla^2 - L^{-2})\phi_L \;=\; -\frac{4\pi G}{c^2}\alpha_{\rm std}\,\rho_{\rm src} \qquad \left(\text{or in the } \phi_L \to -\phi_L \text{ sign convention, } +\frac{\alpha_{\rm std}G}{c^2}\rho_{\rm src} \text{ with the } 4\pi \text{ factor absorbed}\right).$$

4.2.3 2.3. Mapping to the Massive Scalar-Tensor / Dilaton Klein–Gordon Equation

In string-theoretic dilaton gravity or general scalar–tensor gravity (e.g., Damour & Polyakov 1994; Fujii & Maeda 2003; Khoury & Weltman 2004), the field equation of a canonical scalar field of mass m_ϕ with a quadratic self-interaction potential ($V(\varphi) = \frac{1}{2}\mu_\phi^2\varphi^2$, where $\mu_\phi = \frac{m_\phi c}{\hbar}$) is:

$$\left(-\Box_g + \mu_\phi^2\right)\phi_L \;=\; \frac{4\pi G}{c^2}\,\alpha_{\rm std}\,\rho_{\rm src}.$$

From this, the exact dictionary correspondence between the TAL smoothing scale (L) and the effective Compton wavelength and mass of the scalar/dilaton field is:

$$\boxed{\mu_\phi \;\equiv\; \frac{1}{L}, \qquad m_\phi \;=\; \frac{\hbar}{Lc}}$$

- **Physical interpretation:** What the nonlocal formulation of V19/V21 describes as a 3D spherical top-hat windowing of radius L corresponds in V23 Formulation II to the exchange of a **Yukawa-screened scalar/dilaton field with range $\lambda_{\rm Compton} = L$ (i.e., mass $m_\phi = \hbar/(Lc)$)**.

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4.3 3. How Does the Yukawa Solution Recover the 3D Top-Hat Law $\phi_L = \gamma_L \bar{\rho}_L$?

1. **The 3D Yukawa Green's Function:** The fundamental solution of $(-L^2\nabla^2 + 1)G_L(\mathbf{x}-\mathbf{y}) = \delta^{(3)}(\mathbf{x}-\mathbf{y})$ in three-dimensional Euclidean space is:

$$G_L(\mathbf{x}-\mathbf{y}) \;=\; \frac{\exp\!\left(-\frac{|\mathbf{x}-\mathbf{y}|}{L}\right)}{4\pi L^2\,|\mathbf{x}-\mathbf{y}|}, \qquad \int_{\mathbb{R}^3} G_L(\mathbf{r})\,d^3r \;=\; \frac{4\pi}{4\pi L^2}\int_0^\infty r\,e^{-r/L}\,dr \;=\; \frac{L^2}{L^2} \;=\; 1.$$

   Thus, the solution of V23 Eq. (6) for an arbitrary distribution $\rho_{\rm src}(\mathbf{y})$ is a **positive-definite Yukawa smoothing**:

$$\phi_L^{\rm Yuk}(\mathbf{x}) \;=\; \alpha_Y\,\gamma_L \int_{\mathbb{R}^3} G_L(\mathbf{x}-\mathbf{y})\,\rho_{\rm src}(\mathbf{y})\,d^3y \;=\; \alpha_Y\,\frac{G}{c^2}\int_{\mathbb{R}^3}\frac{e^{-|\mathbf{x}-\mathbf{y}|/L}}{4\pi\,|\mathbf{x}-\mathbf{y}|}\,\rho_{\rm src}(\mathbf{y})\,d^3y.$$

2. **Homogeneous / Slowly Varying Density Limit ($\rho_{\rm src}$ = const.):** Since $\int_{\mathbb{R}^3} G_L(\mathbf{r})\,d^3r = 1$, in a homogeneous medium:

$$\phi_L^{\rm Yuk} \;=\; \alpha_Y\,\gamma_L\,\rho_{\rm src}.$$

   Choosing the **unit-normalized kernel** ($\alpha_Y = 1 \iff \alpha_{\rm std} = \frac{1}{4\pi} \approx 0.079577$):

$$\phi_L^{\rm Yuk} \;=\; \gamma_L\,\rho_{\rm src} \quad\Longrightarrow\quad f_L \;=\; e^{-\phi_L^{\rm Yuk}} \;=\; \exp(-\gamma_L\rho_{\rm src}),$$

   which recovers precisely the TAL exponential law examined in Step 1 (`step1_dilaton_form.md`)!

3. **Point-Mass / Compact-Source Limit ($R_* \ll r$):** Outside a compact source of mass $M = \int \rho_{\rm src}\,d^3y$ ($r > R_*$), the solution V23 Eq. (7) simplifies to a classical fifth-force Yukawa potential:

$$\phi_L^{\rm Yuk}(r) \;\propto\; \alpha_Y\,\frac{GM}{c^2 r}\,e^{-r/L}.$$

4.4 4. Checks

4.4.1 4.1. Dimensional Analysis of V23 Eqs. (6) and (7)

- **Left-hand side of V23 Eq. (6):** ϕ_L is dimensionless ($[\phi_L] = 1$), the dimension of the d'Alembert/Laplace operator is $[\Box_g] = [\nabla^2] = \mathrm{m}^{-2}$, and $[L^{-2}] = \mathrm{m}^{-2}$, therefore:

$$\left[\left(-\Box_g + \frac{1}{L^2}\right)\phi_L\right] \;=\; \mathrm{m}^{-2}.$$

- **Right-hand side of V23 Eq. (6):** $[G/c^2] = \mathrm{m\,kg^{-1}}$, $[\alpha_{\rm std}] = 1$, $[\rho_{\rm src}] = \mathrm{kg\,m^{-3}}$, therefore:

$$\left[\frac{4\pi G}{c^2}\alpha_{\rm std}\,\rho_{\rm src}\right] \;=\; \left(\mathrm{m\,kg^{-1}}\right)\cdot\left(\mathrm{kg\,m^{-3}}\right) \;=\; \mathrm{m}^{-2} \quad \checkmark$$

- **Central integral of V23 Eq. (7) ($\phi_L^{\rm Yuk}(0)$):**

$$\left[\alpha_Y\,\frac{G}{c^2}\int_0^{R_*} r'\,e^{-r'/L}\rho_{\rm src}(r')\,dr'\right] \;=\; 1\cdot\left(\mathrm{m\,kg^{-1}}\right)\cdot\mathrm{m}\cdot 1\cdot\left(\mathrm{kg\,m^{-3}}\right)\cdot\mathrm{m} \;=\; \mathrm{m^3\,kg^{-1}\cdot kg\,m^{-3}} \;=\; 1 \quad \text{(dimensionless)} \quad \checkmark$$

4.4.2 4.2. Agreement of SymPy Verifications with the V23 Package (`conservation_Residuals.json`)

In V23, `covariant_closure_50aa2db0.py` (lines 126–148) runs three symbolic checks on this sector, each yielding an exact residual of 0:

1. `sym_yukawa_center_limit_residual = 0`: $\lim_{r\to 0}\frac{L}{r}\sinh(r/L) = 1$, so $\phi_L^{\rm Yuk}(0)$ contains no extra L-multiplier ✓.
2. `sym_yukawa_alphaY_eq_4pi_alphastd_residual = 0`: $\alpha_{\rm std} = \frac{1}{4\pi} \implies \alpha_Y = 4\pi\alpha_{\rm std} = 1 \implies \phi_L = \gamma_L\rho$ in a homogeneous medium ✓.
3. `sym_yukawa_el_source_factor_residual = 0` and `sym_yukawa_el_matter_pressure_residual = 0`: variation of the matter term $\mathcal{L} \supset -e^{-\phi_L}\varepsilon(n)$ yields the source term $-\alpha_Y f_L \varepsilon$, the effective pressure $P_{\rm matter} = f_L P$, and the modified hydrostatic equilibrium $h' = -\nu' + \phi_L'$ ✓.

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5 Step 4: Covariant Stress-Energy Conservation, Scalar-Matter Momentum Exchange, and Modified Hydrostatic Equilibrium ($h' = -\nu' + \phi_L'$)

Source file: step4_covariant_conservation_hydrostatic_en.md

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Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md), step2_conformal_disformal_en.md (step2_conformal_disformal.md), step3_yukawa_helmholtz_eom_en.md (step3_yukawa_helmholtz_eom.md)
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Local Archival Sources:

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_co (Subsections 1.1 and 1.2, lines 47–65, Eqs. 4–5 and Eqs. 6–7; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 2, lines 205–207; DOI: 10.5281/zenodo.23604423)
- V23 SymPy verification script: covariant_closure_50aa2db0.py (lines 72–98 and 143–160) and conservation_Residuals.

5.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 3 (step3_yukawa_helmholtz_eom.md), we derived the Euler–Lagrange equation of the scalar/dilaton field ϕ_L. The next necessary theoretical step is the variation with respect to the metric $g^{\mu\nu}$ and the derivation of **covariant energy–momentum conservation following from the Bianchi identity** ($\nabla_\mu G^{\mu\nu} \equiv 0$) ($\nabla_\mu (T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_X) = 0$), together with the resulting **modified hydrostatic equilibrium (fifth force / scalar-gradient force)**.

1. **Formulation of the Physical Problem in the V22 Monograph (hu_p1.tex, line 206):** > *"At the same time, inserting $f_L(\mathbf{x})$ as an external multiplier into the Tolman–Oppenheimer–Volkoff equations or the clock-frequency formulas leaves behind a residual $\nabla_\mu T^{\mu\nu}_{matter} \neq 0$ unless balanced by a dynamical field sector. Therefore, I formulated the dynamical 'why' as task **50aa2db0**: elevating the nonlocal kernel into a covariant scalar–tensor or nonlocal action principle $S = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_{matter}(\psi, \tilde{g}_{\mu\nu}) + \mathcal{L}_\phi(\phi, \nabla\phi)\right]$ whose Euler–Lagrange equation recovers the profile $\phi(\mathbf{x}) \to \gamma_L \bar{\rho}_L(\mathbf{x})$ in the quasi-static limit while guaranteeing the conservation law $\nabla_\mu(T^{\mu\nu}_{matter} + T^{\mu\nu}_X) = 0$."*
2. **Algebraic X-Sector Reconstruction in Formulation I of V23 (tal_covariant_closure_50aa2db0_EN_2026-10-24.t / HU, lines 47–52, Eqs. 4–5):** > *"In a static spherically symmetric interior, if the auxiliary sector $T^\mu_\nu(X) = diag(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$ absorbs the window stress on the right-hand side of $G^\mu_\nu = \frac{8\pi G}{c^4}(T^\mu_{\nu,matter} + T^\mu_{\nu,X})$ while the fluid elements satisfy $\nabla_\mu T^{\mu\nu}_{matter} = 0$ ($P' = -(\varepsilon + P)\nu'$ in geometrized units, $G = c = 1$, $\nu = \Phi_{eff}/c^2$, $g_{tt} = -e^{2\nu} = -e^{2\Phi_{eff}/c^2}$), then the Bianchi identity $\nabla_\mu G^\mu_r \equiv 0$ uniquely fixes $T^\mu_\nu(X)$ in both V21 modes:"*

$$\textbf{Mass mode } (m' = 4\pi r^2 f_L \varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1-f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1-f_L)\varepsilon\nu', \quad \text{(V23 Eq. 4)}$$

$$\textbf{Gradient mode } (m' = 4\pi r^2 \varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1-f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}\big(P'_{r,X} + P_{r,X}\nu$$

> *"where $A_{geo}(r) = (m+4\pi r^3 P)/[r(r-2m)]$. Symbolic verification performed in **SymPy** (**conservation_Residuals.json**) confirms that $\nabla_\mu T^\mu_r(matter) \equiv 0$ and $\nabla_\mu T^\mu_r(X) \equiv 0$ hold identically."*

3. **Local Scalar–Tensor (Yukawa) Stress–Energy Tensor and Action-Consistent Hydrostatic Equilibrium in Formulation II (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 65):** > *"and the variation $\delta S_\phi/\delta g^{\mu\nu}$ provides the local covariant stress–energy tensor $T^{(\phi)}_{\mu\nu} = \frac{c^4}{8\pi G\tilde{\alpha}_L}\big[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\big((\nabla\phi_L)^2 + L^{-2}\phi_L^2\big)\big]$. [...] The **yukawa_action_el** rows follow from the action $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ ($G = c = 1$; **SymPy: sym_yukawa_el_source_factor** and **_hydrostatic** residual $= 0$): $B\phi'' + (2B/r + B\nu' + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L \varepsilon$, $B = 1 - 2m/r$, $P_{matter} = f_L P$, $h' = -\nu' + \phi'$, $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$, exterior matching $\phi'(R) = -(1/R + 1/L)\phi(R)$."*

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5.2 2. Derivation of Scalar–Matter Momentum Exchange and Modified Hydrostatic Equilibrium

5.2.1 2.1. Derivation of the Effective Matter Pressure ($P_{\mathbf{matter}} = f_L P$)

In the V23 `yukawa_action_el` branch (and in general Einstein-frame scalar–tensor theory where the matter Lagrangian is $\mathcal{L}_{\rm fluid}(n,\phi_L) = -f_L(\phi_L)\,\varepsilon(n) = -e^{-\phi_L}\varepsilon(n)$, with n denoting the baryon number density), the thermodynamic pressure is defined via the Legendre transformation under baryon-number conservation:

$$P_{\rm matter} \;=\; n\,\frac{\partial\big(f_L(\phi_L)\,\varepsilon(n)\big)}{\partial n} \;-\; f_L(\phi_L)\,\varepsilon(n) \;=\; f_L(\phi_L)\Big(n\,\frac{d\varepsilon}{dn} - \varepsilon(n)\Big) \;=\; f_L(\phi_L)\,P(n).$$

Thus the stress–energy tensor of the f_L-weighted ideal fluid is:

$$T^{\mu\nu}_{\rm matter} \;=\; f_L(\phi_L)\left[(\varepsilon+P)\frac{u^\mu u^\nu}{c^2} + P\,g^{\mu\nu}\right] \quad\Longrightarrow\quad \varepsilon_{\rm matter} = f_L\varepsilon, \qquad P_{\rm matter} = f_L P.$$

5.2.2 2.2. Divergence of the Scalar-Field Stress–Energy Tensor ($\nabla^\mu T^{(\phi)}_{\mu\nu}$)

In geometrized units ($G = c = 1$, where $\alpha_Y = 8\pi\tilde{\alpha}_L$, so that $\frac{c^4}{8\pi G\tilde{\alpha}_L} = \frac{1}{\alpha_Y}$), the stress–energy tensor of the scalar field is:

$$T^{(\phi)}_{\mu\nu} \;=\; \frac{1}{\alpha_Y}\left[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\left(g^{\alpha\beta}\nabla_\alpha\phi_L\nabla_\beta\phi_L + \frac{1}{L^2}\phi_L^2\right)\right].$$

Let us compute its covariant divergence ($\nabla^\mu T^{(\phi)}_{\mu\nu}$):

$$\nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; \frac{1}{\alpha_Y}\left[(\Box_g\phi_L)\,\nabla_\nu\phi_L + (\nabla^\mu\phi_L)\,\nabla_\mu\nabla_\nu\phi_L - (\nabla^\alpha\phi_L)\,\nabla_\nu\nabla_\alpha\phi_L - \frac{1}{L^2}\phi_L\nabla_\nu\phi_L\right].$$

Since second covariant derivatives commute on a scalar field ($\nabla_\mu\nabla_\nu\phi_L = \nabla_\nu\nabla_\mu\phi_L$), the second and third terms cancel:

$$\nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; \frac{1}{\alpha_Y}\left(\Box_g\phi_L - \frac{1}{L^2}\phi_L\right)\nabla_\nu\phi_L.$$

Now substitute the exact `yukawa_action_el` Euler–Lagrange field equation derived in Step 3 (`step3_yukawa_helmholtz_eom.md`):

$$\left(-\Box_g + \frac{1}{L^2}\right)\phi_L \;=\; \alpha_Y\,f_L(\phi_L)\,\varepsilon \quad\Longleftrightarrow\quad \Box_g\phi_L - \frac{1}{L^2}\phi_L \;=\; -\alpha_Y\,f_L\,\varepsilon.$$

With this, the divergence of the scalar sector in closed form is:

$$\boxed{\nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; -f_L(\phi_L)\,\varepsilon\,\nabla_\nu\phi_L \;=\; \varepsilon\,\nabla_\nu f_L(\phi_L).}$$

5.2.3 2.3. Joint Conservation ($\nabla^\mu(T^{\mathbf{matter}}_{\mu\nu} + T^{(\phi)}_{\mu\nu}) = 0$) and the Modified Euler/TOV Equation

Because of the contracted Bianchi identity ($\nabla^\mu G_{\mu\nu} \equiv 0$) on the left-hand side of the Einstein equations ($G_{\mu\nu} = 8\pi(T^{\rm matter}_{\mu\nu} + T^{(\phi)}_{\mu\nu})$), the joint divergence of the right-hand side vanishes identically:

$$\nabla^\mu T^{\rm matter}_{\mu\nu} \;+\; \nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; 0 \quad\Longrightarrow\quad \nabla^\mu T^{\rm matter}_{\mu\nu} \;=\; +f_L(\phi_L)\,\varepsilon\,\nabla_\nu\phi_L.$$

In a static spherically symmetric spacetime ($ds^2 = -e^{2\nu(r)}dt^2 + B(r)^{-1}dr^2 + r^2d\Omega^2$, $B(r) = 1 - 2m(r)/r$), the radial ($\nu = r$) covariant divergence of the matter tensor is:

$$\nabla_\mu T^\mu_r(\text{matter}) \;=\; \frac{d}{dr}\big(f_L P\big) \;+\; f_L(\varepsilon+P)\,\nu' \;=\; f_L\,P' \;-\; f_L\,P\,\phi_L' \;+\; f_L(\varepsilon+P)\,\nu',$$

where we used $f_L' = \frac{d}{dr}(e^{-\phi_L}) = -f_L\,\phi_L'$. Adding the divergence of the scalar sector $\nabla_\mu T^\mu_r(\phi) = -f_L\,\varepsilon\,\phi_L'$ gives:

$$\nabla_\mu T^\mu_r(\text{matter}) + \nabla_\mu T^\mu_r(\phi) \;=\; f_L\Big[P' + (\varepsilon+P)\nu' - (\varepsilon+P)\phi_L'\Big] \;=\; 0.$$

Since $f_L = e^{-\phi_L} > 0$, we can divide by $f_L(\varepsilon+P)$, and introducing the specific enthalpy ($dh = \frac{dP}{\varepsilon+P}$) yields:

$$\boxed{h'(r) \;\equiv\; \frac{P'(r)}{\varepsilon(r)+P(r)} \;=\; -\nu'(r) \;+\; \phi_L'(r) \;=\; -\nu'(r) \;-\; \frac{d\ln f_L(r)}{dr}.}$$

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5.3 3. Physical Interpretation in the Scalar–Tensor / Dilaton Framework

The relation $h' = -\nu' + \phi_L'$ derived above is the direct dynamical bridge between scalar–tensor gravity (Damour & Esposito-Farèse 1992; Fujii & Maeda 2003) and the Time Attraction Law:

1. **Decomposition of the Effective Potential:** The net equilibrium force acting on a baryonic fluid element consists of two parts:
   - the standard metric gravitational attraction $-\nu'(r) = -\frac{1}{c^2}\frac{d\Phi_{\rm eff}}{dr}$, and
   - the **scalar / time-coherence gradient force** $+\phi_L'(r) = -\frac{d\ln f_L}{dr}$ (referred to in the scalar–tensor literature as the *fifth force* / dilaton-gradient force).
2. **Weak-Field / Galactic Limit:** In a non-relativistic medium ($P \ll \varepsilon \simeq \rho c^2$), the effective acceleration of a test body or fluid element is:

$$\mathbf{a}_{\rm eff} \;=\; -\nabla\Phi_N \;+\; c^2\,\nabla\phi_L \;=\; -\nabla\left(\Phi_N - c^2\phi_L\right) \;=\; -\nabla\left(\Phi_N + c^2\ln f_L\right).$$

   Because the density around a mass concentration increases inward ($\frac{d\bar\rho_L}{dr} < 0$), $\phi_L(r)$ increases inward ($\phi_L'(r) < 0$, or $\frac{d\ln f_L}{dr} > 0$), so the $+\phi_L'(r)$ term provides a density-gradient-driven contribution in both hydrostatic equilibrium and orbital dynamics.

5.4 4. Checks

1. **SymPy Symbolic Identities in the V23 Package (`covariant_closure_50aa2db0.py`, lines 72–98 and 150–160; `conservation_Residuals.json`):**
   - `sym_yukawa_el_matter_pressure_residual = 0`: verifies the identity $P_{\rm matter} = n\frac{\partial(f_L\varepsilon)}{\partial n} - f_L\varepsilon = f_L P$ ✓.
   - `sym_yukawa_el_hydrostatic_residual = 0`: verifies that substituting the Yukawa EL equation into $\nabla_\mu T^\mu_r(\text{matter}) + \nabla_\mu T^\mu_r(\phi) = 0$ yields $h' = -\nu' + \phi_L'$ exactly ✓.
   - `div_matter_mass_symbolic = 0`, `div_X_mass_symbolic = 0`, `div_matter_grad_symbolic = 0`, `div_X_grad_symbolic = 0`: verifies conservation of the algebraic X-sector in V23 Formulation I Eqs. (4)–(5) ✓.
2. **General Relativistic (GR) Limit ($\phi_L \to 0$, $f_L \to 1$):** When $\phi_L \equiv 0$ ($\phi_L' = 0$, $f_L = 1$), the scalar stress–energy tensor vanishes ($T^{(\phi)}_{\mu\nu} = 0$), the effective pressure reduces to $P_{\rm matter} \to P$, and hydrostatic equilibrium returns to the standard GR Tolman–Oppenheimer–Volkoff equation ($h' = -\nu'$, i.e., $P' = -(\varepsilon + P)\nu'$) ✓.

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6 Step 5: Dictionary of Coupling Constants, Weak-Field Yukawa Fifth Force, and Conformal vs. Disformal PPN Light Deflection

Source file: step5_coupling_dictionary_ppn_screening_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step5_coupling_dictionary_ppn_screening_en.md
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md), step2_conformal_disformal_en.md (step2_conformal_disformal.md), step3_yukawa_helmholtz_eom_en.md (step3_yukawa_helmholtz_eom.md), step4_covariant_conservation_hydrostatic_en.md (step4_covariant_conservation_hydrostatic.md)
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Local Archival Sources:

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_co (Section 1.2, Eqs. 6–7, lines 56–65; Section 3, lines 161–167; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 3, lines 245–255, Eqs. 4–5) and hu_p2.tex (Chapter 4, lines 23–38; DOI: 10.5281/zenodo.23604423)

6.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 2 (step2_conformal_disformal.md), we established the taxonomy of the V19–V23 model branches and showed that one must distinguish between the universal conformal Jordan metric (**Branch II-Conf**), the comoving timelike disformal metric (**Branch II-Disf**), and the effective fluid-density coupling (**Branch II-Fluid**).

1. **The Coupling Constant C, γ_L, and the Compactness Scale in the V22 Monograph (hu_p1.tex, lines 247–254, Eqs. 4–5):**

$$C \equiv \frac{4\pi G}{c^2} = 9.33306\times10^{-27}\,\mathrm{m\,kg^{-1}}, \qquad \gamma_L = \frac{G\,L^2}{c^2} = \frac{C\,L^2}{4\pi} = 7.425915\times10^{-28}\,L^2\,\mathrm{m^3\,kg^{-1}}. \quad \text{(V22 Eqs. 4–5)}$$

> *"When a point mass M is averaged over a sphere of radius L, its mean density is $\bar\rho_L = \frac{3M}{4\pi L^3}$, so the exponent is $\gamma_L\bar\rho_L = \frac{3GM}{4\pi c^2 L} = \frac{3\,C\,M}{(4\pi)^2 L}$, which up to the geometric factor $3/(4\pi)$ recovers the dimensionless compactness scale $GM/(c^2L)$."*

2. **Solar System Screening in Chapter 4 of V22 (hu_p2.tex, lines 32 and 38):** > *"For subgalactic $L \gtrsim 1$ pc, $\gamma_L\bar\rho_L^{\odot} = \frac{3GM_\odot}{4\pi c^2 L} \lesssim 10^{-14}$; planetary orbits remain untouched! [...] when L exceeds the size of planetary orbits, the solar mass $M_\odot$ is diluted over the volume $\frac{4}{3}\pi L^3$, so gradients of f_L along planetary orbits are suppressed below the Cassini and ephemeris bounds."*

3. **The V23 Yukawa Normalization Constants (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 65):** > *"The unit-normalized kernel corresponds to $\alpha_Y = 1$, i.e., $\alpha_{std} = 1/(4\pi) \approx 0.0796$, which recovers the homogeneous top-hat limit ($\phi_L = \gamma_L\rho_{src}$); $\alpha_Y = 1/3$ ($\alpha_{std} = 1/(12\pi)$) is the volume-matched variant."*

6.2 2. Weak-Field Effective Potential, Yukawa Fifth Force, and the Effective Gravitational Constant ($G_{\mathrm{eff}}(r)$)

In Step 4 (step4_covariant_conservation_hydrostatic.md), we derived from the covariant conservation law ($\nabla_\mu(T^{\mu\nu}_{\mathrm{matter}} + T^{\mu\nu}_{(\phi)}) = 0$) the modified hydrostatic / Euler equation:

$$h'(r) = -\nu'(r) + \phi_L'(r), \qquad \mathbf{a}_{\mathrm{eff}}(\mathbf{r}) = -\nabla\Phi_{\mathrm{eff}}(\mathbf{r}) = -\nabla\Big(\Phi_N(\mathbf{r}) - c^2\,\phi_L(\mathbf{r})\Big).$$

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6.2.1 2.1. Exterior Yukawa Potential Around a Compact Source ($r > R_*$)

In the exterior of a spherically symmetric body of mass M and radius R_* ($r > R_*$, in the weak-field approximation), the flat-space Green's function solution written in Step 3 (`step3_yukawa_helmholtz_eom.md`, V23 Eq. 7) takes the form:

$$\phi_L^{\rm Yuk}(r) \;=\; \alpha_Y \,\frac{G}{c^2}\,\frac{L\,e^{-r/L}}{r}\int_0^{R_*} r'\,\sinh\!\left(\frac{r'}{L}\right)\rho_{\rm src}(r')\,dr' \;=\; \frac{\alpha_{\rm std}\,GM_{\rm eff}(L)}{c^2\,r}\,e^{-r/L},$$

where $\alpha_{\rm std} = \frac{\alpha_Y}{4\pi}$, and in the point-source limit ($R_* \ll L$, where $\sinh(r'/L) \simeq r'/L$):

$$M_{\rm eff}(L) \;=\; 4\pi\int_0^{R_*} r'^2\left[\frac{L}{r'}\sinh\!\left(\frac{r'}{L}\right)\right]\rho_{\rm src}(r')\,dr' \;\xrightarrow{R_*\ll L}\; \int_0^{R_*} 4\pi r'^2\rho_{\rm src}(r')\,dr' \;=\; M.$$

Substituting this alongside the Newtonian potential $\Phi_N(r) = -\frac{GM}{r}$ into the effective potential gives:

$$\Phi_{\rm eff}(r) \;=\; \Phi_N(r) \;-\; c^2\,\phi_L^{\rm Yuk}(r) \;=\; -\frac{GM}{r}\Big(1 \;+\; \alpha_{\rm std}\,e^{-r/L}\Big).$$

Differentiating with respect to r yields the distance-dependent effective gravitational constant:

$$\boxed{\;a_{\rm eff}(r) \;=\; -\frac{G_{\rm eff}(r)\,M}{r^2}, \qquad G_{\rm eff}(r) \;=\; G\left[1 \;+\; \alpha_{\rm std}\left(1+\frac{r}{L}\right)e^{-r/L}\right].\;}$$

6.2.2 2.2. Canonical Scalar–Tensor Normalization and the Relation $\alpha_{\rm std} = 2\beta^2$

Let us rewrite the V23 Eq. (8) scalar action:

$$S_\phi \;=\; -\int d^4x\,\sqrt{-g}\,\frac{c^4}{16\pi G}\frac{1}{\tilde\alpha_L}\left(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\right) \;=\; -\int d^4x\,\sqrt{-g}\,\frac{c^4}{2G\,\alpha_Y}\left(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\right)$$

(where $\alpha_Y = 8\pi\tilde\alpha_L = 4\pi\alpha_{\rm std}$, so $\tilde\alpha_L = \frac{\alpha_{\rm std}}{2} = \frac{\alpha_Y}{8\pi}$, and in units $G = c = 1$ the scalar Lagrangian is precisely $-\frac{1}{2\alpha_Y}[(\nabla\phi_L)^2 + \phi_L^2/L^2]$) in terms of the canonical Einstein-frame scalar field φ of Damour–Esposito-Farèse (1992) / Damour–Polyakov (1994), whose kinetic term is $-\frac{c^4}{4\pi G}\,\frac{1}{2}(\nabla\varphi)^2 = -\frac{c^4}{8\pi G}(\nabla\varphi)^2$:

$$\frac{c^4}{2G\,\alpha_Y}(\nabla\phi_L)^2 \;=\; \frac{c^4}{8\pi G\,\alpha_{\rm std}}(\nabla\phi_L)^2 \;\equiv\; \frac{c^4}{8\pi G}\,2(\nabla\varphi)^2 \quad\Longrightarrow\quad \phi_L \;=\; \sqrt{2\alpha_{\rm std}}\,\varphi \;=\; 2\beta\,\varphi.$$

Since the matter-coupling factor is $f_L = e^{-\phi_L} = e^{-2\beta\varphi}$, the exact relation between the dimensionless scalar–matter coupling constant (β) and the V23 parameters $\alpha_{\rm std}, \alpha_Y, \tilde\alpha_L$ is:

$$\boxed{\;\alpha_{\rm std} \;=\; 2\beta^2 \;=\; 2\tilde\alpha_L \;=\; \frac{\alpha_Y}{4\pi}, \qquad \beta \;=\; \sqrt{\frac{\alpha_{\rm std}}{2}} \;=\; \sqrt{\frac{\alpha_Y}{8\pi}}.\;}$$

- For the **unit-normalized top-hat counterpart** ($\alpha_Y = 1$):

$$\alpha_{\rm std} \;=\; \frac{1}{4\pi} \;\approx\; 0.079577, \qquad \beta \;=\; \frac{1}{\sqrt{8\pi}} \;\approx\; 0.199471, \qquad 2\omega_{\rm BD}{+}3 \;=\; \frac{1}{\beta^2} \;=\; 8\pi \;\Longrightarrow\; \omega_{\rm BD} \;=\; 4\pi{-}\frac{3}{2} \;\approx\; 11.066.$$

- For the **volume-matched variant** ($\alpha_Y = 1/3$):

$$\alpha_{\rm std} \;=\; \frac{1}{12\pi} \;\approx\; 0.026526, \qquad \beta \;=\; \frac{1}{\sqrt{24\pi}} \;\approx\; 0.115165, \qquad \omega_{\rm BD} \;=\; 12\pi - \frac{3}{2} \;\approx\; 36.199.$$

6.3 3. Strict Separation of the Three Coupling Branches (Branch II-Conf, Branch II-Fluid, Branch II-Disf) in the PPN and Light-Deflection Sector

In accordance with peer-review requirements, we record that the conformal coupling $\tilde g_{\mu\nu}^{\rm conf}$, the disformal coupling $\tilde g_{\mu\nu}^{\rm disf}$, and the effective fluid coupling $S_m = -\int d^4x\sqrt{-g}\,e^{-\phi_L}\varepsilon(n)$ represent **three distinct physical hypotheses** regarding the electromagnetic/null-geodesic sector, and consequently their PPN behaviors also differ:

1. **Branch II-Conf — Universal Conformal Jordan Coupling ($\tilde g_{\mu\nu}^{\rm conf} = e^{-2\phi_L}g_{\mu\nu}$, $\sqrt{-\tilde g_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$):**

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- In this branch, every matter and electromagnetic field couples universally to the metric $\tilde{g}^{\rm conf}_{\mu\nu} = e^{-2\phi_L} g_{\mu\nu}$.
- Because the Maxwell action is conformally invariant in $D = 4$ dimensions ($\sqrt{-\tilde{g}_{\rm conf}}\,\tilde{g}^{\mu\alpha}_{\rm conf}\tilde{g}^{\nu\beta}_{\rm conf}F_{\mu\nu}F_{\alpha\beta} = \sqrt{-g}\,g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$), photon null geodesics ($\tilde{g}^{\rm conf}_{\mu\nu}dx^\mu dx^\nu = 0 \iff g_{\mu\nu}dx^\mu dx^\nu = 0$) sense only the Einstein-frame metric $g_{\mu\nu}$, and the electromagnetic field does not source the scalar ϕ_L ($T^\mu_\mu(\mathrm{EM}) = 0$).
- The standard scalar–tensor / Brans–Dicke PPN parameter (Will, *Living Rev. Relativ.* **17**, 4, 2014) holds **exclusively for this Branch II-Conf**:

$$\gamma^{\rm conf}_{\rm PPN}(r) \;=\; \frac{1-\alpha_{\rm std}\,e^{-r/L}}{1+\alpha_{\rm std}\,e^{-r/L}}, \qquad \gamma^{\rm conf}_{\rm PPN}(r \ll L) \;=\; \frac{2\omega_{\rm BD}+1}{2\omega_{\rm BD}+3} \;=\; \frac{4\pi-1}{4\pi+1} \;\approx\; 0.8526\ (\alpha_Y = 1), \qquad \gamma^{\rm conf}_{\rm PPN}(r \gg L) \;\to\; 1$$

2. **Branch II-Fluid + H_0 — Effective Baryonic Fluid Coupling with Minimal Photon/Atomic Metric ($S_m = -\int d^4x\sqrt{-g}\,e^{-\phi_L}\varepsilon(n)$, $\Psi_{\rm atomic/EM}$ Coupled to $g_{\mu\nu}$):**
   - Macroscopic baryonic fluid experiences the fifth force ($h' = -\nu' + \phi_L'$), so the effective gravitational constant between two macroscopic bodies is $G_{\rm eff}(r) = G\big[1 + \alpha_{\rm std}(1 + r/L)e^{-r/L}\big]$.
   - However, in the regime $r \gg L$ (for example when L is submillimeter or when on astronomical scales $r_{\rm AU} \gg L$), the Yukawa term decays exponentially ($e^{-r/L} \to 0$), the exterior spacetime is asymptotically Schwarzschild geometry (with mass $M_{\rm ADM}$), and $\gamma_{\rm PPN}(r \gg L) = 1$, $\beta_{\rm PPN}(r \gg L) = 1$.
3. **Branch II-Disf + H_1 — Comoving Timelike Disformal Coupling ($\tilde{g}^{\rm disf}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$, $\sqrt{-\tilde{g}_{\rm disf}} = f_L\sqrt{-g}$):**
   - In this branch, the volume element $\sqrt{-\tilde{g}_{\rm disf}} = e^{-\phi_L}\sqrt{-g}$ coincides exactly with the $e^{-\phi_L}$ weight factor of Branch II-Fluid, but the four-velocity u^μ enters the metric explicitly (which either requires a comoving khronon/aether fixing or yields an additional constraint equation under free u^μ variation).
   - Because the disformal transformation is non-conformal, it breaks conformal invariance in the Maxwell sector: on the one hand, a static magnetic field sources ϕ_L with effective mass density $\rho_B = \frac{|\mathbf{B}|^2}{2\mu_0 c^2}$ (Steps 15 and 20), and on the other hand, light rays propagate on the disformal light cone $\tilde{g}^{\rm disf}_{\mu\nu}k^\mu k^\nu = 0$ with an effective refractive index $n_{\rm refr}(r) = f_L(r)^{-1} = e^{+\phi_L(r)}$ (Step 14).
   - Consequently, light deflection in Branch II-Disf is **not** described by the conformal Brans–Dicke formula $\gamma^{\rm conf}_{\rm PPN} = (1-\alpha_{\rm std}e^{-r/L})/(1+\alpha_{\rm std}e^{-r/L})$, but rather by the refractive deflection angle $\hat{\alpha}_{\rm disf}(b) = \frac{4GM}{c^2 b} - \int_{-\infty}^{\infty}\frac{\partial_\perp\phi_L}{c^2}dz$ derived in Step 14.

6.4 4. Summary Coupling Dictionary (TAL ↔ Scalar–Tensor / Dilaton)

Physical Quantity	Time Attraction Law (V19–V23)	Scalar–Tensor / Dilaton Counterpart	Exact Conversion and Branch-Specific Note
Dimensionless scalar field	$\phi_L(\mathbf{x})$	$2\beta\varphi(\mathbf{x})$ (or $2\phi_{\rm dil}(\mathbf{x})$)	$f_L(\mathbf{x}) = e^{-\phi_L} \leftrightarrow e^{-2\beta\varphi} \leftrightarrow e^{-2\phi_{\rm dil}}$
Fundamental gravitational–temporal constant	$C \equiv \frac{4\pi G}{c^2} = 9.33306 \times 10^{-27}$ m/kg	$8\pi G/c^4$ ($\times c^2/2$)	$\frac{G}{c^2} = \frac{C}{4\pi} = 7.425915 \times 10^{-28}$ m/kg
Scale-dependent density coupling	$\gamma_L = \frac{GL^2}{c^2} = \frac{CL^2}{4\pi}$ (m^3/kg)	$\frac{G}{c^2\mu_\phi^2}$	$[\gamma_L] = \mathrm{m}^3\,\mathrm{kg}^{-1} = [\text{density}]^{-1}$
Smoothing radius / Range	L (3D top-hat / Yukawa scale)	Compton wavelength $\lambda_\phi = \mu_\phi^{-1}$	$m_\phi = \frac{\hbar}{Lc}$ (scalar/dilaton mass)
Action normalization constants	$\alpha_Y = 4\pi\alpha_{\rm std} = 8\pi\tilde{\alpha}_L$	$8\pi\beta^2 = \frac{4\pi}{2\omega_{\rm BD}+3}$	Scalar prefactor in action: $-\frac{c^4}{16\pi G}\frac{1}{\tilde{\alpha}_L} = -\frac{c^4}{2G\alpha_Y}$ ($-\frac{1}{2\alpha_Y}$ in $G = c = 1$)
Conformal branch (Branch II-Conf)	$\tilde{g}^{\rm conf}_{\mu\nu} = e^{-2\phi_L}g_{\mu\nu}$	Jordan–Brans–Dicke metric	$\sqrt{-\tilde{g}_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$; $\alpha_{\rm std} = \frac{\alpha_Y}{4\pi}$, $\omega_{\rm BD} = \frac{4\pi}{\alpha_Y} - \frac{3}{2}$
Disformal branch (Branch II-Disf)	$\tilde{g}^{\rm disf}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$	Bekenstein disformal metric	$\tilde{g}_{00} = f_L^2 g_{00}$, $\tilde{g}_{ij} = g_{ij}$, $\sqrt{-\tilde{g}_{\rm disf}} = f_L\sqrt{-g}$; magnetic source ρ_B and refractive index $n = e^{\phi_L}$

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6.5 5. Checks

1. **Yukawa Limits ($r \ll L$ and $r \gg L$):**
   - $r \gg L$ (strongly screened regime, $e^{-r/L} \to 0$): $G_{\rm eff}(r) \to G$ and $\gamma_{\rm PPN}^{\rm conf}(r) \to 1$, so the model recovers general relativity at all distances $r \gg L$ ✓.
   - $r \ll L$ (unscreened short distance, $e^{-r/L} \to 1$): $G_{\rm eff}(r) \to G(1+\alpha_{\rm std}) = G(1+2\beta^2)$, which matches the standard scalar–tensor gravity two-body interaction formula (Damour & Esposito-Farèse 1992, Eq. 2.12) ✓.
2. **Solar System Dilution Bound of the 3D Top-Hat (V22 `hu_p2.tex`, line 32):** For the solar mass ($M_\odot = 1.98847 \times 10^{30}$ kg) at $L = 1$ pc $= 3.0857 \times 10^{16}$ m:

$$\gamma_L \bar{\rho}_L^{\odot} \;=\; \frac{3GM_\odot}{4\pi c^2 L} \;=\; \frac{3 \times 1.4766 \times 10^3 \text{ m}}{4\pi \times 3.0857 \times 10^{16} \text{ m}} \;\approx\; 1.14 \times 10^{-14} \;\ll\; 10^{-5} \quad \text{(Cassini PPN bound)} \quad \checkmark$$

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7 Step 6: Galactic Rotation Curves (155 / 175 SPARC Galaxies), Positive-Definite 3D Top-Hat / Yukawa Kernel Positivity, and the NGC 3198 ($R = 30.08$ kpc) Negative Control Check

Source file: `step6_sparc_galactic_ngc3198_consistency_en.md`

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step6_sparc_galactic_ngc3198_consistency_en.md`
Previous Steps: `step1_dilaton_form_en.md` (`step1_dilaton_form.md`) – `step5_coupling_dictionary_ppn_screening_en.` (`step5_coupling_dictionary_ppn_screening.md`)
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Local Archival Sources:

- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex` (Chapter 3, lines 256–265, Eq. 6) and `hu_p2.tex` (Chapters 4–5, lines 1–61, Eqs. 7–8; DOI: `10.5281/zenodo.23604423`)
- V17 and V19 Zenodo records: DOI `10.5281/zenodo.19666220` (V17 SPARC 155 / 123 blind sample) and DOI `10.5281/zenodo.23497128` (V19 3D spherical top-hat, 175 SPARC galaxies)
- Local SPARC and NGC 3198 working documents: `docs/README_draft.md` (lines 6–24), `negative_density.md` (lines 1–12 and 55–63), `foundation_mode.md` (lines 13–15 and 50–63)

7.1 1. Verbatim Citations from the V22 Monograph and Local SPARC Documents

In Steps 1–5, we constructed the V23 covariant scalar–tensor / Yukawa framework and verified its weak-field / Solar System (PPN) limit. In this step, we examine **the mathematical positivity of the 3D kernel on the galactic scale (exclusion of negative-density points across all 175 SPARC galaxies)** and the **NGC 3198 ($R = 30.08$ kpc) negative control point**, strictly separating the boost required by observation ($C_{\rm req}$) from the boost actually predicted by the isolated model with fixed coupling $\alpha_{\rm std} = 1/(4\pi)$ ($C_{\rm pred}^{\rm isol}$).

1. **Definition of the 3D Spherical Top-Hat and Exclusion of Negative Density ($\rho < 0$) (`hu_p2.tex`, lines 12–21, Eqs. 7–8):** > *"Instead of pointwise derivatives or linear approximations, I define the source of temporal damping at any point $\mathbf{x} \in \mathbb{R}^3$ as the unweighted three-dimensional top-hat average of the physical mass–energy density $\rho(\mathbf{y}) = T_{00}(\mathbf{y})/c^2 \geq 0$ inside a ball $B(\mathbf{x}, L)$ of radius $L > 0$:"*

$$\bar{\rho}_L(\mathbf{x}) \;\equiv\; \frac{3}{4\pi L^3}\int_{|\mathbf{y}-\mathbf{x}|\leq L} \rho(\mathbf{y})\, d^3y \;=\; \frac{M_{\rm enc}(\mathbf{x};L)}{\frac{4}{3}\pi L^3}, \qquad f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\, \bar{\rho}_L(\mathbf{x})\big). \quad \text{(V22 Eq. 7)}$$

> *"Consequently, the exponential damping factor satisfies the exact analytical bounds: $0 < f_L(\mathbf{x}) \leq 1 \quad \forall\, \mathbf{x} \in \mathbb{R}^3$, $\forall\, L > 0$. Neither negative density, nor sign reversal, nor a metric horizon inversion arising from f_L can occur."*

2. **The SPARC 175 / 155 / 123 Galaxy Sample and the 3D Axisymmetric Disk Model (`hu_p2.tex`, lines 47–55):** > *"The main SPARC catalog contains **175 late-type and dwarf galaxies** with 3.6 μm surface photometry and resolved H I/Hα rotation curves. During my V17 processing, after filtering out 20 systems exhibiting severe kinematic asymmetry or inclination quality flags ($i < 30^\circ$ or $Q = 3$), the **processed rotation-curve sample of 155 galaxies** was obtained. Strict data-hygiene cuts (at least 5 radial measurement points and uninhibited $V_{gas}, V_{disk}, V_{bulge}$ baryonic decomposition) defined the **123-galaxy blind test sample**, encompassing **2596 radial velocity measurements**."* > *"In Version 19, I replaced the thin-disk midplane approximation $\rho_{mid}(r) = \Sigma_{bary}(r)/(2h_z)$ with the exact 3D axisymmetric mass distribution $\rho_{bary}(R,z) = \frac{\Sigma_{bary}(R)}{2h_z}\exp(-|z|/h_z)$ (with scale height $h_z = 0.196\, R_d^{0.633}$) [...]. Across all 175 SPARC galaxies and over the range $L \in [10^{16}, 10^{23}]$ m, the maximum isolated galactic damping is $\max|1 - f_L| = 2.284 \times 10^{-7}$ on the logarithmic grid (and 2.300×10^{-7} under continuous optimization at $L \approx 23.09$ kpc in the massive spiral galaxy **UGC02487**)."*

3. **Exact Clarification of the NGC 3198 ($R = 30.08$ kpc) Velocity Values (`docs/README_draft.md`, lines 6–19 and `foundation_mode.md`, lines 13–14):** > *"At $R = 30.08$ kpc (`NGC3198_rotmod.dat`, $D = 13.8$ Mpc):* > - **`Vobs = 146 ± 2 km/s`**: *Observed rotation speed. '145' in earlier text is consistent with this within 1σ.*

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Neighbouring points: 148 at 28.16 kpc, 147 at 32.14 kpc. > - **`Vbar` ($\Upsilon_{\rm disk}=0.5, \Upsilon_{\rm bul}=0.7$, signed gas) = `70.32 km/s`**: *Baryonic speed with the documented v16 mass-to-light ratio.* > - **`Vbar` ($\Upsilon_{\rm disk}=1.0$) = `88.53 km/s`**: *Baryonic speed computed from the raw SPARC columns* $\sqrt{V_{gas}^2+V_{disk}^2}$. ***'88.5' is a baryonic value with*** $\Upsilon_{disk}=1$***, not an observed speed.*** > *The required boost is* $C_{req}=(V_{obs}/V_{bar})^2$*, which is* 4.311 *for* $\Upsilon=0.5$ *and* 2.720 *for* $\Upsilon=1.0$."

7.2 2. Derivation of the Galactic Circular Velocity and Positivity Check of the 3D Kernel

7.2.1 2.1. Circular Velocity from the Modified Euler Equation ($h' = -\nu' + \phi_L'$)

In Steps 4 and 5 (`step4_covariant_conservation_hydrostatic.md`, `step5_coupling_dictionary_ppn_screening.md`), we derived from the covariant conservation law ($\nabla_\mu(T^{\mu\nu}_{\rm matter}+T^{\mu\nu}_{(\phi)})=0$) that the radial effective acceleration acting on a baryonic element orbiting in the midplane ($z=0$) of a galactic disk is:

$$a_{r,\rm eff}(r) \;=\; -\frac{d\Phi_N(r)}{dr} \;+\; c^2\,\frac{d\phi_L(r)}{dr} \;=\; -\frac{V_{\rm bar}^2(r)}{r} \;-\; c^2\,\frac{d\ln f_L(r)}{dr}.$$

The equilibrium circular velocity ($V_{\rm rot}^2(r) = -r\,a_{r,\rm eff}(r)$) can therefore be written in two ways:

1. **Additive scalar-gradient (V19–V23 canonical scalar–tensor / Yukawa) form:**

$$\boxed{V_{\rm rot}^2(r) \;=\; V_{\rm bar}^2(r) \;-\; c^2\,r\,\frac{d\phi_L(r)}{dr} \;=\; V_{\rm bar}^2(r) \;+\; c^2\,r\,\gamma_L\left|\frac{d\bar\rho_L(r)}{dr}\right|}$$

   (since $\frac{d\bar\rho_L}{dr}<0 \implies \frac{d\phi_L}{dr}<0$, the scalar gradient provides an inward additional attraction).

2. **Multiplicative $C(\rho)$ force-multiplier (V14–V16 phenomenological) form (`hu_p1.tex`, Eq. 6):**

$$V_{\rm rot}^2(r) \;=\; C(\rho(r))\,V_{\rm bar}^2(r), \qquad C(\rho) \;=\; 1 + C_0\exp\!\left(-\frac{\rho}{\rho_c}\right).$$

   The exact algebraic relation between the two formulations at any radius r is:

$$C_{\rm eff}(r) \;\equiv\; \frac{V_{\rm rot}^2(r)}{V_{\rm bar}^2(r)} \;=\; 1 \;-\; \frac{c^2\,r}{V_{\rm bar}^2(r)}\,\frac{d\phi_L(r)}{dr}.$$

7.2.2 2.2. What Does the Result "0 Negative-Density Points Across 175 SPARC Galaxies" Prove, and What Does It Not Prove?

For peer-review clarity, we strictly state the exact meaning of the V19 (`sparc_tophat.py`) test performed across the 175 SPARC galaxies:

- **Failure of the Earlier Spherical Derivative Formula (`negative_density.md`, line 11; `foundation_mode.md`, lines 16–18 and 55–63):** When density is estimated from the numerical derivative of the rotation curve ($\rho_{\rm deriv}(r) = \frac{1}{4\pi G r^2}\frac{d(rV_{\rm bar}^2)}{dr}$), non-monotonic gas and disk profiles cause $\rho_{\rm deriv}(r)<0$ at **239** ($\Upsilon_{\rm disk}=0.5$) and **303** ($\Upsilon_{\rm disk}=1.0$) interior points across the 140 non-holdout SPARC galaxies (of which 22 points fall in the 29.5–31.5 kpc band, closest to 30.37 kpc in **NGC 4013** at $R=30.31$ kpc with $\rho=-1.56\times10^{-24}$ kg/m^3).
- **Mathematical Positivity of the V19 3D Top-Hat and V23 Yukawa Kernels:** Both the 3D spherical top-hat kernel ($K_L(\mathbf{x},\mathbf{y}) = \frac{\Theta(L-|\mathbf{x}-\mathbf{y}|)}{\frac{4}{3}\pi L^3} \ge 0$) and the 3D Yukawa Green's function derived in Step 3 ($G_L(\mathbf{x}-\mathbf{y}) = \frac{e^{-|\mathbf{x}-\mathbf{y}|/L}}{4\pi L^2|\mathbf{x}-\mathbf{y}|} > 0$) **integrate the photometric surface-brightness density $\rho_{\rm bary}(R,z)\ge 0$ with a strictly positive kernel**. Thus, at every radial point of every galaxy:

$$\bar\rho_L(r) > 0, \qquad \phi_L^{\rm Yuk}(r) > 0, \qquad 0 < f_L(r) \le 1 \quad \checkmark$$

- **Critical Methodological Distinction:** This result is a **proof of the mathematical positivity of the 3D integral kernel** on the positive photometric mass model of all 175 SPARC galaxies (`n_negative_density_points = 0`), **not** a rotation-curve fitting success of the constant-$\alpha_Y=1$ ($\alpha_{\rm std}=1/(4\pi)$) Yukawa action! As demonstrated numerically in Section 3 below, the isolated model with fixed coupling $\alpha_{\rm std}=1/(4\pi)$ by itself does not reproduce flat galactic rotation curves.

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7.3 3. Exact Numerical Negative Control: Isolated Fixed-Coupling ($\alpha_{\mathbf{std}} = 1/(4\pi)$) Model vs. NGC 3198 ($R = 30.08$ kpc)

In accordance with peer-review requirements, the table below lists in separate columns for the control point `NGC3198_rotmod.dat` ($R = 30.08$ kpc) the observed velocity ($V_{\rm obs}$), the baryonic velocity ($V_{\rm bar}$), the boost **required** by observation ($C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2$), and the boost ($C^{\rm isol}_{\rm pred}$) and rotation velocity ($V^{\rm isol}_{\rm pred} = \sqrt{C^{\rm isol}_{\rm pred}}\, V_{\rm bar}$) **actually predicted** by the isolated model with fixed $\gamma_L = GL^2/c^2$ ($\alpha_Y = 1$, $\alpha_{\rm std} = 1/(4\pi) \approx 0.07958$):

$\Upsilon_{\rm disk}$, $\Upsilon_{\rm bul}$ Choice	Observed $V_{\rm obs}$ (km/s)	Baryonic $V_{\rm bar}$ (km/s)	Required Boost $C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2$	Predicted Boost $C^{\rm isol}_{\rm pred}$ (Isolated 3D Top-Hat, $\gamma_L = GL^2/c^2$)	Predicted Boost $C^{\rm isol,max}_{\rm pred}$ (Point-Source Yukawa Upper Bound, $1+\alpha_{\rm std}$)	Predicted Velocity $V^{\rm isol}_{\rm pred}$ (Isolated $\alpha_{\rm std} = 1/(4\pi)$, km/s)	Observed Minus Predicted Residual $\Delta V = V_{\rm obs} - V^{\rm isol}_{\rm pred}$
$\Upsilon_{\mathbf{disk}} = 1.0$, $\Upsilon_{\mathbf{bul}} = 0$ (raw SPARC columns)	146.0 ± 2.0	88.53	2.720	$1 + \mathcal{O}(10^{-7}) \approx 1.0000002$	$\le 1 + \frac{1}{4\pi} =$ **1.07958**	88.53–91.99	-54.0 **to** -57.5 **km/s** (does not fit the flat curve!)
$\Upsilon_{\mathbf{disk}} = 0.5$, $\Upsilon_{\mathbf{bul}} = 0.7$ (V16/V17 mass-to-light ratio)	146.0 ± 2.0	70.32	4.311	$1 + \mathcal{O}(10^{-7}) \approx 1.0000002$	$\le 1 + \frac{1}{4\pi} =$ **1.07958**	70.32–73.06	-72.9 **to** -75.7 **km/s** (does not fit the flat curve!)

7.3.1 Physical Conclusion (Negative Control):

1. In an isolated galaxy, the depth of the gravitational potential is $\frac{GM_{\rm gal}}{c^2 R} \sim \frac{V^2_{\rm bar}}{c^2} \sim 10^{-7}$, and the 3D top-hat damping computed with the canonical coupling $\gamma_L = \frac{GL^2}{c^2}$ across all 175 SPARC galaxies satisfies $\max|1 - f_L| \le 2.30 \times 10^{-7}$ (`sparc_tophat_summary.json`).
2. Even in the optimal transition zone $r \sim L$, the Yukawa fifth force with fixed $\alpha_Y = 1$ ($\alpha_{\rm std} = \frac{1}{4\pi} \approx 0.07958$) increases $V^2_{\rm bar}$ by at most a factor of $1 + \alpha_{\rm std} \approx 1.0796$ (yielding 91.99 km/s for $\Upsilon_{\rm disk} = 1.0$ and 73.06 km/s for $\Upsilon_{\rm disk} = 0.5$, instead of the observed 146 km/s).
3. Consequently, the NGC 3198 ($R = 30.08$ kpc) control point proves as an **exact negative control** that the isolated V23 Yukawa model with constant coupling $\alpha_{\rm std} = 1/(4\pi)$ by itself **does not** explain flat galactic rotation curves; doing so requires either the V16/V17 phenomenological midplane boost $C(\rho) = 1 + C_0 e^{-\rho/\rho_c}$ ($C_0 \approx 3.34$–5.09) or a chameleon/symmetron extension with density-dependent coupling ($\beta_{\rm eff}(\rho)$, $\alpha^{\rm gal}_{\rm eff} = 2\beta^2_{\rm eff} \approx 1.72$–$3.31$) going beyond the constant-α_Y V23 action (see Step 11, **Branch II-Cham**).

7.4 4. Checks

1. **NGC 3198 $R = 30.08$ kpc Numerical Identities ($C_{\mathbf{req}}$ vs. $C^{\mathbf{isol}}_{\mathbf{pred}}$):**
   - $V_{\rm bar}(\Upsilon_{\rm disk} = 1.0) = 88.53$ km/s $\implies$ $C_{\rm req} = (146/88.53)^2 = 2.7199 \approx 2.720$, whereas $C^{\rm isol}_{\rm pred} \le 1 + \frac{1}{4\pi} = 1.07958 \implies V^{\rm isol}_{\rm pred} \le 91.99$ km/s < 146 km/s ✓.
   - $V_{\rm bar}(\Upsilon_{\rm disk} = 0.5) = 70.32$ km/s $\implies$ $C_{\rm req} = (146/70.32)^2 = 4.3107 \approx 4.311$, whereas $C^{\rm isol}_{\rm pred} \le 1.07958 \implies V^{\rm isol}_{\rm pred} \le 73.06$ km/s < 146 km/s ✓.
2. **Positivity Check Across All (175) SPARC Galaxies:**
   - Photometric 3D disk model ($\rho_{\rm bary}(R, z) \ge 0$) + 3D spherical top-hat or 3D Yukawa Green's function: number of points with $\rho \le 0$ is **0** across all 175 galaxies (`negative_density.md`, lines 5–7 and `hu_p2.tex`, lines 17–21) ✓.

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8 Step 7: Compact Objects, Curved-Spacetime Scalar-Tensor / Yukawa TOV Equations (Flat-Robin Approximation vs. Curved Exterior ADM Mass), and Static Nonlocal Tangent Stability Consistency (V21 / V23)

Source file: `step7_neutron_star_tov_stability_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step7_neutron_star_tov_stability_en.md`
Previous Steps: `step1_dilaton_form_en.md` (`step1_dilaton_form.md`) – `step6_sparc_galactic_ngc3198_consistency_en` (`step6_sparc_galactic_ngc3198_consistency.md`)
License: CC BY-NC 4.0
Local Archival Sources:

- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex` (Chapter 7, lines 102–124, Eq. 10) and `hu_p3.tex` (Chapter 8, lines 1–28, Eqs. 11–12; DOI: `10.5281/zenodo.23604423`)
- V21 Zenodo record: DOI `10.5281/zenodo.23602501` (Neutron Star Conditional TOV Sensitivity and Radial Pulsation Study)
- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Sections 1.1–1.2 and 2, lines 37–159, Eqs. 4–9, Tables 1–2; DOI: `10.5281/zenodo.23639084`)
- Curved exterior `yukawa_action_el` ADM mass and coupled stability: `curved_yukawa_adm_and_stability.py` and `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md`

8.1 1. Verbatim Citations from the V22 Monograph and the V23 (`50aa2db0`) Manuscript

In this step, we derive the equations of **compact objects and neutron-star TOV stability (V21 / V23)**, strictly separating:

1. the nonlocal static tangent operator V23 Eq. (9) formulated on the **Branch I-A** (V21 nonlocal 3D top-hat `tophat_mass` and `tophat_gradient`) background, and
2. the **Branch II-Fluid** (V23 local `yukawa_action_el`) coupled Einstein–scalar–fluid system, for which we separately compare the **flat-space Robin approximation and surface mass** ($M_{\rm surf} = m(R_*)$) used in Table 1 of the V23 manuscript with the **exact Dirichlet–Neumann slope and asymptotic ADM mass determined from the coupled curved exterior vacuum solution** ($r \in [R_*, \infty)$) ($M_{\rm ADM} = \lim_{r\to\infty} m(r) = M_{\rm surf} + \Delta M^{\rm curved}_{\phi,\rm ext}$, detailed derivation in Step 22, `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md`).
3. **The Two Conditional TOV Modes in V21, the 1.214×10^{-6} Kernel Accuracy, the 769 Stellar Models (Including the 33 Causality Endpoints), and the Methodological Stipulation on $e^{\nu}F$ (`hu_p2.tex`, Chapter 7, lines 107–124):** > *"In our audited code, the measured maximum relative discrepancy between the analytical shell kernel and independent 3D quadrature across all profiles is 1.214×10^{-6}. I then integrated two conditional Tolman–Oppenheimer–Volkoff (TOV) closures:"*

   (A) `mass` mode: $\dfrac{dm_{\rm eff}}{dr} = 4\pi r^2 \rho(r)\, f_L(r),$ **(B) `gradient` mode:** $\dfrac{d\Phi_{\rm eff}}{dr} = f_L(r)\, \dfrac{G\big(m(r) + 4\pi r^3 P(r)/c^2\big)}{c^2 r\big(r - 2Gm(r)/c^2\big)},$ (V22 E

   > *"I integrated 33 equilibrium sequences across three nuclear equations of state — **SLy** (nucleonic), **APR4** (variational nucleonic; reaching the causal sound-speed limit $c_s = c$ at rest-mass density $\rho_0 \approx 1.4458 \times 10^{15}$ **g/cm**^{3}), and **H4** (stiff hyperonic) — over 11 logarithmically spaced smoothing scales $L \in [1\,m, 100\,km]$. The V21 archival baseline table contains **182 equilibrium models**, while the dense diagnostic stability grid consists of **769 stellar models in total** (including all **33 exact causality endpoints** where $c_s/c = 1$)."* > *"When evaluating the TOV equations, I explicitly state: **the appearance of the modified hydrostatic factor $e^{\nu}F$ (or $\Phi'_{\rm eff}$) in the conditional pressure-gradient equation is an algebraic equilibrium assumption, and by itself does not derive how atomic clocks, nuclear transitions, or surface photon redshifts couple to the factor f_L.**"*

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4. **Algebraic X-Sector Stress Reconstruction and the Limitations of Frozen-f_L Chandrasekhar Operators (`hu_p3.tex`, Chapter 8, lines 8–22 and `tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, lines 48–52):** > *"Across the 769 audited neutron-star models, the relative tangential stress remains bounded, with a maximum of* $\max|p_{tX}/P_c| = 0.140$ *(14.0% of the central pressure). I emphasize that this X-sector reconstruction is an algebraic bookkeeping closure of the background Bianchi identity, not yet a dynamical field equation derived from an action principle."*

$$\textbf{Mass mode } (m' = 4\pi r^2 f_L \varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1-f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1-f_L)\varepsilon\nu', \quad \text{(V23 Eq. 4)}$$

$$\textbf{Gradient mode } (m' = 4\pi r^2 \varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1-f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}\big(P'_{r,X} + P_{r,X}\nu$$

5. **Curved-Spacetime Yukawa Action-EL (`yukawa_action_el`) Equations of V23 (`tal_covariant_closure_50aa2db0_` / `HU`, line 65):** > *"The `yukawa_action_el` rows follow from the action* $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ *($G = c = 1$; `SymPy: sym_yukawa_el_source_factor` and `_hydrostatic` residual $= 0$): $B\phi'' + (2B/r + B\nu' + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L \varepsilon$, $B = 1 - 2m/r$, $P_{matter} = f_L P$, $h' = -\nu' + \phi'$, $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$, exterior matching $\phi'(R) = -(1/R + 1/L)\phi(R)$; the exterior tail $\Delta M_{\phi,ext}$ is listed separately (SLy, $L = 10$ km: $3.4\times10^{-3}\,M_\odot$ at $\log_{10}\rho_{0c} = 15$, $9.0\times10^{-3}\,M_\odot$ at $M_{\max}$)."*

6. **The Frozen-f_L Sturm–Liouville Residual and the Nonlocal Static Tangent Operator (`tal_covariant_closure_50` / `HU`, Section 2, lines 122–135, Eqs. 8–9):**

$$\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\, \mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L = 0} = \frac{4\pi r^{5/2}(f_L - 1)\Gamma P \varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big]. \quad \text{(V23 Eq. 8)}$$

$$\delta f_L(r_i) = -\gamma_L f_L(r_i) \sum_j K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j), \qquad \begin{cases} \dfrac{d(\delta h)}{dr} = -\delta\nu'[\delta h, \delta m, \delta f_L[\delta h]] + \omega_0^2\, e^{\lambda - 2\nu}\xi, \\ \dfrac{d(\delta m)}{dr} = 4\pi r^2\Big[f_{\rm mass}\frac{\varepsilon+P}{c_s^2}\delta h + \varepsilon\, \delta f_{\rm mass}[\delta h]\Big]. \end{cases} \quad \text{(V23 Eq. 9)}$$

8.2 2. Derivation of the Curved-Spacetime Scalar-Tensor / Yukawa TOV System (`yukawa_action_el`)

8.2.1 2.1. The Curved-Spacetime d'Alembertian Operator and the Yukawa Field Equation

In a static, spherically symmetric spacetime (in geometrized units $G = c = 1$):

$$ds^2 \;=\; -e^{2\nu(r)}dt^2 + B(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2), \qquad B(r) \;\equiv\; e^{-2\lambda(r)} \;=\; 1 - \frac{2m(r)}{r},$$

the metric determinant is $\sqrt{-g} = e^{\nu(r)}B(r)^{-1/2}r^2\sin\theta$. For a scalar field $\phi_L(r)$ depending only on r, the covariant Laplace–Beltrami / d'Alembert operator is:

$$\Box_g \phi_L \;=\; \frac{1}{\sqrt{-g}}\partial_r\big(\sqrt{-g}\, g^{rr}\, \partial_r\phi_L\big) \;=\; \frac{B^{1/2}}{e^\nu r^2}\frac{d}{dr}\Big(e^\nu B^{1/2} r^2 \phi_L'\Big) \;=\; B\phi_L'' + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi_L'.$$

For the Lagrangian $\mathcal{L}_{\phi+m} = -\frac{1}{2\alpha_Y}\left[g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{\phi_L^2}{L^2}\right] - f_L(\phi_L)\varepsilon(n)$ ($f_L = e^{-\phi_L}$, $\alpha_Y = 4\pi\alpha_{\rm std} = 8\pi\tilde{\alpha}_L = 2\beta^2\cdot 4\pi$), the Euler–Lagrange equation with respect to ϕ_L ($\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_L)}) - \frac{\partial\mathcal{L}}{\partial\phi_L} = 0$, taking into account $\frac{\partial(-e^{-\phi_L}\varepsilon)}{\partial\phi_L} = +f_L\varepsilon$) is:

$$\boxed{B\phi_L'' \;+\; \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi_L' \;-\; \frac{\phi_L}{L^2} \;=\; -\alpha_Y\, f_L(\phi_L)\, \varepsilon(r).}$$

This is precisely the `yukawa_action_el` equation appearing in Section 1.2 of V23 (`tal_covariant_closure_50aa2db0_EN_2026-10` / `HU`, line 65), where the source term is multiplied by the dilaton-like self-consistent weight factor $f_L = e^{-\phi_L}$ (`sym_yukawa_el_source_factor = 0`).

8.2.2 2.2. The Coupled Einstein–Yukawa–Euler (TOV) System, the Curved Exterior Boundary Condition, and the ADM Mass ($M_{\rm ADM}$ vs. $M_{\rm surf}$)

From the tensors $T^\mu_\nu(\text{matter}) = \text{diag}(-f_L\varepsilon, f_L P, f_L P, f_L P)$ and $T^\mu_\nu(\phi)$ derived in Step 4 (`step4_covariant_conservation_hydros` the (tt) and (rr) Einstein equations together with the conservation law $\nabla_\mu(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_{(\phi)}) = 0$ yield the coupled,

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action-closed TOV system:

$$\rho_\phi(r) \;=\; -T^t_t(\phi) \;=\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi_L'(r)\big]^2 + \frac{\phi_L(r)^2}{L^2}\right), \qquad p_{r,\phi}(r) \;=\; T^r_r(\phi) \;=\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi_L'(r)\big]^2 - \frac{\phi_L(r)^2}{L^2}\right),$$

$$\frac{dm(r)}{dr} \;=\; 4\pi r^2\big[f_L(r)\varepsilon(r) + \rho_\phi(r)\big],$$

$$\frac{d\nu(r)}{dr} \;=\; \frac{m(r) + 4\pi r^3\big[f_L(r)P(r) + p_{r,\phi}(r)\big]}{r\big(r - 2m(r)\big)},$$

$$\frac{dh(r)}{dr} \;=\; \frac{P'(r)}{\varepsilon(r) + P(r)} \;=\; -\frac{d\nu(r)}{dr} \;+\; \frac{d\phi_L(r)}{dr}.$$

Critical Methodological Clarification (Point 1 and Point 2 — detailed derivation in Step 22):

1. **Difference Between the Flat-Space Robin Approximation and the True Curved Exterior Dirichlet–Neumann Slope:**
   - The V23 manuscript (the original `yukawa_action_el` row of `covariant_closure_50aa2db0.py`) used as boundary condition at the stellar surface $r = R_*$ the derivative of the flat-space Helmholtz solution ($\phi_L^{\rm flat}(r) = \phi_L(R_*)\frac{R_*}{r}e^{-(r-R_*)/L}$):
   $$\phi'_{L,\rm flat}(R_*) \;=\; -k_{\rm flat}\,\phi_L(R_*), \qquad k_{\rm flat} \;=\; \frac{1}{R_*} + \frac{1}{L}.$$
   - In reality, however, in the exterior vacuum outside the stellar surface ($r \geq R_*$, where $\varepsilon = P = 0$), spacetime is strongly curved ($B(R_*) = 1 - 2M_{\rm surf}/R_* \approx 0.50$–$0.62$). In the exterior vacuum, the exact identity is $B\nu' + B'/2 = \frac{2m(r)}{r^2} - \frac{4\pi r}{\alpha_Y L^2}\phi_L(r)^2$, so the exterior coupled Einstein–scalar system is:
   $$\left(1 - \frac{2m(r)}{r}\right)\phi_L''(r) + \left[\frac{2}{r}\left(1 - \frac{m(r)}{r}\right) - \frac{4\pi r}{\alpha_Y L^2}\phi_L(r)^2\right]\phi_L'(r) - \frac{\phi_L(r)}{L^2} \;=\; 0, \qquad m'(r) \;=\; 4\pi r^2\rho_\phi(r).$$
   Because $B(R_*) < 1$, the effective local decay rate of the mass term ϕ_L/L^2 at the surface is $\sim B(R_*)^{-1/2}/L > 1/L$. As proven in Step 22 (`curved_yukawa_adm_and_stability.py`), for SLy at $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$, the true curved Dirichlet–Neumann slope is $k_{\rm curved} = -\phi'_{L,\rm curved}(R_*)/\phi_L(R_*) = 0.233542$ km^{-1}, which is +24.74% **larger** than the flat Robin approximation $k_{\rm flat} = 0.187216$ km^{-1}!
2. **Surface Mass ($M_{\rm surf}$) vs. Asymptotic ADM Mass ($M_{\rm ADM}$):**
   - In Table 1 of the V23 manuscript, the $M_{\rm max}$ and $R_{1.4}$ columns of the `yukawa_action_el` row were optimized for the surface mass $m(R_*) = M_{\rm surf}$, while the exterior scalar mass $\Delta M_{\phi,\rm ext}$ was listed in a separate column.
   - For a distant observer (and for the Bardeen–Carter–Hawking / Harrison–Wheeler turning-point theorem), the physical gravitational mass is the asymptotic **ADM mass**:
   $$\boxed{M_{\rm ADM} \;\equiv\; \lim_{r\to\infty} m(r) \;=\; M_{\rm surf} \;+\; \Delta M^{\rm curved}_{\phi,\rm ext}, \qquad \Delta M^{\rm curved}_{\phi,\rm ext} \;=\; \int_{R_*}^{\infty} 4\pi r^2\rho_\phi(r)\,dr.}$$
   - On the `yukawa_action_el_curved_adm` sequences recomputed with the curved exterior solution (SLy, $L = 10$ km, $\alpha_Y = 1$):
     - $M_{\rm ADM,max} = \mathbf{2.03321\,M_\odot}$ ($\log_{10}\rho_{0c}(M_{\rm ADM,max}) = \mathbf{15.316404}$, $R(M_{\rm ADM,max}) = 9.8853$ km, $\Delta M^{\rm curved}_{\phi,\rm ext} = 0.00411\,M_\odot$, $M_{\rm surf} = 2.02911\,M_\odot$),
     - the maximum of the conserved baryonic rest mass ($M_B = \int_0^{R_*} 4\pi r^2 B(r)^{-1/2}\rho_0(r)\,dr$) occurs at $\log_{10}\rho_{0c}(M^{\rm quad}_{B,\rm max}) = \mathbf{15.316409}$ ($|\Delta\log_{10}\rho_{0c}| = 4.47\times10^{-6}$ dex agreement with the ADM-mass turning point!),
     - $R(M_{\rm ADM} = 1.4\,M_\odot) = \mathbf{11.5359}$ km (whereas $R(M_{\rm surf} = 1.4\,M_\odot) = 11.5348$ km),
     - and the zero crossing of the coupled $(\delta h, \delta\phi_L)$ dynamical radial pulsation eigenvalue solved on the `yukawa_action_el` background (with the `scalar_m_sign = -1.0` main operator) is $\log_{10}\rho_{0c}(\omega_0^2 = 0) = \mathbf{15.316410}$ ($+\mathbf{5.80\times10^{-6}}$ dex from the turning point $dM_{\rm ADM}/d\rho_c = 0$ and $+1.33\times10^{-6}$ dex from $dM_B/d\rho_c = 0$; at $L = 1$ km, $\mathbf{15.303080}$, -6.86×10^{-6} dex; see Steps 12 and 22).

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8.3 3. Stability Analysis of the V21 Nonlocal 3D Top-Hat Background (Branch I-A): Frozen-f_L Residual vs. $\delta f_L[\delta h]$ Tangent Operator

(Note on separating model branches — Point 4: Subsections 3.1–3.2 below apply specifically to the V21 nonlocal 3D spherical top-hat `tophat_mass` *and* `tophat_gradient` *[**Branch I-A**] backgrounds, whereas the coupled stability of* `yukawa_action_el` *[**Branch II-Fluid**] is presented in Steps 12 and 22.)*

8.3.1 3.1. Algebraic Residual Term of the Frozen-f_L Chandrasekhar Operator (V23 Eq. 8)

In general relativity ($f_L \equiv 1$), taking the derivative $\partial/\partial\rho_c$ along the static equilibrium sequence generates a zero-eigenfrequency ($\omega_0^2 = 0$) radial trial perturbation at the extremal point $dM/d\rho_c = 0$. However, the derivation of the classical second-order Chandrasekhar Sturm–Liouville operator ($\mathcal{L}_{\rm SL}^{\rm matter}[\xi]$) substitutes during integration by parts the GR background identity

$$\lambda' + \nu' \;=\; 4\pi r e^{2\lambda}(\varepsilon + P) \qquad \text{(only when } f_L \equiv 1\text{!)}.$$

When the background is modified according to the V21 `mass` mode, the difference between the Chandrasekhar operator and the linearized hydrostatic operator evaluated at $\delta f_L = 0$ ($\mathcal{H}_{\rm lin}[\xi]\big|_{\delta f_L=0}$) does not vanish, but produces precisely the residual proportional to $(f_L - 1)$ given in V23 Eq. (8):

$$\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\, \mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L=0} \;=\; \frac{4\pi r^{5/2}(f_L-1)\Gamma P \varepsilon e^{2\nu}}{(r-2m)^{5/2}} \Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big].$$

8.3.2 3.2. Incorporating the Nonlocal $\delta f_L[\delta h]$ Tangent Perturbation on the V21 Background (V23 Eq. 9 and Table 2)

When the central density is varied by $\rho_{0c} \to \rho_{0c} + \delta\rho_{0c}$ on the V21 3D top-hat background, the Eulerian perturbation of the energy density is $\delta\varepsilon(r) = \frac{\varepsilon(r)+P(r)}{c_s^2(r)}\,\delta h(r)$, which also perturbs the factor f_L through the nonlocal kernel:

$$\delta f_L(r_i) \;=\; -\gamma_L\, f_L(r_i) \sum_j K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j).$$

Substituting this operator $\delta f_L[\delta h]$ back into the linearized $(\delta h, \delta m)$ boundary-value problem around the V21 background (V23 Eq. 9), at $\omega_0^2 = 0$ the system coincides exactly with the tangent operator of the V21 static equilibrium equations with respect to $\log_{10}\rho_{0c}$. As a result, on the SLy ($L = 10$ km) V21 sequence, the discrepancy between the zero crossing $\omega_0^2 = 0$ and the static turning point $dM/d\rho_c = 0$ (V23 Table 2) is:

- **GR baseline ($L = 0$ km):** $\Delta \log_{10}\rho_{0c} = +7.12 \times 10^{-5}$ dex (discretization baseline),
- **V21 SLy `mass` mode ($L = 10$ km, turning point $\log_{10}\rho_{0c} = 15.29924$):** improves from the frozen-f_L BVP discrepancy of -4.06×10^{-3} dex (and `matter` SL -1.34×10^{-3} dex) to $+\mathbf{6.97 \times 10^{-5}}$ **dex** (a 58.3× and 19.2× improvement, respectively),
- **V21 SLy `gradient` mode ($L = 10$ km, turning point $\log_{10}\rho_{0c} = 15.30106$):** improves from the frozen-f_L BVP discrepancy of -4.39×10^{-3} dex (and `matter` SL -3.57×10^{-3} dex) to $+\mathbf{7.39 \times 10^{-5}}$ **dex** (a 59.3× and 48.3× improvement, respectively).

8.4 4. Checks

1. $f_L \to 1$ **($L \to 0$ or $\alpha_Y \to 0$) GR Limit:**
   - $\phi_L \to 0 \implies \rho_\phi \to 0,\ p_{r,\phi} \to 0,\ f_L \to 1$, the `yukawa_action_el` TOV system recovers the standard GR TOV equations ($M_{\rm max}^{\rm SLy} = 2.04841\,M_\odot$, $M_{\rm max}^{\rm APR4} = 2.18800\,M_\odot$, $M_{\rm max}^{\rm H4} = 2.01385\,M_\odot$), and the V23 Eq. (8) residual term vanishes identically because $(f_L - 1) = 0$ ✓.
2. **Agreement Between Curved Exterior `yukawa_action_el` ADM Mass and Coupled Stability Turning Point (see `step22`):**
   - SLy $L = 10$ km, $\alpha_Y = 1$: $\log_{10}\rho_{0c}(dM_{\rm ADM}/d\rho_c = 0) = 15.316404$, $\log_{10}\rho_{0c}(dM_B/d\rho_c = 0) = 15.316409$, coupled $(\delta h, \delta\phi_L)$ zero crossing $\log_{10}\rho_{0c}(\omega_0^2 = 0) = 15.316410$ ($\Delta_{\rm ADM} = +5.80 \times 10^{-6}$ dex, a 2097× reduction compared to the frozen-ϕ_L shift of $+1.22 \times 10^{-2}$ dex; at $L = 1$ km, 15.303080 vs. 15.303086, -6.86×10^{-6} dex, a 165× reduction) ✓.

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3. **Inertia-Scaling Invariance at the V21 Zero Crossing ($\omega_0^2 = 0$):**
   - Scaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ yields the exact ratio $\omega_{4W}^2/\omega_W^2 = 0.2500$ ($\omega_W^2 = 4.897 \times 10^6\ \mathrm{s}^{-2}$), whereas at the zero crossing $\omega_0^2 = 0$, $F(0)$ is invariant, confirming that the $+6.97 \times 10^{-5}$ dex and $+7.39 \times 10^{-5}$ dex agreement in V21 is a consequence of the exact consistency of the static nonlocal tangent operator ✓.

V25 Update (curved `yukawa_action_el` branch): 4-component exterior BVP ($\delta\nu_R = 1.63 \times 10^{-14}$), 108-entry multi-EOS / multi-point 4×4 Jacobian verification (max rel. err $= 2.74 \times 10^{-5}$; SLy 15.0, $L = 10$ km: $\partial\phi_L'(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}\ \mathrm{km}^{-2}$, rel. diff. 3.98×10^{-10}; interior $r = 6$ km `a_m_sc` $= -1.72833 \times 10^{-4}\ \mathrm{km}^{-3}$, rel. diff. 6.08×10^{-9}), three-method baryonic mass ($M_B^{\mathrm{quad}}, M_B^{\mathrm{ODE}}, M_B^{\mathrm{trapz}}$) and First-Law audit, and nodeless ($n = 0$) fundamental mode and static tangent projection ($|\langle \mathbf{\Psi}_0, \mathbf{\Psi}_{\mathrm{tan}} \rangle| \in [0.9963,\ 0.99999993]$). SLy $L = 10$ km, $\alpha_Y = 1$: M_{ADM} turning point 15.316404, M_B^{quad} turning point 15.316409 (M_B^{trapz}: 15.316389, M_B^{ODE}: 15.316522), main dynamical $\omega_0^2 = 0$ root (`scalar_m_sign=-1.0`) 15.316410 ($+5.80 \times 10^{-6}$ dex; with diagnostic `scalar_m_sign=+1.0`, 15.319129, $+2.72 \times 10^{-3}$ dex). Detailed tables: `step22`, Sections 4 and 7.

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9 Step 8: Microscopic Dirac–Maxwell Clock and Photon Coupling from the Action (H0 vs. Conditional H1), Sirius B Redshift Constraints, and the V20 Laboratory Tests (8558 kg Granite Chamber and 8-Coil Magnetic-Field Experiment)

Source file: step8_microscopic_clock_coupling_siriusb_lab_v20_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step8_microscopic_clock_coupling_siriusb_lab_v20_en.md
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step7_neutron_star_tov_stability_en.md (step7_neutron_star_tov_stability.md)
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Local Archival Sources:

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Section 3, lines 161–190, Table 3; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p3.tex (Chapters 9 and 10, lines 30–91, Eq. 13; DOI: 10.5281/zenodo.23604423)
- V20 and V21 Zenodo records: DOI 10.5281/zenodo.23540741 (V20 8-coil magnetic field and 8558 kg granite chamber) and DOI 10.5281/zenodo.23602501 (V21 Sirius B and NICER conditional H0/H1 analysis)
- Local solenoid working report: /workspace/artifacts/solenoid_update/report_hu.md (lines 17–60)

9.1 1. Verbatim Citations from the V23 (50aa2db0) Manuscript and the V22 Monograph

The last two test items of the test plan in README.md (the second half of Section 3.4 and Section 3.5) concern the derivation of **microscopic clock and photon coupling (H0 vs. conditional H1)** from the covariant scalar–tensor / disformal action, together with its quantitative evaluation on the **Sirius B white dwarf** and on the **two branches of the V20 laboratory program** (8558 kg granite chamber and 8-coil solenoid).

1. **The Two Branches of Microscopic Dirac–Maxwell Clock Coupling in the Action Principle (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 3, lines 162–166):** > *"Let $\Psi_{atomic} = (\psi_e, A_\mu)$ denote the Dirac electron and Maxwell electromagnetic fields of an atomic clock comoving with the four-velocity u^μ of the baryonic medium, coupled to the effective Jordan-frame metric $\tilde{g}_{\mu\nu}[\phi_L]$:* > - **Minimal / purely auxiliary-field coupling ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$ on the Ψ_{atomic} sector $\Longrightarrow$ H0 hypothesis):** *If ϕ_L couples exclusively to the stress–energy tensor of the macroscopic baryonic fluid while atomic electrons and photons propagate on the physical spacetime metric $g_{\mu\nu}$, then the proper time of an atomic clock stationary at the surface $r = R_*$ is $d\tau_{clock} = \sqrt{-g_{00}(R_*)}\,dt = e^{\Phi_{eff}(R_*)/c^2}dt$. The surface redshift is purely metric ($1 + z_{H0} = e^{-\Phi_{eff}(R_*)/c^2}$), with no direct $f_L(R_*)$ multiplier.* > - **Universal disformal lapse or conformal Jordan coupling ($\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2(\phi_L)}{c^2}u_\mu u_\nu \Longrightarrow$ H1 hypothesis):** *If atomic clocks sense the disformal metric $\tilde{g}_{\mu\nu}$, whose comoving time–time component for $g_{00} = -1$, $u_0 = -c$ is $\tilde{g}_{00} = -1 + (1 - f_L^2) = -f_L^2$ (SymPy: sym_disformal_g00_residual $= 0$), then the proper-time element of an atomic transition is $d\tilde{\tau}_{clock} = f_L(R_*)\,e^{\Phi_{eff}(R_*)/c^2}dt$. The coordinate frequency of photons propagating to infinity is conserved, so the direct surface clock multiplier appears: $1 + z_{H1} = e^{-\Phi_{eff}(R_*)/c^2}\,f_L(R_*)^{-1}$."*
2. **Conditional H0 and Conditional H1 Redshift Results for the Sirius B White Dwarf (hu_p3.tex, Chapter 9, lines 53–57):** > *"I tested all six hypotheses against the high-precision HST/STIS Balmer-line measurement of the **Sirius B** white dwarf ($M = 1.018 \pm 0.011\,M_\odot$, $R = 0.008098 \pm 0.000046\,R_\odot$, observed redshift velocity $v_{obs} = 80.65 \pm 0.77$ km/s, GR baseline $v_{GR} \approx 80.42$ km/s), across both non-relativistic ($n = 1.5$) and relativistic ($n = 3$) polytropic interior profiles:* > - **Under H0-gradient lapse (Hydrostatic coupling only):** *[The] maximum velocity shift across all L is $|\Delta v| \leq 1.27$ **m/s** (0.00127 km/s), three orders of magnitude smaller than the ± 770 m/s measurement uncertainty. Thus Sirius B places no constraint on L under H0.* > - **Under conditional H1 (Direct surface clock coupling, $\nu \propto f_L(R_*)$):** *When atomic transitions in the photosphere are*

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*directly multiplied by $f_L(R_*)$, a smoothing sphere of radius $L \sim R_* \approx 5.63 \times 10^6$ m encloses a substantial fraction of the white dwarf's mass, inducing up to $\Delta v \approx +5.56$ km/s of excess redshift and reaching $\max \Delta\chi^2 = 52.3$ (single-factor H1) and $\max \Delta\chi^2 = 72.0$ (H1+gradient lapse). At the 2σ ($\Delta\chi^2 = 4$) level, conditional H1 excludes the intermediate band $L \in [4.08 \times 10^6,\ 3.37 \times 10^7]$ **m** ($n = 1.5$) and $L \in [4.95 \times 10^6,\ 3.37 \times 10^7]$ **m** ($n = 3$), while leaving both laboratory ($L \sim 1$ m) and sub-kilometer neutron-star scales unconstrained!"*

3. **The Two Branches of the V20 Laboratory Program (`hu_p3.tex`, Chapter 10, lines 71–85):** > - **Branch "A" (8558 kg granite chamber):** *"Inner bore radius $R_{in} = 0.25$ m, outer radius $R_{out} = 0.85$ m, vertical height $H = 1.20$ m ($z \in [-0.60, +0.60]$ m), volume $V = \pi(R_{out}^2 - R_{in}^2)H = 3.1102$ m^3, and total granite mass $M_{\textbf{granite}} = 8558$ **kg**."* > - **Branch "B" (8-coil magnetic-field measurement):** *"8 identical solenoids, each of length $l_{coil} = 400$ mm, outer structural **flange diameter** ∅300 **mm**, effective **copper winding diameter** ∅291.7 **mm** ($R_{coil} = 0.14585$ m), $N = 2000$ turns of 0.15 mm enameled copper wire, with DC resistance $R_{DC} = 1789\,\Omega$ per coil, driven by current $I = 14.0$ mA (0.351 W/coil, 2.805 W in total)."*

$$\Delta y_{\rm peak}(L = 0.630957\,\mathrm{m}) \;=\; -1.2017108 \times 10^{-48}. \quad \text{(V22 Eq. 13)}$$

9.2 2. Mathematical Derivation: From Dirac–Maxwell Proper Time to the Clock-Shift Formulas

9.2.1 2.1. How Does the Dirac–Maxwell Action Distinguish Between H0 and the Conditional H1 Hypothesis?

Consider the local Dirac electron (ψ_e, mass m_e) and Coulomb/Maxwell (A_μ) action of a stationary atomic clock ($\mathbf{x} = \text{const.}$, $u^\mu = (c/\sqrt{-g_{00}}, \mathbf{0})$) on the effective clock metric $\tilde{g}_{\mu\nu}$:

$$S_{\rm atomic}[\tilde{g}_{\mu\nu}, \psi_e, A_\mu] \;=\; \int d^4x\, \sqrt{-\tilde{g}} \left[\bar{\psi}_e \left(i\hbar c\, \tilde{e}^\mu_a \gamma^a D_\mu - m_e c^2 \right) \psi_e - \frac{1}{4\mu_0} \tilde{g}^{\mu\alpha} \tilde{g}^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} \right].$$

By the tetrad transformation derived in Step 2 (`step2_conformal_disformal.md`), in the local comoving frame $\tilde{e}^0_{\ 0} = \sqrt{-\tilde{g}_{00}}$, so the energy (Bohr/hyperfine transition frequency $\nu_0 = \Delta E_{\rm atomic}/h$) of stationary Dirac eigenstates ($\psi_e \propto e^{-i\omega_{\rm coord} t}$) is constant in the local clock proper time ($d\tilde{\tau}_{\rm clock} = \sqrt{-\tilde{g}_{00}}\, dt/c$):

$$\nu_{\rm coord}(\mathbf{x}) \;=\; \nu_0 \sqrt{-\tilde{g}_{00}(\mathbf{x})}.$$

Because in a static spacetime ($\partial_t g_{\mu\nu} = 0$, $\partial_t \phi_L = 0$) the coordinate frequency $\omega_{\rm coord}$ of photons is conserved along the Killing vector $\xi^\mu = (\partial_t)^\mu$ between the emitter ($\mathbf{x}_A$) and a distant observer ($\mathbf{x}_B \to \infty$, where $g_{00} \to -1$, $f_L \to 1$), the observed frequency ratio is:

1. **H0 branch ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$ on the $\Psi_{\rm atomic}$ sector — minimal clock coupling):**

$$\sqrt{-\tilde{g}_{00}(\mathbf{x})} \;=\; \sqrt{-g_{00}(\mathbf{x})} \;=\; e^{\Phi_{\rm eff}(\mathbf{x})/c^2} \quad\Longrightarrow\quad \left.\frac{\nu_{\rm obs}}{\nu_0}\right|_{\rm H0} \;=\; e^{\Phi_{\rm eff}(\mathbf{x})/c^2}, \qquad 1 + z_{\rm H0} \;=\; e^{-\Phi_{\rm eff}(\mathbf{x})/c^2}.$$

   In a weak field ($\Phi_{\rm eff} = \Phi_N + \Delta\Phi_{\rm TAL}$), the fractional frequency shift relative to standard Newtonian GR is:

$$\Delta y_{\rm H0}(\mathbf{x}) \;\equiv\; \left(\frac{\nu_{\rm obs}}{\nu_0}\right)_{\rm H0} - \left(\frac{\nu_{\rm obs}}{\nu_0}\right)_{\rm GR} \;\simeq\; \frac{\Delta\Phi_{\rm TAL}(\mathbf{x})}{c^2} \;=\; \mathcal{O}\!\left(\frac{GM}{c^2 R} \cdot \phi_L\right) \;=\; \mathcal{O}\!\left(\frac{G^2}{c^4}\right).$$

   Therefore, the H0 signal is suppressed at second order ($\propto G^2/c^4$): in the V20 granite chamber $\Delta y_{\rm H0} \sim +10^{-48}$ to $+10^{-47}$, and for Sirius B $|\Delta v_{\rm H0}| \le 1.27$ m/s.

2. **Conditional H1 branch ($\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2} u_\mu u_\nu \;\Longrightarrow\; \tilde{g}_{00} = f_L^2 g_{00}$ — disformal lapse / Jordan clock coupling):**

$$\sqrt{-\tilde{g}_{00}(\mathbf{x})} \;=\; f_L(\mathbf{x})\, e^{\Phi_{\rm eff}(\mathbf{x})/c^2} \quad\Longrightarrow\quad \left.\frac{\nu_{\rm obs}}{\nu_0}\right|_{\rm H1} \;=\; f_L(\mathbf{x})\, e^{\Phi_{\rm eff}(\mathbf{x})/c^2}, \qquad 1 + z_{\rm H1} \;=\; \frac{e^{-\Phi_{\rm eff}(\mathbf{x})/c^2}}{f_L(\mathbf{x})}.$$

   In a weak field ($\phi_L \ll 1 \;\Longrightarrow\; f_L = e^{-\phi_L} \simeq 1 - \phi_L$), the excess frequency shift beyond Newtonian GR is measurable at first order ($\propto G/c^2$):

$$\boxed{\Delta y_{\rm H1}(\mathbf{x}) \;\simeq\; f_L(\mathbf{x}) - 1 \;\simeq\; -\phi_L(\mathbf{x}) \;=\; \begin{cases} -\gamma_L\, \bar{\rho}_L(\mathbf{x}) & \text{(Formulation I: 3D top-hat)}, \\ -\phi_L^{\rm Yuk}(\mathbf{x}) & \text{(Formulation II: screened Yukawa)}. \end{cases}}$$

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9.3 3. Quantitative Verification on Sirius B and on the Two-Arm V20 Laboratory Program

9.3.1 3.1. Sirius B ($M = 1.018\,M_\odot$, $R_* = 5586.5$ km, $n = 3$ Polytrope) and the V20 8558 kg Granite Chamber (V23 Table 3)

The exact numerical values from V23 `clock_coupling_H0_H1.csv` (Table 3) across both formulations are:

System	L [m]	$f_L(R_*)$ Top-Hat	$f_L(R_*)$ Yukawa	H0 Prediction	Conditional H1 Top-Hat	Conditional H1 Yukawa ($\alpha_Y = 1$)	Unit
Sirius B ($1.018\,M_\odot$)	1.0×10^3	1.000000	1.000000	80.70055	80.70055	80.70055	km/s
Sirius B ($1.018\,M_\odot$)	1.0×10^4	1.000000	1.000000	80.70055	80.70055	80.70055	km/s
Sirius B ($1.018\,M_\odot$)	1.0×10^6	1.000000	1.000000	80.69839	80.70094	80.74476	km/s
Sirius B ($1.018\,M_\odot$)	5.0×10^6	0.999984	0.999993	80.69487	85.38911	82.85137	km/s
V20 granite (8558 kg)	0.50	1.000000	1.000000	$+3.075 \times 10^{-48}$	-3.318×10^{-25} (-3.318×10^{-25})	-2.008×10^{-25} (-1.943×10^{-25})	Δy
V20 granite (8558 kg)	0.85	1.000000	1.000000	$+1.208 \times 10^{-47}$	-1.303×10^{-24} (-1.303×10^{-25})	-3.373×10^{-25} (-3.101×10^{-25})	Δy
V20 granite (8558 kg)	1.00	1.000000	1.000000	$+1.331 \times 10^{-47}$	-1.436×10^{-24} (-1.436×10^{-25})	-3.781×10^{-25} (-3.407×10^{-25})	Δy

(Note on the V20 rows: in the conditional H1 columns, the first value is the IN shift at $\mathbf{r} = \mathbf{0}$, and the value in parentheses is the IN−OUT differential signal at $D_{out} = 2.0$ m; the Newtonian GR shift is $\Phi_N(IN)/c^2 = -9.269 \times 10^{-24}$, IN−OUT $= -6.094 \times 10^{-24}$. The upper bound of the 103 kg static granite offset under canonical γ_L is $\leq 5.8 \times 10^{-26}$).

9.3.2 3.2. The V20 8-Coil Magnetic-Field Energy Experiment in the Scalar–Tensor / Disformal Interpretation

If the source term of the scalar field also couples to the energy density $T_{00}^{(\mathrm{EM})} = u_B = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0}$ of the electromagnetic energy–momentum tensor ($\rho_B(\mathbf{x}) = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0 c^2}$; although the trace of conformally invariant 4D Maxwell theory vanishes, $T^\mu_\mu(\mathrm{EM}) = 0$, in the medium-anchored disformal coupling $\tilde{g}_{00} = -f_L^2$ the electromagnetic energy component is weighted directly by f_L), then on the conditional H1 branch the ON/OFF modulation of the magnetic field gives:

$$\Delta y_B(\mathbf{0}; L) \;=\; -\gamma_L\, \bar{\rho}_{B,L}(\mathbf{0}) \;=\; -\frac{GL^2}{c^2} \cdot \frac{3}{4\pi L^3} \int_{|\mathbf{y}| \leq L} \frac{|\mathbf{B}(\mathbf{y})|^2}{2\mu_0 c^2}\, d^3y \;=\; -\frac{3G}{8\pi\mu_0 c^4 L} \int_{|\mathbf{y}| \leq L} |\mathbf{B}(\mathbf{y})|^2\, d^3y.$$

- **Geometric peak at $L = 0.630957$ m:** Integrating the Biot–Savart field of the 8 coils (⌀300 mm flange boundary, ⌀291.7 mm effective copper winding diameter, $l_{\mathrm{coil}} = 400$ mm, $N = 2000$, $I = 14.0$ mA), the polarity-even signal $\Delta y_{\mathrm{even}}(L) = \frac{1}{2}[\Delta y(+I) + \Delta y(-I)]$ reaches its extremum precisely at $L = 0.630957$ m ($\Delta y_{\mathrm{peak}} = -1.2017108 \times 10^{-48}$).
- **Zeeman masking and subtraction of the background cross-term:** Polarity reversal $+I \leftrightarrow -I$ cancels the linear Zeeman term and the $2\mathbf{B}_{\mathrm{bg}} \cdot \mathbf{B}_{\mathrm{coil}}$ cross-term; the co-directional quadratic Zeeman shift (at 0.86–43 μT: -4×10^{-20} to -9.9×10^{-17} for Sr, 3.4×10^{-12} to 8.6×10^{-9} for Cs) can be symmetrically reduced to $\mathbf{B}(\mathbf{0}) = \mathbf{0}$ at the center by alternating coil polarities.

9.4 4. Checks

1. **Analytical 3D Top-Hat Value of the V20 Granite Chamber at $L = 1.00$ m:**

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- Because the outer radius of the hollow granite cylinder is $R_{\rm out} = 0.85$ m and its half-height is $H/2 = 0.60$ m, the corner distance from the origin is $r_{\rm corner} = \sqrt{0.85^2 + 0.60^2} = \sqrt{1.0825} \approx 1.0404$ m. At $L = 1.00$ m, the sphere $B(\mathbf{0}, 1\text{ m})$ encloses nearly the entirety of the 8558 kg granite mass:

$$\Delta y_{\rm H1}^{\text{top-hat}}(L = 1.00\text{ m}) \;=\; -\frac{GL^2}{c^2}\,\frac{M_{\rm enc}(<1\text{ m})}{\frac{4}{3}\pi L^3} \;=\; -1.436\times10^{-24} \quad \checkmark$$

2. **Order-of-Magnitude Verification of the Sirius B H1 Velocity Shift at $L = 5.0\times10^6$ m:**

- $1-f_L(R_*) = 1-0.999984 = 1.6\times10^{-5} \implies \Delta v_{\rm H1} \simeq c\,\big(1-f_L(R_*)\big) \approx 3\times10^5$ km/s$\times 1.56\times10^{-5} \approx +4.69$ km/s, so $v_{\rm H1} = 80.69487 + 4.69424 = 85.38911$ km/s (Top-Hat) and 82.85137 km/s (Yukawa), matching Table 3 of V23 ✓.

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10 Step 9: High-Energy Photon Propagation in the Disformal/Conformal Scalar-Tensor Metric, the Effective Refractive Index $n_t(s) = f_L(s)^{-1}$, and the Multi-Source V17 H.E.S.S./MAGIC Gamma-Ray Timing Audit

Source file: `step9_vhe_photon_propagation_hess_dispersion_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step9_vhe_photon_propagation_hess_dispersion_en.md`
Previous Steps: `step1_dilaton_form_en.md` (`step1_dilaton_form.md`) – `step8_microscopic_clock_coupling_siriusb_la` (`step8_microscopic_clock_coupling_siriusb_lab_v20.md`)
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Local Archival Sources:

- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex` (Chapter 6, lines 63–99, Eq. 9 and the 5-source H.E.S.S./MAGIC control table; DOI: `10.5281/zenodo.23604423`)
- V17 Zenodo record: DOI `10.5281/zenodo.19666220` (SPARC 155 galaxies and multi-source H.E.S.S. VHE gamma-ray timing audit)
- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Sections 1 and 3, Eq. 1 and lines 161–166; DOI: `10.5281/zenodo.23639084`)

10.1 1. Verbatim Citations from Chapter 6 of the V22 Monograph (`hu_p2.tex`)

In Step 8 (`step8_microscopic_clock_coupling_siriusb_lab_v20.md`), we derived the stationary proper-time and redshift coupling of a comoving atomic clock ($\Psi_{\rm atomic} = (\psi_e, A_\mu)$) on the effective metric $\tilde{g}_{\mu\nu}$. The next logical step is to derive the **propagation of the Maxwell photon field ($F_{\mu\nu}$) across astronomical and cosmological baseline distances ($s \in [0, D]$)** from the scalar–tensor / disformal action principle, comparing it with the multi-source V17 H.E.S.S./MAGIC gamma-ray timing audit documented in Chapter 6 of the V22 monograph:

1. **Local Refractive Index of the Temporal Carrier ($n_t(s) = f_L(s)^{-1}$), Achromatic Delay, and Dispersion Parameterization (`hu_p2.tex`, Chapter 6, lines 68–74, Eq. 9):** > *"When a VHE gamma-ray photon traverses cosmic voids, filaments, host-galaxy halos, and galactic interstellar structures along a line of sight $s \in [0, D]$, the local refractive index of the temporal carrier is $n_t(s) = f_L(s)^{-1} = \exp(+\gamma_L \bar{\rho}_L(s))$. In the purely metric (geometric) limit, where all photon energies couple identically to the effective lapse factor $f_L(s)$, the arrival time experiences an achromatic Shapiro-like excess delay: $\Delta t_{geom} = c^{-1} \int_0^D \left[\exp(\gamma_L \bar{\rho}_L(s)) - 1\right] ds$. If, however, high-energy (E) photon wave packets undergo finite-wavelength dispersion by scattering across spatial gradients $\nabla \bar{\rho}_L$ on scale L, the parameterized form of the energy-dependent time lag between energy channels E_1 and E_2 is:"*

$$\Delta t(E_1, E_2) \;=\; \tau_n \, \frac{E_2^n - E_1^n}{E_{\rm ref}^n} \int_0^D \gamma_L \, \bar{\rho}_L(s) \, (1 + z(s))^n \, ds, \qquad n \in \{1, 2\}, \quad \text{(V22 Eq. 9)}$$

> *"where τ_n is the dispersion coupling coefficient and $D(z)$ is the redshift-weighted comoving path length."*

2. **The Five-Source V17 H.E.S.S. / MAGIC Control Table (`hu_p2.tex`, Chapter 6, lines 78–91):**

VHE Source	Redshift / Dist.	Energy Band (TeV)	Observed Lag / Bound	V17 Audit Result
PKS 2155-304 (H.E.S.S.)	$z = 0.116$	0.2–4.0 TeV	$\Delta t \approx +28 \pm 12$ s	Marginal $\sim 2\sigma$ lag; dominated by intrinsic SSC emission profile.

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VHE Source	Redshift / Dist.	Energy Band (TeV)	Observed Lag / Bound	V17 Audit Result
Mrk 501 (1997 / H.E.S.S.)	$z = 0.034$	0.15–10 TeV	$\Delta t \approx +240 \pm 60\,\mathrm{s}$	Larger lag at 3.4× lower z violates monotonic distance scaling!
GRB 190114C (MAGIC)	$z = 0.4245$	0.2–1.0 TeV	$\lvert\Delta t/\Delta E\rvert \leq 19\,\mathrm{s/TeV}$	Stringent cosmological bound ruling out large linear vacuum dispersion.
Crab Pulsar (H.E.S.S./MAGIC)	$d \approx 2.0\,\mathrm{kpc}$	0.1–1.5 TeV	$\lvert\Delta t\rvert < 0.4\,\mathrm{ms}$	Tight Galactic-disk bound across dense ISM ($\bar{n}_H \sim 1\,\mathrm{cm}^{-3}$).
Vela Pulsar (H.E.S.S. II)	$d \approx 0.287\,\mathrm{kpc}$	0.02–0.1 TeV	$\lvert\Delta t\rvert < 0.2\,\mathrm{ms}$	Sub-millisecond phase alignment across the dense Vela SNR shell.

3. **Two Main Conclusions of the V17 Multi-Source Audit (`hu_p2.tex`, Chapter 6, lines 95–98):** > - **Rejection of Naive Distance-Proportional Dispersion:** *"Because Mrk 501 ($z = 0.034$) exhibits a larger apparent flare lag than PKS 2155-304 ($z = 0.116$) while GRB 190114C ($z = 0.4245$) and the Crab/Vela pulsars yield tight null bounds, explaining the 28 s lag of PKS 2155-304 via a uniform intergalactic dispersion constant is refuted. Intrinsic blazar synchrotron self-Compton (SSC) acceleration timescales must be modeled first."* > - **Unique Environmental Overdensity Prediction vs. ΛCDM:** *"Unlike Lorentz-invariance-violating (LIV) models that predict a purely redshift-dependent delay $\Delta t \propto \int_0^z \frac{(1+z')\,dz'}{H(z')}$, the Time Attraction Law predicts that any genuine non-geometric timing residual must correlate with the baryonic overdensity integral ($\int \bar{\rho}_L(s)\,ds$) along lines of sight passing through rich galaxy clusters (such as Coma or Perseus) compared to cosmic voids at the same redshift."*

10.2 2. Mathematical Derivation: Light Cone, Effective Refractive Index, and EFT Dispersion in the Scalar-Tensor Framework

10.2.1 2.1. Conformal Invariance vs. Disformal Light Cone: Derivation of the Refractive Index $n_t(s) = f_L(s)^{-1}$

Let us examine the free Maxwell action ($S_{\mathrm{EM}} = -\frac{1}{4\mu_0}\int d^4x\sqrt{-\tilde{g}}\,\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$) under the two metric transformations derived in Step 2 (`step2_conformal_disformal.md`):

1. **Pure Conformal Transformation ($\tilde{g}^{\mathbf{conf}}_{\mu\nu} = f_L^2 g_{\mu\nu}$):** In $D = 4$ spacetime dimensions, $\sqrt{-\tilde{g}^{\mathrm{conf}}} = f_L^4\sqrt{-g}$ and $\tilde{g}^{\mu\alpha}_{\mathrm{conf}}\tilde{g}^{\nu\beta}_{\mathrm{conf}} = f_L^{-4}g^{\mu\alpha}g^{\nu\beta}$, so the factor $f_L^4 f_L^{-4} = 1$ cancels:

$$S_{\mathrm{EM}}\big[f_L^2 g_{\mu\nu}, A_\mu\big] \;=\; S_{\mathrm{EM}}[g_{\mu\nu}, A_\mu]\,.$$

   Therefore, in a purely conformal scalar–tensor coupling, the light cone ($\tilde{g}^{\mathrm{conf}}_{\mu\nu}k^\mu k^\nu = 0 \iff g_{\mu\nu}k^\mu k^\nu = 0$) is identical to that of the background metric.

2. **Medium-Anchored Disformal Clock Metric of V23 ($\tilde{g}^{\mathbf{disf}}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$):** If electromagnetic waves couple to the disformal metric $\tilde{g}^{\mathrm{disf}}_{\mu\nu}$ defined by the four-velocity u^μ of the baryonic medium, then in the local rest frame ($g_{\mu\nu} \simeq \eta_{\mu\nu}$, $u^\mu = (c, \mathbf{0})$) the line element is:

$$d\tilde{s}^2_{\mathrm{disf}} \;=\; -f_L(s)^2c^2dt^2 + ds^2, \qquad ds^2 = dx^2 + dy^2 + dz^2.$$

   In the geometric-optics (eikonal / null-geodesic) limit, $d\tilde{s}^2_{\mathrm{disf}} = 0$, from which the coordinate speed of light of the photon and the effective temporal refractive index of the medium follow:

$$v_{\mathrm{eff}}(s) \;=\; \frac{ds}{dt} \;=\; c\,f_L(s) \;=\; c\,\exp\!\big(-\phi_L(s)\big) \quad\Longrightarrow\quad \boxed{n_t(s) \;\equiv\; \frac{c}{v_{\mathrm{eff}}(s)} \;=\; f_L(s)^{-1} \;=\; \exp\!\big(+\phi_L(s)\big) \;=\; \exp\!\big(+\gamma_L\,\bar{\rho}_L(s)\big).}$$

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Integrating over the path $s \in [0, D]$ directly yields the achromatic Shapiro-like travel-time excess given in line 69 of V22 `hu_p2.tex`:

$$\Delta t_{\rm geom} \;=\; \int_0^D \left(\frac{1}{v_{\rm eff}(s)} - \frac{1}{c}\right) ds \;=\; \frac{1}{c}\int_0^D \Big[\exp\!\big(\gamma_L \bar{\rho}_L(s)\big) - 1\Big]\, ds \;\simeq\; \frac{1}{c}\int_0^D \phi_L(s)\, ds \qquad (\phi_L \ll 1).$$

Because $\Delta t_{\rm geom}$ does not depend on the photon energy $E = \hbar\omega$, in the pure eikonal limit the relative delay between photons of energies E_1 and E_2 emitted simultaneously vanishes: $\Delta t_{\rm geom}(E_1, E_2) = 0$.

10.2.2 2.2. Finite-Wavelength Scalar–Photon EFT Operators and Derivation of the Dispersion Formula V22 Eq. (9)

When the wavelength of high-energy (VHE, $E \sim 0.1$–10 TeV) gamma-ray photons interacts with gradients of the scalar field $\phi_L = \gamma_L \bar{\rho}_L$ (or $\phi_L^{\rm Yuk}$), the leading dimension-$4+n$ ($n \in \{1, 2\}$) scalar–Maxwell operator in the effective field theory (EFT) expansion with the medium four-velocity u^μ is:

$$\Delta\mathcal{L}^{(n)}_{\rm disp} \;=\; -\frac{c\,\tau_n}{4\mu_0}\,\frac{\phi_L(x)}{E^n_{\rm ref}}\, F_{\mu\nu}\left(i\hbar\, u^\alpha \nabla_\alpha\right)^n F^{\mu\nu}.$$

In the plane-wave approximation ($A_\mu \propto e^{ik_\alpha x^\alpha}$, where the local comoving photon energy is $E_{\rm loc}(s) = -\hbar u^\alpha k_\alpha = E_{\rm obs}\,(1 + z(s))$), the modified dispersion relation is:

$$\omega^2 \;=\; c^2k^2\left[1 - \frac{2c\,\tau_n}{n+1}\,\phi_L(s)\,\left(\frac{E_{\rm obs}(1+z(s))}{E_{\rm ref}}\right)^n\right].$$

Integrating the reciprocal of the group velocity ($v_g = \partial\omega/\partial k$) between two distinct observed energy channels ($E_1 < E_2$) yields V22 Eq. (9):

$$\boxed{\Delta t(E_1, E_2) \;=\; t_{\rm arr}(E_2) - t_{\rm arr}(E_1) \;=\; \tau_n\,\frac{E_2^n - E_1^n}{E^n_{\rm ref}}\int_0^D \phi_L(s)\,\big(1 + z(s)\big)^n\, ds \;=\; \tau_n\,\frac{E_2^n - E_1^n}{E^n_{\rm ref}}\int_0^D \gamma_L\,\bar{\rho}_L(s)\,\big(1 + z(s)\big)^n\, ds.}$$

10.3 3. Checks and Physical Implications

1. **Why Does the 5-Source V17 Audit Exclude Universal Vacuum Dispersion ($\phi_L = $ const.) and Support Local $\phi_L(\mathbf{x})$ Coupling?**
   - If dispersion were caused by density-independent vacuum LIV ($\phi_L(s) \equiv$ const.), then $\Delta t/\Delta E$ would increase strictly monotonically with redshift z. Instead, **Mrk 501** ($z = 0.034$, $\Delta t \approx +240 \pm 60$ s) shows an $\sim 8.6\times$ larger lag at $3.4\times$ lower redshift than **PKS 2155-304** ($z = 0.116$, $\Delta t \approx +28 \pm 12$ s), while **GRB 190114C** at $z = 0.4245$ gives $|\Delta t/\Delta E| \leq 19$ s/TeV and the Galactic **Crab** ($|\Delta t| < 0.4$ ms) / **Vela** ($|\Delta t| < 0.2$ ms) pulsars impose tight upper bounds ✓.
2. **Discriminating Prediction of Scalar–Tensor / Yukawa TAL Against LIV Models:**
   - Because the scalar field $\phi_L(s)$ is proportional to the baryonic mass density along the line of sight ($\phi_L = \gamma_L\bar{\rho}_L$ or $\phi_L^{\rm Yuk}$), for sources at identical redshift z the line-of-sight integral $\mathcal{I}_L \equiv \int_0^D \phi_L(s)(1+z(s))^n\, ds$ is proportional to the baryonic overdensity of intervening galaxy clusters (e.g., Coma, Perseus) relative to cosmic voids ✓.
3. **Vacuum Limit ($\bar{\rho}_L \to 0 \implies \phi_L \to 0$):**
   - $\phi_L(s) \to 0 \implies n_t(s) = e^0 = 1$, $\Delta t_{\rm geom} = 0$, and $\Delta t(E_1, E_2) = 0$: light propagation recovers the special/general relativistic vacuum light cone ✓.

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11 Step 10: The Nonlocal Bilateral Kernel Action (Formulation I), Lapse-Weighted Reciprocity Potential ($V_{\mathrm{eff},N}$), and the Moment/EFT Bridge Between Formulations I and II (Yukawa Scalar-Tensor)

Source file: step10_bilateral_kernel_reciprocity_eft_synthesis_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step10_bilateral_kernel_reciprocity_eft_synthesis_en.md
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step9_vhe_photon_propagation_hess_dispersio (step9_vhe_photon_propagation_hess_dispersion.md)
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Local Archival Sources:

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Sections 1.1 and 1.2, lines 37–66, Eqs. 2–3, Table 1; DOI: 10.5281/zenodo.23639084)

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p3.tex (Chapters 11 and 12, lines 94–120; DOI: 10.5281/zenodo.23604423)

11.1 1. Verbatim Citations from the V23 (50aa2db0) Manuscript and the V22 Monograph

In Steps 1–9, we developed Formulation II (the local screened Helmholtz–Yukawa scalar–tensor action) together with its observational and laboratory consequences. In this step, we derive **Formulation I (the nonlocal bilateral kernel action with auxiliary fields ϕ_L, λ_L, unweighted bilateral_eom, and lapse-weighted bilateral_lapse_weighted reciprocal equilibrium)** as well as the **exact moment expansion connecting Formulation I (3D spherical top-hat, Branch I-A/I-B) and Formulation II (Yukawa scalar–tensor, Branch II-Fluid, $\alpha_Y = 1$ vs. $\alpha_Y = 1/3$)**, strictly recording that the second-order gradient moments of the two kernels ($\frac{1}{10}L^2\nabla^2$ vs. $L^2\nabla^2$) differ from each other.

1. **Formulation I of V23: Nonlocal Positive-Definite Kernel Action and Lagrange Multiplier (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 1.1, lines 38–46, Eqs. 2–3):** > *"In Formulation I, a Lagrange multiplier field $\lambda_L(x)$ couples the auxiliary scalar $\phi_L(x)$ to the spatial top-hat average taken on the hypersurface Σ_u orthogonal to the fluid four-velocity u^μ ($h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu/c^2$):"*

$$S_{\mathrm{I},X}[\phi_L, \lambda_L, g_{\mu\nu}] = -\frac{c^4}{8\pi G\gamma_L}\int d^4x\sqrt{-g}\,\lambda_L(x)\left[\phi_L(x) - \gamma_L\int_{\Sigma_u} d^4y\sqrt{-g(y)}\,K_L(x,y)\,\rho_{\mathrm{src}}(y)\right], \quad \text{(V23 Eq. 2)}$$

> *"where $K_L(x,y) = \delta(\tau_u(x) - \tau_u(y))\,\Theta(L - d_\Sigma(x,y))/V_L(x)$ is the positive-definite 3D spherical kernel of radius L and proper volume $V_L = \frac{4}{3}\pi L^3$. Variation with respect to $\lambda_L(x)$, $\phi_L(x)$, and $\rho_{src}(x)$ yields the following Euler–Lagrange closure:"*

$$\frac{\delta S}{\delta\lambda_L(x)} = 0 \implies \phi_L(x) = \gamma_L\bar{\rho}_L(x), \quad f_L(x) = e^{-\phi_L(x)}; \qquad \frac{\delta S}{\delta\phi_L(x)} = 0 \implies \lambda_L(x) = \frac{8\pi G\gamma_L}{c^4}\frac{\partial\mathcal{L}_{\mathrm{eff}}}{\partial\phi_L}(x). \quad \text{(V23 Eq. 3)}$$

2. **Bilateral Kernel Reciprocity ($x \leftrightarrow y$), Lapse-Weighted $V_{\mathrm{eff},N}$, and the Two-Cell Variational Test (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 54):** > *"If λ_L is absorbed into the fluid in the form $\mathcal{L}_{fluid}(n,\phi_L) = -f_L(\phi_L)\,\varepsilon(n)$ and ε is varied both explicitly and inside the kernel, then $V_{eff}(x) = f_L(x) - \gamma_L\int K_L(y,x)f_L(y)\varepsilon(y)\,d^3y$ and the modified equilibrium is $dh/dr = -A(r) - dV_{recip}/dr$, $V_{recip} = \ln V_{eff}$. I solve this as a self-consistent fixed point (**bilateral_eom** rows in Table 1); it differs from the algebraically closed V21 mass mode. Varying the lapse-weighted integral $\int dV\,N f_L\varepsilon$ with $N = e^\nu$ (at fixed kernel and metric) gives $V_{eff,N}(x) = f_L - \frac{\gamma_L}{N(x)}\int dV_y\,N(y)K(y,x)f_L\varepsilon$, which I solve in the lapse equation with $P_{matter} = f_L P$ (**bilateral_lapse_weighted** rows; two-cell test $N = (1; 0.8)$, $\varepsilon = (1; 2)$, $\gamma = 0.1$: $(0.74882; 0.72084)$, unweighted 0.73160). Covariant closure still requires metric variation of the kernel and volume element; neither row is the EL solution of action (2)."*

3. **The $\alpha_Y = 1$ (Unit-Normalized) and $\alpha_Y = 1/3$ (Volume-Matched) Yukawa Normalizations (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 65):** > *"The unit-normalized kernel corresponds to $\alpha_Y = 1$, i.e., $\alpha_{std} = 1/(4\pi) \approx 0.0796$, which recovers the homogeneous top-hat limit ($\phi_L = \gamma_L\rho_{src}$); $\alpha_Y = 1/3$ ($\alpha_{std} = 1/(12\pi)$) is the volume-matched variant."*

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11.2 2. Mathematical Derivation

11.2.1 2.1. Derivation of Bilateral Kernel Reciprocity ($x \leftrightarrow y$) and Lapse-Weighted $V_{\mathrm{eff},N}(x)$

Consider the static energy functional in which the matter Lagrangian $\mathcal{L}_{\mathrm{fluid}}(n, \phi_L) = -f_L(\phi_L)\,\varepsilon(n)$ ($f_L(x) = \exp[-\gamma_L \int dV_y\, K_L(x,y)\varepsilon(y)]$) appears weighted by the lapse factor $N(x) = e^{\nu(x)}$ (with fixed background metric and fixed kernel $K_L(x,y)$):

$$E_{\mathrm{matter}}[\varepsilon] \;=\; \int dV_x\, N(x)\, f_L(x)\, \varepsilon(x), \qquad \phi_L(x) \;=\; \gamma_L \int dV_y\, K_L(x,y)\, \varepsilon(y).$$

Let us vary $E_{\mathrm{matter}}[\varepsilon]$ with respect to the energy density $\varepsilon(x)$! Because ε appears in two places — once as an explicit factor at point x and once in the exponent of $f_L(y) = \exp[-\gamma_L \int dV_x\, K_L(y,x)\varepsilon(x)]$ at every point y whose smoothing ball contains x — the functional derivative by the chain rule and by exchanging the dummy integration variables $x \leftrightarrow y$ (*bilateral kernel reciprocity*) is:

$$\delta E_{\mathrm{matter}} \;=\; \int dV_x\, N(x)\, f_L(x)\, \delta\varepsilon(x) \;+\; \int dV_y\, N(y)\, \varepsilon(y)\, \frac{\partial f_L(y)}{\partial \phi_L(y)}\, \gamma_L \int dV_x\, K_L(y,x)\, \delta\varepsilon(x).$$

Since $\frac{\partial f_L(y)}{\partial \phi_L(y)} = -f_L(y)$, exchanging the integration order of dV_y and dV_x in the second integral and factoring out $N(x)\,\delta\varepsilon(x)$ yields:

$$\delta E_{\mathrm{matter}} \;=\; \int dV_x\, N(x) \underbrace{\left[f_L(x) \;-\; \frac{\gamma_L}{N(x)} \int dV_y\, N(y)\, K_L(y,x)\, f_L(y)\, \varepsilon(y) \right]}_{\equiv\, V_{\mathrm{eff},N}(x)} \delta\varepsilon(x).$$

This derives the two bilateral potentials appearing in Section 1.1 (line 54) of V23:

1. **Lapse-weighted bilateral potential (`bilateral_lapse_weighted`):**

$$\boxed{V_{\mathrm{eff},N}(x) \;=\; f_L(x) \;-\; \frac{\gamma_L}{N(x)} \int dV_y\, N(y)\, K_L(y,x)\, f_L(y)\, \varepsilon(y).}$$

2. **Unweighted bilateral potential (`bilateral_eom`, $N(y)/N(x) \to 1$):**

$$\boxed{V_{\mathrm{eff}}(x) \;=\; f_L(x) \;-\; \gamma_L \int dV_y\, K_L(y,x)\, f_L(y)\, \varepsilon(y).}$$

   In the hydrostatic Euler equation ($\delta\varepsilon/\delta n = (\varepsilon + P)/n = \mu e^h$), the condition of constant chemical potential ($N(r)\, V_{\mathrm{eff}}(r)\, e^{h(r)} = \text{const.}$) after logarithmic differentiation directly gives the modified equilibrium equation:

$$\frac{dh}{dr} \;=\; -\frac{d\nu}{dr} \;-\; \frac{dV_{\mathrm{recip}}}{dr}, \qquad V_{\mathrm{recip}}(r) \;\equiv\; \ln V_{\mathrm{eff}}(r).$$

   Since $\gamma_L \int K_L f_L \varepsilon\, dV_y \;>\; 0$, the bilateral reciprocal term **reduces** V_{eff} in the dense stellar core ($V_{\mathrm{eff}}(0) < f_L(0)$), producing an inward additional compressive force that explains why the `bilateral_eom` and `bilateral_lapse_weighted` branches yield a lower maximum mass than GR ($M_{\max}^{\mathrm{SLy}} = 1.93214\, M_\odot$ and $1.90217\, M_\odot$ at $L = 10$ km, compared to $2.09822\, M_\odot$ in the V21 `mass` mode; V23 Table 1).

11.2.2 2.2. Exact Moment-Expansion Comparison: Where Do the 3D Top-Hat (Branch I) and Yukawa (Branch II) Formulations Agree and Where Do They Differ?

Expanding the source density $\rho_{\mathrm{src}}(\mathbf{x} + \mathbf{u})$ in a Taylor series around point $\mathbf{x}$ inside the 3D spherical top-hat average ($|\mathbf{u}| \le L$) gives:

$$\bar{\rho}_L(\mathbf{x}) \;=\; \frac{3}{4\pi L^3} \int_{|\mathbf{u}| \le L} \left[\rho_{\mathrm{src}}(\mathbf{x}) + u_i \partial_i \rho_{\mathrm{src}}(\mathbf{x}) + \frac{1}{2} u_i u_j \partial_i \partial_j \rho_{\mathrm{src}}(\mathbf{x}) + \dots \right] d^3u.$$

By spherical symmetry, $\int u_i\, d^3u = 0$ and $\int u_i u_j\, d^3u = \frac{1}{3}\delta_{ij} \int_0^L 4\pi u^4\, du = \frac{4\pi}{15} L^5\, \delta_{ij}$, so the gradient expansion of the 3D top-hat operator is:

$$\boxed{\phi_L^{\text{top-hat}}(\mathbf{x}) \;=\; \gamma_L\, \bar{\rho}_L(\mathbf{x}) \;=\; \gamma_L \left(1 \;+\; \frac{L^2}{10} \nabla^2 \;+\; \mathcal{O}(L^4 \nabla^4) \right) \rho_{\mathrm{src}}(\mathbf{x}).}$$

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On the other hand, the formal inverse of the Helmholtz–Yukawa equation of Formulation II ($(-L^2\nabla^2+1)\phi_L^{\rm Yuk} = \alpha_Y\gamma_L\rho_{\rm src}$) for small gradients is:

$$\boxed{\phi_L^{\rm Yuk}(\mathbf{x}) \;=\; \alpha_Y\,\gamma_L\left(1-L^2\nabla^2\right)^{-1}\rho_{\rm src}(\mathbf{x}) \;=\; \alpha_Y\,\gamma_L\left(1 \;+\; L^2\nabla^2 \;+\; \mathcal{O}(L^4\nabla^4)\right)\rho_{\rm src}(\mathbf{x}).}$$

Critical Theoretical Distinction (Point 5):

1. **Zeroth-Moment (Homogeneous $\nabla^2\rho_{\rm src}\to 0$) Limit:** Under the normalization $\alpha_Y = 1$ ($\alpha_{\rm std} = \frac{1}{4\pi} \approx 0.07958$), the 3D top-hat and the Yukawa Green's function agree **exclusively in the zeroth moment** ($\int K_L(\mathbf{u})\,d^3u = \int G_L(\mathbf{u})\,d^3u = 1$, i.e., in a homogeneous medium: $\phi_L^{\rm Yuk} = \gamma_L\rho_{\rm src} = \phi_L^{\text{top-hat}}$)!
2. **Difference in the Second-Order Gradient Moment ($\frac{1}{10}L^2\nabla^2$ vs. $L^2\nabla^2$) and Compact Support:** In an inhomogeneous medium ($\nabla^2\rho_{\rm src}\neq 0$), the two models are **not** the same effective field theory:
   - at fixed L and $\alpha_Y = 1$, the second-order $\nabla^2\rho_{\rm src}$ coefficient differs by a factor of 10 ($\frac{1}{10}L^2$ for the 3D top-hat kernel vs. $1\cdot L^2$ for the Yukawa Green's function; matching the second moment would require $\ell_Y = L/\sqrt{10}$);
   - the 3D top-hat kernel has compact support ($r \le L$, so the exterior stellar halo terminates sharply at $r = R_* + L$), whereas the Yukawa Green's function generates an exponentially decaying ($\propto e^{-r/L}/r$) curved exterior scalar tail over an infinite domain and an ADM mass contribution $\Delta M_{\phi,\rm ext}^{\rm curved}$;
   - the $\alpha_Y = \frac{1}{3}$ ($\alpha_{\rm std} = \frac{1}{12\pi} \approx 0.02653$) variant matches the ratio of the spherical volume $\frac{4}{3}\pi L^3$ to the Yukawa normalization volume $4\pi L^3$ ($\frac{4\pi/3}{4\pi} = \frac{1}{3}$) at the stellar center ($r\to 0$).

11.3 3. Checks

1. **Exact Reproduction of the V23 Two-Cell Variational Test (`bilateral_lapse_weighted` vs. `bilateral_eom`):**
   - Consider two cells of equal volume ($\Delta V_1 = \Delta V_2 = 1/2$, symmetric overlapping kernel $K_{ij} = 1$, so $\bar\varepsilon = \frac{1}{2}(\varepsilon_1+\varepsilon_2)$), where according to line 54 of V23, $N = (1.0;\ 0.8)$, $\varepsilon = (1.0;\ 2.0)$, and $\gamma_L = 0.1$:
     - Mean density: $\bar\varepsilon = \frac{1+2}{2} = 1.5 \implies \phi_L = \gamma_L\bar\varepsilon = 0.15 \implies f_L(1) = f_L(2) = e^{-0.15} = 0.86070798$.
     - **Unweighted (`bilateral_eom`):**

$$V_{\rm eff}(1) = V_{\rm eff}(2) = f_L\big(1-\gamma_L\bar\varepsilon\big) = e^{-0.15}\times(1-0.15) = 0.85\,e^{-0.15} = \mathbf{0.73160178} \approx \mathbf{0.73160} \quad \checkmark$$

     - **Lapse-weighted (`bilateral_lapse_weighted`):** The lapse-weighted source average is $\overline{N\varepsilon} = \frac{1}{2}(N_1\varepsilon_1 + N_2\varepsilon_2) = \frac{1}{2}(1.0\times 1.0 + 0.8\times 2.0) = \frac{2.6}{2} = 1.3$:

$$V_{{\rm eff},N}(1) = f_L\left(1-\frac{\gamma_L\,\overline{N\varepsilon}}{N_1}\right) = e^{-0.15}\left(1-\frac{0.13}{1.0}\right) = 0.87\,e^{-0.15} = \mathbf{0.74881594} \approx \mathbf{0.74882} \quad \checkmark$$

$$V_{{\rm eff},N}(2) = f_L\left(1-\frac{\gamma_L\,\overline{N\varepsilon}}{N_2}\right) = e^{-0.15}\left(1-\frac{0.13}{0.8}\right) = 0.8375\,e^{-0.15} = \mathbf{0.72084293} \approx \mathbf{0.72084} \quad \checkmark$$

   - All three values agree digit for digit with the text of the V23 manuscript (`tal_covariant_closure_50aa2db0_EN_2026-` / `HU`, line 54).

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12 Step 11: Environment-Dependent Scalar-Tensor Screening Extension Hypothesis (Branch II-Cham: $V_{\text{eff}}(\phi_L;\rho)$, Chameleon/Symmetron $\beta_{\text{eff}}(\rho)$), Correlation Length $L_{\text{eff}}(\rho)$, and Multi-Scale EFT Synthesis

Source file: `step11_chameleon_screening_multiscale_eft_synthesis_en.md`

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step11_chameleon_screening_multiscale_eft_synthesis_en.md`
Previous Steps: `step1_dilaton_form_en.md` (`step1_dilaton_form.md`) – `step10_bilateral_kernel_reciprocity_eft_syn` (`step10_bilateral_kernel_reciprocity_eft_synthesis.md`)
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Local Archival Sources:

- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex` (Chapters 1 and 3, lines 171–180 and 256–265, Eq. 6), `hu_p2.tex` (Chapters 4–5, lines 24–61), and `hu_p3.tex` (Chapters 11–12, lines 99–120; DOI: `10.5281/zenodo.23604423`)

- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Sections 1–3; DOI: `10.5281/zenodo.23639084`)

12.1 1. Verbatim Citations and Methodological Delimitation of the Branch II-Cham Extension

In Steps 1–10, under a fixed correlation/Yukawa length L and fixed coupling $\alpha_Y \in \{1, 1/3\}$ ($\alpha_{\text{std}} = \frac{\alpha_Y}{4\pi}$), we derived the equations of V23 Formulation I (3D spherical top-hat, **Branch I-A/I-B**) and Formulation II (local quadratic Helmholtz–Yukawa scalar–tensor, **Branch II-Fluid / II-Conf / II-Disf**). As shown in Step 6 (`step6_sparc_galactic_ngc3198_consistency.md`) via the NGC 3198 ($R = 30.08$ kpc) negative control check, the isolated V23 model with constant coupling $\alpha_{\text{std}} = 1/(4\pi) \approx 0.07958$ yields at most a squared-velocity boost of $C_{\text{pred}}^{\text{isol}} \leq 1 + \alpha_{\text{std}} = 1.0796$, compared to the observed $C_{\text{req}} = 2.720$–4.311.

Methodological Delimitation (Point 5): The density-dependent $\beta_{\text{eff}}(\rho)$ and $V(\phi_L)$ chameleon/symmetron mechanism examined in this step (**Branch II-Cham** in the taxonomy of Step 2) is a **standalone nonlinear extension hypothesis** that goes beyond the quadratic, constant-α_Y Yukawa action of V23 (`50aa2db0`), providing a candidate 4D effective field theory (EFT) framework for open task **(d)** of Chapter 11 of the V22 monograph (`hu_p3.tex`, line 104: *"together with a physical scale-selection mechanism for L"*).

1. **The V5–V16 Chameleon and Horndeski Precedents, and the Force Multiplier $C(\rho)$ (`hu_p1.tex`, lines 172–174 and 256–265, Eq. 6):** > *"V5 – V7: Weak-Field and Chameleon Explorations — I investigated scalar–tensor and chameleon-like screening analogies to connect local density gradients with low-density spatial effective acceleration."* > *"In Versions 14–16 (DOI `10.5281/zenodo.23217189`), during my first phenomenological studies of galactic rotation curves, I parameterized the effective gravitational boost using the following density-dependent force multiplier:"*

$$C(\rho) \;=\; 1 + C_0 \,\exp\!\left(-\frac{\rho}{\rho_c}\right), \qquad V_{\text{rot}}^2(r) \;=\; C(\rho(r))\, V_{\text{bar}}^2(r), \quad \text{(V22 Eq. 6)}$$

   > *"where ρ_c is the critical transition density, and in the early works I set $C_0 = 5$ (so that for $\rho \ll \rho_c$, $C(\rho) \to 6$, whereas at high density $\rho \gg \rho_c \implies C(\rho) \to 1$, recovering the Newtonian limit in the Solar System)."*

2. **Open Theoretical Task (d) of Chapter 11 of the V22 Monograph (`hu_p3.tex`, line 104):** > *"**(d) Self-Consistent Multi-Channel Observational Evaluation:** Once the action principle and matter coupling in (a)–(c) are fixed, galactic rotation curves (including cosmic-background embedding ρ_{bg}), VHE gamma-ray timing across cluster overdensities, Sirius B, and NICER posteriors must be jointly fitted **together with a physical scale-selection mechanism for** L."*

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12.2 2. Mathematical Derivation: The Environment-Dependent Effective Potential $V_{\text{eff}}(\phi_L;\rho)$ and Scale Selection $L_{\text{eff}}(\rho)$

12.2.1 2.1. General Self-Interacting Scalar-Tensor Action and the Effective Potential

Let us generalize the quadratic mass term ($\frac{1}{2L_0^2}\phi_L^2$) of V23 Formulation II (Eq. 8) to a general self-interaction potential $V(\phi_L)$ in the 4-dimensional scalar–tensor action:

$$S \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}R \;-\; \frac{\kappa_\phi}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L \;-\; \kappa_\phi\,V(\phi_L) \;+\; f_L(\phi_L)\,\mathcal{L}_m\right], \qquad \kappa_\phi \;=\; \frac{c^4}{4\pi G\alpha_{\text{std}}} \;=\; \frac{c^4}{G\alpha_Y}, \quad f_L(\phi_L)=e^{-\phi_L}$$

(Note on normalization: $\kappa_\phi/2=\frac{c^4}{2G\alpha_Y}=\frac{c^4}{16\pi G}\frac{1}{\tilde\alpha_L}$, agreeing with the V23 Eq. 8 normalization used in Steps 3 and 7!) In a non-relativistic baryonic medium ($\mathcal{L}_m=-\rho c^2$), the Euler–Lagrange equation for the field ϕ_L is:

$$\Box_g\phi_L \;=\; \frac{\partial V_{\text{eff}}(\phi_L;\rho)}{\partial\phi_L}, \qquad \boxed{V_{\text{eff}}(\phi_L;\rho) \;\equiv\; V(\phi_L) \;+\; \frac{\rho c^2}{\kappa_\phi}\,e^{-\phi_L} \;=\; V(\phi_L) \;+\; \frac{\alpha_Y G}{c^2}\,\rho\,e^{-\phi_L}}$$

(where $\alpha_Y=4\pi\alpha_{\text{std}}=8\pi\tilde\alpha_L$).

12.2.2 2.2. Equilibrium Minimum $\phi_{\min}(\rho)$ and the Density-Dependent Yukawa Length $L_{\text{eff}}(\rho)$

In a locally quasi-homogeneous region of density ρ, the scalar field takes the minimum $\frac{\partial V_{\text{eff}}}{\partial\phi_L}\big|_{\phi_{\min}}=0$, and small perturbations $\delta\phi_L=\phi_L-\phi_{\min}(\rho)$ propagate with an **environment-dependent Yukawa length** ($L_{\text{eff}}(\rho)$) determined by the local curvature:

$$\left.\frac{\partial V_{\text{eff}}}{\partial\phi_L}\right|_{\phi_{\min}} \;=\; V'\big(\phi_{\min}(\rho)\big) \;-\; \frac{\alpha_Y G}{c^2}\,\rho\,e^{-\phi_{\min}(\rho)} \;=\; 0,$$

$$\boxed{L_{\text{eff}}^{-2}(\rho) \;\equiv\; \mu_{\text{eff}}^2(\rho) \;\equiv\; \left.\frac{\partial^2 V_{\text{eff}}(\phi_L;\rho)}{\partial\phi_L^2}\right|_{\phi_L=\phi_{\min}(\rho)} \;=\; V''\big(\phi_{\min}(\rho)\big) \;+\; \frac{\alpha_Y G}{c^2}\,\rho\,e^{-\phi_{\min}(\rho)}.}$$

Let us examine the two fundamental physical cases of this formula:

1. **Quadratic (Constant Bare Mass) Limit of V23 (**Branch II-Fluid**,** $V(\phi_L)=\frac{\phi_L^2}{2L_0^2}$**):** Here $V'(\phi_L)=\phi_L/L_0^2$ and $V''(\phi_L)=L_0^{-2}$. The minimum condition can be written in terms of the Lambert W function ($\phi_{\min}e^{\phi_{\min}}=\alpha_Y\gamma_{L_0}\rho$, so $\phi_{\min}(\rho)\simeq\alpha_Y\gamma_{L_0}\rho$ for small ϕ_L), and the local effective Yukawa length **decreases with increasing density** even from the bare $e^{-\phi_L}\rho c^2$ coupling alone:

$$L_{\text{eff}}^{-2}(\rho) \;=\; L_0^{-2}\Big(1+\phi_{\min}(\rho)\Big) \;\simeq\; L_0^{-2}\Big(1+\alpha_Y\,\gamma_{L_0}\,\rho\Big) \quad\Longrightarrow\quad L_{\text{eff}}(\rho) \;=\; \frac{L_0}{\sqrt{1+\phi_{\min}(\rho)}}.$$

2. **Nonlinear Chameleon / Symmetron Extension Hypothesis (**Branch II-Cham**,** `hu_p1.tex` **V5–V7):** If, going beyond the V23 quadratic model, the bare potential is $V(\phi_L)=\Lambda_0^4\exp(\sigma\phi_L)$ or $V(\phi_L)=M_0^4\phi_L^{-n}$ ($n>0$) and the coupling function has a general nonlinear form $f_{\text{eff}}(\phi_L)=\exp[-A(\phi_L)]$ whose effective slope at the local minimum is $\beta_{\text{eff}}(\rho)=\beta_0\exp\!\Big(-\frac{\rho}{2\rho_c}\Big)$, then the effective gravitational multiplier ($G_{\text{eff}}/G=1+2\beta_{\text{eff}}^2(\rho)$) reproduces the V14–V16 phenomenological force multiplier $C(\rho)$ (`hu_p1.tex`, Eq. 6):

$$\boxed{\frac{G_{\text{eff}}(\rho)}{G} \;=\; 1+2\beta_{\text{eff}}^2(\rho) \;=\; 1+2\beta_0^2\exp\!\left(-\frac{\rho}{\rho_c}\right) \;\equiv\; C(\rho), \qquad C_0 \;\equiv\; 2\beta_0^2 \;=\; \alpha_{\text{std},0}.}$$

   - **Galactic outer regime ($\rho\ll\rho_c\sim10^{-23}$–$10^{-21}$ kg/m^3):** in this nonlinear extension hypothesis, screening is lifted and $2\beta_{\text{eff}}^2(\rho)\to C_0$ (at the NGC 3198 $R=30.08$ kpc point, $C_{\text{req}}-1=1.720$ [$\Upsilon=1.0$] and 3.311 [$\Upsilon=0.5$]).
   - **High-density Solar System and terrestrial regime ($\rho\gg\rho_c$):** the exponential factor suppresses the long-range scalar force multiplier ($C(\rho)-1=C_0e^{-\rho/\rho_c}\to0$), leaving only the short-scale (L) canonical $\gamma_L=GL^2/c^2$ ($\alpha_{\text{std}}=1/(4\pi)$) V23 mode.

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12.3 3. Multi-Scale EFT Overview Across the V17–V23 Investigated Regimes (with Branch Labels)

The table below specifies for every physical system examined which model branch (**Branch I-A**, **Branch II-Fluid**, **Branch II-Disf**, **Branch II-Cham**) governs the corresponding result:

Physical Scale and System	Model Branch (`step2` Taxonomy)	Characteristic Density ρ [kg/m^3]	Investigated L Scale	$\phi_L = \gamma_L \bar{\rho}_L$ (or ϕ_L^{Yuk})	Primary Observable and V17–V23 Audit Outcome
V20 8-coil magnetic field (`step8`, `step20`)	**Branch II-Disf / I-A** (H_1)	$\rho_B = \frac{B^2}{2\mu_0 c^2} \sim 10^{-22}$	$L_{\mathrm{peak}} = 0.630957$ m	$1.2017108 \times 10^{-48}$	Polarity-even $\Delta y_{\mathrm{peak}} = -1.2017108 \times 10^{-48}$; Zeeman + cross-term canceled.
V20 8558 kg granite chamber (`step8`, `step19`)	**Branch I-A / II-Fluid** (H_0 vs. H_1)	2.75×10^3	$L = 0.50$–1.00 m	1.436×10^{-24} (1 m)	H_0: $+1.331 \times 10^{-47}$; conditional H_1: -1.436×10^{-24} (top-hat), -3.781×10^{-25} (Yukawa).
Sirius B white dwarf (`step8`, `step19`)	**Branch I-A / II-Fluid** (H_0 vs. H_1)	$\bar{\rho} \sim 3 \times 10^9$	$L = 1$–5000 km	1.6×10^{-5} (5000 km)	H_0: $\lvert\Delta v\rvert \leq 1.27$ m/s (compatible); H_1: excluded for $L \geq 1$ km ($L \lesssim 10^6$ m for local kernel).
Neutron stars (`step7`, `step18`, `step22`)	**Branch I-A** vs. **Branch II-Fluid**	10^{17}–1.45×10^{18}	$L = 1$–10 km	0.016–0.050 (10 km)	Curved exterior `yukawa_action_el` M_{ADM} and coupled $(\delta h, \delta\phi_L)$ stability (`step22`); V21 nonlocal tangent BVP (`step18`).
Solar System (PPN / Cassini) (`step5`)	**Branch II-Conf** vs. **II-Disf / I-A**	$\bar{\rho}_L^{\odot} \propto L^{-3}$	$L = 1$ AU–1 pc	1.14×10^{-14} (1 pc)	3D top-hat dilution: $\lvert 1 - f_L^{\odot}\rvert \approx 1.14 \times 10^{-14} \ll 10^{-5}$; for $r \gg L$, $\gamma_{\mathrm{PPN}} \to 1$.
SPARC (175/155) and NGC 3198 (`step6`)	**Branch I-A / II-Fluid** (negative control) vs. **II-Cham**	10^{-24}–10^{-20}	$L \sim 1$–23.09 kpc	$\leq 2.300 \times 10^{-7}$ (isolated)	0 points with $\rho \leq 0$ (kernel positivity); NGC 3198 (30.08 kpc): $C_{\mathrm{req}} = 2.720/4.311$, whereas constant α_{std}: $C_{\mathrm{pred}}^{\mathrm{isol}} \leq 1.0796$.

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Physical Scale and System	Model Branch (step2 Taxonomy)	Characteristic Density ρ [kg/m^3]	Investigated L Scale	$\phi_L = \gamma_L \bar{\rho}_L$ (or ϕ_L^{Yuk})	Primary Observable and V17–V23 Audit Outcome
V17 H.E.S.S./MAGIC gamma timing (step9)	**Branch II-Disf** (H_1 EM)	$\rho_{\mathrm{IGM/ISM}} \sim 10^{-27}$–$10^{-21}$	$z = 0.034$–0.4245	$\int_0^D \phi_L(s)\, ds$	Naive z-proportional LIV excluded; prediction correlates with cluster overdensity ($\int \bar{\rho}_L\, ds$).

12.4 4. Checks

1. **Limiting Values of the Correspondence $C(\rho) = 1 + 2\beta^2_{\mathbf{eff}}(\rho)$ (Branch II-Cham):**
   - $\rho \to 0$ (deep galactic outer regime): $2\beta^2_{\mathrm{eff}}(0) = 2\beta_0^2 = C_0 \implies C(0) = 1 + C_0$ ✓.
   - $\rho \gg \rho_c$ (stellar interior, Solar System, laboratory): $2\beta^2_{\mathrm{eff}}(\rho) = C_0 e^{-\rho/\rho_c} \to 0 \implies C(\rho) \to 1$ ✓.
2. **Verification of the Quadratic Yukawa Limit $L_{\mathbf{eff}}(\rho) = L_0/\sqrt{1 + \phi_{\min}(\rho)}$ (Branch II-Fluid):**
   - At low density ($\phi_{\min}(\rho) \to 0$), $L_{\mathrm{eff}}(\rho) \to L_0$, recovering the constant Yukawa length L_0 of V23 Formulation II ✓.

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13 Step 12: Second Variation ($\delta^2 S$) of the Scalar-Tensor / Yukawa Action, Explicit Coefficients of the Coupled $(\xi, \delta\phi_L)$ Radial Pulsation Operator, and Derivation of the Positive-Definite Physical Kinetic Hessian

Source file: `step12_second_variation_coupled_pulsation_hessian_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step12_second_variation_coupled_pulsation_hessian_en.md`
Previous Steps: `step1_dilaton_form_en.md` (`step1_dilaton_form.md`) – `step11_chameleon_screening_multiscale_eft_s` (`step11_chameleon_screening_multiscale_eft_synthesis.md`)
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Local Archival Sources:

- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p3.tex` (Chapter 8, lines 13–23, Eq. 12, and Chapter 11 task **(b)**, line 102; DOI: `10.5281/zenodo.23604423`)
- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Section 1.2, line 65 and Section 2, lines 120–153, Eqs. 8–9, Table 2; DOI: `10.5281/zenodo.23639084`)
- V21 Zenodo record: DOI `10.5281/zenodo.23602501`
- Numerical implementation and verification on the `yukawa_action_el` sequence: `curved_yukawa_adm_and_stability.py`, `curved_yukawa_coupled_stability.csv`, and `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stabilit`

13.1 1. Verbatim Citations from the V22 Monograph and the V23 (`50aa2db0`) Manuscript

In Step 7 (`step7_neutron_star_tov_stability.md`), we derived the curved-spacetime `yukawa_action_el` TOV background equations as well as the **static ($\omega_0^2 = 0$) nonlocal tangent model** ($\delta f_L[\delta h]$) around the V21 background. Both open task **(b)** of Chapter 11 of the V22 monograph (`hu_p3.tex`, line 102) and Section 2 of the V23 manuscript (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, lines 134–136) explicitly distinguish the static tangent model (and its Gram matrix) from the **genuine coupled dynamical pulsation system and physical kinetic Hessian derived from the second variation $\delta^2 S$ of the action**:

1. **Theoretical Task (b) of Chapter 11 of the V22 Monograph (`hu_p3.tex`, line 102):** > *"**(b) Genuine Modified Radial Stability Analysis Derived from the Coupling:** Once task (a) provides the dynamical field equations of the perturbations $\delta\phi$ and $\delta T^{\mu\nu}_X$, the Chandrasekhar radial pulsation operator must be re-derived variationally from the second variation $\delta^2 S$ of the action, eliminating the non-variational metric operator pathology flagged above $L \geq 367\,m$ and lifting the frozen-f ($\delta f_L = 0$) approximation."*
2. **Non-Variational Character of the V21 `gradient/metric` Operator Above $L \geq 367$ m (`hu_p3.tex`, Chapter 8, line 21):** > *"**`gradient/metric` operator (flagged as non-variational above $L \geq 367\,m$):** Substitutes the metric lapse derivative $\nu' = 2\Phi'_{eff}$ into the standard GR curvature terms of $\mathcal{Q}(r)$ without accounting for the perturbation of the X-sector tangential stress δp_{tX}. My audit openly flags this operator as **non-positive-definite and non-variational above $L \geq 367\,m$ across 12 sequences (609 models)**, where uncompensated residual terms produce unphysical eigenvalues $\omega_0^2 < 0$ even along the rising branch $\partial M/\partial\rho_c > 0$."*
3. **Inertia Invariance of the V23 Static Tangent Model and the Difference Between the Gram Matrix and the Physical Kinetic Hessian (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, Section 2, lines 134–136):** > *"Because at $\omega_0^2 = 0$ the problem reduces to $\mathcal{H}_{lin}[\delta h] = 0$, this agreement verifies static nonlocal linearization rather than dynamical inertia: scaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ yields a ratio $\omega_{4W}^2/\omega_W^2 = 0.2500$ ($\omega_W^2 = 4.897 \times 10^6\,s^{-2}$) with the identical shooting function $F(0) = 0.023252$, so the zero crossing is inertia-invariant. The raw BVP W matrix is not symmetric; the symmetrized Galerkin form $\mathbf{M}_{sym} = \mathbf{D}^T \mathrm{diag}(w\,\Delta r)\mathbf{D}$ is symmetric with $\min \mathrm{diag}\,\mathbf{M}_{sym} = 5.68 \times 10^{-9} > 0$; this is a Gram-matrix property, not a proof of positive-definiteness of the physical kinetic Hessian. The `omega0_sq_sym_tangent_s2` column gives the first-order tangent estimate ($\omega^2 \simeq -F(0)/F'(0)$)."*

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13.2 2. Mathematical Derivation: The Second Variation $\delta^2 S$, Elimination of Metric Constraints, and Explicit Coefficients

13.2.1 2.1. Time-Dependent Spherically Symmetric Perturbations and the Metric Constraints $(\delta\lambda, \delta m, \delta\nu')$

Consider the covariant action of V23 Formulation II (`yukawa_action_el`, **Branch II-Fluid**, $G = c = 1$) in a time-dependent, spherically symmetric spacetime ($ds^2 = -e^{2\nu(r,t)}dt^2 + e^{2\lambda(r,t)}dr^2 + r^2 d\Omega^2$, where on the static background $B_0(r) = e^{-2\lambda_0(r)} = 1 - 2m_0(r)/r$):

$$S[g_{\mu\nu}, \phi_L, \Psi_{\rm fluid}] \;=\; \int dt \int_0^\infty dr\, 4\pi r^2 e^{\nu+\lambda} \left[\frac{R}{16\pi} - \frac{1}{2\alpha_Y} \left(-e^{-2\nu}(\dot{\phi}_L)^2 + e^{-2\lambda}(\phi_L')^2 + \frac{\phi_L^2}{L^2} \right) - e^{-\phi_L}\varepsilon(n) \right].$$

Let us introduce small radial Lagrangian fluid displacements $\xi(r,t)$ (for which the enthalpy perturbation is $\delta h(r,t) = \frac{\delta P(r,t)}{\varepsilon_0(r)+P_0(r)}$) and the Eulerian scalar-field perturbation $\delta\phi_L(r,t)$ around the static equilibrium background $(\nu_0(r), \lambda_0(r), \phi_{L,0}(r), \varepsilon_0(r), P_0(r))$:

$$r_{\rm fluid}(r,t) \;=\; r + \xi(r,t), \qquad \phi_L(r,t) \;=\; \phi_{L,0}(r) + \delta\phi_L(r,t).$$

1. **Metric Constraints from the (tr) and (tt) Einstein Equations ($\delta\lambda$ and δm):** Integrating the spherically symmetric (tr) Einstein equation ($\delta G_t^{\,r} = 8\pi\, \delta T_t^{\,r}$) with respect to time in the interior ($0 < r < R_*$) gives:

$$\frac{\partial(\delta\lambda)}{\partial t} \;=\; -4\pi r e^{2\lambda_0} \left[f_{L,0}(\varepsilon_0 + P_0)\, \dot{\xi} - \frac{1}{\alpha_Y} \phi_{L,0}'\, \delta\dot{\phi}_L \right],$$

whereas from the (tt) Einstein equation ($\delta m'(r,t) = 4\pi r^2\, \delta(-T_t^{\,t})$, where $\delta B = -2\delta m/r = -2B_0\, \delta\lambda$) at fixed baryon number ($\delta N_B = 0$), the radial derivative of the mass perturbation in variables $(\delta h, \delta\phi_L, \delta\phi_L', \delta m)$ is:

$$\boxed{\frac{\partial(\delta m)}{\partial r} \;=\; 4\pi r^2 \left[f_{L,0} \frac{\varepsilon_0 + P_0}{c_s^2}\, \delta h - f_{L,0}\varepsilon_0\, \delta\phi_L + \frac{1}{\alpha_Y} \left(B_0 \phi_{L,0}'\, \delta\phi_L' + \frac{\phi_{L,0}}{L^2}\delta\phi_L - \frac{(\phi_{L,0}')^2}{r}\delta m \right) \right].}$$

Equivalently, expressed in terms of the Lagrangian displacement $\xi(r)$ (by substituting the baryon-number conservation constraint $\delta n = -\frac{1}{r^2 e^{\lambda_0}}(r^2 e^{\lambda_0} n_0 \xi)' - n_0 \delta\lambda$ and integrating by parts), the mass perturbation is:

$$\delta m(r) \;=\; -4\pi r^2 f_{L,0}(\varepsilon_0 + P_0)\, \xi(r) + \int_0^r 4\pi \tilde{r}^2 \left[\frac{B_0 \phi_{L,0}'}{\alpha_Y} \delta\phi_L' + \left(\frac{\phi_{L,0}}{\alpha_Y L^2} - f_{L,0}\varepsilon_0 \right) \delta\phi_L + f_{L,0}(\varepsilon_0 + P_0)\phi_{L,0}'\, \xi - \frac{(\phi_{L,0}')^2}{\alpha_Y \tilde{r}} \delta m \right] d\tilde{r}$$

2. **Lapse-Gradient Constraint from the (rr) Einstein Equation ($\delta\nu'$):** Because $\nu_0'(r) = \frac{m_0 + 4\pi r^3 P_{\rm tot,0}}{r(r-2m_0)}$, where $P_{\rm tot,0} = f_{L,0}P_0 + p_{r,\phi,0}$ and $p_{r,\phi,0} = \frac{1}{2\alpha_Y}[B_0(\phi_{L,0}')^2 - \phi_{L,0}^2/L^2]$, the total pressure perturbation is:

$$\delta P_{\rm tot} \;=\; f_{L,0}(\varepsilon_0 + P_0)\, \delta h - f_{L,0} P_0\, \delta\phi_L + \frac{1}{\alpha_Y} \left(B_0 \phi_{L,0}'\, \delta\phi_L' - \frac{\phi_{L,0}}{L^2}\delta\phi_L - \frac{(\phi_{L,0}')^2}{r}\delta m \right),$$

from which the perturbation of the lapse gradient follows:

$$\boxed{\delta\nu'(r) \;=\; \frac{1 + 2r\nu_0'(r)}{r\big(r - 2m_0(r)\big)}\, \delta m(r) \;+\; \frac{4\pi r^3}{r\big(r - 2m_0(r)\big)}\, \delta P_{\rm tot}(r).}$$

13.2.2 2.2. Second-Order Action $\delta^2 S$ and the Positive-Definite Physical Kinetic Hessian ($\mathbf{T_{kin}}$)

After substituting the metric constraints back into the quadratic second-order action ($\delta^2 S = \int dt\, (T_{\rm kin}[\dot{\xi}, \delta\dot{\phi}_L] - U_{\rm pot}[\xi, \delta\phi_L])$), the kinetic part under harmonic time dependence ($\xi(r,t) = \xi(r)e^{i\omega t}$, $\delta\phi_L(r,t) = \delta\phi_L(r)e^{i\omega t}$) is:

$$\boxed{T_{\rm kin}\big[\xi, \delta\phi_L\big] \;=\; \frac{\omega^2}{2} \int_0^{R_*} dr\, \mathcal{W}_\xi(r)\, \xi(r)^2 \;+\; \frac{\omega^2}{2} \int_0^\infty dr\, \mathcal{W}_\phi(r) \left[\delta\phi_L(r)\right]^2,}$$

where the two physical inertia (kinetic) weight functions are:

$$\boxed{\mathcal{W}_\xi(r) \;=\; 4\pi r^2\, e^{3\lambda_0(r) - \nu_0(r)}\, f_{L,0}(r) \big(\varepsilon_0(r) + P_0(r)\big) \;>\; 0 \quad (0 < r < R_*), \qquad \mathcal{W}_\phi(r) \;=\; \frac{4\pi r^2}{\alpha_Y}\, e^{\lambda_0(r) - \nu_0(r)} \;>\; 0 \quad (r > 0).}$$

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13.2.3 2.3. Explicit Coefficients of the Coupled $(\xi, \delta\phi_L)$ and $(\delta h, \delta\phi_L, \delta m)$ Eigenvalue System (Point 3)

The coupled radial pulsation system can be written in two strictly equivalent forms — the second-order Lagrangian $(\xi, \delta\phi_L)$ form and the first-/second-order canonical $(\delta h, \delta\phi_L, \delta m)$ form:

(A) Explicit Second-Order Coupled $(\xi, \delta\phi_L)$ Sturm–Liouville / Klein–Gordon Form:

$$\begin{cases} -\dfrac{d}{dr}\left(\mathcal{P}_\xi(r)\dfrac{d\xi}{dr}\right) + \mathcal{Q}_{\xi\xi}(r)\,\xi(r) + \mathcal{C}_{\xi\phi}[\delta\phi_L]\,(r) \;=\; \omega^2\,\mathcal{W}_\xi(r)\,\xi(r), \\ -\dfrac{d}{dr}\left(\mathcal{P}_\phi(r)\dfrac{d(\delta\phi_L)}{dr}\right) + \mathcal{Q}_{\phi\phi}(r)\,\delta\phi_L(r) + \mathcal{C}_{\phi\xi}[\xi]\,(r) \;=\; \omega^2\,\mathcal{W}_\phi(r)\,\delta\phi_L(r), \end{cases}$$

where the principal-part coefficients are:

$$\mathcal{P}_\xi(r) \;=\; 4\pi r^2\, e^{\nu_0(r)+\lambda_0(r)}\, f_{L,0}(r)\,\Gamma_1(r)\,P_0(r) \;>\; 0, \qquad \mathcal{P}_\phi(r) \;=\; \frac{4\pi r^2}{\alpha_Y}\, e^{\nu_0(r)-\lambda_0(r)} \;>\; 0,$$

the fluid–fluid potential coefficient ($\mathcal{Q}_{\xi\xi}$, with the modified hydrostatic gradient $P_0' = -(\varepsilon_0 + P_0)(\nu_0' - \phi_{L,0}')$ and the combined background pressure/energy $\rho_{\mathrm{tot},0} = f_{L,0}\varepsilon_0 + \rho_{\phi,0}$, $P_{\mathrm{tot},0} = f_{L,0}P_0 + p_{r,\phi,0}$) is:

$$\boxed{\begin{aligned} \mathcal{Q}_{\xi\xi}(r) \;=\; 4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}\Bigg\{ & \frac{4}{r}P_0' + \frac{(P_0')^2}{\varepsilon_0+P_0} - 8\pi e^{2\lambda_0}(\varepsilon_0+P_0)\big(f_{L,0}P_0 + p_{r,\phi,0}\big) \\ & - \frac{d}{dr}\left(\frac{2\Gamma_1 P_0}{r}\right) + (\varepsilon_0+P_0)\left[\phi_{L,0}'' + \big(\nu_0' + \lambda_0' - \phi_{L,0}'\big)\phi_{L,0}'\right]\Bigg\}, \end{aligned}}$$

the scalar–scalar potential coefficient ($\mathcal{Q}_{\phi\phi}$, including the metric back-reaction from $\delta\nu'$ and δB) is:

$$\boxed{\mathcal{Q}_{\phi\phi}(r) \;=\; \frac{4\pi r^2}{\alpha_Y}\, e^{\nu_0+\lambda_0}\left[\frac{1}{L^2} + \alpha_Y\, f_{L,0}(r)\,\varepsilon_0(r) + \frac{8\pi r\, e^{2\lambda_0}}{\alpha_Y L^2}\phi_{L,0}(r)\phi_{L,0}'(r)\right] \;+\; \mathcal{Q}_{\phi\phi}^{\mathrm{metric}}[\delta m],}$$

and the mutually adjoint fluid–scalar cross-coupling operators ($\langle \eta, \mathcal{C}_{\xi\phi}[\delta\phi_L]\rangle = \langle \mathcal{C}_{\phi\xi}[\eta], \delta\phi_L\rangle$) are:

$$\boxed{\begin{aligned} \mathcal{C}_{\xi\phi}[\delta\phi_L](r) \;&=\; 4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}(\varepsilon_0+P_0)\left[-\delta\phi_L'(r) + \delta\nu'\big|_{\delta\phi_L,\delta m[\delta\phi_L]}(r)\right], \\ \mathcal{C}_{\phi\xi}[\xi](r) \;&=\; -\frac{d}{dr}\Big(4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}(\varepsilon_0+P_0)\,\xi(r)\Big) \;+\; 4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}\frac{\varepsilon_0+P_0}{c_s^2}\,\delta h[\xi](r) \;+\; \mathcal{C}_{\phi\xi}^{\mathrm{metric}}[\delta m[\xi]](r). \end{aligned}}$$

(B) Exact Canonical $(\delta h, \delta\phi_L, \delta m)$ System and Moving-Surface Matching Conditions (Implemented in `curved_yukawa_adm_and_stability.py`): For numerical integration and exact handling of polytrope transitions (r_{trans}), the canonical variables $(\delta h, \delta\phi_L, \delta m)$ are optimal (since δh and δm remain continuous even across jump points of c_s^2!):

$$\boxed{\begin{aligned} \frac{d(\delta h)}{dr} \;&=\; -\delta\nu'(r) \;+\; \delta\phi_L'(r) \;+\; \omega^2\, e^{2(\lambda_0-\nu_0)}\,\xi(r), \\ \frac{d(\delta m)}{dr} \;&=\; 4\pi r^2\left[f_{L,0}\frac{\varepsilon_0+P_0}{c_s^2}\,\delta h - f_{L,0}\varepsilon_0\,\delta\phi_L + \frac{1}{\alpha_Y}\left(B_0\phi_{L,0}'\,\delta\phi_L' + \frac{\phi_{L,0}}{L^2}\delta\phi_L - \frac{(\phi_{L,0}')^2}{r}\delta m\right)\right], \\ B_0\,\delta\phi_L'' \;+\; & \left(\frac{2B_0}{r} + B_0\nu_0' + \frac{B_0'}{2}\right)\delta\phi_L' - \left(\frac{1}{L^2} + \alpha_Y f_{L,0}\varepsilon_0 - \omega^2 e^{-2\nu_0}\right)\delta\phi_L \\ & \;=\; -\alpha_Y f_{L,0}\frac{\varepsilon_0+P_0}{c_s^2}\,\delta h + \frac{2\,\delta m}{r}\phi_{L,0}'' - \left[\delta(B\nu') + \frac{\delta B'}{2} - \frac{2\,\delta m}{r^2}\right]\phi_{L,0}'. \end{aligned}}$$

- **Moving-Surface Boundary and Matching Conditions ($r = R_*$):**

  1. At the free surface, the Lagrangian pressure perturbation vanishes: $\Delta P(R_*) = \delta P(R_*) + P_0'(R_*)\xi(R_*) = 0 \implies \delta h(R_*) = -h_0'(R_*)\xi(R_*)$, which together with integrated mass conservation ($\delta m(R_*^-) = -4\pi R_*^2 f_{L,0}(R_*)(\varepsilon_0 + P_0)\big|_{R_*^-}\xi(R_*) = 0$, since for a polytropic crust $\varepsilon_0(R_*^-) = P_0(R_*^-) = 0$) guarantees $\delta m(R_*) \in C^0$.

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2. Across the moving surface $r = R_* + \xi(R_*)$, the scalar field and its normal derivative are continuous:

$$[\![\delta\phi_L + \xi\phi'_{L,0}]\!]_{R_*} = 0, \qquad [\![\delta\phi'_L + \xi\phi''_{L,0}]\!]_{R_*} = 0.$$

Since $\phi_{L,0} \in C^1([0,\infty))$, $[\![\phi'_{L,0}]\!]_{R_*} = 0 \implies [\![\delta\phi_L]\!]_{R_*} = 0$; furthermore, $[\![\phi''_{L,0}]\!]_{R_*} = \frac{\alpha_Y}{B_0(R_*)} f_{L,0}(R_*)\varepsilon_0(R_*^-) = 0$ at the surface of the polytropic crust ($\varepsilon_0(R_*^-) = 0$), so $[\![\delta\phi'_L]\!]_{R_*} = 0$ holds as well!

3. In the curved exterior vacuum ($r \in [R_*, R_{\max}]$), $\varepsilon_0 = P_0 = 0$, so the $(\delta\phi_L, \delta m)$ system is integrated in coupled form out to $R_{\max} = R_* + 12L$, where the asymptotic Yukawa decay condition closes the boundary-value problem, and the asymptotic mass perturbation $\delta M_{\rm ADM}(\omega^2) = \delta m(R_{\max}; \omega^2) = 0$ determines the eigenvalue!

13.3 3. Numerical Verification on the Curved Exterior `yukawa_action_el` Sequence (Point 4)

The script `curved_yukawa_adm_and_stability.py` solves the coupled $(\delta h, \delta\phi_L, \delta m)$ eigenvalue problem ($r \in [0, R_* + 14L]$, with the exact left-hand side sign `scalar_m_sign = -1.0`) directly on the `yukawa_action_el` equilibrium sequences equipped with the curved exterior solution. The results (`curved_yukawa_coupled_stability.csv` and `curved_yukawa_audit_summary.json`) are:

Sequence	$\log_{10}\rho_{0c}(dM_{\rm ADM}/d\rho_c = 0)$	$\log_{10}\rho_{0c}(dM_B/d\rho_c = 0)$	Dynamical $\omega_0^2 = 0$	$\Delta_{\rm ADM}$ Dynamical (dex)	Δ_{M_B} Dynamical (dex)	Quasi-Static $\omega_0^2 = 0$	Frozen $\delta\phi_L = 0$	$\Delta_{\rm ADM}$ Frozen (dex)
SLy, $L = 10$ **km,** $\alpha_Y = 1$	15.316404	15.316409	15.316410	+5.80e-06	+1.33e-06	15.316410	15.328565	+1.22e-02
SLy, $L = 10$ **km,** $\alpha_Y = 1/3$	15.307166	15.307169	15.307170	+4.20e-06	+6.41e-07	15.307170	15.311340	+4.17e-03
SLy, $L = 1$ **km,** $\alpha_Y = 1$	15.303086	15.303082	15.303080	-6.86e-06	-2.61e-06	15.303080	15.304220	+1.13e-03
APR4, $L = 10$ **km,** $\alpha_Y = 1$	15.293022	15.293025	15.293025	+3.57e-06	-1.40e-07	15.293025	15.301564	+8.54e-03
H4, $L = 10$ **km,** $\alpha_Y = 1$	15.227785	15.227787	15.227783	-1.34e-06	-4.05e-06	15.227783	15.251384	+2.36e-02

(GL2 Volterra, $N = 480$, `scalar_m_sign = -1.0`, dynamical scalar wave, finite-difference verified exterior $\partial\phi'_L(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ km^{-2} and interior $r = 6$ km $\partial\phi''_L/\partial m = -1.72833 \times 10^{-4}$ km^{-3} derivatives; N- and $R_{\max}$-convergence on the root: $p_{eff,root} = 4.248$, $\log_{10}\rho_{0c,Rich}(\omega_0^2 = 0) = 15.31640981$, see `step22`, Sections 4 and 7.)

13.4 4. Checks

1. $f_{L,0} \to 1$, $\alpha_Y \to 0$ **(GR Limit):**
   - For $\phi_{L,0} \to 0$ and $\delta\phi_L \to 0$, $\mathcal{W}_\xi(r) \to 4\pi r^2 e^{3\lambda_0 - \nu_0}(\varepsilon_0 + P_0)$, $\mathcal{P}_\xi(r) \to 4\pi r^2 e^{\nu_0 + \lambda_0}\Gamma_1 P_0$, and $\mathcal{Q}_{\xi\xi}(r)$ recovers the classical Chandrasekhar (1964) general-relativistic radial pulsation Sturm–Liouville operator ✓.
2. **Comparison Between the Coupled** $(\delta h, \delta\phi_L, \delta m)$ **Eigenvalue Zero Crossing and the ADM/Baryonic Mass Peaks on the `yukawa_action_el` Branch:**
   - For SLy $L = 10$ km, $\alpha_Y = 1$, the frozen-ϕ_L ($\delta\phi_L = 0$) approximation shifts the zero crossing by $+1.22 \times 10^{-2}$ dex from the $dM_{\rm ADM}/d\rho_c = 0$ turning point (15.328565 vs. 15.316404), whereas the coupled $(\delta h, \delta\phi_L, \delta m)$ operator (`scalar_m_sign = -1.0`) yields $\log_{10}\rho_{0c}(\omega_0^2 = 0) = 15.316410$ ($+5.80 \times 10^{-6}$ dex above the ADM mass peak and $+1.33 \times 10^{-6}$ dex above the baryonic mass peak, a 2097× reduction relative to the frozen-ϕ_L shift; across all 5 sequences the agreement is within $\leq 6.9 \times 10^{-6}$ dex, a 165 × –17593× reduction) ✓.

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3. **Positivity of the Kinetic Weights Above $L \geq 367$ m:**
   - Because $f_{L,0}(r) = e^{-\phi_{L,0}(r)} > 0$ and $\alpha_Y > 0$, $\mathcal{W}_\xi(r) > 0$ and $\mathcal{W}_\phi(r) > 0$ hold across every scale $L \in (0, \infty)$ without sign reversal ✓.

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14 Step 13: Metric Variation of the Curved-Spacetime Bilateral Top-Hat Window and Volume Element ($\delta[\sqrt{h}\,K_L(x,y)]/\delta g^{\mu\nu}$), Synge's World Function, and the $\delta_D(L-d_\Sigma)$ Boundary Surface Stress

Source file: step13_curved_window_metric_variation_synge_world_function_en.md

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Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step12_second_variation_coupled_pulsation_h (step12_second_variation_coupled_pulsation_hessian.md)
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Local Archival Sources:

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Abstract, line 28 and Section 1.1, lines 38–55, Eqs. 2–3; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex (Chapters 4 and 7) and hu_p3.tex (Chapters 8 and 11; DOI: 10.5281/zenodo.23604423)

14.1 1. Verbatim Citations from the V23 (50aa2db0) Manuscript

In Step 10 (step10_bilateral_kernel_reciprocity_eft_synthesis.md), we derived the unweighted (bilateral_eom) and lapse-weighted (bilateral_lapse_weighted) reciprocal equilibrium of V23 Formulation I **at fixed kernel and metric**. However, the abstract (line 28) and Section 1.1 (lines 38–54) of the V23 manuscript explicitly state that one open mathematical step remains for the covariant Euler–Lagrange closure of Formulation I (Eq. 2): **the variation of the curved-space window function $K_L(x,y)$ and the volume element with respect to the metric $g_{\mu\nu}$**:

1. **Covariant Action of V23 Formulation I and the Σ_u Hypersurface Kernel (tal_covariant_closure_50aa2db0_EN_2 / HU, Section 1.1, lines 38–46, Eqs. 2–3):** > *"In Formulation I, a Lagrange multiplier field $\lambda_L(x)$ couples the auxiliary scalar $\phi_L(x)$ to the spatial top-hat average taken on the hypersurface Σ_u orthogonal to the fluid four-velocity u^μ ($h_{\mu\nu}=g_{\mu\nu}+u_\mu u_\nu/c^2$):"*

$$S_{\mathrm{I},X}[\phi_L,\lambda_L,g_{\mu\nu}] = -\frac{c^4}{8\pi G\gamma_L}\int d^4x\sqrt{-g}\,\lambda_L(x)\left[\phi_L(x)-\gamma_L\int_{\Sigma_u} d^4y\sqrt{-g(y)}\,K_L(x,y)\,\rho_{\mathrm{src}}(y)\right], \quad \text{(V23 Eq. 2)}$$

> *"where $K_L(x,y)=\delta(\tau_u(x)-\tau_u(y))\,\Theta(L-d_\Sigma(x,y))/V_L(x)$ is the positive-definite 3D spherical kernel of radius L and proper volume $V_L=\frac{4}{3}\pi L^3$."*

2. **Curved-Space Window and Volume-Element Variation Flagged as Open in the V23 Manuscript (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 28 and 54):** > - **Abstract (line 28):** *"[T]his is not a dynamical proof of **curved-space window variation**, which remains an open question."* > - **Section 1.1 (line 54):** *"Varying the lapse-weighted integral $\int dV\,N f_L\varepsilon$ with $N=e^\nu$ (at fixed kernel and metric) gives $V_{\mathit{eff},N}(x)=f_L-\frac{\gamma_L}{N(x)}\int dV_y\,N(y)K(y,x)f_L\varepsilon$, which I solve in the lapse equation with $P_{matter}=f_LP$ (bilateral_lapse_weighted rows; two-cell test $N=(1;0.8)$, $\varepsilon=(1;2)$, $\gamma=0.1$: $(0.74882;0.72084)$, unweighted 0.73160). **Covariant closure still requires metric variation of the kernel and volume element; neither row is the EL solution of action (2).**"*

14.2 2. Mathematical Derivation: Metric Variation of the Curved-Space Volume Element and the Geodesic Window $d_\Sigma(x,y)$

14.2.1 2.1. Variation of the Hypersurface Integration Measure on Σ_u and the Proper Ball Volume $V_L(x)$

In a static spacetime ($ds^2=-N(x)^2c^2dt^2+h_{ij}(x)dx^idx^j$, where $N(x)=e^{\nu(x)}$ is the lapse function and $h_{ij}(x)$ is the induced 3-metric on the spatial hypersurface Σ_u, with $\sqrt{-g(x)}=N(x)\sqrt{h(x)}$), the kernel integral restricted to the

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simultaneous hypersurface $\tau_u(x) = \tau_u(y)$ is:

$$\bar{\rho}_L(x) \;=\; \int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\mathcal{K}_L(x,y)\,\varepsilon(y), \qquad \mathcal{K}_L(x,y) \;\equiv\; \frac{\Theta\big(L - d_\Sigma(x,y)\big)}{V_L(x)},$$

where $d_\Sigma(x,y)$ is the minimal spatial geodesic arc length connecting points x and y on Σ_u, and the proper volume of the geodesic ball $B_\Sigma(x,L) = \{y \in \Sigma_u : d_\Sigma(x,y) \le L\}$ is:

$$V_L(x) \;=\; \int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\Theta\big(L - d_\Sigma(x,y)\big).$$

By V23 Eq. (3), the Lagrange multiplier evaluated on the solution ($\phi_L = \gamma_L\bar{\rho}_L$, $\mathcal{L}_{\rm eff} = -f_L(\phi_L)\varepsilon$) is:

$$\lambda_L(x) \;=\; \frac{8\pi G\gamma_L}{c^4}\,\frac{\partial\big(-e^{-\phi_L(x)}\varepsilon(x)\big)}{\partial\phi_L(x)} \;=\; \frac{8\pi G\gamma_L}{c^4}\,f_L(x)\,\varepsilon(x).$$

When the action $S_{\rm I} = \int dt \int d^3x\, N(x)\sqrt{h(x)}\,\big[-f_L(x)\varepsilon(x)\big]$ is varied with respect to the spatial metric $h_{ij}(z)$ (at fixed $N(x)$ and $\varepsilon(x)$), three distinct metric contributions appear in the variation $\delta\phi_L(x) = \gamma_L\,\delta\bar{\rho}_L(x)$:

$$\delta\bar{\rho}_L(x) \;=\; \int_{\Sigma_u} d^3y\,\delta\Big[\sqrt{h(y)}\,\mathcal{K}_L(x,y)\Big]\varepsilon(y) \;=\; \delta\bar{\rho}_L^{(\rm vol)}(x) \;+\; \delta\bar{\rho}_L^{(\rm norm)}(x) \;+\; \delta\bar{\rho}_L^{(\rm bdry)}(x).$$

1. **Direct Variation of the Volume Element $\sqrt{h(y)}$ ($\delta\bar{\rho}_L^{(\mathbf{vol})}$):** Since $\delta\sqrt{h(y)} = \frac{1}{2}\sqrt{h(y)}\,h^{ij}(y)\,\delta h_{ij}(y)$:

$$\delta\bar{\rho}_L^{(\rm vol)}(x) \;=\; \frac{1}{2}\int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\mathcal{K}_L(x,y)\,\varepsilon(y)\,h^{ij}(y)\,\delta h_{ij}(y).$$

2. **Variation of the Normalization Volume $V_L(x)$ ($\delta\bar{\rho}_L^{(\mathbf{norm})}$):** Since $\delta\big(V_L(x)^{-1}\big) = -V_L(x)^{-2}\,\delta V_L(x)$, the volume-element variation from the normalization contributes a subtraction weighted by the local mean $\bar{\rho}_L(x)$:

$$\delta\bar{\rho}_L^{(\rm norm)}\big|_{\rm vol}(x) \;=\; -\bar{\rho}_L(x)\,\frac{1}{2}\int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\mathcal{K}_L(x,y)\,h^{ij}(y)\,\delta h_{ij}(y).$$

   The sum of the two volume-element terms is remarkable in its own right: **in a homogeneous medium ($\varepsilon(y) = \bar{\rho}_L(x)$), the two volume-element variations ($\delta\bar{\rho}_L^{(\mathbf{vol})} + \delta\bar{\rho}_L^{(\mathbf{norm})}\big|_{\mathbf{vol}}$) cancel each other identically!**

14.2.2 2.2. Variation of the Geodesic Distance $\delta d_\Sigma(x,y)$ (Synge's World Function) and the Boundary Surface Stress $\delta_D(L - d_\Sigma)$

Let $\gamma_{xy} : [0, d_\Sigma(x,y)] \to \Sigma_u$, $l \mapsto z^k(l)$ denote the minimal spatial geodesic curve connecting points x and y, parameterized by proper arc length l, with unit tangent vector $\hat{t}^k(l) = \frac{dz^k}{dl}$ ($h_{ij}\hat{t}^i\hat{t}^j = 1$). Because of the geodesic equation, the first variation of Synge's spatial world function ($\sigma_\Sigma(x,y) = \frac{1}{2}d_\Sigma(x,y)^2$) at fixed endpoints (x,y) is independent of the path displacement $\delta z^k(l)$ of the geodesic, and reduces purely to the line integral of the metric perturbation δh_{ij} along the geodesic:

$$\boxed{\delta d_\Sigma(x,y) \;=\; \frac{1}{2}\int_0^{d_\Sigma(x,y)} dl\,\hat{t}^i(l)\,\hat{t}^j(l)\,\delta h_{ij}\big(\gamma_{xy}(l)\big) \;=\; \frac{1}{2}\int_{\Sigma_u} d^3z\,\delta^{(3)}\big(z - \gamma_{xy}(l)\big)\,\hat{t}^i(l)\hat{t}^j(l)\,\delta h_{ij}(z)\,dl.}$$

Because the derivative of the Heaviside step function is the Dirac delta ($\frac{\partial}{\partial d_\Sigma}\Theta(L - d_\Sigma) = -\delta_D(L - d_\Sigma)$), the displacement of the window boundary generates a **spherical surface ($d_\Sigma(x,y) = L$) boundary stress**:

$$\boxed{\delta\bar{\rho}_L^{(\rm bdry)}(x) \;=\; -\frac{1}{2V_L(x)}\oint_{d_\Sigma(x,y)=L} dA_y\,\big[\varepsilon(y) - \bar{\rho}_L(x)\big]\int_0^L dl\,\hat{t}^i(l)\hat{t}^j(l)\,\delta h_{ij}\big(\gamma_{xy}(l)\big),}$$

where dA_y is the proper surface element of the geodesic sphere $d_\Sigma(x,y) = L$, and the factor $\varepsilon(y) - \bar{\rho}_L(x)$ arises from combining the boundary variation of $\Theta(L - d_\Sigma)$ in the numerator with that of $V_L(x)$ in the denominator.

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14.2.3 2.3. Evaluation Inside a Spherically Symmetric Neutron Star ($h_{rr} = e^{2\lambda(r)} = B(r)^{-1}$) and Connection to Formulation II (Yukawa)

In a spherically symmetric metric ($ds^2_\Sigma = e^{2\lambda(r)}dr^2 + r^2 d\Omega^2$, where $e^{-2\lambda(r)} = B(r) = 1 - 2m(r)/r$), the radial metric variation $\delta h_{rr}(r) = 2e^{2\lambda(r)}\delta\lambda(r)$ in the (rr) Einstein equation ($G^r{}_r = 8\pi T^r{}_{r,\mathrm{tot}}$) adds the following nonlocal radial window pressure ($P_{r,\mathrm{window}}(r)$) to the `bilateral_lapse_weighted` equations:

$$P_{r,\mathrm{window}}(r) \;=\; -\frac{1}{4\pi r^2 N(r) e^{\lambda(r)}}\,\frac{\delta S_{\mathrm{I},X}}{\delta\lambda(r)}.$$

This expression illuminates two fundamental statements of the V23 manuscript:

1. **Why does `bilateral_lapse_weighted` computed with a flat-space kernel differ from the EL solution of action (2)?** Because the flat-space kernel ($d_{\mathrm{flat}}(x,y) = |\mathbf{x}-\mathbf{y}|$) omits the curved volume-element factor $e^{\lambda(r)} = (1-2m/r)^{-1/2}$ and the geodesic boundary line integral $\delta d_\Sigma(x,y)$.
2. **Why does Formulation II (Helmholtz–Yukawa, `yukawa_action_el`) eliminate the distributional surface boundary stress $\delta_D(L - d_\Sigma)$?** When the sharp top-hat window $\Theta(L - d_\Sigma)$ is replaced by the smooth 3D Helmholtz–Yukawa resolvent ($(-\Box_\Sigma + L^{-2})^{-1}$), the nonlocal boundary stress arising from geodesic line integrals transforms into the **local, continuously differentiable scalar pressure** derived in Step 7 (`step7_neutron_star_tov_stability.md`):

$$p_{r,\phi}(r) \;=\; \frac{1}{2\alpha_Y}\left(e^{-2\lambda(r)}\left[\phi_L'(r)\right]^2 - \frac{\phi_L(r)^2}{L^2}\right),$$

in which the variation of the factor $g^{rr} = e^{-2\lambda(r)} = B(r)$ ($\delta g^{rr} = -2e^{-2\lambda}\delta\lambda$) automatically and algebraically incorporates all metric back-reactions of the curved-space distance measure!

14.3 3. Checks

1. **Homogeneous-Density Limit ($\varepsilon(y) \equiv \varepsilon_0 = $ const.):**
   - If $\varepsilon(y) = \varepsilon_0$ is constant inside the ball $d_\Sigma(x,y) \le L$, then $\bar\rho_L(x) = \varepsilon_0$, so both the volume-element variation ($\varepsilon(y) - \bar\rho_L(x) = 0$) and the $\delta_D(L - d_\Sigma)$ surface boundary stress ($\varepsilon(y)\big|_{d_\Sigma = L} - \bar\rho_L(x) = 0$) **vanish identically**: $\delta\bar\rho_L(x) \equiv 0$ ✓.
2. **Flat-Spacetime ($\lambda(r) \to 0$, $N(r) \to 1$) Limit:**
   - $h_{ij} \to \delta_{ij} \implies d_\Sigma(x,y) \to |\mathbf{x}-\mathbf{y}|$, $V_L(x) \to \frac{4}{3}\pi L^3$, and the variational equation recovers the `bilateral_eom` / `bilateral_lapse_weighted` expressions derived in Step 10 ✓.

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15 Step 14: Conformal vs. Disformal Gravitational Lensing (Null-Geodesic Light Deflection Angle $\hat{\alpha}_{\mathbf{def}}(b)$), NICER Ray-Tracing, and Cosmological Background Embedding ($L_{\mathbf{hor}}^{(m)} \approx 22.94$ Gpc vs. $L_{\mathbf{hor}}^{(b)} \approx 57.98$ Gpc)

Source file: step14_gravitational_lensing_null_geodesics_cosmological_background_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step14_gravitational_lensing_null_geodesics_cosmological_background_en.m
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step13_curved_window_metric_variation_synge (step13_curved_window_metric_variation_synge_world_function.md)
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Local Archival Sources:

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 2, line 199), hu_p2.tex (Chapter 4, line 31; Chapter 5, lines 52–55), and hu_p3.tex (Chapter 9, lines 59–60; Chapter 11, line 104; DOI: 10.5281/zenodo.23604423)

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Section 3, lines 161–186; DOI: 10.5281/zenodo.23639084)

15.1 1. Verbatim Citations from the V22–V23 Manuscripts

In Step 5 (step5_coupling_dictionary_ppn_screening.md), we derived the weak-field PPN parameter $\gamma_{\rm PPN}$, and in Step 9 (step9_vhe_photon_propagation_hess_dispersion.md) we analyzed straight line-of-sight VHE gamma-ray time-of-flight dispersion. In this step, we derive three closely connected results: **(A)** what difference the purely temporal disformal clock metric ($\tilde{g}_{00} = f_L^2 g_{00}$, $\tilde{g}_{ij} = g_{ij}$, **Branch II-Disf**) induces in the **light deflection angle (gravitational lensing, $\hat{\alpha}_{\mathbf{def}}(b)$)** compared to the conformal case $\tilde{g}_{\mu\nu}^{\rm conf} = f_L^2 g_{\mu\nu}$ (**Branch II-Conf**); **(B)** how **cosmological background embedding** ($\gamma_L \rho_{\rm bg} = 1$) enters the exponential factor, computing separately the horizon scales associated with the baryonic density ($\Omega_b = 0.0493 \implies L_{\rm hor}^{(b)} \approx 57.98$ Gpc) and the total cosmic matter density ($\Omega_m \approx 0.315 \implies L_{\rm hor}^{(m)} \approx 22.94$ Gpc); and **(C)** what this implies for **NICER pulsar ray-tracing likelihoods** (PSR J0030+0451, PSR J0740+6620, PSR J0437-4715).

1. **Temporal Refractive Index and Cosmological Background Embedding in the V22 Monograph (hu_p1.tex line 199 and hu_p2.tex line 54):**
   - hu_p1.tex, line 199: > *"I interpret the flow of time as possessing an effective local carrier density (analogous to an optical medium with refractive index $n_t(\mathbf{x}) = f_L(\mathbf{x})^{-1} = \exp(+\gamma_L \bar{\rho}_L(\mathbf{x}))$ or a compressible time fluid)."*
   - hu_p2.tex, Chapter 5, line 54: > *"**Cosmological Background Embedding** ($L \approx 22.94$ **Gpc**): For $\bar{\rho}_L$ on megaparsecs scales to reach $\gamma_L \bar{\rho}_L \sim 1$, the ball $B(\mathbf{x}, L)$ must include the ambient intergalactic and cosmological background density $\rho_{bg} \approx \Omega_b \rho_{crit}$. At uniform cosmic baryon density, the condition $\gamma_L \bar{\rho}_{bg} = 1$ occurs at the formal horizon scale $L \approx 22.94$ Gpc (7.08×10^{26} m)."*
2. **NICER Pulsar Likelihoods in the V22 Monograph (hu_p3.tex, lines 59–60 and 104):**
   - hu_p3.tex, Chapter 9, line 60: > *"The joint log-likelihood computed for the three NICER pulsars (PSR J0030+0451, PSR J0740+6620, PSR J0437-4715) under causal endpoint ($M_{max}(causal)$) mass normalization shows that at $L = 1$ km the shift relative to GR is at most 4.8×10^{-4} **dex** ($\Delta \ln \mathcal{L} \approx 1.1 \times 10^{-3}$). Consequently, **current NICER mass–radius data cannot statistically distinguish the scale L from GR under either hypothesis**."*

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15.2 2. Conformal vs. Purely Temporal Disformal Gravitational Lensing ($\hat{\alpha}_{\mathbf{def}}(b)$)

Consider a static, weak gravitational field in which the line element of the Einstein-frame gravitational metric ($g_{\mu\nu}$) in isotropic coordinates is:

$$ds^2 \;=\; g_{\mu\nu}dx^\mu dx^\nu \;=\; -\left(1+\frac{2\Phi_N(\mathbf{x})}{c^2}\right)c^2dt^2 \;+\; \left(1-\frac{2\Psi_N(\mathbf{x})}{c^2}\right)d\mathbf{x}^2, \qquad \Phi_N(\mathbf{x})=\Psi_N(\mathbf{x})=-\frac{GM(<r)}{r}.$$

15.2.1 2.1. Light Deflection under Purely Conformal Jordan Coupling (Branch II-Conf: $\tilde{g}^{\mathbf{conf}}_{\mu\nu}=f_L^2 g_{\mu\nu}$)

If the Maxwell field couples to the conformally rescaled metric $\tilde{g}^{\rm conf}_{\mu\nu}=f_L(\mathbf{x})^2 g_{\mu\nu}=e^{-2\phi_L(\mathbf{x})}g_{\mu\nu}$, then in $D=4$ spacetime dimensions the conformal invariance $\sqrt{-\tilde{g}^{\rm conf}}\tilde{g}^{\mu\alpha}_{\rm conf}\tilde{g}^{\nu\beta}_{\rm conf}F_{\mu\nu}F_{\alpha\beta}=\sqrt{-g}g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$ ensures that null geodesics of light rays ($\tilde{g}^{\rm conf}_{\mu\nu}dx^\mu dx^\nu=0 \iff g_{\mu\nu}dx^\mu dx^\nu=0$) are **independent of the scalar field** $\phi_L(\mathbf{x})$. The effective optical refractive index entering Fermat's principle ($\delta\int n_{\rm conf}(\mathbf{x})\,dl=0$) is:

$$n_{\rm conf}(\mathbf{x}) \;=\; \sqrt{\frac{g_{rr}}{-g_{00}}} \;\simeq\; 1 \;-\; \frac{\Phi_N(\mathbf{x})+\Psi_N(\mathbf{x})}{c^2} \;=\; 1 \;-\; \frac{2\Phi_N(\mathbf{x})}{c^2}.$$

Consequently, the deflection angle of a light ray with impact parameter b ($\hat{\alpha}_{\rm def}(b)=-\int_{-\infty}^{+\infty}\nabla_\perp \ln n\,dz$) under purely conformal coupling **senses only the baryonic Weyl potential** ($\Phi^{\mathbf{conf}}_{\mathbf{lens}}=\frac{1}{2}(\Phi_N+\Psi_N)=\Phi_N$):

$$\boxed{\hat{\alpha}^{\rm conf}_{\rm def}(b) \;=\; \frac{2}{c^2}\int_{-\infty}^{+\infty}\nabla_\perp\Phi_N\big(\sqrt{b^2+z^2}\big)\,dz \;=\; \frac{4GM_{\rm bary}(<b)}{c^2 b}.}$$

That is, in purely conformal coupling the scalar field (ϕ_L) enhances the circular orbital velocity of non-relativistic stars and gas ($V^2_{\rm rot}(r)=V^2_{\rm bar}(r)-c^2 r\phi_L'(r)$), **but does not contribute to gravitational lensing**!

15.2.2 2.2. Light Deflection in the V23 Purely Temporal Disformal Clock Metric (Branch II-Disf: $\tilde{g}_{00}=f_L^2 g_{00}$, $\tilde{g}_{ij}=g_{ij}$)

By contrast, if electromagnetic waves propagate on the V23 disformal physical metric ($\tilde{g}^{\rm disf}_{\mu\nu}=g_{\mu\nu}+\frac{1-f_L^2}{c^2}u_\mu u_\nu$), then the null-cone equation ($d\tilde{s}^2_{\rm disf}=0$) is:

$$0 \;=\; \tilde{g}^{\rm disf}_{00}\,c^2dt^2 \;+\; \tilde{g}^{\rm disf}_{ij}\,dx^i dx^j \;=\; -f_L(\mathbf{x})^2\left(1+\frac{2\Phi_N(\mathbf{x})}{c^2}\right)c^2dt^2 \;+\; \left(1-\frac{2\Psi_N(\mathbf{x})}{c^2}\right)d\mathbf{x}^2.$$

From this, the effective optical refractive index associated with Fermat's principle (`hu_p1.tex`, line 199: $n_t(\mathbf{x})=f_L(\mathbf{x})^{-1}=e^{+\phi_L(\mathbf{x})}$) is:

$$n_{\rm disf}(\mathbf{x}) \;=\; \frac{c}{v_{\rm phase}(\mathbf{x})} \;=\; f_L(\mathbf{x})^{-1}\sqrt{\frac{1-2\Psi_N/c^2}{1+2\Phi_N/c^2}} \;\simeq\; 1 \;-\; \frac{\Phi_N(\mathbf{x})+\Psi_N(\mathbf{x})}{c^2} \;+\; \phi_L(\mathbf{x}).$$

Substituting the transverse gradient $\nabla_\perp \ln n_{\rm disf}\simeq -\frac{1}{c^2}\nabla_\perp(\Phi_N+\Psi_N-c^2\phi_L)$ into the Fermat integral, the disformal lensing potential and light deflection angle are:

$$\boxed{\Phi^{\rm disf}_{\rm lens}(\mathbf{x}) \;=\; \Phi_N(\mathbf{x}) \;-\; \frac{1}{2}c^2\,\phi_L(\mathbf{x}), \qquad \hat{\alpha}^{\rm disf}_{\rm def}(b) \;=\; \frac{4GM_{\rm bary}(<b)}{c^2 b} \;-\; \int_{-\infty}^{+\infty}\frac{b}{r}\,\frac{d\phi_L(r)}{dr}\,dz.}$$

Since $\frac{d\phi_L}{dr}<0$ for an attractive time potential (density decreases outward), the second term produces a **positive (inward-directed) excess light deflection**: the disformal temporal refractive index ($n_t=e^{+\phi_L}$) around galaxies and galaxy clusters amplifies not only rotation curves, but gravitational lensing as well!

15.3 3. Cosmological Background Embedding: Exact Arithmetic for $\Omega_b=0.0493$ (57.98 Gpc) and $\Omega_m=0.315$ (22.94 Gpc)

Let us decompose the cosmic mass–energy density into a spatially homogeneous FLRW background density ($\rho_{\rm bg}(t)=\Omega_{\rm bg}\,\rho_{\rm crit,0}$) and the local overdensity ($\delta\rho(\mathbf{x})$):

$$\rho(\mathbf{x}) \;=\; \rho_{\rm bg} \;+\; \delta\rho(\mathbf{x}) \quad\Longrightarrow\quad \bar{\rho}_L(\mathbf{x}) \;=\; \frac{3}{4\pi L^3}\int_{|\mathbf{y}-\mathbf{x}|\le L}\big[\rho_{\rm bg}+\delta\rho(\mathbf{y})\big]d^3y \;=\; \rho_{\rm bg} \;+\; \delta\bar{\rho}_L(\mathbf{x}).$$

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15.3.1 3.1. General Formula and Exact Separation of the Two Cosmological Density Parameters (Ω_b vs. Ω_m)

Using the Planck 2018 (TT,TE,EE+lowE+lensing) parameters:

$$H_0 \;=\; 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}} \;=\; 2.184285\times10^{-18}\,\mathrm{s^{-1}}, \qquad R_H \;\equiv\; \frac{c}{H_0} \;=\; 1.37250\times10^{26}\,\mathrm{m} \;=\; 4.4480\,\mathrm{Gpc},$$

$$\rho_{\mathrm{crit},0} \;=\; \frac{3H_0^2}{8\pi G} \;=\; 8.53287\times10^{-27}\,\mathrm{kg/m^3}.$$

The exact general solution of the unit-background-exponent condition ($\gamma_L\rho_{\mathrm{bg}}=1$, where $\gamma_L=\frac{GL_{\mathrm{hor}}^2}{c^2}$) for any density fraction $\Omega_{\mathrm{bg}}\equiv\rho_{\mathrm{bg}}/\rho_{\mathrm{crit},0}$ is:

$$\gamma_L\,\rho_{\mathrm{bg}} \;=\; \frac{GL_{\mathrm{hor}}^2}{c^2}\,\Omega_{\mathrm{bg}}\,\frac{3H_0^2}{8\pi G} \;=\; \frac{3\,\Omega_{\mathrm{bg}}}{8\pi}\left(\frac{L_{\mathrm{hor}}}{c/H_0}\right)^2 \;=\; 1 \quad\Longleftrightarrow\quad \boxed{L_{\mathrm{hor}}(\Omega_{\mathrm{bg}}) \;=\; \sqrt{\frac{8\pi}{3\,\Omega_{\mathrm{bg}}}}\,\frac{c}{H_0} \;=\; \frac{c}{\sqrt{G\,\rho_{\mathrm{bg}}}}.}$$

Observe that Newton's constant G cancels identically from the product $\gamma_L\rho_{\mathrm{bg}}$. To clarify the numerical values cited in line 54 of V22 `hu_p2.tex`, it is essential to separate the purely baryonic (Ω_b) and combined cosmic matter (Ω_m) cases:

Background Density Model	Density Parameter Ω_{bg}	Background Density $\rho_{\mathrm{bg}}=\Omega_{\mathrm{bg}}\rho_{\mathrm{crit},0}$ ($\mathrm{kg/m^3}$)	Dimensionless Factor $\sqrt{\frac{8\pi}{3\Omega_{\mathrm{bg}}}}$	Horizon Scale L_{hor} (m)	Horizon Scale L_{hor} (Gpc)	Note / Source
Purely Baryonic Cosmic Background (Ω_b)	$\Omega_b=0.0493$	4.2067×10^{-28}	13.0306	1.7885×10^{27}	57.96–57.98 **Gpc**	Exact $\gamma_L\rho_{b,0}=1$ solution for $\Omega_b=0.0493$
Cosmic Matter Density Background (Ω_m)	$\Omega_m=$ 0.3147–0.3150	2.6855–2.6879×10^{-27}	5.1583–5.1605	7.080–7.083×10^{26}	22.94–22.95 **Gpc**	Value 7.08×10^{26} m $\approx$ 22.94 Gpc in V22 `hu_p2.tex` line 54 (corresponds to $\Omega_m\approx0.315$!)

(Note on line 54 of V22 `hu_p2.tex`: the manuscript prose verbally mentions $\Omega_b\rho_{crit}$, but the reported numerical value 7.08×10^{26} $m = 22.94$ Gpc arises arithmetically from the cosmic matter density $\Omega_m\approx0.315$, $\rho_{bg}=\frac{1}{\gamma_L(L_{hor})}=2.686\times10^{-27}$ kg/m^3 and the factor $\sqrt{8\pi/(3\Omega_m)}=5.160$, whereas for $\Omega_b=0.0493$ one obtains $\sqrt{8\pi/(3\Omega_b)}=13.031$ and $L_{hor}^{(b)}=1.789\times10^{27}$ $m\approx57.98$ Gpc.)

15.3.2 3.2. Factorization of the Homogeneous Background Factor from Local Dynamical and Lensing Gradients

By Cauchy's multiplicative property (`hu_p1.tex`, Eq. 2: $f(\rho_1+\rho_2)=f(\rho_1)f(\rho_2)$), the exponential time factor factorizes as a product:

$$f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L[\rho_{\mathrm{bg}}+\delta\bar\rho_L(\mathbf{x})]\big) \;=\; \underbrace{e^{-\gamma_L\rho_{\mathrm{bg}}}}_{f_{L,\mathrm{bg}}\ (\text{global constant})}\cdot\underbrace{e^{-\gamma_L\,\delta\bar\rho_L(\mathbf{x})}}_{f_{L,\mathrm{loc}}(\mathbf{x})}.$$

Since $f_{L,\mathrm{bg}}$ is spatially constant ($\nabla f_{L,\mathrm{bg}}=0$), rescaling the global time coordinate $d\tilde t=f_{L,\mathrm{bg}}\,dt$ absorbs the homogeneous background factor into the cosmological background clock, while the logarithmic gradient appearing in local dynamical acceleration ($\mathbf{a}_{\mathrm{eff}}=-\nabla\Phi_N+c^2\nabla\phi_L$) and disformal light deflection ($\hat\alpha_{\mathrm{def}}^{\mathrm{disf}}$) is **independent of the homogeneous background $\rho_{\mathbf{bg}}$**:

$$\nabla\ln f_L(\mathbf{x}) \;=\; -\nabla\phi_L(\mathbf{x}) \;=\; -\gamma_L\,\nabla\delta\bar\rho_L(\mathbf{x}).$$

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15.4 4. Implication for NICER Pulsar Ray-Tracing (PSR J0030+0451, PSR J0740+6620, PSR J0437-4715)

In ray-tracing of X-ray photons emitted from the surface ($r \geq R_*$) of a neutron star of radius R_* (Beloborodov / Pechenick–Ftaclas–Cohen formula) in the exterior vacuum ($\rho(r > R_*) = 0$):

- **For a 3D top-hat kernel ($L \leq 1$ km $\ll R_*$):** Outside the stellar surface ($r > R_* + L$), the top-hat sphere does not intersect the star, so $\bar{\rho}_L(r > R_* + L) = 0 \implies f_L = 1$; exterior light deflection follows exact Schwarzschild geodesics, and at $L = 1$ km the relative shift of the mass–radius curve is only 4.8×10^{-4} dex ($\Delta \ln \mathcal{L} \approx 1.1 \times 10^{-3}$, V22 `hu_p3.tex` line 60).
- **For a Yukawa kernel (`yukawa_action_el`):** The exterior scalar profile in curved exterior spacetime decays exponentially as $r \to \infty$ ($\phi_L(r) \propto r^{-(1+M_{\rm ADM}/L)} e^{-r/L}$), and the asymptotic gravitational mass of the star is absorbed into the ADM mass $M_{\rm ADM} = M_{\rm surf} + \Delta M_{\phi,\rm ext}^{\rm curved}$ derived in Steps 7 and 22 (`step7_neutron_star_tov_stability.md`, `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.m`

15.5 5. Checks

1. **Vacuum and GR Limit ($\phi_L \to 0 \implies f_L \to 1$):**
   - As $\phi_L(\mathbf{x}) \to 0$, $\frac{d\phi_L}{dr} \to 0$, so both the conformal ($\hat{\alpha}_{\rm def}^{\rm conf}$) and disformal ($\hat{\alpha}_{\rm def}^{\rm disf}$) light deflection angles recover the Einstein value $\hat{\alpha}_{\rm GR}(b) = \frac{4GM}{c^2 b}$ ✓.
2. **Numerical Verification of $L_{\rm hor}^{(m)} \approx 22.94$ Gpc and $L_{\rm hor}^{(b)} \approx 57.98$ Gpc:**
   - $\Omega_m = 0.3147 \implies \rho_{m,0} = 2.6855 \times 10^{-27}$ kg/m^3 $\implies \sqrt{8\pi/(3\Omega_m)} = 5.1605 \implies L_{\rm hor}^{(m)} = 7.083 \times 10^{26}$ m $= 22.95$ Gpc (and for $\rho_{\rm bg} = 2.686 \times 10^{-27}$ kg/m^3, 7.08×10^{26} m $= 22.94$ Gpc) ✓.
   - $\Omega_b = 0.0493 \implies \rho_{b,0} = 4.2067 \times 10^{-28}$ kg/m^3 $\implies \sqrt{8\pi/(3\Omega_b)} = 13.0306 \implies L_{\rm hor}^{(b)} = 1.7885 \times 10^{27}$ m $=$ 57.96–57.98 Gpc ✓.

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16 Step 15: Electromagnetic Trace-Freeness ($T^{\mu}_{\mu,\mathbf{EM}} = 0$) vs. Disformal Magnetic Source Coupling ($\rho_B = |\mathbf{B}|^2/(2\mu_0 c^2)$), V20 Polarity-Even Cross-Term Cancellation, and the Methodological Boundary ($\Phi_t \equiv 0$)

Source file: step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
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Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step14_gravitational_lensing_null_geodesics (step14_gravitational_lensing_null_geodesics_cosmological_background.md)
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Local Archival Sources:

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 2, points 3–4, lines 208–221) and hu_p3.tex (Chapter 10, point 2, lines 75–85; DOI: 10.5281/zenodo.23604423)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Sections 1 and 3, lines 33–35, 58–66, and 161–166; DOI: 10.5281/zenodo.23639084)
- README.md (scalar-tensor-yukawa-fit/README.md, Section 3, point 5, lines 60–61)

16.1 1. Verbatim Citations from the V22 (hu_p1.tex, hu_p3.tex) and V23 Manuscripts

In Step 8 (step8_microscopic_clock_coupling_siriusb_lab_v20.md), we derived how a given scalar profile ϕ_L shifts the frequencies of atomic clocks and quartz oscillators (H0 vs. H1), and evaluated the numerical value for the V20 8-coil solenoid ($\Delta y_{\text{peak}} = -1.2017108 \times 10^{-48}$ at $L = 0.630957$ m). Within the scalar–tensor framework, however, there is a fundamental theoretical question highlighted in hu_p1.tex (lines 208–221), hu_p3.tex (lines 75–85), and README.md (line 61): **how can pure magnetic energy density ($\rho_B = \frac{|\mathbf{B}|^2}{2\mu_0 c^2}$) act as a source for the scalar field ϕ_L in a covariant action principle when the trace of the 4-dimensional Maxwell stress–energy tensor vanishes identically ($T^{\mu}_{\mu,\mathbf{EM}} = g^{\mu\nu}T^{\mathbf{EM}}_{\mu\nu} = 0$)?**

1. **Methodological Boundary Between the Audited Physical Core ($\rho = T_{00}/c^2$, $\Phi_t \equiv 0$) and the V13–V16 Perspective Layer (hu_p1.tex, Chapter 2, points 3–4, lines 209–221):**

$$\rho_{\text{eff}} \;=\; \rho \;+\; \Xi\,\varepsilon_0\,\Phi_t \;+\; \beta\,\Phi_B \;+\; \gamma\,\kappa, \qquad T^{\text{total}}_{\mu\nu} \;=\; T^{\text{matter}}_{\mu\nu} \;+\; T^{\text{flux}}_{\mu\nu} \;+\; T^{\text{thought}}_{\mu\nu}, \quad \text{(V22 Eq. 2)}$$

> *"When presenting this book to the academic community, I draw a clear boundary between the two layers:*
> - ***Audited Physical Core (V19–V21):*** *Every numerical result in the SPARC galactic rotation analysis, the H.E.S.S. timing bounds, the neutron-star TOV and pulsation sequences, and the V20 granite-chamber and 8-coil quartz-oscillator laboratory protocol is built exclusively on the standard baryonic and electromagnetic mass–energy density $\rho = T_{00}/c^2$, with $\Phi_t \equiv 0$.*
> - ***Perspective Bio-Coherence Layer (V13–V16):*** *The conscious-coherence terms (Φ_t, $T^{thought}_{\mu\nu}$, EEG/HRV bio-coherence) are retained as an open, not-yet-audited theoretical perspective representing part of the intellectual history of the Time Attraction Law."*

2. **V20 Eight-Coil Magnetic Energy-Density Source and Background Cross-Term Cancellation (hu_p3.tex, Chapter 10, point 2, lines 76 and 84):** > - *"To test whether pure magnetic energy density $u_B(\mathbf{x}) = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0}$ slows the flow of time via $\rho_B(\mathbf{x}) = u_B(\mathbf{x})/c^2$, I designed a symmetric eight-coil arrangement..."* > - *"**Cancellation of the Background Cross-Term via Polarity Reversal:** Since $|\mathbf{B}_{tot}|^2 = |\mathbf{B}_{bg}|^2 + 2\mathbf{B}_{bg}\cdot\mathbf{B}_{coil} + |\mathbf{B}_{coil}|^2$, reversing the current direction ($+I \leftrightarrow -I$) reverses $\mathbf{B}_{coil}$, so both the linear Zeeman sensitivity of the quartz oscillator and the cross-term $2\mathbf{B}_{bg}\cdot\mathbf{B}_{coil}$ cancel out of the polarity-even average Δy_{even}."*

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16.2 2. Why Does a Magnetic Field Not Source a Purely Conformal Brans–Dicke / Dilaton Scalar, and Why Does It Source the V23 Disformal Coupling?

16.2.1 2.1. Electromagnetic Blindness of Purely Conformal Coupling ($T^{\mu}_{\mu,\mathbf{EM}} = 0$ in $D = 4$)

Consider the standard Maxwell action of the free electromagnetic field ($F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$) on a general Jordan metric $\tilde{g}_{\mu\nu}$:

$$S_{\rm EM}\big[\tilde{g}_{\mu\nu}, A_\mu\big] \;=\; -\frac{1}{4\mu_0}\int d^4x\,\sqrt{-\tilde{g}}\,\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}.$$

If the metric depends on the scalar field ϕ_L in a purely **conformal** manner ($\tilde{g}^{\rm conf}_{\mu\nu} = \Omega^2(\phi_L)\,g_{\mu\nu} = f_L^2\,g_{\mu\nu}$, so that $\tilde{g}^{\mu\alpha}_{\rm conf} = f_L^{-2}g^{\mu\alpha}$ and in $D = 4$ dimensions $\sqrt{-\tilde{g}^{\rm conf}} = f_L^4\sqrt{-g}$), then the factors of f_L cancel identically:

$$\sqrt{-\tilde{g}^{\rm conf}}\,\tilde{g}^{\mu\alpha}_{\rm conf}\tilde{g}^{\nu\beta}_{\rm conf}F_{\mu\nu}F_{\alpha\beta} \;=\; \big(f_L^4\sqrt{-g}\big)\big(f_L^{-2}g^{\mu\alpha}\big)\big(f_L^{-2}g^{\nu\beta}\big)F_{\mu\nu}F_{\alpha\beta} \;=\; \sqrt{-g}\,g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}.$$

Accordingly, the variation of the Maxwell action with respect to ϕ_L vanishes identically in purely conformal coupling (which is equivalent to the vanishing of the trace of the Maxwell stress–energy tensor in $D = 4$, $T^{\mu}_{\mu,\rm EM} = g^{\mu\nu}T^{\rm EM}_{\mu\nu} = -u_E - u_B + 3\cdot\frac{1}{3}(u_E + u_B) = 0$):

$$\boxed{\frac{\delta S_{\rm EM}\big[\tilde{g}^{\rm conf}_{\mu\nu}, A_\mu\big]}{\delta\phi_L(x)} \;=\; \frac{\partial\tilde{g}^{\rm conf}_{\mu\nu}}{\partial\phi_L}\,\frac{\delta S_{\rm EM}}{\delta\tilde{g}^{\rm conf}_{\mu\nu}} \;=\; \sqrt{-g}\,f_L^2\,\frac{\partial\ln f_L}{\partial\phi_L}\,g^{\mu\nu}T^{\rm EM}_{\mu\nu} \;=\; 0.}$$

Physical consequence: A purely conformal Jordan–Brans–Dicke scalar field is not sourced at tree level by a static magnetic field **B** at all!

16.2.2 2.2. Exact Magnetic Source Term of the V23 Disformal Clock Metric ($\tilde{g}^{\mathbf{disf}}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$)

Now let us substitute the V23 **temporal disformal metric** (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 165):

$$\tilde{g}^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} \;+\; \frac{1-f_L^2(\phi_L)}{c^2}\,u_\mu u_\nu, \qquad \tilde{g}^{\mu\nu}_{\rm disf} \;=\; g^{\mu\nu} \;-\; \frac{1-f_L^{-2}(\phi_L)}{c^2}\,u^\mu u^\nu, \qquad \sqrt{-\tilde{g}^{\rm disf}} \;=\; f_L(\phi_L)\,\sqrt{-g}.$$

Decomposing the field-strength tensor $F_{\mu\nu}$ into electric ($E_\mu = F_{\mu\nu}u^\nu/c$) and magnetic ($B_\mu = \frac{1}{2}\epsilon_{\mu\nu\alpha\beta}u^\nu F^{\alpha\beta}/c$) field four-vectors comoving with the laboratory four-velocity u^μ:

$$\frac{1}{4}\tilde{g}^{\mu\alpha}_{\rm disf}\tilde{g}^{\nu\beta}_{\rm disf}F_{\mu\nu}F_{\alpha\beta} \;=\; \frac{1}{2}\Big[\tilde{g}^{ij}_{\rm disf}\tilde{g}^{kl}_{\rm disf}F_{ik}F_{jl} \;+\; 2\,\tilde{g}^{00}_{\rm disf}\tilde{g}^{ij}_{\rm disf}F_{0i}F_{0j}\Big] \;=\; \frac{|\mathbf{B}|^2}{2\mu_0}\cdot\mu_0 \;-\; f_L^{-2}(\phi_L)\,\frac{\varepsilon_0|\mathbf{E}|^2}{2}\cdot\mu_0.$$

Substituting this together with the volume element $\sqrt{-\tilde{g}^{\rm disf}} = f_L(\phi_L)\sqrt{-g}$ into the Maxwell action gives:

$$\boxed{S_{\rm EM}\big[\tilde{g}^{\rm disf}_{\mu\nu}, A_\mu\big] \;=\; \int d^4x\,\sqrt{-g}\left[-\,f_L(\phi_L)\,\frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0} \;+\; f_L(\phi_L)^{-1}\,\frac{\varepsilon_0|\mathbf{E}(\mathbf{x})|^2}{2}\right].}$$

In a static magnetic field ($\mathbf{E} = \mathbf{0}$, as in the V20 8-coil solenoid), the Lagrangian density $\mathcal{L}^{\rm disf}_{\rm EM}(\mathbf{E} = \mathbf{0}) = -f_L(\phi_L)\,u_B(\mathbf{x})$ takes the exact same $-f_L(\phi_L)\varepsilon$ form as a static baryonic fluid (`step4_covariant_conservation_hydrostatic.md`: $\mathcal{L}_{\rm matter} = -f_L(\phi_L)\varepsilon_{\rm baryon}$)! Varying the V23 Formulation II (Helmholtz–Yukawa) or Formulation I (bilateral top-hat) action with respect to ϕ_L ($f_L = e^{-\phi_L}$, $\partial f_L/\partial\phi_L = -f_L$) yields:

$$\frac{\partial\big(\mathcal{L}_{\rm matter} + \mathcal{L}^{\rm disf}_{\rm EM}\big)}{\partial\phi_L} \;=\; f_L(\phi_L)\,\Big(\varepsilon_{\rm baryon}(\mathbf{x}) + u_B(\mathbf{x})\Big) \;=\; f_L(\phi_L)\,c^2\Big(\rho_{\rm baryon}(\mathbf{x}) + \rho_B(\mathbf{x})\Big), \qquad \rho_B(\mathbf{x}) \;\equiv\; \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0 c^2}.$$

We have thus derived from a covariant action principle that the V23 disformal clock metric ($\sqrt{-\tilde{g}} = f_L\sqrt{-g}$, $\tilde{g}_{ij} = g_{ij}$) automatically selects the magnetic mass–energy density $T^{\rm EM}_{00}/c^2 = \rho_B = \frac{|\mathbf{B}|^2}{2\mu_0 c^2}$ as the source of the scalar field ϕ_L!

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16.3 3. Covariant Derivation of V20 Polarity-Even ($\Delta y_{\mathbf{even}}$) Background Cross-Term Cancellation

In the laboratory, the net magnetic field is the superposition of the static background field $\mathbf{B}_{\rm bg}(\mathbf{x})$ of the Earth (and laboratory environment) and the 8-coil field $\mathbf{B}_{\rm coil}(\mathbf{x};I) = I\,\mathbf{b}_{\rm coil}(\mathbf{x})$ proportional to the current I (`hu_p3.tex`, line 84):

$$\rho_B(\mathbf{x};I) \;=\; \frac{|\mathbf{B}_{\rm bg}(\mathbf{x}) + I\,\mathbf{b}_{\rm coil}(\mathbf{x})|^2}{2\mu_0 c^2} \;=\; \underbrace{\frac{|\mathbf{B}_{\rm bg}(\mathbf{x})|^2}{2\mu_0 c^2}}_{\rho_{B,\rm bg}(\mathbf{x})\ (I^0)} + \underbrace{I\,\frac{\mathbf{B}_{\rm bg}(\mathbf{x})\cdot\mathbf{b}_{\rm coil}(\mathbf{x})}{\mu_0 c^2}}_{\rho_{B,\rm cross}(\mathbf{x};I)\ (I^1,\ \text{odd})} + \underbrace{I^2\,\frac{|\mathbf{b}_{\rm coil}(\mathbf{x})|^2}{2\mu_0 c^2}}_{\rho_{B,\rm coil}(\mathbf{x};I)\ (I^2,\ \text{even})}\,.$$

Because both the 3D top-hat ($\mathcal{K}_L^{\rm TH}$) and 3D Yukawa ($\mathcal{G}_L^{\rm Yuk}$) operators are linear in the source density, the ON ($I = \pm I_0$) minus OFF ($I = 0$) difference of the scalar potential at the oscillator position $\mathbf{x}_1 = \mathbf{0}$ is:

$$\Delta\phi_L(\pm I_0) \;\equiv\; \phi_L(\mathbf{0};\pm I_0) - \phi_L(\mathbf{0};0) \;=\; \pm I_0\,\gamma_L\bar\rho_{B,\rm cross}(\mathbf{0};1) \;+\; I_0^2\,\gamma_L\bar\rho_{B,\rm coil}(\mathbf{0};1).$$

Even if the oscillator possesses a parasitic linear Zeeman sensitivity ($\kappa_Z\,\mathbf{B}_{\rm coil}(\mathbf{0};\pm I_0) = \pm\kappa_Z I_0\mathbf{b}_{\rm coil}(\mathbf{0})$), the **polarity-even average** of the H1 fractional frequency shift ($\Delta y(\pm I_0) \simeq -\Delta\phi_L(\pm I_0) \pm \kappa_Z I_0 b_{\rm coil}$) (`hu_p3.tex`, lines 79 and 84) is:

$$\boxed{\Delta y_{\rm even}(L) \;\equiv\; \frac{\Delta y(+I_0) + \Delta y(-I_0)}{2} \;=\; -I_0^2\,\gamma_L\bar\rho_{B,\rm coil}(\mathbf{0};1) \;=\; -\frac{GL^2}{c^2}\int d^3y\,K_L(\mathbf{0},\mathbf{y})\,\frac{|\mathbf{B}_{\rm coil}(\mathbf{y};I_0)|^2}{2\mu_0 c^2},}$$

from which both the linear Zeeman term $\pm\kappa_Z I_0$ and the background cross-term $\pm 2\mathbf{B}_{\rm bg}\cdot\mathbf{B}_{\rm coil}$ **cancel identically**!

16.4 4. Checks

1. **Conformal vs. Disformal Maxwell-Trace Check:**
   - Conformal metric ($\tilde g_{\mu\nu} = f_L^2 g_{\mu\nu}$): $\sqrt{-\tilde g}\tilde g^{\mu\alpha}\tilde g^{\nu\beta} = f_L^4 f_L^{-2} f_L^{-2}\sqrt{-g}g^{\mu\alpha}g^{\nu\beta} = \sqrt{-g}g^{\mu\alpha}g^{\nu\beta}$, independent of $\phi_L \implies \delta S_{\rm EM}^{\rm conf}/\delta\phi_L \equiv 0$ ✓.
   - Disformal metric ($\tilde g_{00} = f_L^2 g_{00}$, $\tilde g_{ij} = g_{ij}$): $\sqrt{-\tilde g}\tilde g^{ik}\tilde g^{jl} = f_L\cdot 1\cdot 1\sqrt{-g}g^{ik}g^{jl} \implies -\partial\mathcal{L}_{\rm EM}^{\rm disf}/\partial\phi_L\big|_{\mathbf{E}=\mathbf{0}} = -f_L u_B$ ✓.
2. **Polarity-Reversal ($\pm I_0$) Parity Check:**
   - $\Delta\phi_L(+I_0) + \Delta\phi_L(-I_0) = (+I_0 - I_0)\gamma_L\bar\rho_{B,\rm cross} + 2I_0^2\gamma_L\bar\rho_{B,\rm coil} = 2\,\gamma_L\bar\rho_{B,\rm coil}(I_0)$, which at $L = 0.630957$ m and $I_0 = 14.0$ mA reproduces the audited V20 value $\Delta y_{\rm peak} = -1.2017108\times10^{-48}$ ✓.

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17 Step 16: Derivation of the V21/V23 Algebraic X-Sector Anisotropic Stress Reconstruction (Mass and Gradient Modes, V23 Eqs. 4–5) from the Bianchi Identity and Comparison with the Dynamical Yukawa Scalar Stress Tensor ($T^{\mu}_{\nu}(\phi)$)

Source file: step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.m
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step15_electromagnetic_trace_anomaly_disfor
(step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20.md)
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Local Archival Sources:

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Section 1.1, lines 47–53, Eqs. 4–5 and Fig. 1(a–b); DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 8, point 1, lines 400–408; DOI: 10.5281/zenodo.23604423) and hu_p3.tex (Chapter 8, point 1, lines 6–12)

17.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Steps 4 and 7 (step4_covariant_conservation_hydrostatic.md, step7_neutron_star_tov_stability.md), we derived the local covariant stress–energy tensor ($T^{(\phi)}_{\mu\nu}$) of Formulation II (yukawa_action_el), but we had not yet derived step by step **Equations (4) and (5)** of the V23 manuscript (eq:mass_X and eq:grad_X), the non-zero central limit ($p_{tX}(0) = p_{rX}(0) = -(1-f_c)(\varepsilon_c/3 + P_c)$) in Chapter 8 of the V22 monograph, and the three-tier covariant divergence residual hierarchy (conservation_Residuals.json):

1. **Equations (4)–(5) of V23 and the Three-Tier Bianchi Residual (tal_covariant_closure_50aa2db0_EN_2026-10-2 / HU, Section 1.1, lines 47–53):** > *"In a static spherically symmetric interior, if an auxiliary sector $T^{\mu}_{\nu}(X) = diag(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$ absorbs the window stress on the right-hand side of $G^{\mu}_{\nu} = \frac{8\pi G}{c^4}(T^{\mu}_{\nu,matter} + T^{\mu}_{\nu,X})$ while fluid elements satisfy $\nabla_\mu T^{\mu\nu}_{matter} = 0$ ($P' = -(\varepsilon + P)\nu'$ in geometrized units, $G = c = 1$, $\nu = \Phi_{eff}/c^2$, $g_{tt} = -e^{2\nu} = -e^{2\Phi_{eff}/c^2}$), then the Bianchi identity $\nabla_\mu G^{\mu}_{r} \equiv 0$ uniquely fixes $T^{\mu}_{\nu}(X)$ in both V21 modes:"*

$$\textbf{Mass mode } (m' = 4\pi r^2 f_L \varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1-f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1-f_L)\varepsilon\nu', \quad \text{(V23 Eq. 4)}$$

$$\textbf{Gradient mode } (m' = 4\pi r^2 \varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1-f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}\big(P'_{r,X} + P_{r,X}\nu$$

> *"where $A_{geo}(r) = (m + 4\pi r^3 P)/[r(r-2m)]$. Symbolic verification in **SymPy** (**conservation_Residuals.json**) confirms that $\nabla_\mu T^{\mu}_{r}(matter) \equiv 0$ and $\nabla_\mu T^{\mu}_{r}(X) \equiv 0$ hold identically. In Table 1 and Fig. 1(b), evaluating relations (4)–(5) on numerical TOV grids ($L = 10\,km$, $\log_{10}\rho_{0c} = 15.0$) yields three distinct tiers (**conservation_Residuals.json**): (i) symbolic identities, residual exactly 0; (ii) algebraic X-sector quadrature residual $\max|\nabla_\mu T^{\mu}_{r}(X)|/|\rho_X\nu'| = 4.44\times10^{-16}$ (mass) and $/|P'_{r,X}| = 4.39\times10^{-14}$ (gradient); (iii) divergence of the numerically (spline-) differentiated background $\max|\nabla_\mu T^{\mu}_{r}|/|P'| = 2.70\times10^{-5}$ (mass) and 2.36×10^{-5} (gradient), governed by the finite-difference derivative of $h(r)$, not by the action."*

2. **Central Limit of the Gradient Mode in the V22 Monograph (tal_book_HU_2026-10-24.tex, Chapter 8, point 1, lines 404–408):** > *"The background X-sector differs between the two insertion modes: >*

$$\textbf{Gradient mode:} \quad \rho_X = 0, \quad p_{rX} = -(1-f)\Big(P + \frac{m}{4\pi r^3}\Big), \quad p_{tX} = p_{rX} + \frac{r}{2}\big(p'_{rX} + p_{rX}\nu'\big),$$

> *whose central limit $p_{tX}(0) = p_{rX}(0) = -(1-f_c)(\varepsilon_c/3 + P_c)$ is generally non-zero. In the **mass-source mode**, by contrast, $\rho_X = -(1-f)\varepsilon$, $p_{rX} = 0$, and $p_{tX} = -\frac{r}{2}(1-f)\varepsilon\nu'$ ($p_{tX}(0) = 0$). The value 0.140 in the 36-configuration stress-diagnostics table is a background bookkeeping diagnostic of that configuration, which does not replace a covariant action principle (**50aa2db0** task)."*

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17.2 2. Exact Derivation of V23 Equations (4) and (5) from the Einstein Equations and the Bianchi Identity

In a static, spherically symmetric spacetime ($ds^2 = -e^{2\nu(r)}dt^2 + e^{2\lambda(r)}dr^2 + r^2 d\Omega^2$, in geometrized units $G = c = 1$, where $e^{-2\lambda(r)} \equiv 1 - \frac{2m(r)}{r}$), the non-vanishing mixed components of the Einstein tensor are:

$$G^t{}_t \;=\; -\frac{2m'(r)}{r^2}, \qquad G^r{}_r \;=\; -\frac{2m(r)}{r^3} \;+\; \frac{2}{r}\left(1 - \frac{2m(r)}{r}\right)\nu'(r).$$

Let us write the Einstein equations for the sum of the baryonic perfect fluid ($T^{\mu}_{\nu,\text{matter}} = \text{diag}(-\varepsilon, P, P, P)$) and the anisotropic X-sector ($T^{\mu}_{\nu,X} = \text{diag}(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$):

$$G^t{}_t \;=\; -8\pi\big(\varepsilon + \varepsilon_X\big), \qquad G^r{}_r \;=\; 8\pi\big(P + P_{r,X}\big), \qquad G^{\theta}{}_{\theta} \;=\; G^{\phi}{}_{\phi} \;=\; 8\pi\big(P + P_{t,X}\big).$$

If the fluid satisfies $P'(r) = -(\varepsilon + P)\nu'(r)$, then $\nabla_\mu T^{\mu}_{r}(\text{matter}) = P' + (\varepsilon + P)\nu' \equiv 0$. Because $\nabla_\mu G^{\mu}_{r} \equiv 0$ by the contracted Bianchi identity, the X-sector by itself must also satisfy the radial covariant conservation equation:

$$\boxed{\nabla_\mu T^{\mu}_{r}(X) \;=\; P'_{r,X}(r) \;+\; \big[\varepsilon_X(r) + P_{r,X}(r)\big]\nu'(r) \;+\; \frac{2}{r}\big[P_{r,X}(r) - P_{t,X}(r)\big] \;=\; 0.}$$

17.2.1 2.1. Derivation in the V21/V23 `mass` (Mass-Source) Mode (V23 Eq. 4)

In the `mass` mode, the mass function and lapse gradient are defined by:

$$m'(r) \;=\; 4\pi r^2 f_L(r)\,\varepsilon(r), \qquad \nu'(r) \;=\; A_{\text{geo}}(r) \;\equiv\; \frac{m(r) + 4\pi r^3 P(r)}{r\big(r - 2m(r)\big)}.$$

1. **From the (tt) Einstein Equation (ε_X):**

$$-\frac{2m'}{r^2} = -8\pi f_L \varepsilon = -8\pi(\varepsilon + \varepsilon_X) \quad\Longrightarrow\quad \varepsilon_X(r) \;=\; -\big(1 - f_L(r)\big)\,\varepsilon(r).$$

2. **From the (rr) Einstein Equation ($P_{r,X}$):** Since $\nu' = A_{\text{geo}}$, the expression for $G^r{}_r$ yields $8\pi P(r)$ directly, hence:

$$8\pi P(r) = 8\pi\big(P(r) + P_{r,X}(r)\big) \quad\Longrightarrow\quad P_{r,X}(r) \;\equiv\; 0.$$

3. **From the Radial Bianchi Identity ($P_{t,X}$):** Substituting $P_{r,X} = 0$ and $P'_{r,X} = 0$ into $\nabla_\mu T^{\mu}_{r}(X) = 0$ gives:

$$\varepsilon_X(r)\,\nu'(r) \;-\; \frac{2}{r}P_{t,X}(r) \;=\; 0 \quad\Longrightarrow\quad \boxed{P_{t,X}(r) \;=\; \frac{r}{2}\,\varepsilon_X(r)\,\nu'(r) \;=\; -\frac{r}{2}\big(1 - f_L(r)\big)\,\varepsilon(r)\,\nu'(r),}$$

which is precisely **V23 Equation (4)** (`eq:mass_X`), and since $\nu'(r) = \mathcal{O}(r)$ at the origin, $P_{t,X}(r) = \mathcal{O}(r^2) \xrightarrow{r\to 0} 0$.

17.2.2 2.2. Derivation in the V21/V23 `gradient` Mode (V23 Eq. 5) and Proof of Central Isotropy

In the `gradient` mode, the mass function and lapse gradient are defined by:

$$m'(r) \;=\; 4\pi r^2\,\varepsilon(r), \qquad \nu'(r) \;=\; f_L(r)\,A_{\text{geo}}(r) \;=\; f_L(r)\,\frac{m(r) + 4\pi r^3 P(r)}{r\big(r - 2m(r)\big)}.$$

1. **From the (tt) Einstein Equation (ε_X):**

$$-\frac{2m'}{r^2} = -8\pi\varepsilon = -8\pi(\varepsilon + \varepsilon_X) \quad\Longrightarrow\quad \varepsilon_X(r) \;\equiv\; 0.$$

2. **From the (rr) Einstein Equation ($P_{r,X}$):** Substituting $\nu'(r) = f_L(r)\frac{m+4\pi r^3 P}{r(r-2m)}$ into $G^r{}_r = 8\pi(P + P_{r,X})$ gives:

$$P(r) + P_{r,X}(r) \;=\; \frac{G^r{}_r}{8\pi} \;=\; -\frac{m(r)}{4\pi r^3} + f_L(r)\left(\frac{m(r)}{4\pi r^3} + P(r)\right) \quad\Longrightarrow\quad \boxed{P_{r,X}(r) \;=\; -\big(1 - f_L(r)\big)\left(P(r) + \frac{m(r)}{4\pi r^3}\right).}$$

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3. **From the Radial Bianchi Identity ($P_{t,X}$):** Since $\varepsilon_X = 0$, solving $\nabla_\mu T^\mu_r(X) = P'_{r,X} + P_{r,X}\nu' + \frac{2}{r}(P_{r,X} - P_{t,X}) = 0$ for $P_{t,X}(r)$ yields:

$$\boxed{P_{t,X}(r) \;=\; P_{r,X}(r) \;+\; \frac{r}{2}\Big[P'_{r,X}(r) \;+\; P_{r,X}(r)\,\nu'(r)\Big],}$$

which is precisely **V23 Equation (5)** (`eq:grad_X`).

4. **Central Limit and Isotropy ($r \to 0$):** Near the stellar center ($r \to 0$), $m(r) = \frac{4\pi}{3}\varepsilon_c r^3 + \mathcal{O}(r^5)$, so $\lim_{r\to 0} \frac{m(r)}{4\pi r^3} = \frac{\varepsilon_c}{3}$. Because the derivative of every regular scalar profile vanishes at the origin ($P'(0) = 0$, $f'_L(0) = 0$, $\nu'(0) = 0$), we have $P'_{r,X}(r) = \mathcal{O}(r)$, so the anisotropy term $\frac{r}{2}[P'_{r,X} + P_{r,X}\nu'] = \mathcal{O}(r^2)$ vanishes at the origin, and:

$$\boxed{\lim_{r\to 0} P_{t,X}(r) \;=\; \lim_{r\to 0} P_{r,X}(r) \;=\; -\big(1 - f_c\big)\left(\frac{\varepsilon_c}{3} + P_c\right) \;\neq\; 0, \qquad \lim_{r\to 0}\big[P_{t,X}(r) - P_{r,X}(r)\big] \;=\; 0.}$$

Hence no geometric conical singularity arises at the origin: although the central pressure is non-zero, the stress of the X-sector at the origin is **isotropic** ($P_{r,X}(0) = P_{t,X}(0)$)!

17.3 3. Comparison with the Dynamical Scalar Stress Tensor of Formulation II (`yukawa_action_el`)

In Formulation II of V23 (`yukawa_action_el`, `step4` and `step7`), the X-sector is determined not by algebraic reconstruction, but directly from the covariant action ($S = \int d^4x\sqrt{-g}\big[\frac{R}{16\pi} - \frac{(\nabla\phi_L)^2 + \phi_L^2/L^2}{2\alpha_Y} - f_L(\phi_L)\varepsilon(n)\big]$). Relative to the standard GR source (ε, P), the resulting effective X-sector consists of two dynamical parts:

$$T^\mu_{\nu,X,\mathrm{dyn}} \;=\; \big(T^\mu_{\nu,\mathrm{matter}} - T^\mu_{\nu,\mathrm{GR}}\big) \;+\; T^\mu_\nu(\phi),$$

that is, component by component ($B(r) = 1 - 2m(r)/r$):

$$\begin{aligned}
\varepsilon_{X,\mathrm{dyn}}(r) \;&=\; -\big(1 - f_L(r)\big)\,\varepsilon(r) \;+\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi'_L(r)\big]^2 + \frac{\phi_L(r)^2}{L^2}\right),\\
P_{r,X,\mathrm{dyn}}(r) \;&=\; -\big(1 - f_L(r)\big)\,P(r) \;+\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi'_L(r)\big]^2 - \frac{\phi_L(r)^2}{L^2}\right),\\
P_{t,X,\mathrm{dyn}}(r) \;&=\; -\big(1 - f_L(r)\big)\,P(r) \;-\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi'_L(r)\big]^2 + \frac{\phi_L(r)^2}{L^2}\right).
\end{aligned}$$

Three deep structural relationships between Formulations I and II follow immediately:

1. **Origin of the $-(1-f_L)\varepsilon$ Term in the Mass Mode:** The first term of $\varepsilon_{X,\mathrm{dyn}}(r)$ is the exact dilution term $\varepsilon_X = -(1-f_L)\varepsilon$ of the V21/V23 `mass` mode, supplemented by the positive self-energy density $\rho_\phi = \frac{1}{2\alpha_Y}(B[\phi'_L]^2 + \phi_L^2/L^2)$ of the scalar field.
2. **Purely Gradient Origin of the Anisotropy ($\Pi_\phi \equiv P_{r,X,\mathbf{dyn}} - P_{t,X,\mathbf{dyn}}$):**

$$\boxed{\Pi_\phi(r) \;\equiv\; P_{r,X,\mathrm{dyn}}(r) - P_{t,X,\mathrm{dyn}}(r) \;=\; p_{r,\phi}(r) - p_{t,\phi}(r) \;=\; \frac{B(r)}{\alpha_Y}\big[\phi'_L(r)\big]^2 \;\geq\; 0.}$$

Because $\phi'_L(r) = \mathcal{O}(r)$ near the origin, the dynamical anisotropy also obeys $\Pi_\phi(r) = \mathcal{O}(r^2) \xrightarrow{r\to 0} 0$, so at the origin the pressure is strictly isotropic: $P_{r,X,\mathrm{dyn}}(0) = P_{t,X,\mathrm{dyn}}(0) = -(1 - f_c)P_c - \frac{\phi_L(0)^2}{2\alpha_Y L^2} < 0$.

3. **Why the Euler–Lagrange Equation for ϕ_L Replaces Separate Bianchi Reconstruction:** Whereas in the algebraic V21 modes $P_{t,X}(r)$ had to be reconstructed a posteriori from $\nabla_\mu T^\mu_r(X) = 0$, in the `yukawa_action_el` system the Noether identity $\nabla_\mu\big(T^\mu_{r,\mathrm{matter}} + T^\mu_r(\phi)\big) \equiv 0$ proved in Step 4 (`step4_covariant_conservation_hydrostatic.md`) is satisfied automatically by virtue of the field equation for ϕ_L!

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17.4 4. Checks

1. **Symbolic Bianchi Identity Check for Both V23 Algebraic Modes:**
   - **Mass mode:** $P'_{r,X} + (\varepsilon_X + P_{r,X})\nu' + \frac{2}{r}(P_{r,X} - P_{t,X}) = 0 + \varepsilon_X\nu' + \frac{2}{r}\left(0 - \frac{r}{2}\varepsilon_X\nu'\right) = \varepsilon_X\nu' - \varepsilon_X\nu' \equiv 0$ ✓.
   - **Gradient mode:** $P'_{r,X}+(0+P_{r,X})\nu'+\frac{2}{r}\Big(P_{r,X}-\big[P_{r,X}+\frac{r}{2}(P'_{r,X}+P_{r,X}\nu')\big]\Big) = P'_{r,X}+P_{r,X}\nu'-(P'_{r,X}+P_{r,X}\nu') \equiv 0$ ✓.
2. $f_L \to 1$ **($\phi_L \to 0$) GR Limit:**
   - In the `mass` mode ($\varepsilon_X, P_{r,X}, P_{t,X} \to 0$), in the `gradient` mode ($\varepsilon_X, P_{r,X}, P_{t,X} \to 0$), and in the dynamical `yukawa_action_el` case ($\varepsilon_{X,\mathrm{dyn}}, P_{r,X,\mathrm{dyn}}, P_{t,X,\mathrm{dyn}} \to 0$), the X-sector vanishes identically, recovering the isotropic GR TOV equations ✓.

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18 Step 17: Exact Analytical Derivation of the V23 Eq. (8) Factorized Chandrasekhar–Hydrostatic Difference Identity (eq:factored_diff) and the gradient-Mode metric Inertia-Weight Indefiniteness ($\mathcal{W}_{\mathbf{metric}}(r) \leq 0$, $L \geq 367$ m)

Source file: step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefinit
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step16_anisotropic_x_sector_bianchi_reconst (step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress.md)
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Local Archival Sources:

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Section 2, lines 121–126, Eq. 8 and Table 2 footnote, line 152; DOI: 10.5281/zenodo.23639084)
- V23 verification scripts: /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/covariant_closure_5 (lines 100–120) and radial_modes.py (lines 20–39)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 8, point 2, lines 409–419; DOI: 10.5281/zenodo.23604423)

18.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 16 (step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress.md), we derived the background stress tensor $T^{\mu}_{\nu}(X)$ appearing in V23 Equations (4)–(5). Section 2 of the V23 manuscript (lines 121–126, Eq. 8, and line 152) and Chapter 8 of the V22 monograph (lines 409–418) show how this same background modification causes both frozen-f_L ($\delta f_L = 0$) Sturm–Liouville operators (matter and metric) to depart from the static turning point $dM/d\rho_c = 0$:

1. **V23 Eq. (8) Factorized Chandrasekhar–Hydrostatic Difference Identity (tal_covariant_closure_50aa2db0_EN_2 / HU, Section 2, lines 121–125, Eq. 8):** > *"To understand why both frozen-f_L operators depart from the static mass peak $dM/d\rho_c = 0$, I first determined in **SymPy** the exact difference between the frozen-f_L Chandrasekhar operator $\mathcal{L}_{SL}^{matter}[\xi]$ and the linearized hydrostatic operator $\mathcal{H}_{lin}[\xi] = (\delta P)' + (\delta\varepsilon + \delta P)\nu' + (\varepsilon + P)\delta\nu'\big|_{\delta f_L=0}$ in mass mode:"*

$$\mathcal{L}_{\mathrm{SL}}^{\mathrm{matter}}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\, \mathcal{H}_{\mathrm{lin}}[\xi]\Big|_{\delta f_L=0} = \frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big]. \quad \text{(V23 Eq. 8)}$$

> *"Expression (8) vanishes identically for $f_L = 1$ (GR), because Chandrasekhar's integration by parts substitutes the GR Einstein identity $\lambda' + 2\nu' = 8\pi r e^{\lambda}(\varepsilon + P)$. When $f_L \neq 1$, however, $\lambda' + 2\nu' = 8\pi r e^{\lambda}(f_L\varepsilon + P)$, which generates the $(f_L - 1)$ residual term in (8) AND ALSO omits the nonlocal perturbation $\delta f_L[\delta\varepsilon, \xi]$."*

2. **$\mathcal{W}(r) \leq 0$ Inertia-Weight Indefiniteness of the metric Operator Above $L \geq 367$ m (tal_book_HU_2026-10-24.tex, line 417 and V23 Table 2 footnote, line 152):**
   - tal_book_HU_2026-10-24.tex, line 417: > *"**Approximate metric operator:** Above $L \geq 367$ m across 12 sequences (609 models), the inertia weight $\mathcal{W}(r)$ becomes indefinite ($\mathcal{W} \leq 0$) and yields negative ω_0^2 values along the selected positive-norm branch; determining physical stability remains open (50aa2db0 task)."*
   - tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 152: > *"* In gradient mode at $L = 10$ km, the weight of the approximate frozen-f_L metric operator becomes indefinite near the surface ($\mathcal{W}(r) \leq 0$); the static tangent model (9) uses a positive Gram weight (this is not a proof of kinetic positivity)."*

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18.2 2. Step-by-Step Derivation of the V23 Eq. (8) Factorized Difference Identity (eq:factored_diff)

18.2.1 2.1. Definition of the Two Operators in the Convention of V23 covariant_closure_50aa2db0.py (Lines 101–120)

In the notation of `covariant_closure_50aa2db0.py` (lines 103–118) and `radial_modes.py` (lines 3–7) ($G = c = 1$, $g_{tt} = -e^{\nu_C}$ where $\nu_C \equiv 2\nu$, and $e^{\lambda} \equiv \left(1 - \frac{2m}{r}\right)^{-1} = \frac{r}{r-2m}$, so that $e^{\lambda/2} = \sqrt{\frac{r}{r-2m}}$):

- The background equations in `mass` mode (`sub_mass_full`):

$$m'(r) \;=\; 4\pi r^2 f_L(r)\,\varepsilon(r), \qquad \nu'(r) \;=\; A(r) \;\equiv\; \frac{m + 4\pi r^3 P}{r(r-2m)}, \qquad P'(r) \;=\; -(\varepsilon+P)A(r), \qquad \varepsilon'(r) \;=\; -\frac{(\varepsilon+P)^2}{\Gamma P}A(r).$$

- The **frozen-f_L Chandrasekhar `matter` Sturm–Liouville operator** ($\zeta(r) \equiv r^3 e^{-\nu(r)}\xi(r)$, where $\xi(r) = \Delta r/r$):

$$\mathcal{L}_{\rm SL}^{\rm matter}[\xi] \;\equiv\; -\frac{d}{dr}\left(\Pi(r)\frac{d\zeta}{dr}\right) \;-\; Q_{\rm matter}(r)\,\zeta(r),$$

  where the coefficients (`covariant_closure_50aa2db0.py`, lines 106–108) are:

$$\Pi(r) \;=\; \sqrt{\frac{r}{r-2m}}\,e^{3\nu} r^{-2}\,\Gamma P, \qquad Q_{\rm matter}(r) \;=\; \sqrt{\frac{r}{r-2m}}\,e^{3\nu} r^{-2}\,(\varepsilon+P)\left(\nu'^2 + \frac{4\nu'}{r} - \frac{8\pi r\,P}{r-2m}\right).$$

- The **linearized hydrostatic operator** at $\delta f_L = 0$ (`covariant_closure_50aa2db0.py`, lines 111–117):

$$\mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L=0} \;\equiv\; \frac{d(\delta P)}{dr} \;+\; (\delta\varepsilon + \delta P)\,A(r) \;+\; (\varepsilon+P)\,\delta A(r)\Big|_{\delta f_L=0},$$

  where the Eulerian pressure, energy-density, and mass perturbations are:

$$\Delta P \;=\; -\Gamma P\,e^{\nu} r^{-2}\zeta'(r), \qquad \delta P \;=\; \Delta P - r\xi P'(r), \qquad \delta\varepsilon \;=\; \frac{\varepsilon+P}{\Gamma P}\delta P, \qquad \delta m\Big|_{\delta f_L=0} \;=\; -4\pi r^3\big(f_L\varepsilon + P\big)\xi,$$

  and the linearized variation of the geometric factor is:

$$\delta A\Big|_{\delta f_L=0} \;=\; \frac{\partial A}{\partial m}\,\delta m\Big|_{\delta f_L=0} \;+\; \frac{\partial A}{\partial P}\,\delta P \;=\; \frac{r + 8\pi r^3 P}{r(r-2m)^2}\,\delta m\Big|_{\delta f_L=0} \;+\; \frac{4\pi r^2}{r-2m}\,\delta P.$$

18.2.2 2.2. Why Does a Residual Proportional to $(f_L - 1)$ Arise Under Integration by Parts?

Let us expand the difference $\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}e^{2\nu}\mathcal{H}_{\rm lin}[\xi]\big|_{\delta f_L=0}$:

1. Because $\delta P = -e^{-\nu}\sqrt{\frac{r-2m}{r}}\,\Pi(r)\zeta'(r) + r(\varepsilon+P)A\xi$, the leading second-order derivative term $-\frac{d}{dr}(\Pi\zeta')$ ($\propto \xi''(r)$) **cancels identically** against the leading term of $-\sqrt{\frac{r}{r-2m}}e^{2\nu}\frac{d(\delta P)}{dr}$!
2. When collecting the remaining first-order ($\propto \xi'(r)$) and zeroth-order ($\propto \xi(r)$) terms, the key step is the radial derivative of the metric prefactor $\sqrt{\frac{r}{r-2m}}e^{2\nu}$:

$$\frac{d}{dr}\left(\ln\left[\sqrt{\frac{r}{r-2m}}\,e^{2\nu}\right]\right) \;=\; \frac{1}{2}\lambda'(r) \;+\; 2\nu'(r) \;=\; \frac{rm'(r) - m(r)}{r(r-2m)} \;+\; \frac{2(m + 4\pi r^3 P)}{r(r-2m)} \;=\; \frac{m + 4\pi r^3\big[f_L(r)\varepsilon + 2P\big]}{r(r-2m)}.$$

   In general relativity ($f_L = 1$), the numerator has $f_L\varepsilon = \varepsilon$, and likewise $(\varepsilon + P)$ appears in the mass perturbation $\delta m_{\rm GR} = -4\pi r^3(\varepsilon+P)\xi$, so the difference between the two expressions vanishes identically.
3. When $f_L(r) \neq 1$, however, a shift proportional to $(f_L - 1)\varepsilon$ survives from both the background derivative $m'(r) = 4\pi r^2 f_L\varepsilon$ and the perturbation $\delta m\big|_{\delta f_L=0} = -4\pi r^3(f_L\varepsilon + P)\xi$:
   - The coefficient of the $\xi'(r)$ term (arising from the product of the $r\xi'$ part of $\Delta P = -\Gamma P(3\xi - r\nu'\xi + r\xi')$ and the factor $\frac{1}{2}\lambda'\big|_{f_L} - \frac{1}{2}\lambda'\big|_{f_L=1} = \frac{4\pi r^2(f_L-1)\varepsilon}{r-2m}$) is:

$$\text{Coeff}(\xi') \;=\; -\sqrt{\frac{r}{r-2m}}\,e^{2\nu}\,\Gamma P\,r\cdot\frac{4\pi r^2(f_L-1)\varepsilon}{r-2m}\cdot\frac{r}{r-2m} \;=\; -\frac{4\pi r^{7/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{3/2}}.$$

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- Combining the contributions from $(3 - r\nu')\Delta P$ and $\frac{\partial A}{\partial m}\big(\delta m|_{f_L} - \delta m|_{f_L=1}\big)$, the coefficient of the $\xi(r)$ term is $\frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}(4\pi r^3 P - 3r + 7m)$. Factoring out the common prefactor $\frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}$ yields **V23 Equation (8)** identically:

$$\boxed{\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\, \mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L=0} \;=\; \frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big].}$$

18.3 3. Why Does the Weight $\mathcal{W}_{\rm metric}(r)$ of the `metric` Operator Turn Negative Near the Surface ($L \geq 367$ m, `gradient` Mode)?

By the definition in `radial_modes.py` (lines 11–12 and 30–38), the `metric` operator variant replaces the matter quantities (ε, P) in the Sturm–Liouville coefficients $Q(r)$ and $\mathcal{W}(r)$ with the effective Einstein-tensor sources of the modified background ($\varepsilon_{\rm eff} = \varepsilon + \varepsilon_X$, $P_{r,\rm eff} = P + P_{r,X}$):

$$\mathcal{W}_{\rm metric}(r) \;=\; e^{(3\lambda-\nu_C)/2}\, r^4\, \big[\varepsilon_{\rm eff}(r) \;+\; P_{r,\rm eff}(r)\big].$$

1. **In `mass` Mode (`radial_modes.py`, lines 31–32):** By V23 Eq. (4) derived in Step 16, $\varepsilon_{\rm eff} = \varepsilon + \varepsilon_X = f_L(r)\varepsilon(r) > 0$ and $P_{r,\rm eff} = P(r) > 0$, hence:

$$\varepsilon_{\rm eff}^{\rm mass}(r) + P_{r,\rm eff}^{\rm mass}(r) \;=\; f_L(r)\,\varepsilon(r) \;+\; P(r) \;>\; 0 \qquad \forall\, r \in [0, R_*),$$

   so in `mass` mode the inertia weight remains positive everywhere.

2. **In `gradient` Mode (`radial_modes.py`, lines 33–34):** By V23 Eq. (5) derived in Step 16, $\varepsilon_{\rm eff} = \varepsilon(r)$, whereas $P_{r,\rm eff}(r) = P(r) + P_{r,X}(r) = f_L(r)P(r) - \big(1 - f_L(r)\big)\frac{m(r)}{4\pi r^3}$. Therefore, the effective inertial sum is:

$$\boxed{\varepsilon_{\rm eff}^{\rm grad}(r) + P_{r,\rm eff}^{\rm grad}(r) \;=\; \varepsilon(r) \;+\; f_L(r)\,P(r) \;-\; \big(1 - f_L(r)\big)\,\frac{m(r)}{4\pi r^3}.}$$

   Let us examine the behavior of this expression near the stellar surface ($r \to R_*$):

   - At the surface, matter pressure and energy density vanish: $\lim_{r\to R_*} P(r) = 0$ and $\lim_{r\to R_*} \varepsilon(r) = 0$.
   - By contrast, the mean enclosed density ($\frac{m(R_*)}{4\pi R_*^3} = \frac{M}{4\pi R_*^3} > 0$) is strictly positive, and for any nonlocal smoothing radius $L > 0$ the surface ball reaches into the stellar interior, so $\bar\rho_L(R_*) > 0 \implies 1 - f_L(R_*) > 0$!
   - Consequently, near the surface ($r \to R_*$) the limit of the expression is **strictly negative**:

$$\boxed{\lim_{r\to R_*}\Big[\varepsilon_{\rm eff}^{\rm grad}(r) + P_{r,\rm eff}^{\rm grad}(r)\Big] \;=\; -\big(1 - f_L(R_*)\big)\,\frac{M}{4\pi R_*^3} \;<\; 0 \qquad (\forall\, L > 0)!}$$

   When $L \geq 367$ m, the factor $1 - f_L(R_*)$ and the crustal density profile are wide enough that the outermost Gauss–Legendre points of the $N = 1200$ finite-element quadrature grid (`radial_modes.py`, lines 45–54) fall inside the crustal zone where $\varepsilon(r) + f_L(r)P(r) < (1 - f_L(r))\frac{m(r)}{4\pi r^3}$, making $\mathcal{W}_{\rm metric}(r) \leq 0$ (across 12 sequences, 609 models in `item2_radial_grid.csv`). This proves analytically that the `metric` weight indefiniteness is not a numerical artifact, but a consequence of the `metric` ansatz substituting the static background stress ($P_{r,X} < 0$) as inertial mass, whereas the genuine kinetic weights ($\mathcal{W}_\xi \propto f_L(\varepsilon + P) > 0$ and $\mathcal{W}_\phi \propto \alpha_Y^{-1} > 0$) derived in Step 12 (`step12_second_variation_coupled_pulsation_hessian.md`) from the second variation $\delta^2 S$ of the covariant action are positive-definite for all $r \in (0, R_*)$ and all $L > 0$!

18.4 4. Checks

1. $f_L \to 1$ **(GR) Limit in V23 Equation (8):**

   - If $f_L(r) \equiv 1$, the factor $(f_L - 1)$ makes the right-hand side of V23 Eq. (8) vanish identically (`diff_gr_limit = 0` in line 120 of `covariant_closure_50aa2db0.py`), so $\mathcal{L}_{\rm SL}^{\rm matter}[\xi] \equiv \sqrt{\frac{r}{r-2m}}e^{2\nu}\mathcal{H}_{\rm lin}[\xi]$, and the eigenvalue crossing $\omega_0^2 = 0$ coincides with the static turning point $dM/d\rho_c = 0$ ✓.

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2. **Sign-Reversal Radius ($r_{\mathbf{sign}}$) of $\mathcal{W}_{\mathbf{metric}}(r)$ in `gradient` Mode:**

   - The sign reversal occurs in the crust precisely where $\varepsilon(r_{\rm sign}) + f_L(r_{\rm sign})P(r_{\rm sign}) = \big(1 - f_L(r_{\rm sign})\big)\frac{m(r_{\rm sign})}{4\pi r_{\rm sign}^3}$; as $L \to 0$, $1 - f_L(R_*) \to 0$ so $r_{\rm sign} \to R_*$ (pushed outside the quadrature points), whereas for $L \geq 367$ m we have $r_{\rm sign} < r_{\rm quad,max} < R_*$ ✓.

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19 Step 18: Exact Derivation of the V23 Eq. (9) Nonlocal Static Tangent Operator (eq:coupled_bvp), the 3D Spherical-Shell Intersection Kernel K_{ij}, and the $W \to \kappa W$ Inertia-Scaling Invariance

Source file: step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step17_chandrasekhar_factored_difference_eq
(step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness.md)
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Local Archival Sources:

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (Section 2, lines 127–153, Eq. 9 and Table 2; DOI: 10.5281/zenodo.23639084)
- V23 verification script and results: /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/covariant_ (lines 179–191 and 583–695) and coupled_stability_results.csv
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 7, point 1, line 374 and Chapter 8, point 2, lines 409–419; DOI: 10.5281/zenodo.23604423)

19.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 17 (step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness.md), we derived V23 Equation (8) (eq:factored_diff) and the surface sign reversal $\mathcal{W}_{\rm metric}(r) \le 0$ of the gradient-mode metric operator. Immediately following this in Section 2 of the V23 manuscript comes **V23 Equation (9) (eq:coupled_bvp)**, the nonlocal kernel perturbation K_{ij} aligned to the piecewise-polytrope transition radii $r_{\rm trans}$, the $W \to 4W$ inertia-scaling invariance, the first-order tangent estimate $\omega^2 \simeq -F(0)/F'(0)$, and the symmetrized Galerkin matrix $\mathbf{M}_{\rm sym} = \mathbf{D}^T \mathrm{diag}(w\,\Delta r)\mathbf{D}$:

1. **V23 Eq. (9) Coupled Static Nonlocal Tangent Model and Quasi-Dynamical BVP Test (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 2, lines 127–136, Eq. 9):** > *"I linearize the coupled system (around the V21 dense_profile background, not around the new branches) in terms of the enthalpy perturbation $\delta h(r) = \delta P/(\varepsilon + P)$ and mass perturbation $\delta m(r)$ on a grid aligned with the exact piecewise-polytrope transition radii r_{trans}, incorporating the nonlocal kernel perturbation operator:"*

$$\delta f_L(r_i) = -\gamma_L f_L(r_i) \sum_j K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j), \qquad \begin{cases} \dfrac{d(\delta h)}{dr} = -\delta\nu'[\delta h, \delta m, \delta f_L[\delta h]] + \omega_0^2\, e^{\lambda - 2\nu}\xi, \\ \dfrac{d(\delta m)}{dr} = 4\pi r^2 \Big[f_{\rm mass} \frac{\varepsilon + P}{c_s^2}\delta h + \varepsilon\, \delta f_{\rm mass}[\delta h] \Big]. \end{cases} \tag{V23 Eq. 9}$$

> *"At $\omega_0^2 = 0$, system (9) is the exact tangent operator of the static equilibrium equations with respect to $\log_{10}\rho_{0c}$. As shown in Table 2 and Fig. 3, solving the coupled eigenvalue problem (9) along the SLy $L = 10\,km$ sequences yields a zero crossing within $\sim 7 \times 10^{-5}$ dex of $dM/d\rho_c = 0$ (the pure GR discretization baseline is 7.12×10^{-5} dex). Because at $\omega_0^2 = 0$ the problem reduces to $\mathcal{H}_{lin}[\delta h] = 0$, this agreement verifies static nonlocal linearization rather than dynamical inertia: scaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ yields a ratio $\omega^2_{4W}/\omega^2_W = 0.2500$ ($\omega^2_W = 4.897 \times 10^6\, s^{-2}$) with the identical shooting function $F(0) = 0.023252$, so the zero crossing is inertia-invariant. The raw BVP W matrix is not symmetric; the symmetrized Galerkin form $\mathbf{M}_{sym} = \mathbf{D}^T \mathrm{diag}(w\,\Delta r)\mathbf{D}$ is symmetric with $\min \mathrm{diag}\,\mathbf{M}_{sym} = 5.68 \times 10^{-9} > 0$; this is a Gram-matrix property, not a proof of positive-definiteness of the physical kinetic Hessian. The omega0_sq_sym_tangent_s2 column gives the first-order tangent estimate ($\omega^2 \simeq -F(0)/F'(0)$)."*

2. **3D Spherical-Shell Intersection Integral (tal_book_HU_2026-10-24.tex, Chapter 7, point 1, line 374):** > *"Inside a spherically symmetric star of radius R_*, the exact 3D top-hat average $\bar\rho_L(r)$ at radial coordinate r can be evaluated analytically over the intersection of the ball $B(\mathbf{x}, L)$ with spherical shells $r' \in [|r - L|, r + L]$, yielding $f_L(r) = \exp(-\gamma_L \bar\rho_L(r))$."*

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19.2 2. Exact Derivation of the 3D Spherical-Shell Intersection Quadrature Kernel K_{ij} and the $r_{\mathbf{trans}}$ Transition Grid

19.2.1 2.1. Solid-Angle Fraction of the 3D Spherical Top-Hat Shell Intersection (`build_tophat_matrix`)

Let $\mathbf{x}_i = (0, 0, r_i)$ be the query point ($r_i > 0$), and consider a concentric spherical shell of radius $r_j > 0$ and thickness Δr_j around the stellar center. By the law of cosines, the squared Euclidean distance of a point $\mathbf{y} = (r_j \sin\theta\cos\varphi,\, r_j \sin\theta\sin\varphi,\, r_j\cos\theta)$ on the shell from $\mathbf{x}_i$ is:

$$|\mathbf{y} - \mathbf{x}_i|^2 \;=\; r_i^2 \;+\; r_j^2 \;-\; 2\, r_i\, r_j \cos\theta.$$

The condition $|\mathbf{y} - \mathbf{x}_i| \le L$ is equivalent to:

$$\cos\theta \;\ge\; \mu_{\min}(r_i, r_j; L) \;\equiv\; \frac{r_i^2 + r_j^2 - L^2}{2\, r_i\, r_j}.$$

The surface fraction of the spherical shell of radius r_j falling inside the ball $B(\mathbf{x}_i, L)$ ($\mathrm{frac}(r_i, r_j; L) = \frac{1}{4\pi}\int_{|\mathbf{y}-\mathbf{x}_i|\le L} \sin\theta\, d\theta\, d\varphi = \frac{1}{2}\int_{\max(-1,\mu_{\min})}^{1} d(\cos\theta)$) splits into three geometric regimes:

1. **Fully Enclosed Shell ($r_i + r_j \le L$):** Here $r_i^2 + r_j^2 - L^2 \le -2r_ir_j \implies \mu_{\min} \le -1$, so the entire spherical shell lies inside $B(\mathbf{x}_i, L)$: $\mathrm{frac}(r_i, r_j; L) = 1$.
2. **Partially Intersected Shell ($|r_i - r_j| < L < r_i + r_j$):** Here $-1 < \mu_{\min} < +1$, so the solid-angle fraction of the spherical cap is:

$$\mathrm{frac}(r_i, r_j; L) \;=\; \frac{1 - \mu_{\min}}{2} \;=\; \frac{2r_ir_j - (r_i^2 + r_j^2 - L^2)}{4\, r_i\, r_j} \;=\; \frac{L^2 - (r_i - r_j)^2}{4\, r_i\, r_j}.$$

3. **Exterior Shell ($|r_i - r_j| \ge L$):** Here $\mu_{\min} \ge +1$, so the intersection is empty: $\mathrm{frac}(r_i, r_j; L) = 0$.

Because the volume of the j-th spherical shell is $4\pi r_j^2\, \Delta r_j$ and the volume of the ball $B(\mathbf{x}_i, L)$ is $V_L = \frac{4}{3}\pi L^3$, the entry of the discrete 3D top-hat kernel matrix is (`covariant_closure_50aa2db0.py`, lines 188–190):

$$K_{ij} \;=\; \frac{4\pi r_j^2\, \mathrm{frac}(r_i, r_j; L)\, \Delta r_j}{\frac{4}{3}\pi L^3} \;=\; \frac{3\, r_j^2\, \Delta r_j}{L^3} \begin{cases} 1, & r_i + r_j \le L, \\ \dfrac{L^2 - (r_i - r_j)^2}{4\, r_i\, r_j}, & |r_i - r_j| < L < r_i + r_j, \\ 0, & |r_i - r_j| \ge L. \end{cases}$$

Consequently, the first variation of the nonlocal average $\bar\rho_L(r_i) = \sum_j K_{ij}\, \varepsilon(r_j)$ and of the factor $f_L(r_i) = \exp(-\gamma_L \bar\rho_L(r_i))$ with respect to the enthalpy perturbation $\delta\varepsilon(r_j) = \frac{d\varepsilon}{dh}\big|_{r_j} \delta h(r_j) = \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)} \delta h(r_j)$ gives the first part of V23 Equation (9) identically:

$$\delta f_L(r_i) \;=\; -\gamma_L\, f_L(r_i) \sum_{j=1}^{N_{\mathrm{tot}}} K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j).$$

19.2.2 2.2. Why Must Cell Boundaries Be Aligned with the Transition Radii $r_{\mathbf{trans}}$?

In piecewise-polytrope equations of state (SLy, APR4, H4), the pressure $P(h)$ and energy density $\varepsilon(h)$ are continuous across the polytrope boundaries $h_{\mathrm{tr}} \in \mathrm{EOS}.h_b$, but the adiabatic index Γ_k and hence the squared sound speed $c_s^2(h) = \frac{dP}{d\varepsilon}$ undergo a **discontinuous jump** at the transition radii $h(r_{\mathrm{trans}}) = h_{\mathrm{tr}}$! If a quadrature cell straddled a transition radius r_{trans}, the jump in the integrand $\frac{\varepsilon+P}{c_s^2}\delta h$ would introduce a first-order $\mathcal{O}(\Delta r)$ quadrature error, spoiling the turning-point agreement at $dM/d\rho_c = 0$. Therefore, `solve_coupled_radial_omega2` (`covariant_closure_50aa2db0.py`, lines 604–616) uses Brent's method (`brentq(lambda x: p["h"](x) - h_tr, 0.0, R)`) to locate all transition radii $0 < r_{\mathrm{trans}} < R_*$ to machine precision and embeds them as fixed cell boundaries in the grid.

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19.3 3. Matrix Form of the V23 Eq. (9) Coupled BVP, the $W \to \kappa W$ Scaling Invariance, and Galerkin Symmetrization

19.3.1 3.1. Operator Matrices (S, $\mathbf{D}_M$, $\mathbf{D}_A$, W) and the Fredholm Shooting Function $F(\omega^2)$

In geometrized units ($G = c = 1$, where $\gamma_L = L^2$), the two first-order equations of system V23 (9) for the vector of cell-center enthalpy perturbations $\delta\mathbf{h} = (\delta h(r_1), \ldots, \delta h(r_{N_{\rm tot}}))^T$ (`covariant_closure_50aa2db0.py`, lines 629–660) take the form:

1. **Linearized Mass-Source Matrix (S) and Integrated Mass Perturbation ($\delta\mathbf{m} = \mathbf{D}_M\,\delta\mathbf{h}$):**

$$\frac{d(\delta m)}{dr} \;=\; (\mathbf{S}\,\delta\mathbf{h})_i, \qquad \mathbf{S} \;=\; \begin{cases} \mathrm{diag}\big(4\pi r^2 f_L\big)\,\big[\mathbf{I} - L^2\,\mathrm{diag}(\varepsilon)\,\mathbf{K}\big]\,\mathrm{diag}\left(\dfrac{\varepsilon+P}{c_s^2}\right), & (\texttt{mass mode}),\\[2ex] \mathrm{diag}\left(4\pi r^2\dfrac{\varepsilon+P}{c_s^2}\right), & (\texttt{gradient mode}). \end{cases}$$

   With the trapezoidal lower-triangular integration matrix $\big(\mathbf{M}_{\rm int}\big)_{ij} = \Delta r_j\,\Theta(i-j) + \frac{1}{2}\Delta r_i\,\delta_{ij}$, the mass-perturbation operator is $\mathbf{D}_M \equiv \mathbf{M}_{\rm int}\mathbf{S}$, i.e., $\delta\mathbf{m} = \mathbf{D}_M\,\delta\mathbf{h}$ (automatically satisfying the central boundary condition $\delta m(0) = 0$).

2. **Linearized Hydrostatic Gradient Matrix ($\mathbf{D}_A$):** Since $A(r) = \frac{m+4\pi r^3 P}{r(r-2m)}$, the partial derivatives are $c_{dm}(r) \equiv \frac{\partial A}{\partial m} = \frac{1+2rA}{r(r-2m)}$ and $c_{dh}(r) \equiv (\varepsilon+P)\frac{\partial A}{\partial P} = \frac{4\pi r^3(\varepsilon+P)}{r(r-2m)}$, so the base perturbation is $\mathbf{D}_{A,\rm base} = \mathrm{diag}(c_{dm})\,\mathbf{D}_M + \mathrm{diag}(c_{dh})$, and:

$$\delta\boldsymbol{\nu}' \;=\; \mathbf{D}_A\,\delta\mathbf{h}, \qquad \mathbf{D}_A \;=\; \begin{cases} \mathbf{D}_{A,\rm base}, & (\texttt{mass mode}),\\[2ex] \mathrm{diag}(f_L)\,\mathbf{D}_{A,\rm base} \;-\; L^2\,\mathrm{diag}(f_L A)\,\mathbf{K}\,\mathrm{diag}\left(\dfrac{\varepsilon+P}{c_s^2}\right), & (\texttt{gradient mode}). \end{cases}$$

3. **Adiabatic Mass-Conservation Relation and the Inertia Matrix (W):** For an adiabatic radial displacement, Lagrangian mass conservation $\Delta m = 0 \implies \delta m(r) = -4\pi r^2 f_{\rm mass}(\varepsilon+P)\,r\xi(r)$ ($f_{\rm mass} = f_L$ in `mass` mode and $f_{\rm mass} = 1$ in `gradient` mode), allowing the displacement to be expressed in terms of the mass perturbation:

$$-\omega^2 e^{2(\lambda-\nu)} r\xi(r) \;=\; +\omega^2\,w_{\rm inert}(r)\,\delta m(r), \qquad w_{\rm inert}(r) \;\equiv\; \frac{e^{2(\lambda-\nu)}}{4\pi r^2 f_{\rm mass}(r)\big(\varepsilon(r)+P(r)\big)} \;>\; 0.$$

   Therefore, the inertia matrix is $\mathbf{W}(\kappa) \equiv \kappa\,\mathrm{diag}(w_{\rm inert})\,\mathbf{D}_M$ (where $\kappa \equiv W_{\rm scale} > 0$ is the scaling factor), and integrating the V23 Eq. (9) enthalpy equation with central normalization $\delta h(0) = 1$ yields the discrete Fredholm system (`covariant_closure_50aa2db0.py`, lines 664–665):

$$\boxed{\Big[\mathbf{I} \;+\; \mathbf{M}_{\rm int}\big(\mathbf{D}_A \;+\; \omega^2\,\mathbf{W}(\kappa)\big)\Big]\,\delta\mathbf{h}(\omega^2;\kappa) \;=\; \mathbf{1}, \qquad F(\omega^2;\kappa) \;\equiv\; \delta m(R_*;\omega^2,\kappa) \;=\; \boldsymbol{\Delta r}^T\mathbf{S}\,\delta\mathbf{h}(\omega^2;\kappa) \;=\; 0.}$$

   The surface Birkhoff boundary condition $F(\omega^2;\kappa) = \delta m(R_*;\omega^2,\kappa) = 0$ is equivalent to the free-surface boundary condition $\delta m(R_*) = -4\pi R_*^3 f_{\rm mass}(\varepsilon+P)\xi(R_*)$ and $\Delta P(R_*) = 0$.

19.3.2 3.2. Proof of the $W \to \kappa W$ Inertia-Scaling Invariance and the First-Order Tangent Theorem

Three exact algebraic consequences follow from the Fredholm equation above:

1. **Exact Scaling Identity ($\forall\,\kappa > 0$):** Because $\mathbf{W}(\kappa) = \kappa\,\mathbf{W}(1)$, the combination $\mathbf{D}_A + \omega^2\mathbf{W}(\kappa) = \mathbf{D}_A + (\kappa\omega^2)\mathbf{W}(1)$ depends on the eigenvalue ω^2 and inertia scale κ solely through the product $\tilde\omega^2 \equiv \kappa\,\omega^2$:

$$\delta\mathbf{h}(\omega^2;\kappa) \;\equiv\; \delta\mathbf{h}(\kappa\omega^2;1) \quad\Longrightarrow\quad F(\omega^2;\kappa) \;\equiv\; F(\kappa\omega^2;1).$$

   Consequently, if ω_W^2 is the eigenvalue at $\kappa = 1$ ($F(\omega_W^2;1) = 0$), then under κ-fold inertia scaling the exact eigenvalue is:

$$\boxed{\omega_{\kappa W}^2 \;=\; \frac{\omega_W^2}{\kappa}, \qquad \frac{\omega_{4W}^2}{\omega_W^2} \;=\; \frac{1}{4} \;=\; 0.2500 \quad \text{identically!}}$$

   Specifically, at $\log_{10}\rho_{0c} = 15.30427$ (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 135), $F(0;\kappa) \equiv F(0;1) = 0.023252$ (independent of κ), $\omega_W^2 = 4.897 \times 10^6\ \mathrm{s}^{-2}$, and $\omega_{4W}^2/\omega_W^2 = 0.2500$.

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2. **Inertia Invariance of the Static Turning Point ($\omega_0^2 = 0$):** At $\omega^2 = 0$, the term $\omega^2\mathbf{W}(\kappa)$ drops out, leaving $(\mathbf{I} + \mathbf{M}_{\rm int}\mathbf{D}_A)\delta\mathbf{h}(0) = \mathbf{1}$. Because $\mathbf{D}_A$ and $\mathbf{S}$ are the exact Fréchet derivatives of the nonlocal static TOV equations with respect to the central enthalpy perturbation $\delta h(0)$, the solution $\delta\mathbf{h}(0)$ is proportional to the static sequence derivative $\frac{\partial h(r;\rho_{0c})}{\partial \log_{10}\rho_{0c}}$, and the shooting function is:

$$F(0) \;=\; \delta m(R_*;0) \;=\; \left(\frac{\partial h_c}{\partial \log_{10}\rho_{0c}}\right)^{-1} \frac{dM}{d\log_{10}\rho_{0c}}.$$

Therefore $F(0) = 0 \iff \frac{dM}{d\rho_c} = 0$, explaining why the turning-point offset drops to $\Delta\log_{10}\rho_{0c} = +6.97\times10^{-5}$ dex (`mass`, a 19.2×/58.3× improvement) and $+7.39\times10^{-5}$ dex (`gradient`, a 48.3×/59.3× improvement), matching the pure GR discretization baseline ($+7.12\times10^{-5}$ dex, V23 Table 2).

3. **First-Order Tangent Estimate (`omega0_sq_sym_tangent_s2`) and the Galerkin Gram Matrix ($\mathbf{M}_{\rm sym}$):** First-order Taylor expansion of the root $F(\omega^2) = 0$ around $\omega^2 = 0$ gives $F(0) + F'(0)\,\omega^2 + \mathcal{O}(\omega^4) = 0$, i.e. (`covariant_closure_50aa2db0.py`, lines 680–690):

$$\omega^2_{\rm tangent} \;\simeq\; -\frac{F(0)}{F'(0)}.$$

Although the raw BVP matrix $\mathbf{W} = \mathrm{diag}(w_{\rm inert})\mathbf{D}_M$ is not symmetric (since $\mathbf{D}_M = \mathbf{M}_{\rm int}\mathbf{S}$ is multiplied by a lower-triangular integrator), the symmetrized Galerkin matrix of the quadratic kinetic form expressed in terms of the mass perturbations $\delta\mathbf{m} = \mathbf{D}_M\delta\mathbf{h}$ (`covariant_closure_50aa2db0.py`, lines 685–686) is:

$$\boxed{\mathbf{M}_{\rm sym} \;\equiv\; \mathbf{D}_M^T\,\mathrm{diag}\big(w_{\rm inert}\,\Delta r\big)\,\mathbf{D}_M \;=\; \mathbf{M}_{\rm sym}^T.}$$

Because $w_{\rm inert}(r_i)\,\Delta r_i > 0$ in every cell $i = 1,\dots,N_{\rm tot}$ (in both `mass` and `gradient` modes, since the denominator contains the matter sum $\varepsilon + P > 0$!), for any vector $\mathbf{v} \neq \mathbf{0}$ we have $\mathbf{v}^T\mathbf{M}_{\rm sym}\mathbf{v} = \sum_{i=1}^{N_{\rm tot}} w_{\rm inert}(r_i)\Delta r_i\,(\mathbf{D}_M\mathbf{v})_i^2 \geq 0$, with minimum diagonal entry $\min\mathrm{diag}\,\mathbf{M}_{\rm sym} = 5.68\times10^{-9} > 0$. As emphasized in the V23 manuscript (lines 135 and 152), this is by construction a **Gram-matrix property** ($\mathbf{B}^T\mathbf{B}$ form with $\mathbf{B} = \mathrm{diag}(\sqrt{w_{\rm inert}\Delta r})\mathbf{D}_M$), whereas the positive-definiteness of the genuine dynamical kinetic Hessian was derived in Step 12 (`step12_second_variation_coupled_pulsation_hessian.md`) from the second variation $\delta^2 S$ of the action.

19.4 4. Checks

1. **Normalization of the 3D Spherical-Shell Intersection Kernel K_{ij} ($\sum_j K_{ij} = 1$):**

- When the stellar radius satisfies $R_* > r_i + L$ and the density is homogeneous ($\varepsilon(r_j) \equiv 1$), in the continuum limit:

$$\sum_j K_{ij} \;\longrightarrow\; \frac{3}{L^3}\left[\int_0^{L-r_i} r'^2\,dr' \;+\; \int_{|L-r_i|}^{L+r_i} r'^2\,\frac{L^2-(r_i-r')^2}{4\,r_i\,r'}\,dr'\right] \;=\; 1 \qquad \forall\, r_i \geq 0\;\checkmark.$$

2. **Agreement Between $\omega^2_{\rm tangent} \simeq -F(0)/F'(0)$ and $\omega^2_{\rm BVP}$ Near the Turning Point (`coupled_stability_results.csv`):**

- In row 23 of `coupled_stability_results.csv` (SLy GR, $\log_{10}\rho_{0c} = 15.302445$), $\omega^2_{\rm BVP} = 1.04011\times10^5$ s^{-2} and $\omega^2_{\rm tangent} = 1.03992\times10^5$ s^{-2} (0.018% difference); in row 56 (SLy `mass`, $L = 10$ km, $\log_{10}\rho_{0c} = 15.308269$), $\omega^2_{\rm BVP} = 9.84022\times10^4$ s^{-2} and $\omega^2_{\rm tangent} = 9.83845\times10^4$ s^{-2} (0.018% difference), confirming the accuracy of first-order tangent linearization in the neighborhood of the $\omega_0^2 = 0$ turning point ✓.

19.5 5. Explicit Separation Between the V21 Nonlocal 3D Top-Hat Tangent Model (`step18`) and the V23 Local Curved `yukawa_action_el` Coupled Stability (`step12`, `step22`)

To distinguish the model branches unambiguously, we record the exact scope of the two stability calculations:

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Property	**Branch I-A**: V21 Nonlocal 3D Top-Hat (`step18`, `covariant_closure_50aa2db0.py`)	**Branch II-Fluid**: V23 Curved Exterior `yukawa_action_el` (`step12`, `step22`, `curved_yukawa_adm_and_stability.py`)
Equilibrium Background	V21 `tophat_mass` and `tophat_gradient` (`tp.dense_profile`)	Coupled interior + curved exterior ($r \in [0, R_* + 12L]$) `yukawa_action_el`
Perturbation Variables	$(\delta h, \delta m)$ with nonlocal shell kernel $\delta f_L = -\gamma_L f_L K_{ij} \delta\varepsilon_j$	Coupled $(\delta h, \delta\phi_L, \delta m)$ in the interior and curved exterior vacuum tail
Mass Turning Point (SLy, $L = 10$ km)	$\log_{10}\rho_{0c}(M_{\max}) = 15.29924$ (`mass`), 15.30106 (`gradient`)	$\log_{10}\rho_{0c}(M_{\mathrm{ADM,max}}) = 15.316404$ ($\alpha_Y = 1$), 15.307166 ($\alpha_Y = 1/3$)
$\omega_0^2 = 0$ **Zero Crossing**	15.29931 (`mass`), 15.30113 (`gradient`)	15.316410 ($\alpha_Y = 1$), 15.307170 ($\alpha_Y = 1/3$)
Turning-Point Offset $\Delta\log_{10}\rho_{0c}$	$+6.97 \times 10^{-5}$ dex (`mass`), $+7.39 \times 10^{-5}$ dex (`gradient`)	$+5.80 \times 10^{-6}$ dex ($\alpha_Y = 1$, 2097× reduction), $+4.20 \times 10^{-6}$ dex ($\alpha_Y = 1/3$, 995× reduction; $L = 1$ km: -6.86×10^{-6} dex, 165×)

V25 Update (curved `yukawa_action_el` branch): 4-component exterior BVP ($\delta\nu_R = 1.63 \times 10^{-14}$), finite-difference verified exterior ($\partial\phi_L'(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ km^{-2}) and interior ($r = 6$ km $\partial\phi_L''/\partial m = -1.72833 \times 10^{-4}$ km^{-3}, `scalar_m_sign = -1.0` main operator) derivatives, dynamical scalar wave, GL2 root convergence ($p_{\mathrm{eff,root}} = 4.248$). SLy $L = 10$ km, $\alpha_Y = 1$: M_{ADM} turning point 15.316404, main dynamical $\omega_0^2 = 0$ root 15.316410 ($+5.80 \times 10^{-6}$ dex). Detailed tables: `step22`, Sections 4 and 7.

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20 Step 19: Exact Derivation of the V20 8558 kg Granite-Chamber 3D IN–OUT Differential Quadrature (Three Geometric L-Regimes, Top-Hat vs. Yukawa) and the Sirius B–Sirius A Common Effective-Mass Proxy (`common_background` vs. `fixedM`)

Source file: `step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-26
File Path: `scalar-tensor-yukawa-fit/step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b`
Previous Steps: `step1_dilaton_form_en.md` (`step1_dilaton_form.md`) – `step18_nonlocal_tangent_bvp_shell_kernel_an` (`step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance.md`)
License: CC BY-NC 4.0
Local Archival Sources:

- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Section 3 and Table 3 + footnote, lines 161–189; DOI: `10.5281/zenodo.23639084`)
- V23 verification script and dataset: `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/covariant_` (Part 4, lines 700–885) and `clock_coupling_H0_H1.csv` (rows 1–13)
- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex` (Chapter 9, lines 426–455 and Chapter 10, point 1, lines 463–468; DOI: `10.5281/zenodo.23604423`)

20.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 8 (`step8_microscopic_clock_coupling_siriusb_lab_v20.md`), we derived the Dirac–Maxwell atomic-clock action ($S_{\rm atomic}[\tilde{g}_{\mu\nu}, \psi_e, A_\mu]$) under H0 ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$) and conditional H1 ($\tilde{g}_{00} = f_L^2 g_{00}$). Table 3 of Section 3 of the V23 manuscript (lines 170–189) and Part 4 of `covariant_closure_50aa2db0.py` (lines 700–885), however, contain two additional exact mathematical structures that we now derive step by step:

1. **Footnote to V23 Table 3 on the Granite-Chamber IN–OUT Differential Signal and the Sirius B Common Effective-Mass Proxy (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 188):** > *"V20 rows: H0 column = TAL excess over Newtonian GR at the IN ($\mathbf{r} = \mathbf{0}$) position; H1 columns = IN shift (IN−OUT differential signal, $D_{out} = 2.0\,m$). Newtonian GR: $\Phi_N(IN)/c^2 = -9.269 \times 10^{-24}$, IN−OUT $= -6.094 \times 10^{-24}$; $\sigma_y = 10^{-18}$ is an illustrative reference value. On the Sirius B common effective-mass proxy (fixed $n = 3$ Lane–Emden profile with the H0 effective-mass shift, not a newly solved equilibrium) at $L = 10^3$ km: H0 80.69839, H1 top-hat 80.69878 km/s."*
2. **Geometry and Mass Budget of the 8558 kg Granite Chamber (`tal_book_HU_2026-10-24.tex`, Chapter 10, point 1, line 466):** > *"**Exact Geometry and Mass Budget:** The open cylindrical barrel of dimensions $R_{in} = 0.25\,m$, $R_{out} = 0.85\,m$, $H = 1.20\,m$ has volume $V_{cyl} = \pi(0.85^2 - 0.25^2) \times 1.20 = 2.48814\,m^3$ and mass $M_{cyl} = 6842.39\,kg$; the nominal $V_{tot} = 3.1102\,m^3$ ($M_{tot} = 8558\,kg$) includes an additional $0.6220\,m^3$ (1710.6 kg) of endcap/support granite elements (or an outer radius $R_{out} = 0.942\,m$)."*

20.2 2. 3D IN–OUT Quadrature of the V20 8558 kg Granite Chamber and Derivation of the Three Geometric L-Regimes

20.2.1 2.1. Domain of the Cylindrical Barrel + Endcaps ($\Omega_{\rm granite}$) and the Off-Axis Distance Function

In `covariant_closure_50aa2db0.py` (lines 802–822), the granite chamber of density $\rho_{\rm granite} = 2750\ {\rm kg/m^3}$ and mass $M_{\rm tot} = 8558$ kg ($V_{\rm tot} = M_{\rm tot}/\rho_{\rm granite} = 8558/2750 = 3.112000\ {\rm m^3}$) is constructed in cylindrical coordinates ($\varrho \in [0, R_{\rm out}]$, $z \in [-z_{\rm out}, +z_{\rm out}]$, $\varphi \in [0, 2\pi)$), exploiting $z \leftrightarrow -z$ mirror symmetry, as the union of:

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- **Open Cylindrical Barrel:** $R_{\rm in} = 0.25\ {\rm m} \le \varrho \le R_{\rm out} = 0.85\ {\rm m}$, $|z| \le z_{\rm in} = 0.60\ {\rm m}$ ($H_{\rm in} = 2z_{\rm in} = 1.20\ {\rm m}$), whose volume and mass are:

$$V_{\rm cyl} \;=\; \pi\big(R_{\rm out}^2 - R_{\rm in}^2\big)\,(2z_{\rm in}) \;=\; 2.488141\ {\rm m}^3, \qquad M_{\rm cyl} \;=\; \rho_{\rm granite}\,V_{\rm cyl} \;=\; 6842.39\ {\rm kg}.$$

- **Two Solid Endcaps:** $0 \le \varrho \le R_{\rm out} = 0.85\ {\rm m}$, $z_{\rm in} \le |z| \le z_{\rm out}$, where the endcap thickness $\Delta z_{\rm cap} = z_{\rm out} - z_{\rm in}$ (`covariant_closure_50aa2db0.py`, line 808) is:

$$\Delta z_{\rm cap} \;=\; \frac{V_{\rm tot} - V_{\rm cyl}}{2\pi R_{\rm out}^2} \;=\; \frac{3.112000 - 2.488141}{2\pi \times 0.85^2} \;=\; 0.137426\ {\rm m} \quad\Longrightarrow\quad z_{\rm out} \;=\; 0.60 + 0.137426 \;=\; 0.737426\ {\rm m}.$$

  Thus, the characteristic domain $\Omega_{\rm granite}$ occupied by granite on the half-meridian plane ($\varrho \in [0, R_{\rm out}]$, $z \in [0, z_{\rm out}]$) is:

$$\Omega_{\rm granite} \;=\; \big\{(\varrho, z) \in [0, R_{\rm out}] \times [0, z_{\rm out}] \;:\; (\varrho \ge R_{\rm in}) \;\vee\; (z \ge z_{\rm in})\big\}, \qquad dV \;=\; 2 \cdot \big(2\pi\varrho\, d\varrho\, dz\big),$$

  where the factor of 2 accounts for the sum over the $\pm z$ half-spaces (giving $M_{\rm granite,num} = 8558.23$ kg on an $N_\varrho \times N_z = 500 \times 500$ grid).

The Euclidean distance of a source point $(\varrho, \varphi, \pm z) \in \Omega_{\rm granite}$ from the two clock positions (`covariant_closure_50aa2db0.py`, lines 820 and 834) is:

1. **IN Position ($\mathbf{r_{in}} = (0,0,0)$, Geometric Center of the Chamber):** independent of azimuth:

$$d_{\rm in}(\varrho, z) \;=\; \sqrt{\varrho^2 + z^2}.$$

2. **OUT Position ($\mathbf{r_{out}} = (D_{\rm out}, 0, 0)$, $D_{\rm out} = 2.0$ m in the Midplane):** by the law of cosines:

$$d_{\rm out}(\varrho, z, \varphi) \;=\; \sqrt{\varrho^2 + D_{\rm out}^2 - 2\varrho D_{\rm out}\cos\varphi + z^2},$$

   integrated over $N_\varphi = 256$ uniform azimuthal points ($\varphi_k = \frac{2\pi(k+1/2)}{256}$, $dm_p = dm_g/N_\varphi$). From this, the Newtonian gravitational potential and standard GR redshift difference (`clock_coupling_H0_H1.csv`, rows 8–13) are:

$$\frac{\Phi_N({\rm IN})}{c^2} \;=\; -\frac{G}{c^2}\int_{\Omega_{\rm granite}} \frac{dm_g}{d_{\rm in}} \;=\; -9.268639{\times}10^{-24}, \qquad \frac{\Phi_N({\rm OUT})}{c^2} \;=\; -\frac{G}{c^2}\int_{\Omega_{\rm granite}} \frac{dm_p}{d_{\rm out}} \;=\; -3.174686{\times}10^{-24},$$

$$\Delta y_{\rm GR}({\rm IN} - {\rm OUT}) \;=\; \frac{\Phi_N({\rm IN}) - \Phi_N({\rm OUT})}{c^2} \;=\; -6.093953 \times 10^{-24}.$$

20.2.2 2.2. Exact Proof of the Four Geometric Threshold Distances and Top-Hat vs. Yukawa IN–OUT Behavior

The extremal Euclidean distances of the chamber domain $\Omega_{\rm granite}$ from the IN and OUT points are:

- **At the IN Position ($\mathbf{r_{in}} = 0$):**

$$d_{\rm min}^{\rm IN} \;=\; \min_{(\varrho,z)\in\Omega_{\rm granite}} \sqrt{\varrho^2 + z^2} \;=\; \min(R_{\rm in}, z_{\rm in}) \;=\; \min(0.25,\, 0.60) \;=\; 0.25\ {\rm m},$$

$$d_{\rm max}^{\rm IN} \;=\; \max_{(\varrho,z)\in\Omega_{\rm granite}} \sqrt{\varrho^2 + z^2} \;=\; \sqrt{R_{\rm out}^2 + z_{\rm out}^2} \;=\; \sqrt{0.85^2 + 0.737426^2} \;=\; 1.12530\ {\rm m}.$$

- **At the OUT Position ($D_{\rm out} = 2.00$ m):**

$$d_{\rm min}^{\rm OUT} \;=\; \min_{\Omega_{\rm granite}} d_{\rm out} \;=\; D_{\rm out} - R_{\rm out} \;=\; 2.00 - 0.85 \;=\; 1.15\ {\rm m} \quad (\varphi = 0,\ \varrho = R_{\rm out},\ z = 0),$$

$$d_{\rm max}^{\rm OUT} \;=\; \max_{\Omega_{\rm granite}} d_{\rm out} \;=\; \sqrt{(D_{\rm out} + R_{\rm out})^2 + z_{\rm out}^2} \;=\; \sqrt{2.85^2 + 0.737426^2} \;=\; 2.94385\ {\rm m} \quad (\varphi = \pi,\ \varrho = R_{\rm out},\ z = z_{\rm out}).$$

From these four threshold distances ($0.25\ {\rm m} < 1.1253\ {\rm m} < 1.15\ {\rm m} < 2.9439\ {\rm m}$), the sharp physical contrast between the 3D spherical top-hat ($\phi_L^{\rm top\text{-}hat}(\mathbf{r}) = \frac{3G}{4\pi c^2 L}\int_{d\le L} dm$) and the local Yukawa kernel ($\phi_L^{\rm Yuk}(\mathbf{r}) = \frac{G}{4\pi c^2}\int \frac{e^{-d/L}}{d}\, dm$ at $\alpha_Y = 1$) follows immediately (`clock_coupling_H0_H1.csv`, rows 8–13):

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1. **Cavity Regime ($L < d_{\min}^{\mathbf{IN}} = 0.25$ m, e.g. $L = 0.20$ m):** Both balls $B(\mathbf{0}, L)$ and $B(\mathbf{r}_{\text{out}}, L)$ remain outside the granite walls (inside the inner cavity and outside the chamber, respectively), so $\phi_L^{\text{top-hat}}(\text{IN}) = \phi_L^{\text{top-hat}}(\text{OUT}) = 0$, giving $\Delta y_{\text{H1}}^{\text{top-hat}} = 0$. By contrast, the Yukawa Green's function $e^{-d/L}/d$ senses the walls across the cavity: $\Delta y_{\text{H1}}^{\text{Yuk}}(\text{IN}) = -3.690 \times 10^{-26}$ and $\Delta y_{\text{H1}}^{\text{Yuk}}(\text{IN} - \text{OUT}) = -3.683 \times 10^{-26}$.
2. **Purely Interior Top-Hat Regime ($d_{\min}^{\mathbf{IN}} < L < d_{\min}^{\mathbf{OUT}} = 1.15$ m, e.g. $L = 0.50,\ 0.85,\ 1.00$ m in V23 Table 3):** Because $L < 1.15$ m, the OUT ball $B(\mathbf{r}_{\text{out}}, L)$ **does not intersect the granite chamber** ($d_{\text{out}} \geq 1.15\ \text{m} > L$), so:
$$\boxed{\phi_L^{\text{top-hat}}(\text{OUT}) \equiv 0 \quad \Longrightarrow \quad \Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN} - \text{OUT}) \equiv \Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN}) \qquad \forall\, L < 1.15\ \text{m}!}$$
This is why V23 Table 3 (lines 183–185) has the exact same number before and inside the parentheses in the Top-Hat column: -3.318×10^{-25} (-3.318×10^{-25}) ($L = 0.50$ m), -1.303×10^{-24} (-1.303×10^{-24}) ($L = 0.85$ m), and -1.436×10^{-24} (-1.436×10^{-24}) ($L = 1.00$ m). By contrast, the exponential tail of the Yukawa kernel is non-zero even at $D_{\text{out}} = 2.0$ m ($\phi_L^{\text{Yuk}}(\text{OUT}) = +3.735 \times 10^{-26}$ at $L = 1.00$ m), so the Yukawa differential signal is smaller in magnitude than the absolute IN signal: -3.781×10^{-25} (-3.407×10^{-25}) at $L = 1.00$ m!
3. **Partial Exterior Overlap ($d_{\min}^{\mathbf{OUT}} < L < d_{\max}^{\mathbf{OUT}}$, e.g. $L = 1.50$ m):** Here the IN ball already encloses the entire chamber ($L = 1.50\ \text{m} > d_{\max}^{\text{IN}} = 1.1253$ m), while the OUT ball partially intersects the chamber: $\Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN}) = -1.0115 \times 10^{-24}$, $\Delta y_{\text{H1}}^{\text{top-hat}}(\text{OUT}) = -1.0064 \times 10^{-25}$, yielding $\Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN} - \text{OUT}) = -9.1086 \times 10^{-25}$.
4. **Common Enclosure Regime ($L \geq d_{\max}^{\mathbf{OUT}} = 2.94385$ m, e.g. $L = 10.0$ m):** When $L \geq 2.944$ m, both balls $B(\mathbf{0}, L)$ and $B(\mathbf{r}_{\text{out}}, L)$ enclose the entire $M_{\text{tot}} = 8558$ kg granite chamber! Consequently, the 3D spherical top-hat average at both positions is $\bar{\rho}_L^{\text{top-hat}}(\text{IN}) = \bar{\rho}_L^{\text{top-hat}}(\text{OUT}) = \frac{3M_{\text{tot}}}{4\pi L^3}$, so the **Top-Hat IN−OUT differential signal vanishes identically**:
$$\boxed{\Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN} - \text{OUT})\Big|_{L \geq 2.944\ \text{m}} = 0 \quad (\text{numerically } -5.28 \times 10^{-40} \text{ rounding noise at } L = 10\ \text{m}),}$$
whereas the Yukawa kernel, owing to its $1/d$ weighting, still produces a distinct differential signal at $L = 10.0$ m: $\Delta y_{\text{H1}}^{\text{Yuk}}(\text{IN} - \text{OUT}) = -4.818 \times 10^{-25}$!
5. **Second-Order H0 TAL Excess in the Cavity (`dy_H0_cavity`):** Under H0 there is no direct f_L clock prefactor, so the excess over Newtonian GR arises from the $1 - f_L \simeq \phi_L$ modification of the effective gravitating mass of the chamber (`covariant_closure_50aa2db0.py`, line 852):
$$\Delta y_{\text{H0}}^{\text{excess}}(\text{IN}) = \left|\frac{\Phi_N(\text{IN})}{c^2}\right| \phi_L^{\text{top-hat}}(\text{IN}) = 9.268639{\times}10^{-24}{\times}1.436124{\times}10^{-24} = +1.331091{\times}10^{-47} \quad (L = 1.00\ \text{m}).$$

20.3 3. Derivation of the Sirius B–Sirius A Differential Redshift and the `common_background` vs. `fixedM` Proxy

In `covariant_closure_50aa2db0.py` (lines 710–799), the white dwarf Sirius B ($M_B = 1.018 \pm 0.011\, M_\odot$, $R_B = 0.00803 \pm 0.00011\, R_\odot = 5586.471$ km) and the main-sequence reference star Sirius A ($M_A = 2.042\, M_\odot$, $R_A = 1.711\, R_\odot = 1.19034 \times 10^6$ km) are modeled as $n = 3$ Lane–Emden polytropes:

1. **GR Baseline and Combined Uncertainty (`covariant_closure_50aa2db0.py`, lines 719–723):** The surface compactness is $\mathcal{C}_B = \frac{GM_B}{R_B c^2} = 2.690878 \times 10^{-4}$, giving the exact Schwarzschild redshift and velocity equivalent:
$$z_{\text{GR},B} = \left(1 - 2\mathcal{C}_B\right)^{-1/2} - 1 = 2.6918805 \times 10^{-4}, \qquad v_{\text{GR}} = c\, z_{\text{GR},B} = 80.700550\ \text{km/s}.$$
The parameter uncertainty is $\sigma_{\text{GR}} = v_{\text{GR}}\sqrt{(\sigma_M/M_B)^2 + (\sigma_R/R_B)^2} = 1.4080$ km/s, which combined in quadrature with the observational uncertainty $\sigma_{\text{obs}} = 0.77$ km/s yields $\sigma_{\text{comb}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{GR}}^2} = 1.604806$ km/s (`clock_coupling_H0_H1.csv`, row 2).
2. **Sirius B–Sirius A Differential H1 Clock Shift (`covariant_closure_50aa2db0.py`, lines 767–773):** Because the astrometric gravitational redshift is measured from the spectral-line difference between Sirius B and Sirius A, and under H1 the surface redshift is $1 + z_{\text{H1}} = (1 + z_{\text{metric}})\, f_L(R_*)^{-1} = (1 + z_{\text{metric}})\, e^{+x_*}$ (where $x_* \equiv \gamma_L \bar{\rho}_L(R_*)$), the direct H1 excess velocity between the two stars is:
$$\boxed{\Delta v_{\text{H1,extra}}(L) = \Delta v_B(L) - \Delta v_A(L) = c\,(1 + z_{\text{GR},B})\big(e^{x_B(L)} - 1\big) - c\,\big(e^{x_A(L)} - 1\big).}$$
   - At $L = 1000$ km $= 10^6$ m: $\Delta v_{\text{H1,extra}}^{\text{top-hat}} = +0.000392$ km/s and $\Delta v_{\text{H1,extra}}^{\text{Yuk}} = +0.044215$ km/s.

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- At $L = 5000$ km $= 5 \times 10^6$ m: $\Delta v_{\mathrm{H1,extra}}^{\text{top-hat}} = +4.688557$ km/s ($v_{\mathrm{H1}}^{\text{top-hat}} = 85.38911$ km/s, 2.95σ) and $\Delta v_{\mathrm{H1,extra}}^{\mathrm{Yuk}} = +2.150821$ km/s ($v_{\mathrm{H1}}^{\mathrm{Yuk}} = 82.85137$ km/s, 1.37σ).

3. **Difference Between Fixed-Mass (`fixedM`) and the Common Effective-Mass Proxy (`common_background`) (`covariant_closure_50aa2db0.py`, lines 758–766 and 794–798; V23 line 188):**

   - If the gravitating mass $M_B = 1.018\,M_\odot$ of the star is held fixed from orbital astrometry (`fixedM`), the exterior Schwarzschild metric is unchanged, so $v_{\mathrm{H0,fixedM}} \equiv v_{\mathrm{GR}} = 80.70055$ km/s, and $v_{\mathrm{H1,fixedM}}(L) = v_{\mathrm{GR}} + \Delta v_{\mathrm{H1,extra}}(L)$ (80.70094 km/s at $L = 10^3$ km).
   - If instead one evaluates the `mass`-mode effective mass reduction at a fixed $n = 3$ Lane–Emden baryonic profile $\rho_B(r)$ (`common_background` proxy):

$$\frac{\Delta M_{\mathrm{eff}}(L)}{M_B} \;=\; \frac{\int_0^{R_B} 4\pi r^2 \rho_B(r)\,\big[f_L(r) - 1\big]\,dr}{\int_0^{R_B} 4\pi r^2 \rho_B(r)\,dr} \;<\; 0,$$

   then the H0 prediction is $v_{\mathrm{H0,mass}}(L) = v_{\mathrm{GR}}\big(1 + \frac{\Delta M_{\mathrm{eff}}(L)}{M_B}\big)$ (80.69839 km/s at $L = 10^3$ km and 80.69487 km/s at $L = 5 \times 10^3$ km), while the H1 value built on that same common effective-mass proxy is:

$$v_{\mathrm{H1,common}}(L) \;=\; v_{\mathrm{H0,mass}}(L) \;+\; \Delta v_{\mathrm{H1,extra}}(L) \;=\; 80.698385 + 0.000392 \;=\; 80.69878\ \text{km/s} \quad (\text{at } L = 10^3\ \text{km, Top-H}$$

20.4 4. Checks

1. **Ratio of Top-Hat and Yukawa ϕ_L(IN) at Large L ($L = 10.0$ m $\gg d_{\mathrm{max}}^{\mathrm{IN}} = 1.125$ m):**

   - When $L \gg d_{\mathrm{max}}^{\mathrm{IN}}$, the entire mass M_{tot} lies inside the ball of volume $\frac{4}{3}\pi L^3$ in the 3D spherical top-hat average, so $\phi_L^{\text{top-hat}}(\mathrm{IN}) = \frac{GL^2}{c^2}\frac{3M_{\mathrm{tot}}}{4\pi L^3} = \frac{3GM_{\mathrm{tot}}}{4\pi c^2 L} = 1.517259 \times 10^{-25}$ (at $L = 10$ m), whereas in the Yukawa integral $e^{-d_{\mathrm{in}}/L} \approx 1 - d_{\mathrm{in}}/L$, yielding:

$$\phi_L^{\mathrm{Yuk}}(\mathrm{IN})\Big|_{L \gg d_{\mathrm{max}}^{\mathrm{IN}}} \;=\; \frac{G}{4\pi c^2}\int_{\Omega_{\mathrm{granite}}} \frac{e^{-d_{\mathrm{in}}/L}}{d_{\mathrm{in}}}\,dm_g \;\approx\; \frac{1}{4\pi}\left|\frac{\Phi_N(\mathrm{IN})}{c^2}\right| - \frac{GM_{\mathrm{tot}}}{4\pi c^2 L} \;=\; 7.3757\times10^{-25} - 0.5058\times10^{-25} \;\approx\; 6.87\times$$

   in close agreement with the exact quadrature value (6.888137×10^{-25}) in row 13 of `clock_coupling_H0_H1.csv` ✓.

2. **Verification of the Sirius B `common_background` vs. `fixedM` Difference at $L = 10^3$ km:**

   - $v_{\mathrm{H1,fixedM}}^{\text{top-hat}} - v_{\mathrm{H1,common}}^{\text{top-hat}} = 80.70094147 - 80.69877704 = +0.00216443$ km/s $\equiv v_{\mathrm{GR}} - v_{\mathrm{H0,mass}}$ ($80.70054975 - 80.69838532 = +0.00216443$ km/s) to machine precision ✓.

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21 Step 20: Exact Derivation of the V20 Eight-Coil Elliptic Biot–Savart Loop Integral, the $r = L/t$ Compactified Exterior Quadrature (3.7×10^{-16} / 7.4×10^{-16} Agreement), and the $L = 0.95$ m Switching Identity

Source file: step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-26
File Path: scalar-tensor-yukawa-fit/step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrat
Previous Steps: step1_dilaton_form_en.md (step1_dilaton_form.md) – step19_in_out_differential_clock_quadrature (step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy.md)
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Local Archival Sources:

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 10, point 2, lines 470–485 and Appendix, lines 557–618; DOI: 10.5281/zenodo.23604423)
- V20 (10.5281/zenodo.23540741) verification script and dataset: /workspace/ta-v18/V20/make_revision3_checks.py (lines 1–77) and /workspace/ta-v18/V20/revision3_checks.json (lines 123–197)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure (line 28; DOI: 10.5281/zenodo.23639084)

21.1 1. Verbatim Citations from the V22 and V20 Manuscripts

In Step 15 (step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20.md), we derived why the trace-free Maxwell tensor generates a source $\rho_B(\mathbf{x}) = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0 c^2}$ in the V23 disformal metric, and in Step 19 we closed the 3D IN–OUT quadrature of V20 Branch "A" (8558 kg granite chamber). Point 2 of Chapter 10 (lines 470–485) and the Appendix (lines 557–618) of the V22 monograph, together with V20 make_revision3_checks.py (lines 25–75), contain the exact elliptic-integral magnetic-field evaluation of V20 Branch "B" (the eight-coil magnetic-energy experiment), the two exterior quadrature paths (Path A and Path B), and the $L = 0.95$ m interior/exterior switching identity:

1. **Parameters of the 8-Coil Arrangement and the $L = 0.630957$ m Grid Maximum (tal_book_HU_2026-10-24.tex, Chapter 10, point 2, lines 473–478):** > *"**Audited 8-Coil Hardware Parameters:** 8 identical solenoids, each of length $l_{coil} = 400$ mm, outer structural ∅300 **mm flange diameter**, effective ∅291.7 **mm copper winding diameter** ($R_{coil} = 0.14585$ m), $N = 2000$ turns of 0.15 mm enameled copper wire, DC resistance $R_{DC} = 1789\,\Omega$ per coil, driven by a current $I = 14.0$ mA (0.351 W/coil, 2.805 W in all)." > "**Maximum on the Sampled 25-Point L-Grid ($L = 0.630957$ m):** Evaluating the magnetic energy density of the 8 coils on the 25-point logarithmic grid $L \in [0.1, 3.0]$ m, the conditional polarity-even shift $\Delta y(L) = \frac{1}{2}[\Delta y(+I) + \Delta y(-I)]$ attains its discrete maximum on the grid at $L = 0.630957$ **m** (not a continuous extremum fit):*

$$\Delta y_{\text{grid max}}(L = 0.630957\,\text{m}) \;=\; -1.2017108 \times 10^{-48}. \quad \text{(V22 Eq. 14)}$$

> Here I separate: (i) the 3.7×10^{-16} agreement of the two deterministic integration routines at $L = 3.0$ m, (ii) the −0.0722% Sobol quadrature cross-check, (iii) the 1.23×10^{-3} matching residual at $L = 0.95$ m, and (iv) the actual measurement sensitivity."

2. **The $r = L/t$ Compactified Exterior Integral (Paths A and B) and the $L = 0.95$ m Interior/Exterior Switch (tal_book_HU_2026-10-24.tex, Appendix, lines 581 and 615):**
   - Line 581: > *"The new Path A uses panel integration plus an analytic tail to infinity, while Path B uses the direct substitution $r = L/t$. In the 2.6–3.01 m range, the targeted rerun shows the two paths to agree to a **relative precision of about** 10^{-15}."*
   - Line 615: > *"**B. The new, smaller matching residual is reported and genuine.** The report already notes that at $L = 0.95$ m the interior/exterior switch is not strictly continuous. The archived values are 2.17738947 **and** 2.17597607 **J/A²**, with difference 0.00141340 **J/A²**, i.e. 0.0649% of the interior value. This agrees algebraically with the difference of the total-energy estimates; it is not a persistence of the old large 3 m error."*

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21.2 2. Exact Elliptic Biot–Savart Field of Circular Current Loops and the 8-Coil Superposition

21.2.1 2.1. Off-Axis Magnetic Field of a Single Circular Current Loop of Radius $a = D_{\mathbf{turn}}/2$ (`make_revision3_checks.py`, Lines 30–35)

Consider a circular current loop of radius $a = D_{\mathrm{turn}}/2 = 0.2917/2 = 0.14585$ m lying in the xy-plane and carrying current I. In cylindrical coordinates ($\varrho = \sqrt{x^2+y^2}$, z), the azimuthal vector potential $A_\varphi(\varrho, z)$ from the Biot–Savart law is:

$$A_\varphi(\varrho, z) \;=\; \frac{\mu_0 I\, a}{4\pi} \int_0^{2\pi} \frac{\cos\phi'\, d\phi'}{\sqrt{a^2+\varrho^2+z^2-2a\varrho\cos\phi'}}.$$

With the substitution $\phi' = \pi - 2\theta$ ($\cos\phi' = 2\sin^2\theta - 1$) and the elliptic parameter (`make_revision3_checks.py`, lines 31–32):

$$m(\varrho, z; a) \;\equiv\; \frac{4a\varrho}{(a+\varrho)^2+z^2} \;\in\; [0,1), \qquad d(\varrho, z; a) \;\equiv\; \sqrt{(a+\varrho)^2+z^2}, \qquad q(\varrho, z; a) \;\equiv\; (a-\varrho)^2+z^2,$$

the components of $\mathbf{B} = \nabla \times (A_\varphi \hat{\boldsymbol{\varphi}}) = -\frac{\partial A_\varphi}{\partial z}\hat{\boldsymbol{\varrho}} + \frac{1}{\varrho}\frac{\partial(\varrho A_\varphi)}{\partial \varrho}\hat{\mathbf{z}}$ can be expressed in closed form in terms of the elliptic integrals of the first and second kind, $K(m) = \int_0^{\pi/2}(1 - m\sin^2\theta)^{-1/2} d\theta$ and $E(m) = \int_0^{\pi/2}(1 - m\sin^2\theta)^{1/2} d\theta$ (`make_revision3_checks.py`, lines 33–34):

$$\boxed{\begin{aligned} B_z(\varrho, z; a, I) \;&=\; \frac{\mu_0 I}{2\pi\, d(\varrho, z; a)} \left[K(m) \;+\; \frac{a^2-\varrho^2-z^2}{q(\varrho, z; a)}\, E(m) \right], \\ B_\varrho(\varrho, z; a, I) \;&=\; \frac{\mu_0 I\, z}{2\pi\, \varrho\, d(\varrho, z; a)} \left[-K(m) \;+\; \frac{a^2+\varrho^2+z^2}{q(\varrho, z; a)}\, E(m) \right] \qquad (\varrho > 0,\;\; B_\varrho(0, z) = 0). \end{aligned}}$$

21.2.2 2.2. Superposition of the $N_c = 8 \times N = 2000$ Loops and the Derby–Olbert Current-Sheet Limit (`make_revision3_checks.py`, Lines 25–51)

In the V20 layout, the centers of the $N_c = 8$ solenoids lie on a circle of radius $R_{\mathrm{ax}} = 0.45$ m at vertical coordinate $z_c = 0.200 - 0.300 = -0.100$ m:

$$\mathbf{c}_k \;=\; \left(R_{\mathrm{ax}} \cos\frac{2\pi k}{8},\; R_{\mathrm{ax}} \sin\frac{2\pi k}{8},\; z_c \right), \qquad k = 0, 1, \ldots, 7.$$

Each coil has length $\ell = 0.400$ m and contains $N = 2000$ turns at midpoint positions $z_n = \left(\frac{n+1/2}{N} - \frac{1}{2}\right)\ell$ ($n = 0, \ldots, N-1$). At an arbitrary point $\mathbf{P} = (x, y, z)$, the net magnetic field is:

$$\mathbf{B}_{\mathrm{loops}}(\mathbf{P}; I_0) \;=\; \sum_{k=0}^{7} \sum_{n=0}^{N-1} \left[B_\varrho(\varrho_k, z - z_c - z_n; a, I_0)\, \frac{(x-c_{k,x})\hat{\mathbf{x}} + (y-c_{k,y})\hat{\mathbf{y}}}{\varrho_k} \;+\; B_z(\varrho_k, z - z_c - z_n; a, I_0)\, \hat{\mathbf{z}} \right],$$

where $\varrho_k = \sqrt{(x-c_{k,x})^2 + (y-c_{k,y})^2}$. As $N \to \infty$ at fixed surface current density NI_0/ℓ, the midpoint sum $\frac{1}{N}\sum_{n=0}^{N-1} f(z_n) = \frac{1}{\ell}\int_{-\ell/2}^{+\ell/2} f(z')\, dz' + \mathcal{O}(N^{-2})$ converges to second order to the continuous Derby–Olbert cylindrical current-sheet integral (`fields.coils_B`). Accordingly, at $I_0 = 14.0$ mA, the two independent evaluations (`revision3_checks.json`, lines 124–136) give:

- **At the Main Clock ($\mathbf{P}_{\mathbf{main}} = (0, 0, 0)$):** $|\mathbf{B}_{\mathrm{loops}}| = 11.63290030\ \mu$T, $|\mathbf{B}_{\mathrm{sheet}}| = 11.63289980\ \mu$T (relative difference -4.28×10^{-8}).
- **At the Reference Clock ($\mathbf{P}_{\mathbf{ref}} = (1.60, 0, 0)$ m):** $|\mathbf{B}_{\mathrm{loops}}| = 0.430539231\ \mu$T, $|\mathbf{B}_{\mathrm{sheet}}| = 0.430539228\ \mu$T (relative difference -7.56×10^{-9}).

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21.3 3. Differential Magnetic Energy ($\Delta W(L)$), the $r = L/t$ Compactified Exterior Quadrature, and the $L = 0.95$ m Identity

21.3.1 3.1. Expression of the Polarity-Even Fractional Frequency Shift in Terms of $\Delta W(L)$ (`make_revision3_checks.py`, Lines 53–58)

By definition, the magnetic energy normalized to unit current ($I = 1$ A) inside a ball $B(\mathbf{P}, L)$ of radius L centered at $\mathbf{P}$ is:

$$W_{\rm in}(\mathbf{P}; L) \;\equiv\; \frac{1}{2\mu_0 I_0^2} \int_{|\mathbf{x}-\mathbf{P}|\le L} |\mathbf{B}(\mathbf{x}; I_0)|^2 \, d^3x \qquad [\mathrm{J/A^2}],$$

and the differential enclosed energy between the main clock ($\mathbf{P}_{\rm main} = \mathbf{0}$) and the reference clock ($\mathbf{P}_{\rm ref}$) is $\Delta W(L) \equiv W_{\rm in}(\mathbf{P}_{\rm main}; L) - W_{\rm in}(\mathbf{P}_{\rm ref}; L)$. Because the 3D top-hat averaged magnetic mass-density difference is $\Delta\bar{\rho}_{B,L} = \frac{I_0^2\,\Delta W(L)}{c^2\,V_L}$ (where $V_L = \frac{4}{3}\pi L^3$), the conditional H1 clock-shift sensitivity coefficient ($k_g(L)$, [$\mathrm{A^{-2}}$]) and the signal at $I_0 = 14$ mA (`make_revision3_checks.py`, lines 55–57; `revision3_checks.json`, lines 139–145) are:

$$\boxed{k_g(L) \;=\; -\frac{GL^2}{c^2}\frac{\Delta W(L)}{c^2\,\frac{4}{3}\pi L^3} \;=\; -\frac{3G\,\Delta W(L)}{4\pi c^4 L}, \qquad \Delta y(L) \;=\; k_g(L)\, I_0^2.}$$

At the grid peak on the 25-point logarithmic grid ($L_p = 10^{-0.2}$ m $= 0.6309573444801932$ m):

$$\Delta W(L_p) \;=\; 1.9611445207162488\ \mathrm{J/A^2} \quad\Longrightarrow\quad k_g(L_p) \;=\; -6.131177672606294\times10^{-45}\ \mathrm{A^{-2}},$$
$$\Delta y(L_p) \;=\; k_g(L_p)\times(0.014\ \mathrm{A})^2 \;=\; -1.2017108238308338\times10^{-48}.$$

21.3.2 3.2. Why Is the Exterior Complementary Integral Advantageous at Large L, and How Does the $r = L/t$ Compactification Work?

When $L > L_{\rm jump} = 0.95$ m is large enough that both the main-clock ball $B(\mathbf{P}_{\rm main}, L)$ and the reference-clock ball $B(\mathbf{P}_{\rm ref}, L)$ enclose the intense near-coil zone, directly subtracting the two interior integrals ($W_{\rm in,main}(L) - W_{\rm in,ref}(L)$) would subtract two nearly equal numbers. In Euclidean space, however, the magnetic energy integrated over all of $\mathbb{R}^3$ is **translation-invariant**:

$$W_{\rm tot} \;\equiv\; \frac{1}{2\mu_0 I_0^2}\int_{\mathbb{R}^3} |\mathbf{B}(\mathbf{x}; I_0)|^2\, d^3x \;=\; W_{\rm in}(\mathbf{P}_{\rm main}; L) + W_{\rm ext}(\mathbf{P}_{\rm main}; L) \;=\; W_{\rm in}(\mathbf{P}_{\rm ref}; L) + W_{\rm ext}(\mathbf{P}_{\rm ref}; L),$$

where $W_{\rm ext}(\mathbf{P}; L) \equiv \frac{1}{2\mu_0 I_0^2}\int_{|\mathbf{x}-\mathbf{P}|\ge L} |\mathbf{B}(\mathbf{x}; I_0)|^2\, d^3x$ is the exterior (complementary) energy outside the ball. Because the exact total energy cancels ($W^{\rm exact}_{\rm tot,main} - W^{\rm exact}_{\rm tot,ref} \equiv 0$), the interior energy difference equals the negative difference of the smooth, singularity-free exterior integrals:

$$\boxed{\Delta W_{\rm out}(L) \;\equiv\; -\Big[W_{\rm ext}(\mathbf{P}_{\rm main}; L) \;-\; W_{\rm ext}(\mathbf{P}_{\rm ref}; L)\Big] \;=\; \frac{1}{2\mu_0 I_0^2}\int_{r\ge L} r^2\, dr \int_{S^2} d\Omega \left[|\mathbf{B}(\mathbf{P}_{\rm ref} + r\hat{\mathbf{n}})|^2 - |\mathbf{B}(\mathbf{P}_{\rm main} + r\hat{\mathbf{n}})|^2\right].}$$

V20 `energy_integrator.py` evaluates this exterior integral over the half-line $[L, \infty)$ along two independent paths (`make_revision3_checks.py`, lines 62–68; `tal_book_HU_2026-10-24.tex`, line 581):

- **Path "B" (`dW_B`): Exact $r = L/t$ Compactification Without Finite Cutoff.** Introducing the dimensionless variable $t \equiv L/r \in (0, 1]$, the differential is $dr = -\frac{L}{t^2}\,dt$ and the volume measure is $r^2\,dr = -\frac{L^3}{t^4}\,dt$. Thus the infinite domain $[L, \infty)$ maps onto the compact interval $(0, 1]$:

$$\boxed{\Delta W_{\rm B}(L) \;=\; \frac{L^3}{2\mu_0 I_0^2}\int_0^1 \frac{dt}{t^4}\int_{S^2} d\Omega \left[\left|\mathbf{B}(\mathbf{P}_{\rm ref} + \tfrac{L}{t}\hat{\mathbf{n}})\right|^2 \;-\; \left|\mathbf{B}(\mathbf{P}_{\rm main} + \tfrac{L}{t}\hat{\mathbf{n}})\right|^2\right].}$$

  Let us inspect the behavior of the integrand as $t \to 0^+$ ($r = L/t \to \infty$): because the net charge of the 8 closed current loops is zero, the far magnetic field is dipolar at leading order: $|\mathbf{B}(r\hat{\mathbf{n}})| = \mathcal{O}(r^{-3}) = \mathcal{O}(t^3/L^3)$, so the energy density is $|\mathbf{B}|^2 = \mathcal{O}(t^6/L^6)$, and the difference between the two shifted balls is $|\mathbf{B}(\mathbf{P}_{\rm ref} + \frac{L}{t}\hat{\mathbf{n}})|^2 - |\mathbf{B}(\mathbf{P}_{\rm main} + \frac{L}{t}\hat{\mathbf{n}})|^2 = \mathcal{O}(t^7/L^7)$. Multiplied by the Jacobian factor $\frac{L^3}{t^4}$, the integrand vanishes at $t \to 0^+$ **as a regular polynomial ($\mathcal{O}(t^2)$–$\mathcal{O}(t^3)$)**! Consequently, $n_t = 32$ (or 64) point Gauss–Legendre quadrature on $(0, 1]$ achieves machine precision while simultaneously proving the far-field scaling:

$$\Delta W(L) \;\propto\; L^{-3}\ (\text{or } L^{-5} \text{ under quadrupole/shift symmetry}) \quad\Longrightarrow\quad |\Delta y(L)| \;\propto\; \frac{\Delta W(L)}{L} \;\propto\; L^{-4}\text{–}L^{-6}.$$

- **Path "A" (`dW_A`): Panel Integral + Analytic Asymptotic Multipole Tail.** At $L = 3.0$ m, the relative difference between Path "A" (`dW_A = 0.0023313149428584538 J/A^2`) and Path "B" (`dW_B = 0.0023313149428584555 J/A^2`) is -7.44×10^{-16} (`revision3_checks.json`, lines 164–165; the 3.7×10^{-16} agreement of the two deterministic routines in `tal_book_HU_2026-10-24.tex`, line 478).

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21.3.3 3.3. Exact Algebraic Identity of the $L_{\mathbf{jump}} = 0.95$ m Interior/Exterior Switching Jump (`make_revision3_checks.py`, Lines 70–74)

When the interior and exterior quadratures evaluated at finite angular resolution $n_{\mathrm{hi}} = 24000$ are added together, the numerical total energies centered on the two clock positions (`revision3_checks.json`, lines 186–195) are:

$$W^{\mathrm{num}}_{\mathrm{tot,main}} \;=\; W^{\mathrm{num}}_{\mathrm{in,main}}(L) + W^{\mathrm{num}}_{\mathrm{ext,main}}(L) \;=\; 2.237861258956951\ \mathrm{J/A^2},$$

$$W^{\mathrm{num}}_{\mathrm{tot,ref}} \;=\; W^{\mathrm{num}}_{\mathrm{in,ref}}(L) + W^{\mathrm{num}}_{\mathrm{ext,ref}}(L) \;=\; 2.2364478629240048\ \mathrm{J/A^2}.$$

Subtracting the two equations yields an exact algebraic identity valid for every L (including the switching point $L_{\mathrm{jump}} = 0.95$ m):

$$\boxed{\Delta W^{\mathrm{num}}_{\mathrm{in}}(0.95\ \mathrm{m}) \;-\; \Delta W^{\mathrm{num}}_{\mathrm{out}}(0.95\ \mathrm{m}) \;\equiv\; W^{\mathrm{num}}_{\mathrm{tot,main}} \;-\; W^{\mathrm{num}}_{\mathrm{tot,ref}} \;=\; 0.001413396032946\ \mathrm{J/A^2}}$$

($2.17738947 - 2.17597607 = 0.00141340$ J/A^2, i.e., 0.06491% of the interior value; the algebraic difference check residual is `identity_check_abs` $= 8.88 \times 10^{-16}$ J/A^2).

21.4 4. Checks

1. **Consistency of the DC Coil Resistance with the Measured Winding Diameter $D_{\mathbf{turn}} = 291.7$ mm (`tal_book_HU_2026-10-24.tex`, Line 577):**
   - For $N = 2000$ turns of copper wire of diameter $d_{\mathrm{wire}} = 0.15$ mm ($A_{\mathrm{wire}} = \frac{\pi}{4} d^2_{\mathrm{wire}} = 1.767146 \times 10^{-8}$ m^2, $\rho_{\mathrm{Cu}}(20°\mathrm{C}) = 1.7241 \times 10^{-8}\ \Omega\,\mathrm{m}$) and mean winding diameter $D_{\mathrm{turn}} = 0.2917$ m: $L_{\mathrm{wire}} = \pi D_{\mathrm{turn}} N = 1832.805$ m, giving $R_{\mathrm{DC,calc}} = 1788.160\ \Omega$, which differs by only -0.0470% from the nominal $1789\ \Omega$ ✓.
2. **Verification of the Peak Signal via Formula (1) at $L_p = 0.63095734448$ m:**
   - $k_g(L_p) = -\frac{3\times 6.67430\times 10^{-11}\times 1.961144520716}{4\pi\times(299792458)^4\times 0.63095734448} = -6.1311776726 \times 10^{-45}\ \mathrm{A^{-2}}$, and $\Delta y(L_p) = k_g(L_p) \times 0.014^2 = -1.2017108238 \times 10^{-48}$, reproducing the main V20/V22 result ✓.

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22 Step 21: Exact Derivation of the 7-Segment Piecewise-Polytrope Log-Enthalpy (h) TOV Transformation, the $s = r^2$ Even-Parity Central Regularization (`Audit Fix 01`), and the `gradient`-Mode Exterior Vacuum-Halo Lapse Integral ($\Delta\nu_{\mathbf{ext}}$) over $r \in [R_*, R_* + L]$

Source file: `step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-27
File Path: `scalar-tensor-yukawa-fit/step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_`
Hungarian Original: `step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse.md`
Previous Steps: `step1_dilaton_form_en.md` – `step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_c`
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Local Archival Sources:

- V22 Book Edition: `/workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex` (Chapter 7, lines 102–120; DOI: `10.5281/zenodo.23604423`)
- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Section 1.3 and Table 1, lines 109–156; DOI: `10.5281/zenodo.23639084`)
- V21/V23 profile and TOV integrator: `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_prof` (lines 14–108) and `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/stellar_tal_v21.py` (lines 33–136; DOI: `10.5281/zenodo.23602501`)

22.1 1. Verbatim Citations from the V22 and V23 Manuscripts and the `tal_profiles.py` Module

In Steps 7, 16, 17, and 18 we used the V21/V23 neutron-star equilibrium profiles (`dense_profile`, SLy, APR4, and H4 equations of state, continuum Bianchi residuals 1.82×10^{-11} and 1.59×10^{-11}), without yet deriving the three analytical transformations in `stellar_tal_v21.py` and `tal_profiles.py` that guarantee this accuracy and central regularity:

1. **The 3 Equations of State (SLy, APR4, H4), the Causal $c_s = c$ Bound, and the Two Conditional TOV Modes (`hu_p2.tex`, Chapter 7, lines 108–115):**

   (A) `mass` mode: $\dfrac{dm_{\rm eff}}{dr} = 4\pi r^2 \rho(r)\, f_L(r),$ **(B) `gradient` mode:** $\dfrac{d\Phi_{\rm eff}}{dr} = f_L(r)\, \dfrac{G\big(m(r) + 4\pi r^3 P(r)/c^2\big)}{c^2 r\big(r - 2Gm(r)/c^2\big)},$ (V22 C

   > *"I integrated 33 equilibrium sequences across three nuclear equations of state — **SLy** (nucleonic), **APR4** (variational nucleonic; reaches the causal sound-speed threshold $c_s = c$ at rest-mass density $\rho_0 \approx 1.4458 \times 10^{15}$ **g/cm**^{3}), and **H4** (stiff hyperonic) — across 11 logarithmically spaced smoothing scales $L \in [1\,m, 100\,km]$. The V21 archival baseline table contains **182 equilibrium models**, whereas the dense diagnostic stability grid comprises **769 stellar models** in total (including all **33 exact causal endpoints** where $c_s/c = 1$)."*

2. **The $s = r^2$ Even-Parity Central Mapping and the `gradient`-Mode Exterior Lapse Integral (`tal_profiles.py`, lines 53–56 and 91–93):**

   - Lines 53–56: > *"# Profile mapping, regular at the centre (audit fix 01, run 2): h and mu = m/r^3 are even, analytic functions of r, so they are interpolated in s = r^2 with their exact centre values h(0)=hc, mu(0)=(4pi/3) eps_c f_m. Then h'(r)=2r dh/ds -> 0, m=r^3 mu, A=r(mu+4pi P)/(1-2 mu r^2) are regular. (Run 1 interpolated h(r), m(r) in r with a single node at r=0 -> wrong slopes near the centre.)"*
   - Lines 91–93: > *"def exterior_lapse(cur, L, M, R, nq=160): \"\"\"nu(R) = -int_R^inf f(r) M/(r(r-2M)) dr with f=exp(-L^2 rhobar_L) from the converged profile; f=1 beyond R+L. Returns nu(R) and diagnostic dict.\"\"\""*

3. **Continuum Bianchi Conservation Residuals in V23 Table 1 (`tal_covariant_closure_50aa2db0_HU_2026-10-24.t` lines 144–147):**

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- `tophat_mass`: 1.82×10^{-11}, `tophat_gradient`: 1.59×10^{-11} (exact symbolic residual `0.0`).

22.2 2. Read et al. (2009) 7-Segment Piecewise-Polytrope Log-Enthalpy (h) Closure and the h-Transformation of the TOV System

22.2.1 2.1. Thermodynamic Consistency, a_i Continuity Recursion, and the $c_s^2/c^2 = 1$ Causal Threshold (`stellar_tal_v21.py`, lines 33–69)

The Read et al. (2009) parameterization consists of 4 crust segments $(\Gamma_0, \ldots, \Gamma_3)$ and 3 core segments $(\Gamma_4, \Gamma_5, \Gamma_6 = g_1, g_2, g_3$, with dividing densities $\rho_{0,4} = \rho_{\rm match}$, $\rho_{0,5} = 10^{14.7}$ g/cm^3, $\rho_{0,6} = 10^{15}$ g/cm^3). In the i-th segment $(\rho_0 \in [\rho_{0,i-1}, \rho_{0,i}])$, the pressure (in c^2-scaled CGS units) is $P(\rho_0) = K_i \rho_0^{\Gamma_i}$. Pressure continuity at the segment boundaries enforces:

$$K_{i+1} \;=\; K_i\, \rho_{0,i}^{\Gamma_i - \Gamma_{i+1}}, \qquad \rho_{\rm match} \;=\; \left(\frac{K_3}{K_4}\right)^{\frac{1}{\Gamma_4 - \Gamma_3}}.$$

Integrating the first law of thermodynamics at constant entropy per baryon ($d(\varepsilon/\rho_0) = -P\, d(1/\rho_0) = \frac{P}{\rho_0^2} d\rho_0$) yields the total mass-energy density (`stellar_tal_v21.py`, lines 47–49 and 67):

$$\varepsilon(\rho_0) \;=\; (1 + a_i)\,\rho_0 \;+\; \frac{P(\rho_0)}{\Gamma_i - 1}, \qquad a_{i+1} \;=\; a_i \;+\; \frac{K_i}{\Gamma_i - 1}\rho_{0,i}^{\Gamma_i - 1} \;-\; \frac{K_{i+1}}{\Gamma_{i+1} - 1}\rho_{0,i}^{\Gamma_{i+1} - 1} \quad (a_0 = 0).$$

The specific relativistic log-enthalpy ($dh \equiv \frac{dP}{\varepsilon + P}$) can then be integrated and inverted analytically (`stellar_tal_v21.py`, lines 50 and 60–68):

$$\boxed{h(\rho_0) \;=\; \ln\!\left(\frac{\varepsilon + P}{\rho_0}\right) \;=\; \ln\!\left(1 + a_i + \frac{\Gamma_i}{\Gamma_i - 1} K_i\, \rho_0^{\Gamma_i - 1}\right) \iff \rho_0(h) \;=\; \left[\frac{\left(e^h - 1 - a_i\right)(\Gamma_i - 1)}{\Gamma_i K_i}\right]^{\frac{1}{\Gamma_i - 1}}.}$$

The squared adiabatic sound speed in each segment is:

$$\frac{c_s^2(h)}{c^2} \;=\; \left(\frac{\partial P}{\partial \varepsilon}\right)_s \;=\; \frac{\Gamma_i\, P(h)}{\varepsilon(h) + P(h)} \;=\; (\Gamma_i - 1)\,\frac{e^h - 1 - a_i}{e^h}.$$

Because $\Gamma_i > 2$ in the high-density core (for APR4, $\Gamma_5 = 3.445$, $\Gamma_6 = 3.348$), $c_s^2(h)/c^2$ increases strictly monotonically with h, and the causal threshold ($c_s^2/c^2 = 1$) is located by `brentq` to 10^{-15} accuracy at $h_{\rm causal}$ (for APR4, $\rho_0(h_{\rm causal}) \approx 1.4458 \times 10^{15}$ g/cm^3, `hu_p2.tex`, line 115).

22.2.2 2.2. Rewriting the TOV Equations with h as Independent Variable and Central Taylor Startup (`tal_profiles.py`, lines 38–50)

Because the hydrostatic balance equation is $P'(r) = (\varepsilon + P)h'(r) = -(\varepsilon + P) f_{\rm grad}(r) A_{\rm GR}(r)$, the $(\varepsilon + P)$ factor cancels identically:

$$\frac{dh}{dr} \;=\; -f_{\rm grad}(r)\, A_{\rm GR}(r) \;=\; -f_{\rm grad}(r)\,\frac{m(r) + 4\pi r^3 P(h)}{r\big(r - 2m(r)\big)},$$

where $f_{\rm grad}(r) = f_L(r)$ in `gradient` mode and 1 in `mass` (as well as GR) mode, while $f_{\rm mass}(r) = f_L(r)$ in `mass` mode and 1 in `gradient` mode. Since $h(r)$ decreases strictly monotonically inside the star from central h_c down to surface $h(R_*) = 0$, the ODE system for $(r(h), m(h))$ reads (`tal_profiles.py`, lines 43–48):

$$\boxed{\frac{dr}{dh} \;=\; -\frac{r(r - 2m)}{\big(m + 4\pi r^3 P(h)\big)\, f_{\rm grad}(r)}, \qquad \frac{dm}{dh} \;=\; 4\pi r^2 \varepsilon(h)\, f_{\rm mass}(r)\,\frac{dr}{dh}.}$$

Why is integrating in h advantageous? When r is the independent variable, the stellar surface $P(R_*) = 0$ must be found by event detection where $\varepsilon(P) \to 0$ varies steeply. With h as the independent variable, the surface is a fixed integration endpoint ($h_{\rm surf} = 10^{-10}$) at which $R_* = r(h_{\rm surf})$ and $M = m(h_{\rm surf})$ are read out directly without root-finding overshoot! To avoid the $r = 0$ coordinate singularity, the integration starts at $r_0 = 10^{-4}$ km using the second-order central Taylor expansion (`tal_profiles.py`, lines 41–42):

$$m(r_0) \;=\; \frac{4\pi}{3}\varepsilon_c f_m\, r_0^3 + \mathcal{O}(r_0^5), \qquad h(r_0) \;=\; h_c - \delta h, \qquad \delta h \;=\; \frac{2\pi}{3}(\varepsilon_c f_m + 3P_c)\, f_g\, r_0^2 + \mathcal{O}(r_0^4).$$

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22.3 3. Even-Parity $s = r^2$ Central Regularization (`Audit Fix O1`, `tal_profiles.py`, lines 53–79)

22.3.1 3.1. Why Direct Spline Interpolation in r Causes Central Singularities, and How $s = r^2$ Resolves It

Any spherically symmetric scalar field that is smooth at the origin ($C^\infty(\mathbb{R}^3)$) — such as the log-enthalpy $h(\mathbf{x})$, the mean interior mass density $\mu(\mathbf{x}) \equiv \frac{m(r)}{r^3}$, or the 3D top-hat exponent $x(\mathbf{x}) \equiv \gamma_L \bar{\rho}_L(r)$ — can contain only powers of $|\mathbf{x}|^2 = x^2 + y^2 + z^2 = r^2 \equiv s$ in its Taylor expansion, i.e., it is an **even function of** r:

$$h(r) \;=\; h_s(r^2), \qquad \mu(r) \;\equiv\; \frac{m(r)}{r^3} \;=\; \mu_s(r^2), \qquad x(r) \;=\; x_s(r^2),$$

with $h_s(0) = h_c$ and $\mu_s(0) = \frac{4\pi}{3}\varepsilon_c f_m$. If $h(r)$ and $m(r)$ are interpolated directly in r via PCHIP splines (`Run 1`), the cubic Hermite polynomial at the single node $r = 0$ enforces neither the exact value of $h''(0)$ nor $m'(0) = m''(0) = 0$, injecting numerical noise and $\mathcal{O}(1/r)$ errors into $\frac{m(r)}{r^3}$ and $\frac{h'(r)}{r}$ near the center!

By contrast, `tal_profiles.py` (lines 64–79, `Audit Fix O1`) constructs `PchipInterpolator` objects on the grid $s = r^2 \in [0, R_*^2]$:

$$h_s(s), \qquad \mu_s(s) \;=\; \frac{m(\sqrt{s})}{s^{3/2}} \quad (\mu_s(0) = \mu_c), \qquad x_s(s) \;=\; x(\sqrt{s}).$$

By the chain rule ($\frac{d}{dr} = 2r\frac{d}{ds}$), every central derivative and curvature factor is **analytically regular** as $r \to 0^+$ (`tal_profiles.py`, lines 66–79):

$$\begin{aligned}
h'(r) \;&=\; 2r\,\frac{dh_s}{ds}(r^2) \;=\; \mathcal{O}(r), & m(r) \;&=\; r^3\,\mu_s(r^2) \;=\; \mathcal{O}(r^3),\\
\frac{d\mu}{dr}(r) \;&=\; 2r\,\frac{d\mu_s}{ds}(r^2) \;=\; \mathcal{O}(r), & \frac{d\ln f_L}{dr}(r) \;&=\; -2r\,\frac{dx_s}{ds}(r^2) \;=\; \mathcal{O}(r),\\
A_{\rm GR}(r) \;&=\; \frac{m(r) + 4\pi r^3 P(r)}{r\big(r - 2m(r)\big)} \;=\; \frac{r\big(\mu_s(r^2) + 4\pi P(r)\big)}{1 - 2r^2\,\mu_s(r^2)} \;=\; \mathcal{O}(r).
\end{aligned}$$

Consequently, in the ratios $\frac{h'(r)}{r} = 2\frac{dh_s}{ds}(r^2)$ and $\frac{A_{\rm GR}(r)}{r} = \frac{\mu_s(r^2)+4\pi P(r)}{1-2r^2\mu_s(r^2)}$ entering the Chandrasekhar operator and the Bianchi identity, the factor r **cancels algebraically** at $r = 0$! This regularization reduces the continuum Bianchi conservation residual in V23 Table 1 to 1.82×10^{-11} (`tophat_mass`) and 1.59×10^{-11} (`tophat_gradient`).

22.4 4. `gradient`-Mode Exterior Vacuum-Halo Lapse Integral ($\Delta\nu_{\rm ext}$) over $r \in [R_*, R_* + L]$ and the Interior Bernoulli Integral (`tal_profiles.py`, lines 80–106)

22.4.1 4.1. Why the $f_L(r) < 1$ Domain Extends Beyond the Stellar Surface R_* out to $R_* + L$

Outside the star ($r > R_*$), the local matter density vanishes identically ($\varepsilon(r) = 0$, $P(r) = 0$), so $m(r) \equiv M$ and $A_{\rm GR}(r) = \frac{M}{r(r-2M)}$ for all $r \geq R_*$.

- **In `mass` mode (and in GR):** the exterior lapse gradient $\nu'(r) = A_{\rm GR}(r) = \frac{M}{r(r-2M)}$ carries no $f_L(r)$ factor. Thus the exterior spacetime is exact Schwarzschild geometry, and the surface lapse is (`tal_profiles.py`, line 81):

$$\nu_{\rm mass}(R_*) \;=\; \nu_{\rm GR}(R_*) \;=\; -\int_{R_*}^{\infty} \frac{M}{r(r-2M)}\,dr \;=\; \frac{1}{2}\ln\!\left(1 - \frac{2M}{R_*}\right).$$

- **In `gradient` mode:** the lapse gradient is $\nu'(r) = f_L(r)\,A_{\rm GR}(r)$ everywhere. Although $\varepsilon(r) = 0$ outside the star, a 3D top-hat ball $B(\mathbf{x}, L)$ of radius L centered at an exterior point $r \in (R_*, R_* + L)$ **still intersects the outer layer** $[r - L, R_*]$ **of the star** (via the spherical-cap intersection formula derived in Step 18)! Consequently, a **nonlocal vacuum-halo shell** of thickness L ($r \in [R_*, R_* + L]$) forms around the star, within which:

$$x(r) \;=\; L^2\bar{\rho}_L(r) \;=\; \frac{3}{4r\,L}\int_{r-L}^{R_*} s\,\varepsilon(s)\,\big[L^2 - (r-s)^2\big]\,ds \;>\; 0 \quad\Longrightarrow\quad f_L(r) \;=\; e^{-x(r)} \;<\; 1 \qquad (R_* \leq r < R_* + L),$$

and only for $r \geq R_* + L$ does $x(r) \equiv 0 \implies f_L(r) \equiv 1$!

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22.4.2 4.2. Derivation of the Finite-Range Integral $\Delta\nu_{\mathbf{ext}}$ and the Interior $\nu(r)$ Profile (`tal_profiles.py`, lines 86–105)

Integrating the `gradient`-mode equation $\nu'(r) = f_L(r)\, A_{\rm GR}(r)$ from asymptotically flat infinity ($\nu(\infty) = 0$) inward to the stellar surface $r = R_*$, and using $f_L(r) = 1 - (1 - e^{-x(r)})$ where $1 - e^{-x(r)} \equiv 0$ for all $r \geq R_* + L$, splits the improper integral into the analytical Schwarzschild value plus a finite correction over the compact interval $[R_*, R_* + L]$ (`tal_profiles.py`, lines 91–105):

$$\boxed{\begin{aligned}\nu_{\rm grad}(R_*) \;&=\; -\int_{R_*}^{\infty} e^{-x(r)}\,\frac{M}{r(r-2M)}\,dr \;=\; -\int_{R_*}^{\infty}\frac{M}{r(r-2M)}\,dr \;+\; \int_{R_*}^{R_*+L}\big(1-e^{-x(r)}\big)\,\frac{M}{r(r-2M)}\,dr \\ &=\; \frac{1}{2}\ln\!\left(1-\frac{2M}{R_*}\right) \;+\; \Delta\nu_{\rm ext}, \qquad \Delta\nu_{\rm ext} \;\equiv\; \int_{R_*}^{R_*+L}\big(-\,\mathrm{expm1}(-x(r))\big)\,\frac{M}{r(r-2M)}\,dr \;>\; 0,\end{aligned}}$$

which `exterior_lapse()` evaluates via 200-point Gauss–Legendre quadrature on $[R_*, R_* + L]$. In the stellar interior ($0 \leq r \leq R_*$), both `mass` and `gradient` modes satisfy the relativistic Bernoulli identity $\nu'(r) = -h'(r)$, yielding the interior lapse profile without any additional numerical quadrature (`tal_profiles.py`, lines 86–87):

$$\boxed{\nu(r) \;=\; \nu(R_*) \;-\; \big[h(r) - h(R_*)\big], \qquad \forall\, r \in [0, R_*].}$$

22.5 5. Checks

1. **Physical Verification of the Sign and Limits of $\Delta\nu_{\mathbf{ext}} > 0$:**
   - Because $x(r) > 0$ on $r \in [R_*, R_* + L)$ and $r > 2M$, the integrand $(1 - e^{-x(r)})\frac{M}{r(r-2M)} > 0$, so $\Delta\nu_{\rm ext} > 0$, meaning `gradient` mode softens the surface gravitational potential depth ($|\nu_{\rm grad}(R_*)| < |\nu_{\rm GR}(R_*)|$). As $L \to 0^+$, the integration interval length $L \to 0$ and $\Delta\nu_{\rm ext} \to 0$, continuously recovering the Schwarzschild limit ✓.
2. **Verification of the Derivative of the Interior Bernoulli Integral ($\nu(r) = \nu(R_*) - [h(r) - h(R_*)]$):**
   - In both modes, $\frac{d\nu}{dr} = -\frac{dh}{dr} = -2r\frac{dh_s}{ds}(r^2) = f_{\rm grad}(r) A_{\rm GR}(r)$, which gives $\nu'(0) = 0$ at $r = 0$ and $\nu(R_*)$ at $r = R_*$ ✓.

22.6 6. Explicit Comparison with the V23 Local Yukawa (`yukawa_action_el`, Branch II-Fluid) Branch (Point 5)

For audit clarity, we strictly separate the properties of the V21 phenomenological 3D top-hat branch (`tophat_mass`, `tophat_gradient` — **Branch I-A**) derived above from the V23 local Yukawa-screened Einstein–scalar–fluid branch (`yukawa_action_el` — **Branch II-Fluid**, see `step4`, `step7`, and `step22`):

Physical Property	V21 `tophat_mass` / `tophat_gradient` (**Branch I-A**, Step 21)	V23 `yukawa_action_el` (**Branch II-Fluid**, Steps 4, 7, and 22)
Interior Hydrostatic / Bernoulli Equation ($0 \leq r \leq R_*$)	$h'(r) = -\nu'(r) \implies \nu(r) = \nu(R_*) - [h(r) - h(R_*)]$ (f_L enters directly into $\nu' = f_L A_{\rm GR}$ or $m' = 4\pi r^2 f_L \varepsilon$)	$h'(r) = -\nu'(r) + \phi_L'(r) \implies \nu(r) - \phi_L(r) = \nu(R_*) - \phi_L(R_*) - [h(r) - h(R_*)]$ (distinct scalar fifth force ϕ_L')
Structure of Exterior Domain ($r > R_*$)	Compact-support 3D top-hat halo on $r \in [R_*, R_* + L]$; for $r \geq R_* + L$, $x(r) \equiv 0$, $f_L(r) \equiv 1$, exact Schwarzschild vacuum	Exponentially decaying gravitating scalar tail over $r \in [R_*, \infty)$: $\rho_\phi = \frac{1}{2\alpha_Y}[B(\phi_L')^2 + \phi_L^2/L^2] > 0$
Surface Mass ($M_{\rm surf}$) vs. ADM Mass ($M_{\rm ADM}$)	$m'(r > R_*) = 0 \implies M_{\rm ADM} \equiv M_{\rm surf} = m(R_*)$	$m'(r > R_*) = 4\pi r^2 \rho_\phi(r) > 0 \implies M_{\rm ADM} = M_{\rm surf} + \Delta M_{\phi,\rm ext}^{\rm curved} > M_{\rm surf}$ (see `step22`)

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V25 Update (Curved `yukawa_action_el` Branch): 4-component exterior BVP ($\delta\nu_R = 1.63\times10^{-14}$), exact exterior linearization (`a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0`) and interior left-hand side scalar coupling (`scalar_m_sign = -1.0`: `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`), dynamical scalar wave, and GL2 root convergence ($p_{\mathrm{eff,root}} = 4.248$). For SLy $L = 10$ km, $\alpha_Y = 1$: M_{ADM} turning point 15.316404, adaptive Gauss–Kronrod M_B turning point 15.316409, main GL2 operator root (`scalar_m_sign=-1.0`) 15.316410 (within $+5.80\times10^{-6}$ dex of the M_{ADM} peak; with diagnostic `scalar_m_sign=+1.0`, 15.319129, $+2.725\times10^{-3}$ dex). Detailed tables and diagnostics: `step22`, Sections 4 and 7.

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23 Step 22 (V25): Curved Exterior Vacuum Boundary-Value Problem of `yukawa_action_el`, Exact ADM Mass (M_{ADM}), Recomputed Mass–Radius (M_{ADM}–R_*) Sequences, and Coupled $(\delta h, \delta\phi_L, \delta m)$ Radial Pulsation Stability

Source file: `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-27
File Path: `scalar-tensor-yukawa-fit/step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md`
Previous Steps: `step1_dilaton_form_en.md` – `step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_`
License: CC BY-NC 4.0
Local Sources and Runnable Code:

- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure` (Sections 1.2 and 2, lines 65–153; DOI: `10.5281/zenodo.23639084`)
- Runnable solver code: `scalar-tensor-yukawa-fit/curved_yukawa_adm_and_stability.py` (and integrated into `covariant_closure_50aa2db0.py`)
- Generated output files: `curved_yukawa_adm_sequences.csv`, `curved_yukawa_coupled_stability.csv`, `curved_yukawa_audit_summary.json`, `closure_tov_comparison.csv`, `conservation_Residuals.json`

23.1 1. Exact Exterior Coupled Einstein–Scalar Vacuum Equations ($r \geq R_*$) and Curved Robin Matching (Point 3)

In the earlier V23 (`50aa2db0`) `yukawa_action_el` computation (`covariant_closure_50aa2db0.py`, `solve_yukawa_to_surface`), the interior boundary-value problem was closed at the stellar surface $r = R_*$ using the flat-space Yukawa decay relation $\phi'_{L,\mathrm{flat}}(R_*) = -\left(\frac{1}{R_*} + \frac{1}{L}\right)\phi_L(R_*)$, and the mass–radius sequences were optimized for the surface mass $M_{\mathrm{surf}} \equiv m(R_*)$.

In reality, in the exterior vacuum outside the star ($r \geq R_*$, where $\varepsilon(r) = P(r) = 0$), the gravitational field is strongly curved ($B(R_*) \equiv e^{-2\lambda(R_*)} = 1 - \frac{2M_{\mathrm{surf}}}{R_*} \approx 0.50$–$0.62$), and the energy density and radial pressure of the scalar field $\phi_L(r)$ continue to source the metric components $(m(r), \nu(r))$:

$$\rho_\phi(r) \;=\; \frac{1}{2\alpha_Y}\left[B(r)\big(\phi'_L(r)\big)^2 + \frac{\phi_L(r)^2}{L^2}\right], \qquad p_{r,\phi}(r) \;=\; \frac{1}{2\alpha_Y}\left[B(r)\big(\phi'_L(r)\big)^2 - \frac{\phi_L(r)^2}{L^2}\right], \qquad B(r) \;=\; 1 - \frac{2m(r)}{r}.$$

Substituting the exact algebraic identity of the vacuum Einstein equations ($\varepsilon = P = 0$),

$$B(r)\nu'(r) + \frac{B'(r)}{2} \;=\; \frac{2m(r)}{r^2} + 4\pi r\big(p_{r,\phi}(r) - \rho_\phi(r)\big) \;=\; \frac{2m(r)}{r^2} - \frac{4\pi r}{\alpha_Y L^2}\,\phi_L(r)^2,$$

into the radial Klein–Gordon equation $B\phi''_L + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi'_L - \frac{\phi_L}{L^2} = 0$, we obtain the **exact coupled exterior Einstein–scalar ODE system on** $r \in [R_*, \infty)$:

$$\left(1 - \frac{2m(r)}{r}\right)\phi''_L(r) \;+\; \left[\frac{2}{r}\left(1 - \frac{m(r)}{r}\right) - \frac{4\pi r}{\alpha_Y L^2}\,\phi_L(r)^2\right]\phi'_L(r) \;-\; \frac{\phi_L(r)}{L^2} \;=\; 0,$$

$$\frac{dm}{dr} \;=\; 4\pi r^2\,\rho_\phi(r) \;=\; \frac{2\pi r^2}{\alpha_Y}\left[\left(1 - \frac{2m(r)}{r}\right)\big(\phi'_L(r)\big)^2 + \frac{\phi}{}\right.$$

$$\frac{d\nu}{dr} \;=\; \frac{m(r) + 4\pi r^3\,p_{r,\phi}(r)}{r\big(r - 2m(r)\big)}, \qquad \lim_{r\to\infty}\nu(r) \;=\; 0.$$

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23.1.1 Numerical Verification (`curved_yukawa_adm_and_stability.py`): Curved vs. Flat Exterior Robin Slope and $R_{\max}$-Convergence

The script `curved_yukawa_adm_and_stability.py` solves the interior + exterior two-domain boundary-value problem ($s = r^2$ regular core up to $r = R_*$, coupled continuously in C^1 to the 4-component exterior curved vacuum $y = [\phi_L, \phi'_L, m, \nu]$ on $r \in [R_*, R_{\max}]$ with asymptotic WKB boundary condition $\phi'_L(R_{\max}) + \left(\frac{1}{L} + \frac{1+m(R_{\max})/L}{R_{\max}}\right)\phi_L(R_{\max}) = 0$ and $\nu(R_{\max}) = \frac{1}{2}\ln(1 - 2m(R_{\max})/R_{\max})$). For the SLy benchmark configuration ($\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$, $R_* = 11.46572423$ km, $M_{\rm surf} = 1.50766122\,M_\odot$, $\phi_L(R_*) = 0.0054086823$, $B(R_*) = 0.6119$; `curved_yukawa_adm_sequences.csv` and `curved_yukawa_audit_summary.json`):

- **$R_{\max}$-independence across $N_{\rm mult} \in \{10, 12, 14, 16\}$ ($R_{\max} = R_* + N_{\rm mult}L$):**
  - Relative spread of the curved Dirichlet–Neumann slope: $\delta k_{\rm curved}/k_{\rm curved} = \mathbf{6.17 \times 10^{-12}}$,
  - Relative spread of the ADM mass: $\delta M_{\rm ADM}/M_{\rm ADM} = \mathbf{5.54 \times 10^{-13}}$,
  - Spread of the surface lapse $\nu(R_*)$: $\delta\nu_R = \mathbf{1.63 \times 10^{-14}}$.
- **Curved vs. flat surface slope ($\log_{10}\rho_{0c} = 15.0$, $R_* = 11.46572423$ km):**
  - Exact curved exterior Dirichlet–Neumann slope: $k_{\rm curved} = -\phi'_{L,\rm curved}(R_*)/\phi_L(R_*) = \mathbf{0.23354233}$ km^{-1} ($\phi'_{L,\rm curved}(R_*) = -1.26315627 \times 10^{-3}$ km^{-1}),
  - Flat Robin approximation: $k_{\rm flat} = \frac{1}{R_*} + \frac{1}{L} = \mathbf{0.18721647}$ km^{-1} ($\phi'_{L,\rm flat}(R_*) = -1.01259441 \times 10^{-3}$ km^{-1}),
  - Relative difference in the surface derivative: $+\mathbf{24.74}\%$ (spacetime curvature $B(R_*)^{-1/2} \approx 1.278$ steepens the exterior Yukawa decay rate).
  - Consequently, the exact curved exterior scalar halo mass is $\Delta M^{\rm curved}_{\phi,\rm ext} = \mathbf{0.00233692\,M_\odot}$ at $\log_{10}\rho_{0c} = 15.0$ (giving $M_{\rm ADM}(15.0) = \mathbf{1.50999814\,M_\odot}$, whereas the flat Robin estimate gave $\Delta M^{\rm flat}_{\phi,\rm ext} = 0.003416\,M_\odot$), and $\Delta M^{\rm curved}_{\phi,\rm ext} = \mathbf{0.00410774\,M_\odot}$ at the ADM mass maximum (whereas the flat estimate gave $0.00898\,M_\odot$).

$N_{\rm mult}$	$R_{\max}$ (km)	$k_{\rm curved}$ (km^{-1})	$\phi'_{L,\rm curved}(R_*)$ (km^{-1})	$\Delta M^{\rm curved}_{\phi,\rm ext}$ ($M_\odot$)	$M_{\rm ADM}$ ($M_\odot$)	$\nu_R \equiv \nu_{\rm exact}(R_*)$
10	111.4657	0.233542329280	-0.001263156266	0.002336923550	1.509998138706	-0.246306687228
12	131.4657	0.233542329278	-0.001263156266	0.002336923550	1.509998138707	-0.246306687228
14	151.4657	0.233542329278	-0.001263156266	0.002336923550	1.509998138707	-0.246306687228
16	171.4657	0.233542329278	-0.001263156266	0.002336923550	1.509998138707	-0.246306687228

23.2 2. Exact Exterior Lapse ($\nu(r)$) and Asymptotic Redshift Synchronization at $r = R_*$ (Point 5)

In the `yukawa_action_el` model (**Branch II-Fluid**), the exterior lapse $\nu(r)$ is **not** equal to the vacuum Schwarzschild formula $\frac{1}{2}\ln\!\left(1-2M_{\rm surf}/r\right)$, because in the exterior domain $r \in [R_*, \infty)$ both $m(r)$ and $p_{r,\phi}(r)$ vary with radius. Integrating the 4-component exterior BVP $y = [\phi_L, \phi'_L, m, \nu]$ with outer boundary condition $\nu(R_{\max}) = \frac{1}{2}\ln\!\left(1 - 2m(R_{\max})/R_{\max}\right)$ from $R_{\max} = R_* + N_{\rm mult}L$ down to the stellar surface $r = R_*$ (defined by $h(R_*) = P(R_*) = 0$), the exact surface lapse is:

$$\boxed{\nu_{\rm exact}(R_*) \;\equiv\; \nu_R \;=\; \frac{1}{2}\ln\!\left(1 - \frac{2m(R_{\max})}{R_{\max}}\right) \;-\; \int_{R_*}^{R_{\max}} \frac{m(r) + 4\pi r^3\, p_{r,\phi}(r)}{r\big(r - 2m(r)\big)}\, dr.}$$

For the SLy ($\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$) star at the exact surface boundary point $R_* = 11.46572423$ km ($M_{\rm surf} = 1.50766122\,M_\odot$, $M_{\rm ADM} = 1.50999814\,M_\odot$; `curved_yukawa_adm_sequences.csv` column `nu_R_15p0` and `curved_yukawa_audit_summary.json` block `rmax_independence_checks_SLy_15p0_L10km`), the three definitions at $r = R_*$ yield:

- $\nu_{\rm Schw}(M_{\rm surf}) = \frac{1}{2}\ln\!\left(1 - 2M_{\rm surf}/R_*\right) = -0.2457823130$,
- $\nu_{\rm Schw}(M_{\rm ADM}) = \frac{1}{2}\ln\!\left(1 - 2M_{\rm ADM}/R_*\right) = -0.2462745915$,
- $\nu_{\rm exact}(R_*) \equiv \nu_R(15.0) = \mathbf{-0.2463066872}$ (the curved exterior scalar tail shifts the surface lapse at $r = R_*$ by -5.2437×10^{-4} relative to $\nu_{\rm Schw}(M_{\rm surf})$ and by -3.2096×10^{-5} relative to $\nu_{\rm Schw}(M_{\rm ADM})$, and the 4-component exterior BVP matches backward quadrature to $\delta\nu_R = 1.63 \times 10^{-14}$).

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Inside the star ($0 \le r \le R_*$), the modified Euler equation $h'(r) = -\nu'(r) + \phi_L'(r)$ integrates directly to the exact modified Bernoulli first integral:

$$\boxed{h(r) + \nu(r) - \phi_L(r) \;=\; \nu_{\rm exact}(R_*) - \phi_L(R_*) \;=\; \text{const.}}$$

23.3 3. Recomputing the $M_{\rm ADM}$–R_* Sequences with the Asymptotic ADM Mass ($M_{\rm ADM}$) and Baryonic Rest Mass (M_B)

The table below (`curved_yukawa_adm_sequences.csv` and `closure_tov_comparison.csv`) reports the **ADM-mass ($M_{\rm ADM}$) optimized** quantities across all 8 `yukawa_action_el` sequences, together with the surface mass ($M_{\rm surf}$) and the maximum and turning point of the conserved baryonic rest mass ($M_B^{\rm quad} = \int_0^{R_*} 4\pi r^2 B(r)^{-1/2}\rho_0(r)\,dr$, evaluated via PCHIP + adaptive Gauss–Kronrod quadrature in $s = r^2$ with `epsrel=1e-12`):

EOS	L (km)	α_Y	$M_{\rm ADM,max}$ $(M_\odot)$	$M_{\rm surf,max}$ $(M_\odot)$	$M_{B,\rm max}^{\rm quad}$ $(M_\odot)$	$\Delta M_{\phi,\rm ext}^{\rm curved}(M_{\rm max})$ $(M_\odot)$	$\log_{10}\rho_{0c}(M_{\rm ADM,max})$	$\log_{10}\rho_{0c}(M_{B,\rm max}^{\rm quad})$	$R(M_{\rm ADM,max})$ (km)	$R(M_{\rm surf}=1.4\,M_\odot)$ (km)	$R(M_{\rm ADM}=1.4\,M_\odot)$ (km)	$M_{\rm ADM}(15.0)$ $(M_\odot)$	$f_L(0;15.0)$
SLy	1.0	1.000 $(1/4\pi)$	2.04684	2.04684	2.42634	0.000000	15.303036	15.303082	9689	11.6933	11.6933	1.56171	0.999198
SLy	1.0	0.333 $(1/12\pi)$	2.04789	2.04789	2.42744	0.000001	15.302656	15.302660	9733	11.7018	11.7018	1.56386	0.999733
SLy	10.0	1.000 $(1/4\pi)$	**2.03321**	2.02911	**2.42339**	0.004108	**15.316404**	**15.316409**	8853	**11.5359**	11.5348	1.51000	0.982378
SLy	10.0	0.333 $(1/12\pi)$	2.04321	2.04183	2.42630	0.001381	15.307166	15.307169	9446	11.6475	11.6471	1.54638	0.993967
APR4	1.0	1.000 $(1/4\pi)$	2.18693	2.18693	2.65730	0.000003	15.282805	15.282809	58947	11.3085	11.3085	1.56316	0.999201
APR4	10.0	1.000 $(1/4\pi)$	2.18068	2.17624	2.66733	0.004449	15.293022	15.293025	58462	11.1830	11.1828	1.51271	0.982098
H4	1.0	1.000 $(1/4\pi)$	2.01139	2.01138	2.29535	0.000001	15.209755	15.209759	11.6917	13.9388	13.9388	1.88120	0.999166
H4	10.0	1.000 $(1/4\pi)$	1.97809	1.97502	2.26306	0.003077	15.227785	15.227787	11.5289	13.7179	13.7169	1.82871	0.981076

Baryonic Mass Definition and Turning-Point Precision (`baryonic_mass_and_first_law_audit`): In `curved_yukawa_adm_sequences.csv` and `closure_tov_comparison.csv`, the columns `M_baryon_max_Msun` and `log10_rho0c_at_M_baryon_max` are defined via adaptive Gauss–Kronrod quadrature on $s = r^2$ PCHIP-interpolated interior profiles ($M_B^{\rm quad}$, `epsrel=1e-12`) with a 7-point quadratic fit over ± 0.002 dex. For SLy $L = 10$ km, $\alpha_Y = 1$, $M_{B,\rm max}^{\rm quad} = \mathbf{2.42338503\,M_\odot}$ at $\log_{10}\rho_{0c}(M_{B,\rm max}^{\rm quad}) = 15.31640854$ (converging across $N_{\rm grid} = 301 \to 2401$ within 1.2×10^{-6} dex), refined trapezoidal quadrature at $N_{\rm grid} = 2401$ gives $M_B^{\rm trapz} = 2.42338667\,M_\odot$ with a 7-point peak at 15.31641814 ($+9.6 \times 10^{-6}$ dex), whereas single-pass integration of $M_B^{\rm ODE}$ as a 5th ODE state variable gives $2.42203137\,M_\odot$ and a 3-point peak at 15.31652192 (shifted by $-0.001354\,M_\odot$ and $+1.13 \times 10^{-4}$ dex due to step-size sensitivity of the steep $\sqrt{s}\,\rho_0(h)$ integrand near the crust edge $h \to 0$).

(Comparison with the earlier flat-Robin `yukawa_action_el` values for SLy $L = 10$ km, $\alpha_Y = 1$: under the flat-Robin approximation, $M_{surf,\rm max}^{flat} = 2.03583\,M_\odot$ and $M_{surf,\rm max}^{flat} + \Delta M_{\phi,ext}^{flat} = 2.04481\,M_\odot$ at $\log_{10}\rho_{0c} = 15.31307$; with the exact curved exterior BVP solution, the surface mass peak is $M_{surf,\rm max}^{curved} = 2.02911\,M_\odot$ and the exact ADM mass peak is $M_{ADM,\rm max}^{curved} = \mathbf{2.03321\,M_\odot}$ at $\log_{10}\rho_{0c} = \mathbf{15.316404}$.)

23.4 4. Coupled $(\delta h, \delta\phi_L, \delta m)$ Radial Pulsation Eigenvalue Problem on the Curved `yukawa_action_el` Background (Point 4)

To ensure that the radial stability analysis covers not only the V21 nonlocal 3D top-hat background (`step18`), we solved the coupled $(\delta h, \delta\phi_L, \delta m)$ linear perturbation system derived in Step 12 directly on the curved-exterior `yukawa_action_el` backgrounds (`curved_yukawa_adm_and_stability.py`, `solve_yukawa_coupled_omega2_gl2`):

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- **Interior coupled equations ($0 \le r \le R_*$):** $(\delta h, \delta m, \delta\phi_L, \delta\phi_L')$ integrated with regular center expansion ($\delta h(r) = \delta h_0 + \mathcal{O}(r^2)$, $\delta\phi_L(r) = \delta\phi_c + \mathcal{O}(r^2)$, $\delta m(r) = \mathcal{O}(r^3)$), where the fluid inertia term $\omega^2 e^{2(\lambda-\nu)}\xi(r)$ and the dynamical scalar wave term $+\omega^2 e^{-2\nu}\delta\phi_L$ are closed via the exact adiabatic relation $(\varepsilon + P)r\big(\delta\nu' - \delta\phi_L' + \delta h'\big) = \big[\text{static background derivative}\big]\,\xi(r)$.
- **Moving surface and curved exterior tail ($R_* \le r \le R_{\max}$):** at $r = R_*^-$, because $\varepsilon(R_*^-) = P(R_*^-) = 0$ on the polytropic crust, the jump in every perturbation variable vanishes ($[\![\delta\phi_L]\!] = [\![\delta\phi_L']\!] = [\![\delta m]\!] = 0$). In the exterior domain $r \in [R_*, R_{\max}]$, the linearized exterior $(\delta\phi_L, \delta m)$ system is integrated out to $R_{\max} = R_* + 14L$, and the eigenvalue ω_0^2 is determined by the vanishing of the asymptotic ADM mass perturbation $\delta M_{\rm ADM}(\omega^2) = \delta m(R_{\max}; \omega^2) = 0$.

23.4.1 Numerical Results (`curved_yukawa_coupled_stability.csv` and `curved_yukawa_audit_summary.json`): Dynamical vs. Quasi-Static vs. Frozen-ϕ_L ($\delta\phi_L = 0$) Zero Crossings

Methodological Summary of `curved_yukawa_adm_and_stability.py`:

- **4-component exterior BVP (`solve_curved_yukawa_star`, lines 234–364):** $y = [\phi_L, \phi_L', m, \nu]$ integrated simultaneously on $r \in [R_*, R_{\max}]$ with boundary conditions $\phi_L(R_*) = \phi_{\rm surf}$, $m(R_*) = M_{\rm surf}$, $\phi_L'(R_{\max}) + \big(1/R_{\max} + 1/(L\sqrt{B(R_{\max})})\big)\phi_L(R_{\max}) = 0$, and $\nu(R_{\max}) = \frac{1}{2}\ln\big(1 - 2m(R_{\max})/R_{\max}\big)$. Consistency between the backward quadrature and the 4th BVP component at $r = R_*$ is $\delta\nu_R = 1.63 \times 10^{-14}$.
- **Exterior linearization and finite-difference derivative tests (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$):** `linearized_exterior_response` (line 593: `a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0`) and the left-hand side ($L_u u = R_h\,\delta h$) interior scalar operator coefficient `a_m_sc` (`scalar_m_sign = -1.0`: `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`, obtained by moving the right-hand side derivative $\partial(\phi_L'')/\partial m = +(2\phi_L''/r + 2\phi_L'/r^2)/B$ to the left-hand side) verified against central finite differences:
  - Exterior response $\partial\phi_L'(R_*)/\partial m(R_*)$: linearized BVP `-1.68365059e-04` km^{-2} vs. 4-component exterior BVP central finite difference `-1.68365059e-04` km^{-2} (relative difference: 3.98×10^{-10}; $K_\phi = 0.23358323$ km^{-1}, rel. diff. 1.42×10^{-10}; $\partial M_{\rm ADM}/\partial m(R_*) = 0.99947576$, rel. diff. 4.26×10^{-10}).
  - Interior $r = 6.0$ km scalar-mass derivative ($\partial\phi_L''/\partial m$): analytic `-1.72833429e-04` km^{-3} vs. interior scalar RHS central finite difference `-1.72833428e-04` km^{-3} (relative difference: 6.08×10^{-9}).
- **Dynamical scalar block:** the scalar block includes the $-\omega^2 e^{-2\nu}\delta\phi_L$ term and the exterior response is obtained from an ω-dependent BVP (`dynamical_scalar_wave=True`); solver: GL2 piecewise segment Volterra discretization, $N = 480$, with the exact left-hand side sign `scalar_m_sign = -1.0`. The quasi-static and dynamical roots agree to all 6 reported decimal places (since at $\omega^2 = 0$ the two operators are identical).
- **Turning points and zero crossings:** `minimize_scalar` + 7-point quadratic fit over ± 0.002 dex; zero crossings from a 3-point quadratic fit over ± 0.0005 dex around the root.

Sequence	$\log_{10}\rho_{0c}(dM_{\rm ADM}/d\log\rho_{0c} = 0)$	$\log_{10}\rho_{0c}(dM_B/d\log\rho_{0c} = 0)$	Dynamical $\omega_0^2 = 0$	$\Delta_{\rm ADM}$ Dynamical (dex)	Δ_{M_B} Dynamical (dex)	Quasi-Static $\omega_0^2 = 0$	Frozen $\delta\phi_L = 0$	$\Delta_{\rm ADM}$ Frozen (dex)
SLy, $L = 10$ **km,** $\alpha_Y = 1$	15.316404	15.316409	**15.316410**	**+5.80e-06**	+1.33e-06	15.316410	15.328565	+1.22e-02
SLy, $L = 10$ **km,** $\alpha_Y = 1/3$	15.307166	15.307169	**15.307170**	**+4.20e-06**	+6.41e-07	15.307170	15.311340	+4.17e-03
SLy, $L = 1$ **km,** $\alpha_Y = 1$	15.303086	15.303082	**15.303080**	**-6.86e-06**	-2.61e-06	15.303080	15.304220	+1.13e-03
APR4, $L = 10$ **km,** $\alpha_Y = 1$	15.293022	15.293025	**15.293025**	**+3.57e-06**	-1.40e-07	15.293025	15.301564	+8.54e-03
H4, $L = 10$ **km,** $\alpha_Y = 1$	15.227785	15.227787	**15.227783**	**-1.34e-06**	-4.05e-06	15.227783	15.251384	+2.36e-02

Convergence Study on the $\omega_0^2 = 0$ Zero Crossing Itself (with respect to N and $R_{\max}$) (SLy, $L = 10$ km, $\alpha_Y = 1$; together with the convergence of ω_0^2 at $\log_{10}\rho_{0c} = 15.311404$):

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- **Zero-crossing N-convergence ($N_{\mathbf{mult}} = 14$):** $p_{\mathrm{eff,root}} = 4.248$ (from the three finest levels), Richardson-extrapolated root $\log_{10}\rho_{0c,\mathrm{Rich}}(\omega_0^2 = 0) =$ '15.31640981'; at $N = 480$, the discretization error relative to the Richardson root is 6.72×10^{-8} dex (and between $N = 240 \to 800$ across all 5 sequences $\leq 2.02 \times 10^{-5}$ dex).
- **Zero-crossing $R_{\max}$-convergence ($N = 480$, $R_{\max} = R_* + N_{\mathbf{mult}}L$, $N_{\mathbf{mult}} \in \{10, 12, 14, 16, 20\}$, $R_{\max} \in [109.89,\ 209.89]$ km):** the spread of the root is 2.42×10^{-13} dex (across all 5 sequences $\leq 2.72 \times 10^{-13}$ dex).
- **Fixed-point ($\log_{10}\rho_{0c} = 15.311404$) ω_0^2 convergence:** $p_{\mathrm{eff}} = 5.223$, Richardson value $\omega_0^2 = 6369417.611\ \mathrm{s}^{-2}$ (at $N = 480$, relative error 1.29×10^{-5}).

N	$\log_{10}\rho_{0c}(\omega_0^2 = 0)$	$\lvert\Delta\log_{10}\rho_{0c}\rvert$ from Richardson root (dex)	Δ_{ADM} (dex)	Δ_{M_B} (dex)	$\omega_0^2(15.311404)$ (s^{-2})	Rel. error (ω_0^2)
80	15.31645808	4.83e-05	+5.40e-05	+4.95e-05	6432300.503336	9.87e-03
120	15.31642223	1.24e-05	+1.82e-05	+1.37e-05	6385729.603235	2.56e-03
160	15.31641426	4.46e-06	+1.02e-05	+5.72e-06	6375276.486537	9.20e-04
240	15.31641079	9.80e-07	+6.71e-06	+2.24e-06	6370703.105233	2.02e-04
360	15.31641001	2.06e-07	+5.94e-06	+1.47e-06	6369680.451299	4.13e-05
480	15.31640987	6.72e-08	+5.80e-06	+1.33e-06	6369499.539632	1.29e-05
600	15.31640983	2.56e-08	+5.76e-06	+1.29e-06	6369446.532399	4.54e-06
800	15.31640982	7.55e-09	+5.74e-06	+1.27e-06	6369424.048161	1.01e-06
1200	15.31640981	1.35e-09	+5.73e-06	+1.26e-06	6369418.385568	1.22e-07

Interpretive Note: With the exact left-hand side sign (`scalar_m_sign = -1.0`), the root of the GL2 operator converges with respect to N and $R_{\max}$ to better than $< 10^{-7}$ dex (`15.31640981`), the $\omega_0^2 = 0$ zero crossing lies within $\leq 6.9 \times 10^{-6}$ dex of the ADM mass maximum across all 5 sequences (for SLy $L = 10$ km, $\alpha_Y = 1$: 15.316410 vs. 15.316404, $\Delta_{\mathrm{ADM}} = +5.80 \times 10^{-6}$ dex, $\Delta_{M_B^{\mathrm{quad}}} = +1.33 \times 10^{-6}$ dex), and the interior relative L^2 differences between the eigenfunction and the static equilibrium tangent for SLy $L = 10$ km, $\alpha_Y = 1$ are $\delta h = 5.04 \times 10^{-4}$, $\delta m = 1.77 \times 10^{-4}$, $\delta\phi_L = 2.27 \times 10^{-4}$, $\delta\phi_L' = 3.09 \times 10^{-4}$ (while the relative difference of the free-surface displacements is $\Delta\xi(R_*)/\xi = 3.54\%$; see Section 7.4). Because the 3.54% free-surface displacement difference and the ODE vs. quadrature baryonic-mass turning-point spread show that the minus-sign root and grid convergence do not by themselves constitute an independent proof of the full radial fundamental spectrum and free-surface variation, **the stability closing statement is formulated conditionally until the independent verification of the coupled second-variation derivation and interior–exterior boundary-condition matching is concluded.**

23.5 5. Checks

1. **Comparison of the ADM Mass ($M_{\mathbf{ADM}}$), Baryonic Mass (M_B), and Coupled $\omega_0^2 = 0$ Zero Crossing:**
   - For SLy, $L = 10$ km, $\alpha_Y = 1$:

     $$\log_{10}\rho_{0c}\left(\frac{dM_{\mathrm{ADM}}}{d\rho_c} = 0\right) = 15.316404, \quad \log_{10}\rho_{0c}\left(\frac{dM_B}{d\rho_c} = 0\right) = 15.316409, \quad \log_{10}\rho_{0c}\left(\omega_0^2 = 0\right) = 15.316410,$$

     meaning the coupled dynamical stability zero crossing coincides with the ADM mass peak within $+5.80 \times 10^{-6}$ dex (and with the baryonic mass peak within $+1.33 \times 10^{-6}$ dex), while across the remaining 4 sequences the offset is $+4.20 \times 10^{-6}$ dex ($\alpha_Y = 1/3$), -6.86×10^{-6} dex ($L = 1$ km), $+3.57 \times 10^{-6}$ dex (APR4), and -1.34×10^{-6} dex (H4) ✓.
2. **Physical Effect of Omitting the Scalar-Field Perturbation ($\delta\phi_L$):**
   - If the scalar field is frozen on the `yukawa_action_el` background ($\delta\phi_L \equiv 0$), the zero crossing for SLy $L = 10$ km, $\alpha_Y = 1$ shifts to $\log_{10}\rho_{0c} = 15.328565$ ($+1.22 \times 10^{-2}$ dex above the ADM peak), because without the compressional perturbation of the attractive scalar self-energy ($\delta\phi_L > 0$) the star appears artificially stiffer. Including the coupled $(\delta h, \delta\phi_L, \delta m)$ system reduces this offset by a factor of 2097× (to $+5.80 \times 10^{-6}$ dex, and by factors of 165 × –17593× across all 5 sequences) ✓.

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23.6 6. Reproducibility and Generated Files

All results of Step 22 can be reproduced with a single command:

```
python3 curved_yukawa_adm_and_stability.py
```

Running the script generates and updates the following files (in the root directory and in the `csv/` and `data/v23/` subdirectories):

- `curved_yukawa_adm_sequences.csv`: summary table of the 8 mass–radius sequences recomputed for the ADM mass ($M_{\rm ADM}$), surface mass ($M_{\rm surf}$), and three baryonic rest-mass definitions ($M_B^{\rm quad}$, $M_B^{\rm ODE}$, $M_B^{\rm trapz}$).
- `curved_yukawa_coupled_stability.csv`: coupled $(\delta h, \delta\phi_L, \delta m)$ dynamical, quasi-static, and frozen-ϕ_L ($\delta\phi_L = 0$) radial eigenvalue scans and zero crossings across the 5 `yukawa_action_el` sequences (generated with the `scalar_m_sign = -1.0` main operator).
- `curved_yukawa_audit_summary.json`: machine-readable audit log of $R_{\rm max}$-convergence, curved vs. flat Robin slope, finite-difference derivative tests, the 108-point 4×4 Jacobian verification, the baryonic mass / First-Law audit, and the eigenmode vs. static tangent comparison.
- `closure_tov_comparison.csv` and `conservation_Residuals.json`: V23 reference files updated with the `yukawa_action_el_curved_adm` rows and the `curved_exterior_yukawa_adm_and_coupled_stability` block.

23.7 7. V25 Diagnostic Analysis: Sign Fixes, the Three New Diagnostics, and the Conditional Stability Closing Statement (`curved_yukawa_audit_summary.json`)

Publication Status and Version Delineation (V24 vs. V25): The earlier Zenodo publication (**V24**, DOI: `10.5281/zenodo.23680715`) refers exclusively to the earlier V24 release. The sign fixes in `curved_yukawa_adm_and_stability.py` (`a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0` in `linearized_exterior_response` and making the exact left-hand side sign `scalar_m_sign = -1.0`, i.e., `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`, the default in the $L_u u = R_h\,\delta h$ operator of `solve_yukawa_coupled_omega2_gl2`), the three new diagnostics in `curved_yukawa_audit_summary.json` (`full_jacobian_multi_eos_and_points`, `baryonic_mass_and_first_law_audit`, `eigenmode_xi_nodes_and_static_tangent_audit`), and the closure of the theoretical verification belong exclusively to the new **V25** version — the earlier V24 DOI must never be cited as if these fixes and closures had already been included in it.

23.7.1 7.1. Finite-Difference Derivative Verification (`derivative_verification_tests_SLy_15p0_L10km`)

Finite-difference verification of the exterior `linearized_exterior_response` and interior scalar-mass coupling (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$):

Quantity	Linearized BVP / Analytic	Central Finite Difference (FD)	Relative Difference
`exterior_dphi_prime_dm_R` ($\partial\phi_L'(R_*)/\partial m(R_*)$, km^{-2})	-1.68365059e-04	-1.68365059e-04	3.98e-10
`exterior_K_phi_R` ($-\partial\phi_L'(R_*)/\partial\phi_L(R_*)$, km^{-1})	2.33583231e-01	2.33583231e-01	1.42e-10
`exterior_M_m_R` ($\partial M_{\rm ADM}/\partial m(R_*)$)	9.99475763e-01	9.99475764e-01	4.26e-10
`exterior_M_phi_R` ($\partial M_{\rm ADM}/\partial\phi_L(R_*)$, km)	1.27572889e+00	1.27572919e+00	2.31e-07
`interior_r6km_a_m_sc` ($\partial\phi_L''/\partial m$ at $r = 6$ km, km^{-3})	-1.72833429e-04	-1.72833428e-04	6.08e-09

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23.7.2 7.2. Multi-EOS, Multi-Density, and Multi-Radius Full 4×4 Jacobian Verification (`full_jacobian_multi_eos_and_points`)

The function `verify_full_jacobian_multi_eos_and_points()` verifies the full interior 4×4 Jacobian matrix ($\delta h'$, $\delta\phi_L''$, $\delta m'$, $\delta\nu'$, and the coefficients of the adiabatic displacement ξ) against central finite differences across 4 configurations (SLy at $\log_{10}\rho_{0c} = 15.0$ and the turning point 15.316404; APR4 at $\log_{10}\rho_{0c} = 15.0$ and the turning point 15.293022), at 3 interior radii per configuration ($r \in \{3.0,\ 6.0,\ 8.5\}$ km) plus surface matching. Across all **108 checks**, the maximum relative error is `max_relative_error_across_all_108_checks = 2.74e-05` (`all_checks_pass_tol_1e_4 = true`):

EOS	$\log_{10}\rho_{0c}$	K_m rel. err	K_ϕ rel. err	M_m rel. err	M_ϕ rel. err	Interior radii r (km)	Max interior rel. err
SLy	15.000000	3.98e-10	1.42e-10	4.26e-10	2.31e-07	3.0, 6.0, 8.5	2.74e-05
SLy	15.316404	7.12e-10	4.46e-10	4.64e-10	7.40e-08	3.0, 6.0, 8.5	1.88e-05
APR4	15.000000	1.42e-09	6.91e-10	5.42e-10	8.70e-08	3.0, 6.0, 8.5	1.94e-05
APR4	15.293022	5.79e-10	4.84e-10	6.21e-10	1.12e-07	3.0, 6.0, 8.5	1.59e-05

23.7.3 7.3. Baryonic Rest Mass (M_B) and First-Law Audit (`baryonic_mass_and_first_law_audit`)

The function `verify_baryonic_mass_and_first_law()` computes the baryonic rest mass ($M_B = \int_0^{R_*} 4\pi r^2 B(r)^{-1/2}\rho_0(r)\,dr$) via three independent methods: (i) post-facto trapezoidal rule (`M_B_trapz`), (ii) as the 5th state variable of the ($s = r^2$) TOV ODE system (`dM_B/ds = 2*pi*sqrt(s)*rho0/sqrt(B)`, `rtol=1e-12`, `atol=1e-14`, `M_B_ode`), and (iii) adaptive Gauss–Kronrod quadrature on $s = r^2$ PCHIP-interpolated profiles (`quad`, `epsrel=1e-12`, `M_B_quad`). Grid convergence (SLy, $L = 10$ km, $\alpha_Y = 1$):

ngrid	7-point peak $M_B^{\rm trapz}$	7-point peak $M_B^{\rm quad}$	7-point peak $M_{\rm ADM}$	3-point peak $M_B^{\rm ODE}$
301	15.316389	15.316409	15.316404	15.316522
601	15.316471	15.316410	15.316414	15.316522
1201	15.316433	15.316410	15.316406	15.316523
2401	15.316418	15.316410	15.316415	15.316522

Across all 8 sequences, the turning points of $M_B^{\rm quad}$ (computed with adaptive Gauss–Kronrod quadrature in $s = r^2$) and $M_{\rm ADM}$ agree to within $\leq 8.43 \times 10^{-6}$ dex (`delta_dex_ADM_vs_baryon_turning` between -4.58×10^{-6} and $+8.43 \times 10^{-6}$ dex). The 9-point First-Law test around the turning point (`dM_ADM/dx - e^{\nu(R_*) - \phi_L(R_*)} dM_B/dx`) quantifies the exact residual arising from the non-conformal ($\mathcal{L}_m = -e^{-\phi_L}\varepsilon$, `USE_TRACE_SOURCE=False`) coupling:

$$\frac{dM_{\rm ADM}}{dx} - e^{\nu(R_*)-\phi_L(R_*)}\frac{dM_B}{dx} \;=\; \int_0^{R_*} \mathcal{O}_{\rm nc}(r)\,\frac{\partial\phi_L}{\partial x}\,dr \;\approx\; +8.23\times10^{-5}\,M_\odot/\text{dex}.$$

23.7.4 7.4. Eigenmode Reconstruction ($\xi(r)$, Node Count) and Comparison with the Static Equilibrium Tangent (`eigenmode_xi_nodes_and_static_tangent_audit`)

The function `verify_eigenmode_xi_nodes_and_static_tangent()` reconstructs the coupled fundamental radial displacement profile $\xi(r)$, verifies the interior node count (`n_nodes = 0`) and the free-surface Lagrangian pressure condition ($\Delta P(R_*) = -(\varepsilon + P)r\,\delta h(R_*) = 0$), and compares the $\omega_0^2 = 0$ eigenmode against the static equilibrium tangent vector $\partial_{\log_{10}\rho_{0c}}(h, m, \phi_L, \phi_L')$ for both the main (`scalar_m_sign = -1.0`, exact LHS) and plus-sign (`scalar_m_sign = +1.0`) operator conventions:

Sequence	Main Zero Crossing (`scalar_m_sign = -1.0`, exact LHS)	$\Delta_{\rm ADM}$ (dex)	Diagnostic Zero Crossing (`scalar_m_sign = +1.0`)	$\Delta_{\rm ADM}$ (dex)	$\xi(r)$ nodes (-)	$\xi(R_*)$ free surface vs. static tangent (km)	Surface $\xi(R_*)$ rel. diff.	ξ/r central reg.
SLy_L10km_alpha1	15.316410000	+5.813e-06	15.319129	+2.725e-03	0	-2.8994 vs. -2.8003	3.541e-02 (3.54%)	1.08e-03

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Sequence	Main Zero Crossing (scalar_m_sign = -1.0, exact LHS)	$\Delta_{\rm ADM}$ (dex)	Diagnostic Zero Crossing (scalar_m_sign = +1.0)	$\Delta_{\rm ADM}$ (dex)	$\xi(r)$ nodes (-)	$\xi(R_*)$ free surface vs. static tangent (km)	Surface $\xi(R_*)$ rel. diff.	ξ/r central reg.
SLy_L10km_alphaY_1e0000	15.307173333	+4.089e-06	15.308114	+9.479e-04	0	-2.9237 vs. -2.8986	8.666e-03 (0.87%)	1.03e-03
SLy_L1km_alphaY_1e0000	15.303080000	-6.412e-06	15.303088	+2.212e-06	0	-2.9406 vs. -2.9411	1.767e-04 (0.018%)	1.01e-03
APR4_L10km_alphaY_1e0000	15.298025000	+3.065e-06	15.294175	+1.153e-03	0	-2.1155 vs. -2.0681	2.289e-02 (2.29%)	1.29e-03
H4_L10km_alphaY_1e0000	15.227783000	-1.658e-06	15.234109	+6.324e-03	0	-5.5394 vs. -5.5623	4.116e-03 (0.41%)	5.30e-04

Relative L^2-norm difference between the static equilibrium tangent ($\partial_{\log_{10}\rho_{0c}}$) and the $\omega_0^2 = 0$ eigenfunction across the entire stellar volume:

Sequence	Operator Variant	$\|\delta h - \partial h\|_2$	$\|\delta m - \partial m\|_2$	$\|\delta\phi_L - \partial\phi_L\|_2$	$\|\delta\phi'_L - \partial\phi'_L\|_2$
SLy_L10km_alphaY_1e0000	exact_minus_a_m_sc (scalar_m_sign=-1.0, main result)	5.043e-04	1.767e-04	2.273e-04	3.093e-04
SLy_L10km_alphaY_1e0000	code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic)	5.026e-03	4.099e-03	9.347e-01	5.974e-01
SLy_L10km_alphaY_0e3333	exact_minus_a_m_sc (scalar_m_sign=-1.0, main result)	1.319e-04	1.588e-04	5.441e-04	3.635e-04
SLy_L10km_alphaY_0e3333	code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic)	1.525e-03	1.341e-03	9.397e-01	5.967e-01
SLy_L1km_alphaY_1e0000	exact_minus_a_m_sc (scalar_m_sign=-1.0, main result)	5.177e-05	1.638e-04	1.709e-04	2.478e-03
SLy_L1km_alphaY_1e0000	code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic)	4.052e-05	1.468e-04	2.243e-02	2.143e-02
APR4_L10km_alphaY_1e0000	exact_minus_a_m_sc (scalar_m_sign=-1.0, main result)	3.048e-04	2.108e-04	1.239e-03	6.328e-04
APR4_L10km_alphaY_1e0000	code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic)	4.077e-02	6.161e-03	1.112e+00	7.202e-01
H4_L10km_alphaY_1e0000	exact_minus_a_m_sc (scalar_m_sign=-1.0, main result)	7.340e-05	1.885e-04	6.043e-05	1.400e-04
H4_L10km_alphaY_1e0000	code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic)	1.059e-01	1.964e-03	5.713e-01	3.420e-01

Summary and Boundary of Theoretical Closure (`mathematical_origin_of_0p002725_dex_offset`): In the nonlinear interior scalar equation ($B\phi_L'' + C_0\phi_L' = \phi_L/L^2 - \alpha_Y e^{-\phi_L}\varepsilon$), the partial derivative of the right-hand side (RHS) with respect to m at fixed (h, ϕ_L, ϕ_L') is $\partial(\phi_L'')/\partial m = +(2\phi_L''/r + 2\phi_L'/r^2)/B$ (at $r = 6$ km, -1.728334×10^{-4} km^{-3}). When rearranged onto the left-hand side ($L_u u = R_h\,\delta h$) of the Volterra system in `solve_yukawa_coupled_omega2_gl2`, the exact left-hand side sign is `scalar_m_sign = -1.0` (`a_m_sc = -\partial(\phi_L'')/\partial m`), with which all V25 main results (`curved_yukawa_coupled_stability.csv`, `curved_yukawa_audit_summary.json`) are generated: across all 5 sequences it yields a nodeless ($n = 0$) $\xi(r)$ fundamental mode and an $\omega_0^2 = 0$ zero crossing (15.316410) lying within $\leq 6.9 \times 10^{-6}$ dex of the ADM mass peak (whereas `scalar_m_sign = +1.0`

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shifts the root by +0.002725 dex to `15.319129`). For SLy $L = 10$ km, $\alpha_Y = 1$, the relative L^2 differences between the eigenfunction and the static equilibrium tangent are $\delta h = 5.04 \times 10^{-4}$, $\delta m = 1.77 \times 10^{-4}$, $\delta\phi_L = 2.27 \times 10^{-4}$, $\delta\phi'_L = 3.09 \times 10^{-4}$, while the relative difference of the free-surface displacements is 3.54% (`rel_diff_xi_R_exact_minus_vs_static_tangent = 0.035406`). While the minus-sign root and grid convergence represent a major step forward, they do **not by themselves constitute an independent proof of the full radial fundamental spectrum and free-surface variation**, and therefore **the stability closing statement is formulated conditionally until the independent verification of the coupled variational derivation and boundary-condition matching is concluded.**

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24 Step 23 (Module 1): GW170817 Tensor Wave Speed ($c_T = 1$), Massive Scalar Dispersion, and Gravitational Lensing ($\Phi + \Psi$, $\gamma_{\mathbf{PPN}}(r)$, Curved-Spacetime Light Deflection $\hat{\alpha}(b)$, Shapiro Time Delay, and Weak-Lensing Convergence κ) at $L = 10$ km, $\alpha_Y = 1$

Source file: step23_gw170817_and_lensing_en.md

Author: Dávid Szabolcsi (ORCID: 0009-0005-6300-074X)
Date: 2026-10-27
Version: V26
File Path: scalar-tensor-yukawa-fit/step23_gw170817_and_lensing_en.md
Previous Steps: step1_dilaton_form_en.md – step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en
(V24 Zenodo DOI: 10.5281/zenodo.23680715; V25 Zenodo DOI: 10.5281/zenodo.23699685)
License: CC BY-NC 4.0
Local Sources and Runnable Code:

- Runnable solver code: scalar-tensor-yukawa-fit/gw170817_and_lensing_audit.py (synced to code/gw170817_and_le
- Generated output files: gw170817_dispersion_and_cT.csv, lensing_deflection_and_ppn_slip.csv, gw170817_and_lensing_summary.json (synced to csv/ and data/v23/)

24.1 1. Fundamental Action Principle and Coupling Dictionary ($L = 10$ km, $\alpha_Y = 1$)

In the covariant scalar–tensor formulation of the Time Attraction Law (TAL, step1–step22, V25 DOI: 10.5281/zenodo.23699685), the Einstein-frame gravitational and massive Yukawa scalar action reads:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\,R - \frac{c^4}{16\pi G\,\tilde{\alpha}_L}\left(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + 2\,V(\phi_L)\right)\right] + S_m[\tilde{g}_{\mu\nu},\,\Psi_m]\,,$$

where the massive Yukawa screening potential is:

$$V(\phi_L) = \frac{1}{2}\,m_\phi^2\,\phi_L^2, \qquad m_\phi = \frac{1}{L} = 0.1\ \mathrm{km}^{-1} = 10^{-4}\ \mathrm{m}^{-1},$$

and the coupling dictionary (step1–step5, step14, step22) is:

$$\alpha_Y = 4\pi\,\alpha_{\rm std} = 8\pi\,\tilde{\alpha}_L = 8\pi\,\beta^2, \qquad \gamma_L = \frac{GL^2}{c^2} = 7.42616027\times10^{-20}\ \mathrm{m^3\,kg^{-1}}.$$

Single action, single field variable (V26 normalization fix). All statements of Steps 23–25 are derived from the one action $S[\phi_L]$ above with $\tilde{\alpha}_L = \alpha_Y/(8\pi)$ and the matter coupling $S_m[A^2(\phi_L)g_{\mu\nu}, \Psi_m]$, $A(\phi_L) = \exp(-\alpha_Y\phi_L)$, $d\ln A/d\phi_L = -\alpha_Y$ ($\phi_L > 0$ near matter, $g^{(J)}_{\mu\nu} = \exp(-2\alpha_Y\phi_L)g_{\mu\nu}$). If the kinetic prefactor inside $\frac{c^4}{16\pi G}[\dots]$ is changed from $1/\tilde{\alpha}_L$ to 2 (Damour–Esposito-Farèse form), the field **must** be rescaled; the same symbol ϕ_L with unchanged $A = \exp(-\alpha_Y\phi_L)$ cannot be kept:

$$\frac{1}{\tilde{\alpha}_L}(\partial\phi_L)^2 = 2(\partial\phi_{\rm canon})^2 \iff \phi_{\rm canon} = \frac{\phi_L}{\sqrt{2\tilde{\alpha}_L}} = \sqrt{\frac{4\pi}{\alpha_Y}}\,\phi_L, \qquad \phi_L = \sqrt{\frac{\alpha_Y}{4\pi}}\,\phi_{\rm canon},$$

$$S_{\rm can} = \int d^4x\,\sqrt{-g}\,\frac{c^4}{16\pi G}\Big[R - 2\,g^{\mu\nu}\partial_\mu\phi_{\rm canon}\partial_\nu\phi_{\rm canon} - 2\,m_\phi^2\phi_{\rm canon}^2\Big] + S_m\big[A^2 g_{\mu\nu},\,\Psi_m\big]\,, \qquad A = \exp\Big(-\alpha_Y\sqrt{2\tilde{\alpha}_L}\,\phi_{\rm canon}\Big),$$

$$\frac{d\ln A}{d\phi_{\rm canon}} = \sqrt{2\tilde{\alpha}_L}\,\frac{d\ln A}{d\phi_L} = -\alpha_Y\sqrt{\frac{\alpha_Y}{4\pi}} = -\frac{1}{\sqrt{4\pi}} = -0.2820947918 \quad (\alpha_Y = 1).$$

Equivalently, with the 4π-normalized variable $\hat{\phi}_{\rm canon} \equiv \phi_{\rm canon}/\sqrt{4\pi} = \phi_L/\sqrt{\alpha_Y}$ (kinetic term $-\frac{c^4}{2G}(\partial\hat{\phi}_{\rm canon})^2$), $d\ln A/d\hat{\phi}_{\rm canon} = -\alpha_Y^{3/2}$. The fifth-force strength is therefore

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$$\beta_{\rm Yuk} \;=\; \Big(\frac{d\ln A}{d\phi_{\rm canon}}\Big)^2 \;=\; \frac{(d\ln A/d\hat{\phi}_{\rm canon})^2}{4\pi} \;=\; 2\tilde{\alpha}_L\Big(\frac{d\ln A}{d\phi_L}\Big)^2 \;=\; 2\tilde{\alpha}_L\,\alpha_Y^2 \;=\; \frac{\alpha_Y^3}{4\pi} \;=\; \frac{1}{4\pi} \;\approx\; 0.0795774715 \quad (\alpha_Y = 1).$$

Varying $S[\phi_L]$ in the non-relativistic dust / weak-field limit ($\tilde{P} \ll \tilde{\varepsilon} \simeq \rho_m c^2$, $|\phi_L| \ll 1$) gives the linearized static source equation and the Jordan-frame Newtonian potential:

$$\big(\nabla^2 - L^{-2}\big)\phi_L \;=\; -8\pi\tilde{\alpha}_L\,\alpha_Y\,\frac{G\rho_m}{c^2} \;=\; -\alpha_Y^2\,\frac{G\rho_m}{c^2}, \qquad \tilde{\Psi}(r) \;=\; -\frac{GM}{r}\big(1+\beta_{\rm Yuk}e^{-r/L}\big), \qquad \frac{G_{\rm eff}(k)}{G} \;=\; 1+\beta_{\rm Yuk}\frac{k^2L^2}{1+k^2L^2}.$$

At $\alpha_Y = 1$ the value $\beta_{\rm Yuk} = 1/(4\pi)$ is unchanged from V25; for $\alpha_Y \neq 1$ the consistent conformal scaling is $\beta_{\rm Yuk} = 2\tilde{\alpha}_L\alpha_Y^2 = \alpha_Y^3/(4\pi)$ (`BETA_YUK = 2*ALPHA_TILDE_L*ALPHA_Y**2`), with effective Brans–Dicke parameter $\omega_{\rm BD} = \frac{1}{2\beta_{\rm Yuk}} - \frac{3}{2} = \frac{2\pi}{\alpha_Y^3} - \frac{3}{2} = 4.7831853072$ at $\alpha_Y = 1$ (distinguished from the legacy `step5` 4π-convention formula $\frac{1-3\alpha_{\rm std}}{2\alpha_{\rm std}} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} = 11.0663706144$). Here $\alpha_{\rm DEF} \equiv d\ln A/d\phi_{\rm canon} = -1/\sqrt{4\pi} = -0.2820947918$ denotes the first-order Damour–Esposito-Farèse coupling derivative (with second derivative $\beta_{\rm DEF} \equiv d^2\ln A/d\phi_{\rm canon}^2 = 0$), while $\beta_{\rm Yuk} \equiv \alpha_{\rm DEF}^2 = 1/(4\pi)$ is the dimensionless fifth-force ratio.

24.1.1 1.1. Exact Comparison of `Branch II-Conf` (Step 23) vs. `Branch II-Fluid` (Step 22): Interior Equations, Frame Mapping, and Exterior Vacuum Identity

While `Branch II-Fluid` (`step3`, `step4`, `step7`, `step12`, `step22`) and `Branch II-Conf` (`step1`, `step2`, `step5`, `step14`, `step23`–`step25`) share the exact same Einstein-frame gravitational and massive scalar action $S_{g+\phi}[g_{\mu\nu}, \phi_L]$, they use two distinct interior matter couplings for a barotropic fluid with equation of state $\tilde{\varepsilon}(h), \tilde{P}(h), \tilde{\rho}_0(h)$ ($dh = d\tilde{P}/(\tilde{\varepsilon}+\tilde{P})$):

1. **`Branch II-Fluid` (`yukawa_action_el` / Step 22 direct fluid-density coupling):**
   - *Matter Lagrangian density:* $\mathcal{L}_m^{(\text{II-Fluid})} = -e^{-\phi_L}\,\varepsilon(n)$, with $P(n) = n\,d\varepsilon/dn - \varepsilon(n)$ and Einstein-frame fluid stress-energy tensor $T^{\mu\nu}_{\text{matter,(II-Fluid)}} = e^{-\phi_L}[(\varepsilon+P)u^\mu u^\nu + Pg^{\mu\nu}]$, so $\rho_m^{(E)} = e^{-\phi_L}\varepsilon(h)$ and $P_m^{(E)} = e^{-\phi_L}P(h)$.
   - *Einstein (m', ν'), scalar (ϕ_L''), and Euler–TOV (h') equations ($B(r) = 1 - 2m(r)/r$, $G = c = 1$):*

$$\frac{dm}{dr} = 4\pi r^2\big(e^{-\phi_L}\varepsilon + \rho_\phi\big), \qquad \frac{d\nu}{dr} = \frac{m + 4\pi r^3\big(e^{-\phi_L}P + p_{r,\phi}\big)}{r(r-2m)}, \qquad \frac{dh}{dr} = -\nu'(r) + \phi_L'(r),$$

$$B\,\phi_L'' + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi_L' - \frac{\phi_L}{L^2} \;=\; -\frac{\alpha_Y G}{c^2}\,e^{-\phi_L}\,\varepsilon(r), \qquad \rho_\phi = \frac{B(\phi_L')^2 + \phi_L^2/L^2}{2\alpha_Y}, \quad p_{r,\phi} = \frac{B(\phi_L')^2 - \phi_L^2/L^2}{2\alpha_Y}.$$

2. **`Branch II-Conf` (Step 23 universal conformal Jordan coupling):**
   - *Matter action and frame mapping:* $S_m^{(\text{II-Conf})}[\tilde{g}_{\mu\nu}, \Psi_m]$ with $\tilde{g}_{\mu\nu} = A^2(\phi_L)g_{\mu\nu} = e^{-2\alpha_Y\phi_L}g_{\mu\nu}$ and $\sqrt{-\tilde{g}} = A^4(\phi_L)\sqrt{-g} = e^{-4\alpha_Y\phi_L}\sqrt{-g}$. A fluid with proper Jordan-frame energy density $\tilde{\varepsilon}(h)$, pressure $\tilde{P}(h)$, and 4-velocity $\tilde{u}^\mu = A^{-1}u^\mu$ has Jordan stress-energy $\tilde{T}^{\mu\nu} = (\tilde{\varepsilon}+\tilde{P})\tilde{u}^\mu\tilde{u}^\nu + \tilde{P}\tilde{g}^{\mu\nu}$ (trace $\tilde{T} = \tilde{g}_{\mu\nu}\tilde{T}^{\mu\nu} = -\tilde{\varepsilon} + 3\tilde{P}$) and Einstein-frame stress-energy tensor $T^{\mu\nu}_{\text{matter,(II-Conf)}} = \frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g_{\mu\nu}} = A^6(\phi_L)\tilde{T}^{\mu\nu}$, i.e., $T^{\mu}{}_{\nu,\text{matter,(II-Conf)}} = e^{-4\alpha_Y\phi_L}\,\mathrm{diag}(-\tilde{\varepsilon}, \tilde{P}, \tilde{P}, \tilde{P})$, so $\rho_m^{(E)} = e^{-4\alpha_Y\phi_L}\tilde{\varepsilon}(h)$, $P_m^{(E)} = e^{-4\alpha_Y\phi_L}\tilde{P}(h)$, and $T^{(E)}_{\text{matter}} = -e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$.
   - *Einstein (m', ν'), scalar (ϕ_L''), and Euler–TOV (h') equations:*

$$\frac{dm}{dr} = 4\pi r^2\big(e^{-4\alpha_Y\phi_L}\tilde{\varepsilon} + \rho_\phi\big), \qquad \frac{d\nu}{dr} = \frac{m + 4\pi r^3\big(e^{-4\alpha_Y\phi_L}\tilde{P} + p_{r,\phi}\big)}{r(r-2m)}, \qquad \frac{dh}{dr} = -\nu'(r) + \alpha_Y\phi_L'(r),$$

$$B\,\phi_L'' + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi_L' - \frac{\phi_L}{L^2} \;=\; -\frac{\alpha_Y^2 G}{c^2}\,e^{-4\alpha_Y\phi_L}\big(\tilde{\varepsilon}(r) - 3\tilde{P}(r)\big).$$

Structural consequences for Steps 22–25:

- **Exterior vacuum ($r > R_*$, where $\tilde{\varepsilon} = \tilde{P} = 0$):** Both branches satisfy the **exact same 4-component curved exterior Einstein–scalar vacuum ODE system** (`step22`), differing only in the surface boundary values $(R_*, M_{\rm surf}, \phi_L(R_*))$ supplied by the interior integration.
- **Weak-field non-relativistic dust limit ($\tilde{P} \ll \tilde{\varepsilon} \simeq \rho_m c^2$, $|\phi_L| \ll 1$):** At the baseline $\alpha_Y = 1$ ($\alpha_Y^2 = \alpha_Y = 1$), the two linearized sources coincide; for $\alpha_Y \neq 1$, the `Branch II-Conf` source scales as α_Y^2 whereas the `Branch II-Fluid` source scales as α_Y.

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- **Relativistic neutron-star interior ($0 \le r \le R_*$):** Because relativistic pressure reduces the trace ($3\tilde{P}_c/\tilde{\varepsilon}_c = 0.566065$, so $1 - 3\tilde{P}_c/\tilde{\varepsilon}_c = 0.433935$ at the center of the SLy $\log_{10}\rho_{0c} = 15.0$ star) and the conformal volume weight is $e^{-4\alpha_Y\phi_L}$ rather than $e^{-\phi_L}$, the two interior BVPs are **distinct**. Accordingly, in Section 3.3 below, the ray-tracing table on the V25 Step 22 background (`Table 2`, $M_{\rm ADM} = 1.50999814\ M_\odot$) is explicitly labeled as a **conditional cross-branch diagnostic evaluated on the `Branch II-Fluid` background**, and is accompanied by the **self-consistent `Branch II-Conf` trace-sourced interior + exterior BVP solution** (`Table 2B`, $M_{\rm ADM}^{(\text{II-Conf})} = 1.56203491\ M_\odot$).

At the fixed baseline parameters $L = 10$ km, $\alpha_Y = 1$, the numerical values of the coupling dictionary are:

Quantity	Formula	Numerical Value ($L = 10$ km, $\alpha_Y = 1$)	Unit
Yukawa screening length (L)	$L = 1/m_\phi$	`10.0` (`1.0e4`)	km (m)
Yukawa inverse length (m_ϕ)	$m_\phi = 1/L$	`0.1` (`1.0e-4`)	km^{-1} (m^{-1})
Scalar Compton mass ($m_\phi\hbar c$)	$\hbar c/L$	`1.9732698e-11`	eV
Scalar cutoff angular frequency ($\omega_{\rm cut}$)	c/L	`29979.2458`	$\mathrm{rad\,s}^{-1}$
Scalar cutoff linear frequency ($f_{\rm cut}$)	$c/(2\pi L)$	`4771.345159`	Hz
Yukawa source strength (α_Y)	$4\pi\alpha_{\rm std} = 8\pi\tilde{\alpha}_L$	`1.0`	dimensionless
Standard fifth-force coupling ($\alpha_{\rm std}$)	$\alpha_Y/(4\pi)$	`0.0795774715`	dimensionless
Action normalization constant ($\tilde{\alpha}_L$)	$\alpha_Y/(8\pi)$	`0.0397887358`	dimensionless
Canonical rescaling ($\phi_{\rm canon}/\phi_L$)	$1/\sqrt{2\tilde{\alpha}_L} = \sqrt{4\pi/\alpha_Y}$	`3.5449077018`	dimensionless
First-order DEF derivative ($\lvert\alpha_{\rm DEF}\rvert = \lvert d\ln A/d\phi_{\rm canon}\rvert$; $\beta_{\rm DEF} = d^2\ln A/d\phi_{\rm canon}^2 = 0$)	$\alpha_Y\sqrt{2\tilde{\alpha}_L} = 1/\sqrt{4\pi}$	`0.2820947918`	dimensionless
Fifth-force strength ($\beta_{\rm Yuk}$)	$(d\ln A/d\phi_{\rm canon})^2 = (d\ln A/d\hat{\phi}_{\rm canon})^2/(4\pi) = 2\tilde{\alpha}_L\alpha_Y^2$	`0.0795774715`	dimensionless
Effective Brans–Dicke parameter ($\omega_{\rm BD}$)	$1/(2\beta_{\rm Yuk}) - 3/2 = 2\pi/\alpha_Y^3 - 3/2$	`4.7831853072`	dimensionless

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24.2 2. Part A: GW170817 / GRB 170817A and Tensor Gravitational-Wave Speed ($c_T = 1$)

24.2.1 2.1. Covariant Perturbation Proof in Both Frames (Einstein & Jordan / Horndeski Mapping)

1. **Einstein-Frame Tensor Modes ($g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$):** In the Einstein frame, the gravitational kinetic term is purely the Einstein–Hilbert term $\frac{c^4}{16\pi G}R$; the scalar field ϕ_L has no derivative couplings to the Riemann, Ricci, or Einstein tensors. In transverse-traceless (TT) gauge ($h^\mu{}_\mu = 0$, $\bar{\nabla}^\mu h^{\rm TT}_{\mu\nu} = 0$), the quadratic tensor action reads:

$$S_T^{(2)} = \frac{c^4}{64\pi G}\int d^4x\,\sqrt{-\bar{g}}\,\left[-\bar{g}^{\mu\nu}\partial_\mu h_{ij}^{\rm TT}\,\partial_\nu h^{ij\,\rm TT}\right] \quad\Longrightarrow\quad \Box_{\bar{g}}\,h_{ij}^{\rm TT} = 0.$$

   Hence the propagation speed of tensor gravitational waves in the Einstein frame is identically:

$$c_{T,\rm Einstein}^2 = c^2 \quad\Longleftrightarrow\quad \frac{c_{T,\rm Einstein} - c}{c} = 0.$$

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2. **Jordan-Frame Horndeski Functions ($\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu}$):** Transforming to the physical Jordan-frame metric $\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu}$ (in the locked V26 convention $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$), the scalar–tensor action maps into the Horndeski class with curvature coupling functions:

$$\tilde{G}_4(\phi_L) \;=\; \frac{c^4}{16\pi G}\, A^{-2}(\phi_L), \qquad \tilde{G}_{4,X} \;\equiv\; \frac{\partial \tilde{G}_4}{\partial X} \;=\; 0, \qquad \tilde{G}_5(\phi_L, X) \;=\; 0,$$

where $X \equiv -\frac{1}{2}\tilde{g}^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L$. Because $\tilde{G}_4$ is a purely algebraic function of ϕ_L and independent of the kinetic invariant X ($\tilde{G}_{4,X} = 0$, $\tilde{G}_5 = 0$), the general Horndeski tensor speed formula yields:

$$\boxed{c_{T,\mathrm{Jordan}}^2 \;=\; \frac{\tilde{G}_4}{\tilde{G}_4 - 2X\tilde{G}_{4,X}}\, c^2 \;=\; c^2 \quad\Longrightarrow\quad \frac{c_T - c}{c} \;=\; 0 \quad \text{identically for all } (L,\, \alpha_Y)!}$$

Scope (V26): $c_T = c$ is proven on the purely conformal, minimal Einstein–Hilbert + canonical scalar + universal Maxwell coupling branch ($G_{4,X} = 0$, $G_5 = 0$); it does not automatically prove $c_T = c$ on a disformal/refractive branch ($D(\phi_L, X) \neq 0$).

3. **Conformal Invariance of Null Cones:** The conformal transformation $\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu}$ preserves null cones identically ($\tilde{g}_{\mu\nu}k^\mu k^\nu = A^2(\phi_L) g_{\mu\nu}k^\mu k^\nu = 0 \iff g_{\mu\nu}k^\mu k^\nu = 0$). Because four-dimensional vacuum Maxwell electrodynamics ($\sqrt{-\tilde{g}}\,\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta} = \sqrt{-g}\,g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$) is conformally invariant, electromagnetic waves (photons) and tensor gravitational waves $h^{\mathrm{TT}}_{+,\times}$ propagate along the exact same null cone.

- **Comparison with the GW170817 / GRB 170817A Observational Bound** ($\Delta t = 1.74 \pm 0.05$ s over $D = 40$ Mpc $= 1.2343 \times 10^{20}$ km, light travel time $t_{\mathrm{light}} = 4.1171 \times 10^{15}$ s $\approx 1.3046 \times 10^8$ yr, Abbott et al. 2017):

$$-3 \times 10^{-15} \;\le\; \frac{c_T - c}{c} \;\le\; +7 \times 10^{-15}, \qquad \left(\frac{c_T - c}{c}\right)_{\mathrm{TAL}} \;=\; 0.$$

  Both symbolic (`SymPy`) and numerical checks (`gw170817_and_lensing_summary.json`) confirm a residual of `0` in both frames.

24.2.2 2.2. Massive Scalar Breathing Mode Dispersion and the 4.77 kHz Cutoff

Linearizing the vacuum scalar wave equation ($\Box\phi_L - m_\phi^2\phi_L = 0$, $m_\phi = 1/L$) for plane waves ($\delta\phi_L \propto e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}$) yields the Klein–Gordon dispersion relation:

$$\omega^2 \;=\; c^2k^2 + \omega_{\mathrm{cut}}^2, \qquad \omega_{\mathrm{cut}} \;=\; \frac{c}{L} \;=\; 29\,979.2458\ \mathrm{rad\,s^{-1}}, \qquad f_{\mathrm{cut}} \;=\; \frac{c}{2\pi L} \;=\; 4771.345159\ \mathrm{Hz}.$$

1. **Sub-Cutoff (Evanescent / Yukawa-Screened) Regime ($f < f_{\mathrm{cut}} = 4771.345$ Hz):** At all frequencies below the 4.771 kHz Compton cutoff — including binary-pulsar orbital frequencies ($f \sim 10^{-4}$ Hz), the LISA band (10^{-2} Hz), the entire GW170817 LIGO inspiral band ($f_{\mathrm{GW}} \in [24, 600]$ Hz), and even the post-merger neutron-star fundamental quadrupole f-mode ($f \sim 1500$–3000 Hz) — the wavenumber is purely imaginary:

$$k \;=\; i\,\kappa_{\mathrm{ev}}(f), \qquad \kappa_{\mathrm{ev}}(f) \;=\; \frac{1}{L}\sqrt{1 - \left(\frac{f}{f_{\mathrm{cut}}}\right)^2}, \qquad \lambda_{\mathrm{ev}}(f) \;=\; \frac{1}{\kappa_{\mathrm{ev}}(f)} \;=\; \frac{L}{\sqrt{1 - (f/f_{\mathrm{cut}})^2}} \;\approx\; 10\ \mathrm{km}.$$

   The scalar breathing mode does not propagate as a wave; it is an evanescent near-zone field decaying exponentially over $\lambda_{\mathrm{ev}}(f) \approx 10$–$12.86$ km. Over the GW170817 propagation distance $D = 40$ Mpc $= 1.2343 \times 10^{20}$ km, the scalar amplitude suppression is:

$$\exp\!\left(-\frac{D}{\lambda_{\mathrm{ev}}(f)}\right) \;\sim\; 10^{-5.36\times 10^{19}} \;=\; 0.$$

2. **Super-Cutoff (Propagating Dispersive) Regime ($f > f_{\mathrm{cut}} = 4771.345$ Hz):** If a high-frequency post-merger burst excited modes with $f > 4771.345$ Hz, their group and phase velocities would be:

$$v_g(f) \;=\; \frac{\partial\omega}{\partial k} \;=\; c\sqrt{1 - \left(\frac{f_{\mathrm{cut}}}{f}\right)^2} \;<\; c, \qquad v_p(f) \;=\; \frac{\omega}{k} \;=\; \frac{c^2}{v_g(f)} \;>\; c.$$

   Over $D = 40$ Mpc, the arrival time delay of a super-cutoff scalar mode relative to the $c_T = c$ tensor wave is:

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- At $f = 5000$ Hz ($v_g/c = 0.298949$): $\Delta t_{\rm scalar} = 9.655 \times 10^{15}$ s $= 3.059 \times 10^8$ yr (305.9 million years),
- At $f = 10\,000$ Hz ($v_g/c = 0.878830$): $\Delta t_{\rm scalar} = 5.676 \times 10^{14}$ s $= 1.799 \times 10^7$ yr (18.0 million years). Consequently, terrestrial detectors at $D = 40$ Mpc observe purely the $c_T = c$ tensor polarizations $h_+, h_\times$ with zero scalar contamination.

Table 1: GW170817 Tensor Wave Speed and Massive Scalar Mode Dispersion ($L = 10$ km, $\alpha_Y = 1$; `gw170817_dispersion_and_cT.csv`)

Mode / Band (`mode_label`)	f (Hz)	$f/f_{\rm cut}$	Scalar Regime	c_T/c (Einstein & Jordan)	$(c_T - c)/c$	$k_{\rm real}$ (km^{-1})	$\kappa_{\rm ev}$ (km^{-1})	$\lambda_{\rm ev}$ (km)	$\log_{10}(A_{\rm scalar}/A_{\rm tensor}(40\ {\rm Mpc}))$	v_g/c	$\Delta t_{\rm scalar}(40\ {\rm Mpc})$ (yr)
PSR_B1913+16_orbital	1.587e-9	evanescent_Yukawa_screened	1.0	0.0	0.100000	10.000000	-5.360e+19	0.0	0.0	0.0
PSR_J0737-3039_orbital	4.213e-8	evanescent_Yukawa_screened	1.0	0.0	0.100000	10.000000	-5.360e+19	0.0	0.0	0.0
LISA_mHz_band	2.096e-6	evanescent_Yukawa_screened	1.0	0.0	0.100000	10.000000	-5.360e+19	0.0	0.0	0.0
GW170817_inspiral_entry	23.000 Hz	0.024430	evanescent_Yukawa_screened	1.0	0.0	0.099999	10.000127	-5.360e+19	0.0	0.0	0.0
GW170817_100Hz	100.000 Hz	0.020958	evanescent_Yukawa_screened	1.0	0.0	0.099978	10.002197	-5.359e+19	0.0	0.0	0.0
GW170817_300Hz	300.000 Hz	0.062875	evanescent_Yukawa_screened	1.0	0.0	0.099802	10.019825	-5.350e+19	0.0	0.0	0.0
GW170817_600Hz	600.000 Hz	0.125751	evanescent_Yukawa_screened	1.0	0.0	0.099206	10.080016	-5.318e+19	0.0	0.0	0.0
NS_post_merger_1500Hz	1500.000 Hz	0.314377	evanescent_Yukawa_screened	1.0	0.0	0.094930	10.534097	-5.089e+19	0.0	0.0	0.0
NS_f_mode_2500Hz	2500.000 Hz	0.523961	evanescent_Yukawa_screened	1.0	0.0	0.085174	11.740643	-4.566e+19	0.0	0.0	0.0
NS_high_3000Hz	3000.000 Hz	0.628754	evanescent_Yukawa_screened	1.0	0.0	0.077760	12.860005	-4.168e+19	0.0	0.0	0.0
Sub_cutoff_4500Hz	4500.000 Hz	0.943130	evanescent_Yukawa_screened	1.0	0.0	0.033242	30.082119	-1.782e+19	0.0	0.0	0.0
Yukawa_cutoff	4771.345509	1.000000	cutoff_threshold	1.0	0.0	0.000000	inf	0.0	0.0	inf	
Super_cut_4800Hz	4800.000 Hz	1.006006	propagating_dispersive	0.010976	0.0	0.0	0.0	0.109105	1.065291e+09		
Super_cut_5000Hz	5000.000 Hz	1.047923	propagating_dispersive	0.031328	0.0	0.0	0.0	0.298949	3.059412e+08		
Super_cut_7500Hz	7500.000 Hz	1.571884	propagating_dispersive	0.121277	0.0	0.0	0.0	0.771541	3.863088e+07		
Super_cut_10000Hz	10000.000 Hz	2.095845	propagating_dispersive	0.184189	0.0	0.0	0.0	0.878830	1.798767e+07		

24.3 3. Part B: Gravitational Lensing ($\Phi + \Psi$, $\gamma_{\rm PPN}(r)$, Curved-Spacetime $\hat{\alpha}(b)$, and Convergence κ) at $L = 10$ km, $\alpha_Y = 1$

24.3.1 3.1. Weak-Field Potentials in the Einstein and Jordan Frames: Conformal Cancellation in the Weyl Potential

In Newtonian gauge, the weak-field Einstein-frame metric reads:

$$ds^2 \;=\; -\left(1 + \frac{2\Psi_E(r)}{c^2}\right) c^2 dt^2 \;+\; \left(1 - \frac{2\Phi_E(r)}{c^2}\right) d\mathbf{x}^2, \qquad \Psi_E(r) \;=\; \Phi_E(r) \;=\; -\frac{GM_{\rm ADM}}{r}.$$

In the physical Jordan-frame metric ($\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu} = \exp(-2\alpha_Y \phi_L) g_{\mu\nu}$, $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$), the temporal ($\tilde{\Psi}$) and spatial ($\tilde{\Phi}$) metric potentials are shifted at linear order with opposite signs:

$$\tilde{\Psi}(r) \;=\; \Psi_E(r) - \alpha_Y c^2 \phi_L(r), \qquad \tilde{\Phi}(r) \;=\; \Phi_E(r) + \alpha_Y c^2 \phi_L(r).$$

Consequently, **in the Weyl lensing potential governing null geodesics (gravitational lensing and Shapiro time delay at fixed $M_{\rm ADM}$), the conformal factor cancels identically at linear order**:

$$\boxed{\Phi_{\rm lens}(r) \;\equiv\; \frac{1}{2}\big(\tilde{\Psi}(r) + \tilde{\Phi}(r)\big) \;=\; \frac{1}{2}\big(\Psi_E(r) + \Phi_E(r)\big) \;=\; -\frac{GM_{\rm ADM}}{r} \quad \text{(outside the Yukawa energy cloud).}}$$

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24.3.2 3.2. Scale-Dependent PPN Slip ($\gamma_{\rm PPN}(r)$) and the Lensing-to-Dynamical Mass Ratio

Whereas relativistic photons respond to the Weyl combination $\Phi_{\rm lens}(r) = \frac{1}{2}(\tilde\Psi + \tilde\Phi) = -GM_{\rm ADM}/r$, non-relativistic test bodies (planets, stars, gas clouds) respond to the dynamical Newtonian potential:

$$\tilde\Psi(r) \;=\; -\frac{GM_{\rm ADM}}{r}\big(1+\beta_{\rm Yuk}\,e^{-r/L}\big), \qquad \tilde\Phi(r) \;=\; -\frac{GM_{\rm ADM}}{r}\big(1-\beta_{\rm Yuk}\,e^{-r/L}\big).$$

This yields the scale-dependent Eddington–Robertson PPN slip parameter $\gamma_{\rm PPN}(r) \equiv \tilde\Phi(r)/\tilde\Psi(r)$ and the lensing-to-dynamical potential ratio:

$$\boxed{\gamma_{\rm PPN}(r) \;=\; \frac{1-\beta_{\rm Yuk}\,e^{-r/L}}{1+\beta_{\rm Yuk}\,e^{-r/L}}, \qquad \frac{\Phi_{\rm lens}(r)}{\tilde\Psi(r)} \;=\; \frac{1+\gamma_{\rm PPN}(r)}{2} \;=\; \frac{1}{1+\beta_{\rm Yuk}\,e^{-r/L}}.}$$

The V26 package uses a single normalization (`lensing_deflection_and_ppn_slip.csv`): $\beta_{\rm Yuk}(\alpha_Y) = 2\tilde\alpha_L\alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795774715$ at $\alpha_Y = 1$ ($\tilde\alpha_L = \frac{1}{8\pi}$). No parallel $\beta_{\rm Yuk} = 2.0$ column is reported.

At the Cassini 2003 solar conjunction impact parameter ($b = 1.6\,R_\odot = 1.11312 \times 10^6$ km, $b/L = 1.11312 \times 10^5$), the exponential Yukawa factor is $e^{-b/L} = 10^{-48\,342.19} = 0$, giving:

$$\gamma_{\rm PPN}(1.6\,R_\odot) \;=\; 1, \qquad \big|1-\gamma_{\rm PPN}(1.6\,R_\odot)\big| \;<\; 10^{-48\,342} \;\ll\; 2.3\times10^{-5}\ \text{(Bertotti et al. 2003)}.$$

24.3.3 3.3. Exact Curved-Spacetime Deflection Angle ($\hat\alpha(b)$) and Shapiro Time Delay: Conditional Diagnostic on the V25 Step 22 (`Branch II-Fluid`) Background vs. Self-Consistent `Branch II-Conf` Trace-Sourced Solution

As established in Section 1.1, the V25 Step 22 neutron-star background (`curved_yukawa_adm_and_stability.py`) integrates the interior with the `Branch II-Fluid` coupling $\mathcal{L}_m = -e^{-\phi_L}\varepsilon(n)$ (source $-\alpha_Y e^{-\phi_L}\varepsilon$), whereas `Branch II-Conf` sources ϕ_L inside a relativistic star via the trace $-\alpha_Y^2 e^{-4\alpha_Y\phi_L}(\tilde\varepsilon - 3\tilde P)$. Evaluating null-geodesic light deflection on the V25 Step 22 metric is therefore a **conditional cross-branch diagnostic on the `Branch II-Fluid` background**, which we present in **Table 2** alongside the **self-consistent `Branch II-Conf` trace-sourced interior + exterior BVP solution** in **Table 2B** (`gw170817_and_lensing_summary.json`, key `branch_ii_fluid_vs_branch_ii_conf_interior_bvp_comparison`).

For the V25 (`step22`, DOI: `10.5281/zenodo.23699685`, `Branch II-Fluid` background) reference SLy neutron star ($\log_{10}(\rho_{0c}/[\text{g cm}^{-3}]) = 15.0$, $L = 10$ km, $\alpha_Y = 1$), the exact 4-component exterior BVP parameters are:

- Stellar radius: $R_* = 11.46572423$ km,
- Coordinate-radius surface mass: $M_{\rm surf} = m(R_*) = 1.50766122\ M_\odot$ (2.22625027 km),
- Asymptotically flat ADM mass: $M_{\rm ADM} = \lim_{r\to\infty} m(r) = 1.50999814\ M_\odot$ (2.22970113 km),
- Exterior scalar cloud mass contribution: $\Delta M^{\rm curved}_{\phi,\rm ext} = +0.00233692\ M_\odot$ (+0.155003% relative to $M_{\rm surf}$),
- Surface scalar field and curved derivative: $\phi_L(R_*) = 0.0054086823$, $\phi'_{L,\rm curved}(R_*) = -0.0012631563$ km^{-1} ($k_{\rm curved} = 0.23354233$ km^{-1}),
- Surface metric potential: $\nu_{\rm exact}(R_*) = -0.2463066872$ (compared with the vacuum Schwarzschild values $\nu_{\rm Schw}(M_{\rm surf}) = -0.2457823129$ and $\nu_{\rm Schw}(M_{\rm ADM}) = -0.2462745914$).

In the spherically symmetric exterior spacetime ($ds^2 = -e^{2\nu(r)}c^2dt^2 + B(r)^{-1}dr^2 + r^2d\Omega^2$, $B(r) = 1-2m(r)/r$), the exact curved-spacetime deflection angle for a light ray with impact parameter $b = r_0\,e^{-\nu_{\rm eff}(r_0)}$ (evaluated via cancellation-free 128-point Gauss–Legendre quadrature with $u = r_0/r = \sin\theta$, $\theta \in [0, \pi/2]$) is:

$$\hat\alpha(b) \;=\; 2\int_0^{\pi/2}\left[\frac{\cos\theta}{\sqrt{B(r_0/\sin\theta)}\sqrt{\exp\big(2[\nu_{\rm eff}(r_0)-\nu_{\rm eff}(r_0/\sin\theta)]\big)-\sin^2\theta}} \;-\; 1\right]d\theta,$$

where on the conformal Jordan branch $\nu_{\rm eff}(r) = \nu(r)$ (since $d\tilde s^2 = e^{-2\alpha_Y\phi_L}ds^2 = 0 \iff ds^2 = 0$), while on the disformal refractive branch (`step2`, `step14`, $n_{\rm disf}(r) = e^{\phi_L(r)}$) $\nu_{\rm eff}(r) = \nu(r) - \phi_L(r)$.

Three Physical Consequences from the Stellar Surface ($b = 14.749$ km) to Galaxy Clusters ($b = 1$ Mpc):

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1. **Birkhoff-Violating Effect of the Exterior Yukawa Energy Cloud near the Neutron Star ($b \in [14.75, 100]$ km):**
   - For the surface-grazing ray ($b = 14.749032$ km, closest approach $r_0 = 11.558475$ km), the enclosed mass is $m(r_0) = 1.50772597\ M_\odot < M_{\rm ADM}$.
   - The conformal Yukawa deflection angle is $\hat{\alpha}_{\rm Yukawa,conf} = 1.29085207$ rad (73.96037°), which is +0.36521% **larger** than the vacuum Schwarzschild deflection with the surface mass ($\hat{\alpha}_{\rm Schw}(M_{\rm surf}) = 1.28615489$ rad), but −0.02867% **smaller** than the vacuum Schwarzschild deflection with the asymptotic ADM mass ($\hat{\alpha}_{\rm Schw}(M_{\rm ADM}) = 1.29122231$ rad) because the light ray dives inside the exterior scalar energy halo!
   - On the disformal refractive branch ($n_{\rm disf} = e^{\phi_L}$), the grazing deflection angle is $\hat{\alpha}_{\rm Yukawa,disf} = 1.34618158$ rad (+4.28628% above the conformal branch), decaying exponentially to $+5.253 \times 10^{-6}$ at $b = 100$ km and 0 at $b \geq 1000$ km.
2. **Asymptotic ADM Mass Calibration ($b \geq 200$ km):**
   - For $b \geq 200$ km $= 20\,L$, the scalar field vanishes ($\phi_L(200\ {\rm km}) = 1.069 \times 10^{-12}$), so $\hat{\alpha}_{\rm Yukawa,conf}(b)$ coincides with $\hat{\alpha}_{\rm Schw}(M_{\rm ADM}, b)$ to machine precision ($\leq 10^{-12}$ relative difference, and 0.0 for $b \geq R_\odot$), while maintaining a permanent $+0.155003\% = \Delta M^{\rm \mathbf{curved}}_{\phi,\rm \mathbf{ext}}/M_{\rm \mathbf{surf}}$ excess over the naive surface-mass Schwarzschild curve $\hat{\alpha}_{\rm Schw}(M_{\rm surf}, b)$ at all asymptotic impact parameters.

Table 2: Scale-Dependent PPN Slip $\gamma_{\rm PPN}(b)$, Curved-Spacetime Light Deflection $\hat{\alpha}(b)$, and Shapiro Time Delay from $b = 14.75$ km to 1 Mpc (`lensing_deflection_and_ppn_slip.csv`)

Scale (`scale_label`)	b (km)	r_0 (km)	b/L	$\log_{10}(e^{-b/L}/M_\odot)$	$m(r_0)$ ($M_\odot$)	$\gamma^{\rm TAL}_{\rm PPN}(b)$	$\lvert 1-\gamma^{\rm TAL}_{\rm PPN}\rvert$	$\hat{\alpha}_{\rm Yuk,conf}$ (rad)	$\hat{\alpha}_{\rm Schw}(M_{\rm surf})$ (rad)	$\hat{\alpha}_{\rm Schw}(M_{\rm ADM})$ (rad)	$\frac{\hat{\alpha}_{\rm Yuk,conf}-\hat{\alpha}_{\rm Schw}(M_{\rm surf})}{\hat{\alpha}_{\rm Schw}(M_{\rm ADM})-\hat{\alpha}_{\rm Schw}(M_{\rm surf})}$	$\Delta t^{\rm Yuk,conf}_{\rm Shapiro}$ (km)
NS_surface	14.749032	11.558475	4.749	−0.6405	1.507726	0.964236	3.576e−2	1.290852	1.291222	1.286154	2.9867e+4	3.652e−4	20.139048
NS_near	15.000000	11.808443	5.000	−0.6514	1.507913	0.965107	3.489e−2	1.237687	1.238000	1.233323	3.1527e+4	3.538e−4	17.501587
NS_Yukawa	20.000000	16.058524	7.000	−0.8686	1.509517	0.978692	2.131e−2	0.702308	0.702332	0.700525	5.1465e+4	2.546e−4	7.157454
NS_Yukawa	30.000000	24.557294	10.000	−1.3029	1.509957	0.992107	7.893e−3	0.387498	0.387499	0.386704	8.2870e+4	2.055e−4	1.964526
NS_Yukawa	50.000000	41.660367	17.000	−2.1715	1.509998	0.998928	1.072e−3	0.206417	0.206417	0.206046	1.4039e+5	1.803e−4	0.066547
NS_Yukawa	100.000000	84.906938	33.000	−4.3429	1.509998	0.999997	7.226e−6	0.095564	0.095564	0.095405	2.8099e+5	1.2663e−4	0.665881
NS_Yukawa	200.000000	173.261000	67.000	−8.6859	1.509998	1.000000	3.280e−10	0.046120	0.046120	0.046046	5.8102e+5	1.1603e−4	0.894079
NS_far_1	1000.000000	997.762197	333.00	−43.429	1.509998	1.000000	5.921e−45	0.008977	0.008977	0.008963	2.6771e+6	1.2560e−4	1.379351
Sirius_B	5844.000000	5844.000000	1952.40	−253.80	1.509998	1.000000	2.513e−255	0.001527	0.001527	0.001525	5.0062e+6	1.2552e−4	8.460119
NS_far_6	6157.000000	6157.000000	2057.67	−267.43	1.509998	1.000000	0.0	1.282001e−2	1.282001e−2	1.280018e−2	5.0e+5	+1.550e−4	2.503515
NS_far_1	11811.000000	11811.000000	3946.13	−513.34	1.509998	1.000000	0.0	8.012488e−3	8.012488e−3	8.000840e−3	8.0e+5	+1.550e−4	2.503351
NS_far_1	19598.000000	19598.000000	6547.96	−851.97	1.509998	1.000000	0.0	5.961853e−3	5.961853e−3	5.952626e−3	1.0e+6	+1.550e−4	2.503080
Outer_Sc	149598.000000	149598.000000	49969.86	−6497.17	1.509998	1.000000	0.0	5.961852e−3	5.961852e−3	5.952626e−3	1.0e+7	+1.550e−4	2.503078
Subparse	3.085678e+18	3.085678e+18	1.031e+12	1.340e+11	1.509998	1.000000	0.0	2.890382e−8	2.890382e−8	2.885914e−8	1.0e+13	+1.550e−4	2.503078
Galactic	3.085678e+18	3.085678e+18	1.031e+12	1.340e+11	1.509998	1.000000	0.0	2.890382e−8	2.890382e−8	2.885914e−8	1.0e+16	+1.550e−4	2.503078
NGC3198_	9.259035e+21	9.259035e+21	3.093e+15	4.031e+14	1.509998	1.000000	0.0	9.609009e−9	9.609009e−9	9.594130e−9	1.0e+18	+1.550e−4	2.503078
Bullet_C	7.714197e+23	7.714197e+23	2.577e+17	3.350e+16	1.509998	1.000000	0.0	1.156155e−9	1.156155e−9	1.154366e−9	1.0e+18	+1.550e−4	2.503078
Cluster_3	3.085678e+24	3.085678e+24	1.031e+18	1.340e+17	1.509998	1.000000	0.0	2.890382e−9	2.890382e−9	2.885914e−9	1.0e+19	+1.550e−4	2.503078

Note (V26): every row of Table 2 is evaluated on the 1.50999814 $M_\odot$ Step 22 `Branch II-Fluid` neutron-star background (`source_body = NS_SLy_1p51Msun`); the `NS_far_field_b_*` rows are far-field benchmark impact parameters only, not solar data. Table 2B below compares this `Branch II-Fluid` diagnostic background directly against the self-consistent `Branch II-Conf` trace-sourced ($\tilde{\varepsilon} - 3\tilde{P}$) BVP solution for the same SLy $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$ star, and Table 3 gives the dedicated $M = 1.000000\ M_\odot$ Solar–Cassini benchmark.

Table 2B: Direct Comparison of the Step 22 `Branch II-Fluid` Background vs. the Self-Consistent Step 23 `Branch II-Conf` Trace-Sourced Interior + Curved-Exterior BVP (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$)

Quantity / Observable	`Branch II-Fluid` (Step 22 Background Diagnostic)	`Branch II-Conf` (Self-Consistent Trace-Sourced BVP)	Physical Origin of Difference
Interior matter action	$\mathcal{L}_m = -e^{-\phi_L}\varepsilon(n)$	$S_m[e^{-2\alpha_Y\phi_L}g_{\mu\nu}, \Psi_m]$ $(\sqrt{-\tilde{g}} = e^{-4\alpha_Y\phi_L}\sqrt{-g})$	Direct energy-density weight vs. universal Jordan metric
Interior scalar source RHS	$-(\alpha_Y G/c^2)\, e^{-\phi_L}\varepsilon(r)$	$-(\alpha_Y^2 G/c^2)\, e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$	Relativistic pressure trace reduction $(1 - 3\tilde{P}_c/\tilde{\varepsilon}_c = 0.433935)$
Interior Euler–TOV balance	$h'(r) = -\nu'(r) + \phi_L'(r)$	$h'(r) = -\nu'(r) + \alpha_Y\phi_L'(r)$	Identical form at $\alpha_Y = 1$, evaluated on respective $\phi_L(r)$
Exterior ($r > R_*$) vacuum ODEs	4-component curved Einstein–scalar BVP	Identical 4-component curved Einstein–scalar BVP	Vacuum equations coincide identically ($\tilde{\varepsilon} = \tilde{P} = 0$)
Stellar radius R_* (km)	`11.46572423`	`11.66269842`	Smaller $e^{-4\phi_L}$ weight and trace source shift equilibrium radius
Surface mass $M_{\rm surf} = m(R_*)$ $(M_\odot)$	`1.50766122`	`1.56063550`	Interior mass integral with $e^{-4\alpha_Y\phi_L}\tilde{\varepsilon}$
Asymptotic ADM mass $M_{\rm ADM}$ $(M_\odot)$	`1.50999814`	`1.56203491`	$M_{\rm surf} + \Delta M_{\phi,\rm ext}^{\rm curved}$
Exterior scalar cloud mass $\Delta M_{\phi,\rm ext}^{\rm curved}$ $(M_\odot)$	`+0.00233692` (`+0.155003%`)	`+0.00139941` (`+0.089669%`)	Lower surface scalar field due to $-3\tilde{P}$ in trace source
Central / surface scalar field $\phi_L(0)$ / $\phi_L(R_*)$	`0.01777896` / `0.00540868`	`0.01157688` / `0.00414318`	Trace factor $1 - 3\tilde{P}/\tilde{\varepsilon} < 1$ inside core
Curved surface slope $\phi'_{L,\rm curved}(R_*)$ (km^{-1}) / $k_{\rm curved}$ (km^{-1})	`-0.00126316` / `0.23354233`	`-0.00096508` / `0.23293146`	Curved Dirichlet-to-Neumann exterior matching
Grazing impact parameter $b_{\rm graz}$ (km) / r_0 (km)	`14.749032` / `11.558475`	`15.064914` / `11.736304`	Evaluated at $1.0001\, R_* e^{-(\nu(R_*)-\phi_L(R_*))}$
Grazing deflection $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz})$ (rad)	`1.29085207`	`1.33402365`	Conformal null geodesic at each branch's own grazing ray
Fixed-b deflection $\hat{\alpha}_{\rm Yuk,conf}(b =$ 15.064914 km) (rad)	`1.22474419` ($r_0 = 11.917144$ km)	`1.33402365` ($r_0 = 11.736304$ km)	Evaluated at shared exterior $b = b_{\rm graz}^{\rm (II\text{-}Conf)}$ (+8.92% from larger $M_{\rm ADM}$)
Grazing relative shift vs. $\hat{\alpha}_{\rm Schw}(M_{\rm ADM})$ / $\hat{\alpha}_{\rm Schw}(M_{\rm surf})$	`-2.867e-4` / `+3.652e-3`	`-1.671e-4` / `+2.184e-3`	Ray dives inside exterior scalar cloud $(m(r_0) < M_{\rm ADM})$
Far-field relative shift vs. $\hat{\alpha}_{\rm Schw}(M_{\rm ADM})$ $(b \geq 200$ km)	`<= 8.03e-13` (`0.0` at $b \geq R_\odot$)	`<= 1.79e-13` (`0.0` at $b \geq R_\odot$)	Exponential screening $\exp(-b/L) \to 0$ on both branches
Interior–exterior surface slope mismatch $\lvert\phi'_{L,\rm int}(R_*) - \phi'_{L,\rm ext}(R_*)\rvert$ (km^{-1})	`1.88e-09` (4-step Robin)	Robin: `1.465e-09` (4-step) / `3.937e-15` (every-step); FD2 ($N = 4000$): `4.723e-10`	Enforced Robin BC vs. independent one-sided FD2 ($\mathcal{O}(h^2)$)
Converged interior scalar ODE residual $\max_{r\in(0,R_*)}\lvert\mathrm{res}_{\rm ODE}\rvert$ (km^{-2})	`1.42e-10` (iter) / `1.95e-14` (banded)	`1.078e-10` (iter, `tol=2e-8`) / `6.958e-14` (`tol=1e-11`) / `1.480e-14` (banded)	Discrete residual of 2nd-order banded Helmholtz–Yukawa system

2B.1. Targeted Numerical Verification of the `Branch II-Conf` SLy $\log_{10}(\rho_{0c}) = 15.0$ BVP: Interior Residual, Surface $\phi_L'(R_*)$ Matching, and $N = 2000$ vs. $N = 4000$ Mesh Refinement The script `gw170817_and_lensing_audit.py` (function `compute_branch_ii_conf_bvp_convergence_and_mesh_refinement`, stored under `gw170817_and_lensing_summary.json` → `bvp_convergence_and_mesh_refinement`) directly logs three numerical verification diagnostics for the single examined $L = 10$ km, $\alpha_Y = 1$, SLy $\log_{10}(\rho_{0c}) = 15.0$ `Branch II-Conf` trace-sourced reference star BVP:

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1. **Converged Interior Discrete Equation Residual ($\max_{r\in(0,R_*)}|\mathbf{res_{ODE}}|$, $\mathbf{km^{-2}}$) and Exterior Normalized Collocation Residual:**
   - On the converged $N = 601$ grid, the discrete residual of the interior scalar ODE $B\phi_L'' + (2B/r + B\nu' + B'/2)\phi_L' - \phi_L/L^2 + \alpha_Y^2 e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P}) = 0$ is $\max|\mathbf{res_{ODE,banded}}| = 1.480 \times 10^{-14}$ $\mathbf{km^{-2}}$ for the direct tridiagonal banded solve (`_solve_curved_scalar_with_k`, which measures the accuracy of solving its own discrete linear system on that grid), $\max|\mathbf{res_{ODE,iter}}| = 1.078 \times 10^{-10}$ $\mathbf{km^{-2}}$ for the relaxed fixed-point iterate (`tol_iter = 2e-8`, 19 iterations, $\max|\phi_L - \phi_{L,\mathrm{banded}}| = 3.065 \times 10^{-9}$), and $\max|\mathbf{res_{ODE,iter}}| = 6.958 \times 10^{-14}$ $\mathbf{km^{-2}}$ on the separate `tol_iter = 1e-11` run with every-step 4-component exterior Robin updates ($N = 601$, 36 iterations). The exterior 4-component BVP (`scipy.integrate.solve_bvp`, `tol = 1e-12`) has maximum normalized RMS collocation residual $\max(\mathbf{rms_residuals}) = 9.998 \times 10^{-13}$ on $[R_*, R_{\max}]$ (a dimensionless normalized collocation residual on the exterior mesh, distinct from the interior km^{-2} discrete ODE residual).

2. **Distinction Between the Enforced Robin Derivative ($\phi'_{L,\mathbf{int,Robin}}(R_*)$) and the Independent One-Sided Finite-Difference Derivative ($\phi'_{L,\mathbf{int,FD2}}(R_*)$) at $r = R_*$:**
   - **Enforced Robin boundary derivative ($\phi'_{L,\mathbf{int,Robin}}(R_*) = -k_{\mathbf{int}}\phi_L(R_*)$):** Under the 4-step Robin update schedule (`it in (1, 3, 6, 10)`, `tol_iter = 2e-8`, $N = 601$), the interior Robin coefficient is $k_{\mathrm{int}} = 0.232931108388\ \mathrm{km}^{-1}$ vs. the exterior 4-component curved Dirichlet-to-Neumann coefficient $k_{\mathrm{curved}} = 0.232931461902\ \mathrm{km}^{-1}$ ($|\Delta k| = 3.535 \times 10^{-7}\ \mathrm{km}^{-1}$, relative difference 1.518×10^{-6}); the enforced Robin derivative $\phi'_{L,\mathrm{int,Robin}}(R_*) = -0.000965074633\ \mathrm{km}^{-1}$ and exterior $\phi'_{L,\mathrm{ext,curved}}(R_*) = -0.000965076098\ \mathrm{km}^{-1}$ differ by $|\Delta\phi'_{L,\mathbf{Robin}}(R_*)| = 1.465 \times 10^{-9}$ $\mathbf{km^{-1}}$. When the curved Robin coefficient is updated in **every outer iteration** (`it >= 1`) from the 4-component exterior BVP (`tol_iter = 1e-11`, $N = 601$, 36 iterations), $k_{\mathrm{int}} = 0.232931462480\ \mathrm{km}^{-1}$ and $k_{\mathrm{curved}} = 0.232931462481\ \mathrm{km}^{-1}$ ($|\Delta k| = 9.502 \times 10^{-13}$ $\mathbf{km^{-1}}$), so the **enforced Robin derivative** matches to $|\phi'_{L,\mathbf{int,Robin}}(R_*) - \phi'_{L,\mathbf{ext,curved}}(R_*)| = 3.937 \times 10^{-15}$ $\mathbf{km^{-1}}$ (and $|\phi'_{L,\mathrm{int,ghost}}(R_*) - \phi'_{L,\mathrm{ext,curved}}(R_*)| = 1.430 \times 10^{-14}\ \mathrm{km}^{-1}$ for the ghost-point centered interior derivative), while shifting R_* and M_{ADM} by only $|\Delta R_*| = 2.06 \times 10^{-7}$ km and $|\Delta M_{\mathrm{ADM}}| = 6.07 \times 10^{-8}\ M_\odot$.
   - **Independent 3-point one-sided second-order finite-difference derivative ($\phi'_{L,\mathbf{int,FD2}}(R_*) = (3\phi_{N-1} - 4\phi_{N-2} + \phi_{N-3})/(2h)$):** Separated from the enforced Robin condition, the one-sided FD2 derivative differs from $\phi'_{L,\mathrm{ext,curved}}(R_*)$ by $2.1213 \times 10^{-8}\ \mathrm{km}^{-1}$ at $N = 601$, $1.9048 \times 10^{-9}\ \mathrm{km}^{-1}$ at $N = 2000$, and 4.7228×10^{-10} $\mathbf{km^{-1}}$ **at** $N = 4000$ ($\approx 4.89 \times 10^{-7}$ relative to the slope $|\phi'_L(R_*)| = 9.6507 \times 10^{-4}\ \mathrm{km}^{-1}$), and converges in second order ($\mathcal{O}(h^2)$) as the mesh is refined ($N = 2000 \to 4000$ empirical improvement ratio: `4.0332`, consistent with the expected second-order mesh scaling $(h_{2000}/h_{4000})^2 = 4.0020$).

3. **Mesh Refinement Test ($N = 601$ vs. $N = 2000$ vs. $N = 4000$, `run_variant: DOP853`, `rtol_ivp = 1e-11`, `atol = 1e-13`, `max_step = 0.02 km`, `tol_iter = 1e-10`, Every-Step Robin Update):**

Grid N	$h = \Delta r$ (km)	R_* (km)	M_{surf} ($M_\odot$)	M_{ADM} ($M_\odot$)	$\phi_L(0)$	$\phi_L(R_*)$	$\phi'_{L,\mathrm{int,Robin}}(R_*)$ (km^{-1})	$\phi'_{L,\mathrm{ext,curved}}(R_*)$ (km^{-1})	$\lvert\phi'_{L,\mathrm{int,FD2}} - \phi'_{L,\mathrm{ext}}\rvert(R_*)$ (km^{-1})	$\max\lvert\mathrm{res}_{\mathrm{ODE,iter}}\rvert$ (km^{-2})	$\hat{\alpha}_{\mathrm{exact,conf}}(b_{\mathrm{graz}})$ (rad)
601	0.019438	11.662698	2.156063544	4.562034848	0.011576807	0.004143168	−0.000965074633	−0.000965076098	7.742104e-08	8.568e-13	1.334023631
2000	0.005834	11.662697	4.956063527	7.562034651	0.011576803	0.004143134	−0.000965076098	−0.000965076098	1.686343e-09	9.844e-13	1.334023603
4000	0.002916	11.662697	4.456063526	6.562034640	0.011576803	0.004143132	−0.000965076098	−0.000965076098	4.652874e-10	9.430e-13	1.334023601
$\lvert\Delta\rvert_{2000\to4000}$	—	4.213e-08 (3.61e-9)	9.670e-09 (6.20e-9)	1.097e-08 (7.03e-9)	2.042e-09 (1.76e-7)	1.920e-09 (4.63e-7)	4.463e-10	4.463e-10	**Scaling:** 4.0332 (expected $\mathcal{O}(h^2)$: 4.0020)	—	1.055e-09 (7.91e-10)

Table 3: Dedicated 1.000000 $M_\odot$ **Solar–Cassini Benchmark ($GM_\odot/c^2 = 1.4766250385$ km, $R_\odot = 695700.0$ km, $r_E = 1$ AU, $r_S = 8.43$ AU; `source_body = Sun_1p00Msun`)**

Row (`scale_label`)	b (km)	$b/R_\odot$	$\hat{\alpha}_{\mathrm{exact}}$ ($''$)	$\hat{\alpha}_{\mathrm{1PN}}$ ($''$)	$\Delta t_{\mathrm{Shapiro}}(10\,r_\oplus)$ (μs)	$\Delta t_{\mathrm{Cassini,rt}}$ (μs)	$\Delta t_{\mathrm{Cassini,rt}}$ with geometric -1 (μs)	$\log_{10}\lvert\gamma_{\mathrm{PPN}} - 1\rvert$
Solar_1Msun_1p005Rsun	6.957000e+05	1.0050	1.751201	1.751190	67.8831	280.9434	261.2415	-30214.67
Solar_1Msun_Cassini_1p06_Rsun	7.378200e+05	1.0600	1.094498	1.094494	67.8830	262.4234	242.7215	-48342.99
Solar_1Msun_Rsun	6.957000e+05	1.0000	0.175119	0.175119	67.8829	—	—	-302139.47

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Row (scale_label)	b (km)	$b/R_\odot$	$\hat{\alpha}_{\rm exact}$ ($''$)	$\hat{\alpha}_{\rm 1PN}$ ($''$)	$\Delta t_{\rm Shapiro}(10\,r_s)$ (μs)	$\Delta t_{\rm Cassini,rt}$ (μs)	$\Delta t_{\rm Cassini,rt}$ with geometric -1 (μs)	$\log_{10}\lvert\gamma_{\rm PPN}-1\rvert$
Solar_1Msun_1AU	1.495979e+08	215.0322	0.008144	0.008144	67.8829	—	—	-6496953.77

24.3.4 3.4. Weak-Lensing Convergence ($\kappa(\theta)$) on Galactic and Cluster Scales: Bridge to Module 2 (Bullet Cluster)

The weak-lensing convergence ($\kappa(\theta) = \frac{1}{2}\nabla^2_\theta\psi(\theta)$) is governed by the line-of-sight integral of the Weyl potential ($\Phi_{\rm lens} = \frac{1}{2}(\tilde{\Psi}+\tilde{\Phi})$):

$$\kappa(\mathbf{b}_\perp) \;=\; \frac{1}{2\,c^2}\,\frac{D_L D_{LS}}{D_S}\int_{-\infty}^{+\infty}\nabla^2_\perp\big(\tilde{\Psi}(\mathbf{b}_\perp,z)+\tilde{\Phi}(\mathbf{b}_\perp,z)\big)\,dz \;=\; \frac{\Sigma_{\rm lens}(\mathbf{b}_\perp)}{\Sigma_{\rm crit}}, \qquad \Sigma_{\rm crit} \;=\; \frac{c^2}{4\pi G}\,\frac{D_S}{D_L D_{LS}}.$$

Let us evaluate what the fixed parameter pair $L = 10$ km, $\alpha_Y = 1$ predicts on galactic (~ 30 kpc, SPARC NGC 3198) and galaxy-cluster (~ 250 kpc, Bullet Cluster 1E 0657-558) scales (`gw170817_and_lensing_summary.json`):

1. **Behavior of the Local Fixed-$L = 10$ km Yukawa Component on kpc–Mpc Scales:**
   - **Exterior field of compact sources (stars, remnants):** At $b = 1$ kpc ($b/L = 3.086\times10^{15}$) and $b = 250$ kpc ($b/L = 7.714\times10^{17}$), the exponential screening factor $e^{-b/L} = 10^{-3.35\times10^{17}} = 0$ exponentially suppresses to zero any local $L = 10$ km Yukawa tail.
   - **Equilibrium local field inside diffuse baryonic media (galactic disk or X-ray intracluster plasma, $\rho_{\rm ICM} \sim 10^{-22}$ kg/m^3):** In a quasi-homogeneous background of density ρ, the local Helmholtz–Yukawa equilibrium field is $\phi_L^{\rm local}(L = 10\text{ km}) = \alpha_Y\gamma_L\rho$:
     - In the outer disk of NGC 3198 ($\rho_{\rm gal} \sim 10^{-24}$ kg/m^3, $R = 30.08$ kpc):

       $$\phi_L^{\rm local}(10\text{ km}) \;=\; 7.426\times10^{-44}, \qquad \frac{|\Phi_N|}{c^2} \;\approx\; \frac{V_{\rm bar}^2}{c^2} \;=\; 5.499\times10^{-8} \;\;\Longrightarrow\;\; \frac{c^2\phi_L^{\rm local}(10\text{ km})}{|\Phi_N|} \;=\; 1.350\times10^{-36}.$$

     - In the hot intracluster medium of the Bullet Cluster ($\rho_{\rm ICM} \sim 10^{-22}$ kg/m^3, $b = 250$ kpc):

       $$\phi_L^{\rm local}(10\text{ km}) \;=\; 7.426\times10^{-42} \;\;\Longrightarrow\;\; \kappa_{\rm lens}^{\rm local\ L=10\ km}(\mathbf{b}_\perp) \;=\; \kappa_{\rm baryon}(\mathbf{b}_\perp) \quad \left(\frac{\kappa_{\rm lens}^{\rm local\ L=10\ km}}{\kappa_{\rm baryon}} = 1.000000000000\right)$$

2. **Physical Conclusion and Boundary Condition for Module 2 (Bullet Cluster 1E 0657-558):**
   - The local $L = 10$ km, $\alpha_Y = 1$ Yukawa interaction is strictly a **strong-field / compact-object ($r \lesssim 100$ km) fifth force** that:
     - satisfies the GW170817 $c_T = c$ bound identically ($(c_T - c)/c = 0$),
     - suppresses scalar dipolar and monopolar radiation across binary-pulsar and LIGO inspiral frequencies via the 4771.35 Hz mass cutoff,
     - satisfies the Solar-System Cassini bound at $b = 1.6\,R_\odot$ with $|1-\gamma_{\rm PPN}| = 0 < 2.3\times10^{-5}$,
     - and simultaneously produces **zero non-GR contribution** on kiloparsec and megaparsec scales ($b/L \sim 10^{16}$–10^{18}), neither to galactic rotation curves (where NGC 3198 at $R = 30.08$ kpc requires a dynamical boost $C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2 = 4.311$ at $\Upsilon_* = 0.5$; `step6`) nor to the Bullet Cluster weak-lensing convergence map (where the galaxy subcluster peak exhibits $\kappa_{\rm obs}/\kappa_{\rm baryon,gal} \approx 0.232/0.0258 \approx 9.0$ at $\Sigma_{\rm crit} \approx 3.10\times10^9\ M_\odot\,{\rm kpc}^{-2}$).
   - For a macroscopic clock field ($\phi_{\rm macro}$, such as the chameleon/nonlocal $L_{\rm eff}(\rho) \sim 10$–$300$ kpc branch analyzed in `step6`, `step11`, `step14`) to enhance weak-lensing convergence ($\kappa > \kappa_{\rm baryon}$), **two conditions must hold simultaneously**:
     1. The metric coupling cannot be purely conformal ($\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$), because the conformal factor cancels in the Weyl potential $(\tilde{\Psi}+\tilde{\Phi})/2$; it requires a **disformal / refractive coupling** ($\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu} + B(\phi)\partial_\mu\phi\partial_\nu\phi$ or effective optical metric potential $\Phi_{\rm lens}^{\rm eff} = \Phi_N - \frac{1}{2}c^2\phi_{\rm macro}$, `step14`); and
     2. The effective range of the scalar field must be environment-dependent ($L_{\rm eff}(\rho) \propto \rho^{-1/2}$) or nonlocal, since the fixed $L = 10$ km Yukawa kernel vanishes identically for $b \gg 10$ km.

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25 Step 24 — Bullet Cluster (1E 0657-558) Gravitational Lensing Convergence $\kappa(\theta)$ and Gas vs. Galaxy Peak Offset Analysis at $L = 10$ km, $\alpha_Y = 1$ (step24_en, V26)

Source file: step24_bullet_cluster_en.md

25.1 I. Physical and Methodological Framework: Synthetic 1D Illustrative Collision-Axis Test of the Bullet Cluster (1E 0657-558)

In step23_gw170817_and_lensing_en.md, we proved that within the canonical V24 (10.5281/zenodo.23680715) and V25 (10.5281/zenodo.23699685) short-range scalar–tensor Yukawa baseline ($L = 10$ km $= 10^4$ m, $\alpha_Y = 1$, $\beta_{\rm Yuk} = 2\tilde{\alpha}_L\alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795775$), the tensor gravitational-wave propagation speed on the purely conformal minimal branch is identically $c_T = c$ ($\Delta c_T/c = 0$), and under the conformal Jordan metric ($\tilde{g}_{\mu\nu} = A^2(\phi_L)g_{\mu\nu} = e^{-2\alpha_Y\phi_L}g_{\mu\nu}$) the linear Weyl lensing potential satisfies $\Phi_{\rm lens} = \frac{1}{2}(\tilde{\Psi} + \tilde{\Phi}) = \Phi_N$. In Step 24 (step24), we apply this framework to a **synthetic 1D illustrative collision-axis model** of the **Bullet Cluster (1E 0657-558)** (Markevitch et al. 2004, *ApJ* 606, 819; Clowe et al. 2006, *ApJ* 648, L109; Bradač et al. 2006, *ApJ* 652, 937).

25.1.1 1.1. Cosmological and Lensing Parameters (1E 0657-558)

- **Lens redshift:** $z_L = 0.296$, angular diameter distance $D_L = 910.0$ Mpc ($1' \leftrightarrow 264.7$ kpc).
- **Characteristic background source redshift:** $z_S \approx 0.8$–1.0 (fiducial value $z_S = 0.85$).
- **Critical surface mass density:**

$$\Sigma_{\rm crit} = \frac{c^2 D_S}{4\pi G D_L D_{LS}} = 3.10 \times 10^9\ M_\odot\,{\rm kpc}^{-2} = 0.6474\ {\rm g\,cm}^{-2}.$$

- **Supersonic merger shock velocity:** $v_{\rm coll} \approx 4500$ km/s $\approx 0.01501\,c$, driving ram-pressure stripping that spatially separates the collisional X-ray intracluster medium (ICM) from the collisionless galaxies.

25.1.2 1.2. Four-Component Baryonic Model Along the Collision Axis ($x \in [-800, +800]$ kpc)

Motivated by the characteristic mass and scale ratios from Chandra X-ray surface brightness maps and HST/VLT/Magellan optical imaging, the baryonic mass distribution along the collision axis ($y = 0$, $x \in [-800, +800]$ kpc) is represented by four analytic components with aperture-normalized ($R_{\rm ap} = 650.0$ kpc) King / modified-Hubble ($\beta = 1$) surface density profiles $\Sigma_i(R) = \Sigma_{0,i}(1 + R^2/R_{c,i}^2)^{-1}$, where $\Sigma_{0,i} = \frac{M_i}{\pi R_{c,i}^2 \ln(1+R_{\rm ap}^2/R_{c,i}^2)}$:

Component ID	Physical Description	Centroid $x_{c,i}$ (kpc)	Aperture Mass M_i ($M_\odot$)	Core Radius $R_{c,i}$ (kpc)	Central $\Sigma_{0,i}$ ($M_\odot$/kpc^2)	Central $\kappa_{0,i} = \Sigma_{0,i}/\Sigma_{\rm crit}$
gal_main	Main Cluster Collisionless Galaxies (Stars + Compact Remnants)	-350.0	5.50e12	120.0	3.9329e7	0.012687
gas_main	Main Cluster Collisional X-ray ICM Gas (Chandra, ram-pressure stripped)	-170.0	2.60e13	150.0	1.3638e8	0.043995

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Component ID	Physical Description	Centroid $x_{c,i}$ (kpc)	Aperture Mass M_i ($M_\odot$)	Core Radius $R_{c,i}$ (kpc)	Central $\Sigma_{0,i}$ ($M_\odot$/kpc^2)	Central $\kappa_{0,i} = \Sigma_{0,i}/\Sigma_{\rm crit}$
`gas_sub`	Subcluster ("Bullet") Shocked X-ray ICM Gas	`+140.0`	`6.50e12`	`75.0`	`8.4628e7`	`0.027300`
`gal_sub`	Subcluster ("Bullet") Collisionless Galaxies (160 kpc ahead of shock)	`+300.0`	`1.80e12`	`65.0`	`2.9296e7`	`0.009450`

- **Total baryonic mass budget in the aperture:**

$$M_{\rm gas,tot} = 3.25\times10^{13}\ M_\odot, \qquad M_{\rm gal,tot} = 7.30\times10^{12}\ M_\odot, \qquad M_{\rm bary,tot} = 3.98\times10^{13}\ M_\odot, \qquad \frac{M_{\rm gas,tot}}{M_{\rm gal,tot}} = 4.4521.$$

- **Observationally motivated peak positions and synthetic 1D reference curve (`kappa_synth_target_1D`):** In weak + strong lensing reconstructions (Clowe et al. 2006; Bradač et al. 2006), the convergence peaks lie near the collisionless galaxy centroids ($x^{\rm obs}_{\kappa,\rm main} \approx -350.0$ kpc, $\kappa^{\rm obs}_{\rm peak,main} \approx 0.320$; $x^{\rm obs}_{\kappa,\rm sub} \approx +300.0$ kpc, $\kappa^{\rm obs}_{\rm peak,sub} \approx 0.220$) rather than the dominant X-ray gas centroids ($x = -170$ kpc and $+140$ kpc). **Methodological Scope and Caveat (V26):** The `kappa_synth_target_1D` column in `bullet_cluster_convergence_profile.csv` is **not an observational data series**, but a **self-constructed 1D synthetic reference curve**:

$$\kappa_{\rm synth}(x) = \kappa_{\rm GR,bary}(x) + \frac{w_{\rm synth,main}\,\Sigma_{\rm gal,main}(x) + w_{\rm synth,sub}\,\Sigma_{\rm gal,sub}(x)}{\Sigma_{\rm crit}}, \qquad (w_{\rm synth,main} = 22.9765,\ w_{\rm synth,sub} = 21.3095),$$

built from the model's own four King profiles with two galaxy weights tuned to reproduce the illustrative peak heights at $x = -350$ kpc ($\kappa = 0.320$) and $x = +300$ kpc ($\kappa = 0.220$). Accordingly, the metric reported in `synth_mse_norm_vs_target`,

$$M_{\rm synth} \equiv \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\kappa(x_i) - \kappa_{\rm synth}(x_i)}{\sigma_{\rm ref}}\right]^2 \qquad (\sigma_{\rm ref} = 0.020,\ N = 321),$$

is a **normalized mean-squared deviation from this synthetic 1D target curve**, **NOT** an observational χ^2_ν fit to Clowe et al. (2006) or Bradač et al. (2006) data. A genuine observational χ^2 fit requires a published 2D convergence map, full error covariance matrix, pre-registered spatial selection, and proper subtraction of fitted degrees of freedom ($N_{\rm dof} = N_{\rm bins} - p$).

25.2 II. Test 1 — Pure Local Single-Scale Yukawa Model at Fixed $L = 10$ km, $\alpha_Y = 1$ (Strict Cluster-Scale Falsification / No-Go Result)

We first evaluate the exact 3D scalar field $\phi_L^{\rm local}(\mathbf{r})$ and projected 2D lensing convergence $\kappa_{\rm local}(x)$ produced by the V24/V25 local Yukawa model at fixed $L = 10$ km $= 3.24077929 \times 10^{-16}$ kpc and $\alpha_Y = 1$.

25.2.1 2.1. Algebraic Contact Limit on Cluster Scales ($R_c \sim 65$–150 kpc $\sim (2.0$–$4.6)\times10^{17}\,L$)

In the V23–V25 sign convention ($\phi_L > 0$ near matter sources), the static 3D Helmholtz–Yukawa field equation is:

$$\left(\nabla^2 - \frac{1}{L^2}\right)\phi_L(\mathbf{r}) = -\alpha_{\rm std}\,\frac{4\pi G}{c^2}\,\rho_b(\mathbf{r}), \qquad \alpha_{\rm std} = \frac{\alpha_Y}{4\pi} \approx 0.0795775.$$

Because the cluster core radii satisfy $R_{c,i} \in [65, 150]$ kpc, the dimensionless scale ratio is:

$$\frac{R_{c,i}}{L} \in \left[2.006\times10^{17},\ 4.628\times10^{17}\right] \implies \left(\frac{L}{R_{c,i}}\right)^2 \in \left[4.668\times10^{-36},\ 2.486\times10^{-35}\right].$$

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Spatial derivatives ($|\nabla^2\phi_L| \sim R_c^{-2}|\phi_L|$) are suppressed by more than 35 orders of magnitude relative to the mass term $-L^{-2}\phi_L$, reducing the 3D scalar field to the algebraic contact limit:

$$\left|\phi_L^{\rm local}(\mathbf{r})\right| \simeq \alpha_{\rm std}\, L^2\, \frac{4\pi G}{c^2}\, \rho_b(\mathbf{r}) \in \left[10^{-44},\ 3.086\times 10^{-42}\right].$$

For individual compact objects and stars inside the galaxies, each star's $L = 10$ km Yukawa halo decays as $e^{-r/(10\ \rm km)}$, falling below $e^{-10} \approx 4.5\times 10^{-5}$ within 100 km ($< 1.5\times 10^{-4}\, R_\odot$, or $< 3\times 10^{-18}$ kpc) and vanishing as $\exp(-10^{17}) = 0$ on kiloparsec scales.

25.2.2 2.2. Conformal vs. Disformal Lensing Convergence at $L = 10$ km

1. **Conformal Jordan Branch (`step14` Branch II-Conf, `step23`):** Under $\tilde g_{\mu\nu} = A^2(\phi_L) g_{\mu\nu} = e^{-2\alpha_Y \phi_L} g_{\mu\nu}$, the scalar field cancels identically out of the linear Weyl potential ($\Phi_{\rm lens}^{\rm conf} = \Phi_N$), while the exterior scalar energy cloud of compact objects ($\Delta M_{\phi,\rm ext}^{\rm curved}/M_{\rm surf} = +1.55\times 10^{-3}$ for neutron stars, and $\lesssim 10^{-16}$ for main-sequence stars) is already absorbed into the stars' asymptotic ADM masses $M_{\rm gal}$. Consequently:

$$\Delta\kappa^{\rm conf}_{\rm local\ \ L=10\ km}(x) = 0 \quad \text{(identically)}.$$

2. **Disformal Refractive Clock Branch (`step14` Branch II-Disf):** If on a separate disformal optical branch $\Phi_{\rm lens}^{\rm disf} = \Phi_N - \frac{1}{2}c^2\phi_L$, integrating along the line of sight $z \in (-\infty, +\infty)$ and applying the 2D transverse Laplacian yields the 2D Fourier filter $\frac{k_\perp^2}{k_\perp^2 + L^{-2}} \simeq L^2 k_\perp^2 \leftrightarrow -L^2\nabla_\perp^2$, giving:

$$\Delta\kappa^{\rm disf}_{\rm local\ \ L=10\ km}(x) = -\frac{\alpha_{\rm std}}{2}\, L^2\, \nabla_\perp^2 \kappa_{\rm GR,bary}(x), \qquad \max_x \left|\frac{\Delta\kappa^{\rm disf}_{\rm local}}{\kappa_{\rm GR,bary}}\right| = 2.171\times 10^{-36} \quad (\text{and } < 10^{-40} \text{ in the outer halo}).$$

25.2.3 2.3. Transparent Scientific Verdict for Test 1

On both metric coupling branches:

$$\kappa_{\rm local\ \ L=10\ km}(x) = \kappa_{\rm GR,bary}(x) = \frac{\Sigma_{\rm gas}(x) + \Sigma_{\rm gal}(x)}{\Sigma_{\rm crit}}.$$

- The convergence peaks remain locked to the **X-ray gas centroids**: in the 4-component global superposition they sit at $x^{\rm local}_{\rm peak,main} = -175.44$ kpc ($\kappa^{\rm local}_{\rm peak,main} = 0.04503$) and $x^{\rm local}_{\rm peak,sub} = +137.01$ kpc ($\kappa^{\rm local}_{\rm peak,sub} = 0.03697$), or in the shock-separated / steeper-outer-gas subcluster-pair decomposition at $x^{\rm local}_{\rm peak,main} = -170.7$ kpc and $x^{\rm local}_{\rm peak,sub} = +142.3$ kpc ($\kappa^{\rm local}_{\rm peak,main} = 0.0450$).
- The main peak amplitude is 7.11× **smaller** than the $\kappa = 0.320$ reference peak, and the peaks are **displaced by** 157.7–179.3 **kpc** from the galaxy centroids (−350 kpc and +300 kpc) toward the collisional X-ray gas ($M_{\rm synth} = 32.089$).
- **Conclusion: At fixed $L = 10$ km and $\alpha_Y = 1$, the short-range Yukawa fifth force is purely a compact-object/neutron-star effect and produces zero macroscopic modification to Bullet Cluster lensing ($\Delta\kappa/\kappa_{\rm bary} \le 2.17\times 10^{-36}$; $< 10^{-40}$ in the outer halo).**

25.3 III. Test 2 — Phenomenological Macroscopic Extensions on the Synthetic 1D Test: What It Takes to Move the $\kappa(x)$ Peak from the X-ray Gas to the Galaxies

We now examine quantitatively within the synthetic 1D King model (`step11`, `step14` Branch II-Disf: $\Phi_{\rm lens} = \Phi_N - \frac{1}{2}c^2\phi_{\rm macro}$, with illustrative cluster Helmholtz scale $L_{\rm macro} = 300.0$ kpc) the two macroscopic sourcing cases.

25.3.1 3.1. Case 2A (`Macro_unscreened_bulk_baryon`) — Static Macroscopic Kernel Depending Only on Smoothed Bulk Baryon Density ($\bar\rho_b = \rho_{\rm gas} + \rho_{\rm gal}$)

Suppose a macroscopic scalar field is sourced universally by total baryon density with a uniform coupling $\eta_{\rm bulk}$:

$$\left(\nabla^2 - L_{\rm macro}^{-2}\right)\phi_{\rm macro}(\mathbf{r}) = -\frac{4\pi G}{c^2}\,\eta_{\rm bulk}\left[\rho_{\rm gas}(\mathbf{r}) + \rho_{\rm gal}(\mathbf{r})\right], \qquad \widetilde\Sigma_{\phi,\rm bulk}(\mathbf{k}_\perp) = \eta_{\rm bulk}\,\frac{k_\perp^2}{k_\perp^2 + L_{\rm macro}^{-2}}\,\widetilde\Sigma_b(\mathbf{k}_\perp).$$

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Calibrating $\eta_{\rm bulk} = 8.6523$ boosts the main convergence peak by 7.11× to match $\kappa_{\rm peak,main} = 0.3200$. **However**, because $M_{\rm gas,tot}/M_{\rm gal,tot} = 4.4521$, the convergence peaks **remain locked to the X-ray gas**:

$x^{\rm bulk}_{\rm peak,main} = -176.02$ kpc (offset from galaxies: +173.98 kpc), $x^{\rm bulk}_{\rm peak,sub} = +138.35$ kpc (offset from galaxies: −161.65 kpc

($M_{\rm synth} = 20.723$). This demonstrates that **no static modified-gravity or scalar–tensor kernel sourced solely by smoothed bulk baryon density ($\rho_{\rm gas} + \rho_{\rm gal}$) can shift the convergence peaks from the dominant X-ray gas to the collisionless galaxies.**

25.3.2 3.2. Case 2B (`Macro_collisionless_compact_clock`) — Phenomenological Collisionless / Compact-Well Sourcing Hypothesis

For a macroscopic field to center its lensing convergence peaks on the collisionless galaxies without cold dark matter (CDM) particles, the macroscopic sector would need one of two physical properties:

1. **Hypothesis 2B-1 — Compactness / Deep-Potential-Well Clock Sourcing:** In the diffuse X-ray ICM plasma ($n_e \sim 10^{-3}$ cm^{-3}), small-scale metric-gradient variance vanishes ($\langle(\nabla\nu)^2\rangle_{\rm micro} \approx 0$), whereas in the galaxy component baryons are bound inside stellar and compact-remnant potential wells. If the effective source of a macroscopic clock field is weighted by small-scale gradient energy or local compactness, it couples predominantly to the collisionless ($\sigma_{\rm coll} \approx 0$) stellar/compact-remnant population.
2. **Hypothesis 2B-2 — Non-Equilibrium / Retarded Scalar Memory:** If the macroscopic field has a dynamical relaxation timescale $\tau_{\rm relax} \gtrsim 150$ Myr longer than the core-passage time $t_{\rm cross} = \Delta x/v_{\rm coll} \approx 180$ kpc/(4500 km/s) $\approx$ 39.1 Myr, the scalar halo can travel ballistically with the collisionless galaxies during the supersonic merger.

25.3.3 3.3. Model-Dependent Threshold Calculation for $\eta_{\rm gal}/\eta_{\rm gas}$ and Methodological Caveats

With phenomenological macroscopic refractive coupling weights $\eta_{\rm gal}$ and $\eta_{\rm gas}$ in the chosen 1D King geometry ($R_{c,i} \in [65, 150]$ kpc) at $L_{\rm macro} = 300.0$ kpc:

1. **Peak-Dominance Threshold ($\kappa(-350\ \mathbf{kpc}) \geq \kappa(-170\ \mathbf{kpc})$ and $\kappa(+300\ \mathbf{kpc}) \geq \kappa(+140\ \mathbf{kpc})$):**
   - Direct surface-density ratio (without Helmholtz filtering):
   $$\frac{\Sigma_{\rm gas,main}(-170)}{\Sigma_{\rm gal,main}(-350)} = 3.4592, \qquad \frac{\Sigma_{\rm gas,sub}(+140)}{\Sigma_{\rm gal,sub}(+300)} = 2.8895.$$
   - Including $L_{\rm macro} = 300$ kpc Helmholtz–Yukawa filtering (at reference $\eta_{\rm gas} = 1$):
   $$\left(\frac{\eta_{\rm gal}}{\eta_{\rm gas}}\right)_{\rm dom,main} = 5.1204, \qquad \left(\frac{\eta_{\rm gal}}{\eta_{\rm gas}}\right)_{\rm dom,sub} = 5.0972.$$
2. **< 5.0 kpc Centroid-Offset Threshold ($|x_{\mathbf{peak,main}} - (-350\ \mathbf{kpc})| \leq 5.0\ \mathbf{kpc}$):**
   - To prevent the X-ray gas tail in this 1D King profile from pulling the main convergence peak by more than 5.0 kpc away from the galaxy centroid at −350.0 kpc, the required ratio is:
   $$\left(\frac{\eta_{\rm gal}}{\eta_{\rm gas}}\right)_{|\Delta x| \leq 5\ \rm kpc} \geq 25.6650.$$
3. **Calibrated Case 2B Illustrative Model (`Macro_collisionless_compact_clock`, $\eta_{\rm gas} = 0$, $\eta_{\rm gal,main} = 31.7448$, $\eta_{\rm gal,sub} = 24.1556$):**
   - Main cluster peak: $x_{\rm peak,main} = -347.86$ kpc (centroid offset from galaxies: $+2.14\ \mathbf{kpc} < 5\ \mathbf{kpc}$), $\kappa_{\rm peak,main} = 0.32011$ ($\kappa(-350\ {\rm kpc}) = 0.32000$).
   - Subcluster peak: $x_{\rm peak,sub} = +299.45$ kpc (centroid offset from galaxies: $-0.55\ \mathbf{kpc} < 5\ \mathbf{kpc}$), $\kappa_{\rm peak,sub} = 0.22002$ ($\kappa(+300\ {\rm kpc}) = 0.22000$).
   - Deviation from the synthetic 1D target curve (`kappa_synth_target_1D`): RMS($\Delta\kappa$) = 0.02559, normalized synthetic mean-squared deviation $M_{\rm synth} = 1.637$ (compared with $M_{\rm synth} = 32.089$ for the local $L = 10$ km model and $M_{\rm synth} = 20.723$ for the Case 2A bulk-baryon model).
4. **Strict Theoretical and Methodological Delineation (V26):**

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- The peak-dominance ratio $\eta_{\rm gal}/\eta_{\rm gas} \geq 5.1204$ and the < 5 kpc centroid-offset ratio $\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$ are **model-dependent thresholds specific to the chosen 1D King geometry ($R_{c,i} = 65$–150 kpc) and Helmholtz scale $L_{\rm macro} = 300$ kpc**, not a general model-independent physical theorem.
- Setting $\eta_{\rm gas} = 0$ and fitting two separate galactic weights ($\eta_{\rm gal,main} = 31.7448 \neq \eta_{\rm gal,sub} = 24.1556$) are **phenomenological, illustrative assumptions that are not derived from the covariant scalar–tensor action**.

25.4 IV. Synthetic 1D Illustrative Peak-Offset and Convergence Profile Tables

25.4.1 4.1. Summary of Peak Positions, Amplitudes, and Deviations from the Synthetic 1D Target Curve Across All Four Models (`bullet_cluster_peak_offsets_summary.csv` — Synthetic 1D Illustrative Test)

Model (`model_name`)	$x_{\rm peak,main}$ (kpc)	$\kappa_{\rm peak,main}$	$x_{\rm peak,sub}$ (kpc)	$\kappa_{\rm peak,sub}$	$\Delta x_{\rm main,gal}$ (kpc)	$\Delta x_{\rm main,gas}$ (kpc)	$\Delta x_{\rm sub,gal}$ (kpc)	$\Delta x_{\rm sub,gas}$ (kpc)	RMS $\Delta\kappa_{\rm synth}$	$M_{\rm synth}$ (`synth_mse_norm`
GR_baryon_only	-175.44	0.04503	+137.01	0.03697	+174.56	-5.44	-162.99	-2.99	0.11329	32.089
TAL_local_Yukawa_DM_only	-175.44	0.04503	+137.01	0.03697	+174.56	-5.44	-162.99	-2.99	0.11329	32.089
Macro_unscreened_DM_only	-176.02	0.03202	+138.35	0.26531	+173.98	-6.02	-161.65	-1.65	0.09104	20.723
Macro_collisionless_2comp	-147.86	0.32011	+299.45	0.22002	+2.14	-177.86	-0.55	+159.45	0.02559	1.637

(Note: $M_{synth} = \frac{1}{N}\sum_{i=1}^{N}[(\kappa(x_i) - \kappa_{synth}(x_i))/\sigma_{ref}]^2$ is the normalized mean-squared deviation from the self-constructed 1D `kappa_synth_target_1D` reference curve with $\sigma_{ref} = 0.020$, not an observational χ^2_ν.)

25.4.2 4.2. Projected Surface Density and $\kappa(x)$ Profile at Representative Collision-Axis Positions (`bullet_cluster_convergence_profile.csv` — Synthetic 1D Illustrative Test)

x (kpc)	$\Sigma_{\rm gas}$ ($M_\odot$/kpc^2)	$\Sigma_{\rm gal}$ ($M_\odot$/kpc^2)	$\kappa_{\rm GR,bary}$	$\kappa_{\rm local}$ L=10 km	$\Delta\kappa_{\rm local,disf}$	$\kappa_{\rm Macro,2A}$ (bulk)	$\kappa_{\rm Macro,2B}$ (clock)	$\kappa_{\rm synth,1D}$ (`kappa_synth_targ`
-600.0	1.4233e+07	6.8239e+06	0.006793	0.006793	-5.740e-40	0.010277	0.028049	0.062017
-450.0	2.8832e+07	2.1245e+07	0.016154	0.016154	+2.251e-42	0.076874	0.165259	0.188492
-350.0	5.2453e+07	3.5918e+07	0.028507	0.028507	+1.240e-38	0.178665	0.320000	0.320000
-260.0	9.3503e+07	2.3192e+07	0.037644	0.037644	+8.951e-39	0.254340	0.203216	0.225535
-175.0	1.2766e+08	1.1934e+07	0.045031	0.045031	+2.869e-38	0.320000	0.104232	0.141178
-170.0	1.2794e+08	1.1514e+07	0.044984	0.044984	+2.882e-38	0.319487	0.100405	0.137693
-80.0	9.9460e+07	6.7119e+06	0.034249	0.034249	+6.544e-39	0.219456	0.048876	0.087474
+0.0	7.2882e+07	5.0651e+06	0.025144	0.025144	-3.132e-39	0.138610	0.027722	0.064214
+70.0	7.9995e+07	4.8621e+06	0.027373	0.027373	+3.408e-40	0.166192	0.029235	0.063429
+140.0	1.0829e+08	6.1784e+06	0.036924	0.036924	+8.015e-38	0.265185	0.049209	0.080507
+220.0	5.5598e+07	1.3197e+07	0.022192	0.022192	-2.748e-39	0.134507	0.088322	0.110826
+300.0	2.6688e+07	3.0558e+07	0.018466	0.018466	+3.559e-38	0.110815	0.220000	0.220000
+450.0	1.1510e+07	5.4294e+06	0.005464	0.005464	-9.488e-40	0.006652	0.021648	0.042188
+600.0	6.7045e+06	1.8771e+06	0.002768	0.002768	-1.602e-40	-0.004653	0.000158	0.015925

25.5 V. Reproducibility and Generated Artifacts

- **Numerical audit script:** `bullet_cluster_lensing_audit.py` (synced to `code/`).
- **1D collision-axis synthetic convergence profile (321 grid points, $x \in [-800, +800]$ kpc):** `bullet_cluster_convergence_profile.csv` (synced to root, `code/`, `csv/`, and `data/v23/`).
- **Peak-offset and normalized synthetic deviation ($M_{\rm synth}$) summary table:** `bullet_cluster_peak_offsets_summary` (synced to root, `code/`, `csv/`, and `data/v23/`).

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- **Structured JSON summary:** `bullet_cluster_audit_summary.json` (synced to root, `code/`, `csv/`, and `data/v23/`).

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26 Step 25 (V26): Cosmic Microwave Background (CMB) and Cosmological Background — Separating Numerically Integrated FLRW Background Quantities ($H(z)$, $r_s(z_*)$, $D_M(z_*)$, θ_*) from Fixed Phase/Literature Peak-Height Inputs, Negligibility of the Local $L = 10$ km Yukawa $G_{\mathrm{eff}}(a,k)$ Modification, and the Conditional Negative-ρ_Λ Separate-Branch Hypothesis

Source file: `step25_cmb_and_cosmology_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-27
File Path: `scalar-tensor-yukawa-fit/step25_cmb_and_cosmology_en.md`
Previous Steps: `step1_dilaton_form_en.md` – `step24_bullet_cluster_en.md`
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Local Sources and Runnable Code:

- Runnable Python script: `scalar-tensor-yukawa-fit/cmb_and_cosmology_audit.py`
- Generated output files: `cosmological_background_models.csv`, `cmb_geff_and_acoustic_scales.csv`, `cmb_and_cosmology_summary.json` (and copies in `code/`, `csv/`, and `data/v23/`)
- Referenced theoretical steps: `step14_gravitational_lensing_null_geodesics_cosmological_background_en.md` (Section 3: homogeneous FLRW attractor), `step17_macroscopic_clock_field_closure_sparc_compatibility_en.md` (Section 4: cosmological background and limit), `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability` `step23_gw170817_and_lensing_en.md`, `step24_bullet_cluster_en.md`

26.1 1. Objective and V26 Methodological Delineations

In Steps 23 and 24 we established that the local finite-range ($L = 10$ km, $m_\phi c^2 \approx 1.973 \times 10^{-11}$ eV, $\alpha_Y = 1$, $\beta_{\mathrm{Yuk}} = 1/(4\pi) \approx 0.079577$) `yukawa_action_el` Einstein–Hilbert + canonical scalar branch closed in Step 22 satisfies the GW170817 $c_T = c$ bound and Solar-System PPN/Cassini constraints on the purely conformal, universally coupled Maxwell branch, while its local $L = 10$ km Yukawa fifth force is exponentially suppressed by $\exp(-r/L) \sim 10^{-10^{18}}$ on galaxy-cluster scales ($R \sim 100$–1000 kpc).

In this Step 25 (**V26**), we audit the cosmological (FLRW) background and Cosmic Microwave Background (CMB) acoustic scales via `cmb_and_cosmology_audit.py`, enforcing **three strict V26 methodological and theoretical delineations**:

1. **Explicit separation of numerically integrated FLRW background quantities from fixed analytic phase-shift and literature peak-height inputs (Review Point 2):**
   - The script `cmb_and_cosmology_audit.py` **numerically integrates** (via adaptive Gauss–Kronrod quadrature, `epsrel=1e-12`) the homogeneous FLRW Hubble function $H(z) = H_0 E(z)$, the present-day deceleration parameter q_0, the turnaround scale factor $a_{\max}$ (z_{turn}) where applicable, the comoving sound horizon $r_s(z_*)$ at recombination ($z_* = 1089.80$), the comoving line-of-sight and angular distances $\chi(z_*)$ and $D_M(z_*)$, the physical angular diameter distance $D_A(z_*)$, the acoustic angle $\theta_* = r_s(z_*)/D_M(z_*)$, the acoustic multipole scale $\ell_A = \pi/\theta_*$, and the linear perturbation fifth-force modification $\Delta G_{\mathrm{eff}}(a,k)/G$ of the local $L = 10$ km Yukawa mode.
   - By contrast, the multipole columns $\ell_m^{\mathrm{est}} = \ell_A(m - \phi_m)$ (`l1_analytic_phase_est`, `l2_analytic_phase_est`, `l3_analytic_phase_est`) are **analytic phase-shift template estimates** evaluated from fixed external phase shifts (ϕ_1, ϕ_2, ϕ_3, Hu & Sugiyama 1996 / Doran & Lilley 2002), and the peak-height ratio columns (`H2_H1_lit_benchmark_input`, `H3_H2_lit_benchmark_input`) are **fixed external literature benchmark inputs** (`0.458` and `0.985` assigned to every $\Omega_c > 0$ row, and `0.312` and `0.415` assigned to every $\Omega_c = 0$ row), **not** outputs of a coupled Einstein–Boltzmann solver (CAMB/CLASS) or a spectrum fit (`boltzmann_spectrum_computed = False`).
2. **Exact scope of Planck 2018 compatibility and the input Ω_c/Ω_b ratio (Review Point 2):**

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- The ratio $\Omega_c h^2/\Omega_b h^2 = 0.1200/0.02237 = 5.364327224$ ($\Omega_c/\Omega_b = 0.264157/0.049243$) is **the input Planck 2018 CDM-to-baryon density ratio**, **not** a theoretical prediction derived from the TAL clock sector.
- Claims of "Planck 2018 compatibility" in this step are restricted **strictly** to the fact that the local $L = 10$ km Yukawa branch produces a negligible homogeneous background energy density ($\Omega_{\phi,\mathrm{can}}(z_*) \approx 9.85 \times 10^{-38}$) and a negligible linear perturbation fifth-force correction ($\Delta G_{\mathrm{eff}}/G = 2.50 \times 10^{-36}$ at the first acoustic peak $\ell_1 = 220$ and 4.65×10^{-34} in the Silk damping tail $\ell = 3000$), lying more than 31–33 orders of magnitude below the Planck 2018 sensitivity threshold and therefore leaving an input ΛCDM background undisturbed. Proving that a macroscopic clock sector can replace particle CDM across the acoustic peaks requires a closed covariant action and a full coupled Einstein–Boltzmann perturbation calculation.

3. **Conditional separate-branch status of negative $\rho_\Lambda = -5.96\times10^{-27}$ kg/m^3 and its positive $\rho_{\mathrm{clock,cosmo}} = +2|\rho_\Lambda|$ compensation (Review Point 3):**
   - In the minimal canonical action of Steps 22–23, the scalar potential is $V(\phi_L) = +\frac{1}{2}m_\phi^2\phi_L^2 \geq 0$, so a homogeneous static scalar field ($\partial_\mu\phi_L = 0$) carries a **positive** energy density in its canonical stress–energy tensor $T^{(\phi)}_{\mu\nu}$ ($\rho_{\phi,\mathrm{can}} = +V(\phi_L)/c^2 \geq 0$).
   - The negative vacuum energy density $\rho_\Lambda = -V(\phi_{\mathrm{cosmo}})/c^2 = -5.96 \times 10^{-27}$ kg/m^3 ($\Omega_\Lambda = -0.6851$) discussed in earlier manuscripts and its compensating positive cosmological clock background $\rho_{\mathrm{clock,cosmo}} = +2|\rho_\Lambda| = +1.192 \times 10^{-26}$ kg/m^3 do not follow from the minimal `yukawa_action_el` field equations of Steps 22–23. They are therefore treated strictly as a **conditional separate-branch hypothesis** (`conditional_separate_branch_hypothesis`) until an explicit action-level origin, stress–energy tensor, Bianchi identity, and initial condition are provided for both the minus sign and the compensating positive term. The compensated background ($\rho_{\mathrm{DE,eff}} = \rho_\Lambda + \rho_{\mathrm{clock,cosmo}} = +5.96 \times 10^{-27}$ kg/m^3) recovers the input ΛCDM background by algebraic construction rather than as an independent cosmological prediction.

26.2 2. Cosmological (FLRW) Behavior of the Local $L = 10$ km Yukawa Branch and Linear $G_{\mathrm{eff}}(a,k)$ Modification

26.2.1 2.1. Homogeneous Quasi-Static Attractor and Positive Canonical Energy Density

In a spatially flat or curved FLRW metric ($ds^2 = -c^2dt^2 + a^2(t)\gamma_{ij}dx^idx^j$), the unified Step 23 action $S[\phi_L]$ ($\tilde{\alpha}_L = \alpha_Y/(8\pi)$, $A = \exp(-\alpha_Y\phi_L)$) gives for the homogeneous mode $\bar{\phi}_L(t)$ the wave-operator form (both sides in m^{-2}):

$$\Big(-\frac{1}{c^2}\partial_t^2 - \frac{3H}{c^2}\partial_t - \frac{1}{L^2}\Big)\bar{\phi}_L \;=\; -\frac{\alpha_Y^2 G}{c^4}\,\bar{\varepsilon}_m \;=\; -\frac{\alpha_Y^2 G}{c^2}\,\bar{\rho}_m.$$

Here the **mass density** $\bar{\rho}_m(t)$ [kg m^{-3}] and the **energy density** $\bar{\varepsilon}_m(t) \equiv \bar{\rho}_m(t)c^2$ [J m^{-3}] are kept distinct. Multiplying by $-c^2$, the homogeneous time ODE has SI dimension s^{-2} on both sides:

$$\ddot{\bar{\phi}}_L + 3H(t)\dot{\bar{\phi}}_L + \frac{c^2}{L^2}\bar{\phi}_L \;=\; \alpha_Y^2\, G\,\bar{\rho}_m(t) \;=\; \frac{\alpha_Y^2 G}{c^2}\,\bar{\varepsilon}_m(t) \quad [\mathrm{s}^{-2}], \qquad \omega_\phi \;\equiv\; \frac{c}{L} \;=\; 2.99792458 \times 10^4\ \mathrm{s}^{-1}.$$

Dimension check: $[G\bar{\rho}_m] = (\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2})(\mathrm{kg\,m}^{-3}) = \mathrm{s}^{-2} = [\ddot{\bar{\phi}}_L] = [(c^2/L^2)\bar{\phi}_L]$ ($\bar{\phi}_L$ dimensionless). (The former right-hand side $(4\pi G\alpha_Y/c^2)\bar{\rho}_m$ had dimension m^{-2} and was wrong.) The V25 standard bookkeeping field $\bar{\phi}_{L,\mathrm{std}} \equiv \bar{\phi}_L/(4\pi)$ obeys the same equation with source $\alpha_{\mathrm{std}}G\bar{\rho}_m = (\alpha_Y/4\pi)G\bar{\rho}_m$ (at $\alpha_Y = 1$). Because the scalar mass frequency exceeds the present-day Hubble rate by 22 orders of magnitude (44 in ω_ϕ^2/H_0^2) and the recombination Hubble rate ($H(z_*) = 23\,179.39\,H_0 \approx 5.063 \times 10^{-14}$ s^{-1}) by 17 orders of magnitude:

$$\frac{\omega_\phi}{H_0} \;=\; 1.372497 \times 10^{22}, \qquad \frac{\omega_\phi^2}{H^2(z_*)} \;=\; 3.506055 \times 10^{35} \;\gg\; 1,$$

the homogeneous field adiabatically tracks the quasi-static attractor ($\gamma_L = GL^2/c^2 = 7.42616027 \times 10^{-20}$ m^3 kg^{-1}, $\bar{\rho}_m(z) = \Omega_{m,0}\,\frac{3H_0^2}{8\pi G}(1+z)^3$); both sides are identically equal:

$$\boxed{\bar{\phi}_{\mathrm{attr,unit}}(z) \;=\; \frac{\alpha_Y^2 GL^2}{c^2}\bar{\rho}_m(z) \;=\; \alpha_Y^2\gamma_L\bar{\rho}_m(z) \;=\; \frac{3}{8\pi}\,\alpha_Y^2\,\Omega_{m,0}\Big(\frac{H_0 L}{c}\Big)^2(1+z)^3,}$$

$$\boxed{\bar{\phi}_{\mathrm{attr,V25_std}}(z) \;=\; \frac{\bar{\phi}_{\mathrm{attr,unit}}}{4\pi} \;=\; \alpha_{\mathrm{std}}\gamma_L\bar{\rho}_m(z) \;=\; \frac{3}{32\pi^2}\,\alpha_Y\,\Omega_{m,0}\Big(\frac{H_0 L}{c}\Big)^2(1+z)^3 \quad (\alpha_Y = 1).}$$

Values from `cmb_and_cosmology_summary.json` (`quasi_static_attractor_epochs`, `attractor_background_model_today`) ($H_0 = 67.4$ km/s/Mpc, $\Omega_{m,0} = 0.314820$, $H_0L/c = 7.285991 \times 10^{-23}$, $\bar{\rho}_{m,0} = 2.686310 \times 10^{-27}$ kg/m^3, $\rho_{\mathrm{crit},0} = 8.532855 \times 10^{-27}$ kg/m^3, $\alpha_Y = 1$, $L = 10$ km):

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- **Today ($z = 0$):** `phi_attr_unit_kernel` = LHS $\alpha_Y^2 \gamma_L \bar{\rho}_m = \mathbf{1.994897 \times 10^{-46}}$, RHS $\frac{3}{8\pi}\Omega_{m,0}(H_0 L/c)^2(1+z)^3 = \mathbf{1.994897 \times 10^{-46}}$; `phi_attr_V25_std` = LHS $\mathbf{1.587488 \times 10^{-47}}$, RHS $\mathbf{1.587488 \times 10^{-47}}$.
  - Positive canonical energy density (standard normalization): $\rho_{\phi,\mathrm{can,std}} = \frac{c^2}{8\pi G L^2}\bar{\phi}^2_{\mathrm{attr,V25_std}} = \mathbf{1.350258 \times 10^{-76}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,std}} = \rho_{\phi,\mathrm{can,std}}/\rho_{\mathrm{crit},0} = \mathbf{1.582423 \times 10^{-50}}$.
  - Positive canonical energy density (unit-kernel (V25 ϕ_L) normalization): $\rho_{\phi,\mathrm{can,unit}} = \frac{c^2}{16\pi G \tilde{\alpha}_L L^2}\bar{\phi}^2_{\mathrm{attr,unit}} = \mathbf{2.679455 \times 10^{-73}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,unit}} = \mathbf{3.140162 \times 10^{-47}}$.
- **At recombination ($z_* = 1089.80$):** `phi_attr_unit_kernel` = LHS $\alpha_Y^2 \gamma_L \bar{\rho}_m = \mathbf{2.589142 \times 10^{-37}}$, RHS $\frac{3}{8\pi}\Omega_{m,0}(H_0 L/c)^2(1+z)^3 = \mathbf{2.589142 \times 10^{-37}}$; `phi_attr_V25_std` = LHS $\mathbf{2.060373 \times 10^{-38}}$, RHS $\mathbf{2.060373 \times 10^{-38}}$.
  - Positive canonical energy density (standard normalization): $\rho_{\phi,\mathrm{can,std}} = \frac{c^2}{8\pi G L^2}\bar{\phi}^2_{\mathrm{attr,V25_std}} = \mathbf{2.274509 \times 10^{-58}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,std}} = \rho_{\phi,\mathrm{can,std}}/\rho_{\mathrm{crit}}(z_*) = \mathbf{4.961230 \times 10^{-41}}$.
  - Positive canonical energy density (unit-kernel (V25 ϕ_L) normalization): $\rho_{\phi,\mathrm{can,unit}} = \frac{c^2}{16\pi G \tilde{\alpha}_L L^2}\bar{\phi}^2_{\mathrm{attr,unit}} = \mathbf{4.513540 \times 10^{-55}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,unit}} = \mathbf{9.845073 \times 10^{-38}}$. The physical energy density of the action $S[\phi_L]$ is the unit-kernel value; the standard value is kept for continuity with V25. Both are positive ($V \geq 0$) and negligible for the CMB.

26.2.2 2.2. Scale-Dependent Effective Gravitational Constant $G_{\mathrm{eff}}(a,k)$ and Gravitational Slip at CMB Wavenumbers

In linear cosmological perturbation theory, the effective gravitational constant governing comoving wavenumber k ($k_{\mathrm{phys}}(a) = k/a$) density perturbations on the local Yukawa branch ($\beta_{\mathrm{Yuk}} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.07957747$) is:

$$\boxed{\frac{G_{\mathrm{eff}}(a,k)}{G} = 1 + \beta_{\mathrm{Yuk}} \frac{k^2_{\mathrm{phys}}(a)\, L^2}{1 + k^2_{\mathrm{phys}}(a)\, L^2} = 1 + 2\tilde{\alpha}_L \alpha_Y^2 \frac{(kL/a)^2}{1 + (kL/a)^2} = 1 + \frac{\alpha_Y^3}{4\pi} \frac{(kL/a)^2}{1 + (kL/a)^2} \underset{kL/a \ll 1}{\approx} 1 + \frac{1}{4\pi}\Big(\frac{kL}{a}\Big)^2.}$$

$\Delta G_{\mathrm{eff}}/G = \beta_{\mathrm{Yuk}}(kL/a)^2/[1 + (kL/a)^2]$: the factor $2\tilde{\alpha}_L$ is already contained in $\beta_{\mathrm{Yuk}} = 2\tilde{\alpha}_L \alpha_Y^2 = 1/(4\pi)$ (Step 23, Section 1), so there is no extra factor of 2; this is exactly `delta_geff = BETA_YUK * k_phys_L_sq` in `cmb_and_cosmology_audit.py` (2.5015×10^{-36} at $\ell_1 = 220$, 4.6516×10^{-34} at $\ell = 3000$). The table below (`cmb_geff_and_acoustic_scales.csv`) summarizes the numerically evaluated fractional fifth-force modifications $\Delta G_{\mathrm{eff}}/G \equiv G_{\mathrm{eff}}/G - 1$, the anisotropic Weyl slip ($\eta_{\mathrm{slip}} - 1 \approx 2\,\Delta G_{\mathrm{eff}}/G$), and the late-time integrated Sachs–Wolfe (ISW) modification across representative cosmological and CMB scales:

Scale label (`scale_label`)	Multipole ℓ	k (Mpc^{-1})	$k_{\mathrm{phys}}(z_*)\,L$	$\Delta G_{\mathrm{eff}}/G(z_*)$	$\log_{10}(\Delta G_{\mathrm{eff}}/G)_{z_*}$	$\Delta G_{\mathrm{eff}}/G(z=0)$	$\log_{10}(\Delta G_{\mathrm{eff}}/G)_{z=0}$	Margin below Planck 2×10^{-2} bound (orders)
CMB_quadrupole_ISW_ell2	2.0	0.000144	5.097e-20	2.067e-40	-39.68	1.738e-46	-45.76	37.99
CMB_ISW_ell10	10.0	0.000721	2.548e-19	5.168e-39	-38.29	4.344e-45	-44.36	36.59
CMB_ISW_ell20	20.0	0.001442	5.097e-19	2.067e-38	-37.68	1.738e-44	-43.76	35.99
LSS_horizon_k98-4h	98.4	0.000067	2.383e-20	4.518e-41	-40.35	3.797e-47	-46.42	38.65
CMB_recombination_horizon	100.0	0.007209	2.548e-18	5.168e-37	-36.29	4.344e-43	-42.36	34.59
CMB_first_acoustic_peak	**220.0**	**0.015860**	**5.607e-18**	**2.502e-36**	**-35.60**	**2.102e-42**	**-41.68**	**33.90**
CMB_second_acoustic_peak	537.5	0.038753	1.370e-17	1.493e-35	-34.83	1.255e-41	-40.90	33.13
CMB_third_acoustic_peak	810.6	0.058452	2.066e-17	3.398e-35	-34.47	2.856e-41	-40.54	32.77
LSS_BAO_scale_94-9.1h	94.1	0.067400	2.383e-17	4.518e-35	-34.35	3.797e-41	-40.42	32.65
CMB_Silk_damping_ell1500	1500.0	0.108138	3.823e-17	1.163e-34	-33.93	9.773e-41	-40.01	32.24
CMB_Silk_damping_tail_ell	**3000.0**	**0.216277**	**7.645e-17**	**4.652e-34**	**-33.33**	**3.909e-40**	**-39.41**	**31.63**
LSS_nonlinear_scale_1.1	952.1	0.674000	2.383e-16	4.518e-33	-32.35	3.797e-39	-38.42	30.65
LSS_small_scale_6.8e-1k0h	9531.1	6.740000	2.383e-15	4.518e-31	-30.35	3.797e-37	-36.42	28.65

(Note on wavenumber definitions: at the exact projection wavenumber $k_1 = \ell_1/D_M(z_*) = 0.015860$ Mpc^{-1} of the first acoustic peak $\ell_1 = 220.0$, $\Delta G_{\mathrm{eff}}/G(z_*) = \mathbf{2.5015 \times 10^{-36}}$, while at the rounded reference wavenumber $k_1 = 0.016$ Mpc^{-1}, $\Delta G_{\mathrm{eff}}/G(z_*) = \mathbf{2.5458 \times 10^{-36}}$; similarly, at $\ell = 3000$ ($k = 0.216277$ Mpc^{-1}), $\Delta G_{\mathrm{eff}}/G(z_*) =$

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$\mathbf{4.6516 \times 10^{-34}}$, *and at rounded* $k = 0.22$ *Mpc*$^{-1}$ *it is* $\mathbf{4.8131 \times 10^{-34}}$. *Both sets of numbers are recorded in* `cmb_and_cosmology_summary.json`.*)*

26.3 3. Numerically Integrated FLRW Background Models vs. Fixed Phase-Shift and Literature Peak-Height Inputs

26.3.1 3.1. Equations Actually Integrated by `cmb_and_cosmology_audit.py`

For a given parameter set $(\Omega_b, \Omega_c, \Omega_\gamma, \Omega_\nu, \Omega_{\Lambda,\mathrm{eff}}, \Omega_K)$, `cmb_and_cosmology_audit.py` integrates the Friedmann equation:

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_K(1+z)^2 + \Omega_{\Lambda,\mathrm{eff}}, \qquad q_0 = \frac{\Omega_{m,0}}{2} + \Omega_{r,0} - \Omega_{\Lambda,\mathrm{eff}}.$$

The baryon–photon sound speed and the comoving sound horizon at recombination ($z_* = 1089.80$) are integrated via:

$$c_s(z) = \frac{c}{\sqrt{3\left(1 + R_b(z)\right)}}, \qquad R_b(z) = \frac{3\Omega_b}{4\Omega_\gamma(1+z)}, \qquad r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)\,dz}{H_0 E(z)}.$$

The comoving line-of-sight distance, curvature-corrected comoving angular distance, and acoustic scale are:

$$\chi(z_*) = \frac{c}{H_0}\int_0^{z_*} \frac{dz}{E(z)}, \qquad D_M(z_*) = \begin{cases} \dfrac{c}{H_0\sqrt{\Omega_K}} \sinh\!\left(\dfrac{\sqrt{\Omega_K}H_0}{c}\chi(z_*)\right) & (\Omega_K > 0), \\ \chi(z_*) & (\Omega_K = 0), \end{cases} \qquad \theta_* = \frac{r_s(z_*)}{D_M(z_*)}, \qquad \ell_A = \frac{\pi}{\theta_*}.$$

26.3.2 3.2. Split Summary Tables of the 8 Cosmological Models (`cosmological_background_models.csv`)

In accordance with Review Point 2, we present the 8 cosmological models in **two separate tables**: **Table 3.2.A** contains exclusively the **numerically integrated FLRW background and acoustic-scale quantities** computed by `cmb_and_cosmology_audit.py`, whereas **Table 3.2.B** reports the **analytic multipole estimates using fixed external phase shifts** (ϕ_1, ϕ_2, ϕ_3) alongside the **fixed external literature peak-height benchmark inputs** (`boltzmann_spectrum_computed = False`).

Table 3.2.A: Numerically Integrated FLRW Background and Acoustic-Scale Quantities

Model ID (`model_id`)	$\rho_{\mathrm{DE,eff}}$ (kg/m^3)	Ω_b	Ω_c	$\Omega_{\Lambda,\mathrm{eff}}$	Ω_K	Reaches $a = 1$?	q_0	$a_{\max}$ (z_{turn})	$r_s(z_*)$ (Mpc)	$\chi(z_*)$ (Mpc)	$D_M(z_*)$ (Mpc)	$100\,\theta_*$ (rad)	$\ell_A = \pi/\theta_*$
Planck18_LCDM_physical	−5.96e-27	0.04924	0.26416	0.68510	0.00000	True	−0.5276	—	144.4423	13871.12	13871.12	1.041326	301.6932
TAL_local_best_fit	−5.96e-27	0.04924	0.26416	0.68510	0.00000	True	−0.5276	—	144.4423	13871.12	13871.12	1.041326	301.6932
TAL_unconstrained_negative_lambda	−5.96e-27	0.04924	0.26416	−0.68510	1.37020	True	+0.8426	1.5175 (−0.3410)	144.3662	11073.14	34916.42	0.413467	759.8258
TAL_unconstrained_negative_lambda_flat	−5.96e-27	0.04924	0.26416	−0.68510	0.00000	False	—	0.7718 (+0.2957)	144.4423	16367.75	16367.75	0.882484	355.9943
TAL_compensated_max1	−5.96e-27	0.04924	0.26416	0.68510	0.00000	True	−0.5276	—	144.4423	13871.12	13871.12	1.041326	301.6932
TAL_baryon_only_flat	−8.10e-27	0.04924	0.00000	+0.94926	0.00000	True	−0.9238	—	195.1104	27512.60	27512.60	0.709167	442.9974
TAL_baryon_only_closing_diagnostic	−8.10e-27	0.04924	0.00000	+0.94926	0.00000	True	−0.9238	—	209.1445	27512.60	27512.60	0.760177	413.2712
TAL_baryon_only_open_no_lambda	0.00e+00	0.04924	0.00000	0.00000	+0.94926	True	+0.0254	—	194.9249	18037.73	18631.37	1.043113	301.1911.9746

Table 3.2.B: Analytic Phase-Shift Multipole Estimates (ϕ_m) and Fixed External Literature Peak-Height Inputs (`boltzmann_spectrum_computed = False`)

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Model ID (model_id)	Input phase shifts (ϕ_1, ϕ_2, ϕ_3)	l1_analytic_phase_est	l2_analytic_phase_est	l3_analytic_phase_est	$\Delta\ell_1^{\rm est}$ vs. Planck 220.0	H2_H1_lit_benchmark_input (fixed input)	H3_H2_lit_benchmark_input (fixed input)	boltzmann_spectru
Planck18_LCDM_reference	(0.2709, 0.2188, 0.3131)	219.95	537.38	810.62	-0.05	0.458 *(Planck/lit. $\Omega_c > 0$ input)*	0.985 *(Planck/lit. $\Omega_c > 0$ input)*	False
TAL_local_Yukawa_2700Mpc_sector	(0.2709, 0.2188, 0.3131)	219.95	537.38	810.62	-0.05	0.458 *(Planck/lit. $\Omega_c > 0$ input)*	0.985 *(Planck/lit. $\Omega_c > 0$ input)*	False
TAL_uncompensated_negative_rhoLambda_flat	(0.2709, 0.2188, 0.3131)	558.96	1353.41	2041.58	+333.96	0.458 *(fixed $\Omega_c > 0$ input)*	0.985 *(fixed $\Omega_c > 0$ input)*	False
TAL_uncompensated_negative_rhoLambda_flat	(0.2709, 0.2188, 0.3131)	259.54	634.10	956.52	+39.54	0.458 *(fixed $\Omega_c > 0$ input)*	0.985 *(fixed $\Omega_c > 0$ input)*	False
TAL_compensated_macro_clock_background	(0.2709, 0.2188, 0.3131)	219.95	537.38	810.62	-0.05	0.458 *(Planck/lit. $\Omega_c > 0$ input)*	0.985 *(Planck/lit. $\Omega_c > 0$ input)*	False
TAL_baryon_only_flat_Lambda	(0.3150, 0.2620, 0.3550)	303.45	769.93	1171.73	+83.45	0.312 *(literature $\Omega_c = 0$ input)*	0.415 *(literature $\Omega_c = 0$ input)*	False
TAL_baryon_only_scaling_diagnostic	(0.2709, 0.2188, 0.3131)	301.30 *(or 152.02 at fixed $D_M^{\Lambda CDM}$)*	736.13	1110.42	+81.30 (-67.98)	0.312 *(literature $\Omega_c = 0$ input)*	0.415 *(literature $\Omega_c = 0$ input)*	False
TAL_baryon_only_open_no_Lambda	(0.3150, 0.2620, 0.3550)	1309.70	3323.01	5057.17	+1089.70	0.312 *(literature $\Omega_c = 0$ input)*	0.415 *(literature $\Omega_c = 0$ input)*	False

26.4 4. Theoretical Assessment of Negative $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 and Positive $\rho_{\rm clock,cosmo}$ Compensation (Review Point 3)

1. **Why the canonical scalar energy density is positive in the minimal action of Steps 22–23:**
   - In the Einstein-frame action specified in Step 23 (`step23_gw170817_and_lensing_en.md`, Section 2.1), the scalar potential is $V(\phi_L) = +\frac{1}{2}m_\phi^2\phi_L^2 \geq 0$. The time–time component of the standard variational stress–energy tensor $T_{\mu\nu}^{(\phi)} = \frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}((\nabla\phi_L)^2 + L^{-2}\phi_L^2)\right]$ for a spatially homogeneous field ($\nabla_i\phi_L = 0$) is:

$$\rho_{\phi,\rm can} \;=\; \frac{1}{c^2}T_{00}^{(\phi)} \;=\; \frac{c^2}{8\pi G\tilde{\alpha}_L}\left[\frac{\dot{\phi}_L^2}{2c^2} + \frac{\phi_L^2}{2L^2}\right] \;\geq\; 0.$$

   - Consequently, the minimal canonical action of Steps 22–23 does **not** generate a negative vacuum energy density; the sign $\rho_\Lambda = -V(\phi_{\rm cosmo})/c^2 = -5.96 \times 10^{-27}$ kg/m^3 can only be introduced as a hypothesis belonging to a separate macroscopic/cosmological branch.
2. **Consequences of an uncompensated negative $\rho_\Lambda = -5.96\times10^{-27}$ kg/m^3 ($\Omega_\Lambda = -0.685101$) on a separate branch:**
   - **In flat space ($\Omega_K = 0$, `TAL_uncompensated_negative_rhoLambda_flat`):** today one would have $E^2(z = 0) = \Omega_{r,0} + \Omega_{m,0} + \Omega_\Lambda = -0.370202 < 0$, meaning the universe would have halted and recollapsed at $a_{\rm max} = (\Omega_{m,0}/|\Omega_\Lambda|)^{1/3} = \mathbf{0.771762}$ ($z_{\rm turn} = \mathbf{0.295737}$), never reaching $a = 1$.

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- **With spatial curvature closure ($\Omega_K = 1-\Omega_{m,0}-\Omega_\Lambda = +1.370202$, `TAL_uncompensated_negative_rhoLambda`):** the universe reaches $a = 1$ (and turns around in the future at $a_{\max} = 1.517519$), but strongly decelerates today ($q_0 = +\mathbf{0.842603}$, vs. the observed accelerating $q_0 \approx -0.53$), and the large positive curvature stretches the comoving angular distance to $D_M(z_*) = 34\,916.42$ Mpc, shifting the acoustic scale to $100\,\theta_* = \mathbf{0.413462}$ ($\ell_A = 759.83$, $\ell_1^{\rm est} \approx 553.96$).

3. **Exact status of the compensated macroscopic clock background (`TAL_compensated_macro_clock_background`):**

- If, on the conditional separate branch, one postulates a homogeneous cosmological clock condensate with energy density $\rho_{\rm clock,cosmo} = +2|\rho_\Lambda| = +1.192 \times 10^{-26}$ kg/m^3 ($\Omega_{\rm clock,cosmo} = +1.370202$, $w_{\rm clock} = -1$), the net effective dark-energy density is:

$$\rho_{\rm DE,eff} = \rho_\Lambda + \rho_{\rm clock,cosmo} = -5.96\times10^{-27} + 1.192\times10^{-26} = +5.96\times10^{-27}\ \text{kg/m}^3 \quad (\Omega_{\rm DE,eff} = +0.685101).$$

- This equality restores the input Planck 2018 ΛCDM background ($q_0 = -0.5276$, $100\,\theta_* = 1.041320$, $\ell_A = 301.6932$) **by algebraic construction**; it is **not** an independent cosmological prediction until a joint covariant action, stress–energy tensor, Bianchi identity, and dynamical attractor equation are derived for both $\rho_\Lambda < 0$ and $\rho_{\rm clock,cosmo} = +2|\rho_\Lambda|$.

26.5 5. Why the Local $L = 10$ km Yukawa Sector Does Not Replace Cosmological Cold Dark Matter (Ω_c)

If cosmological cold dark matter were omitted ($\Omega_c = 0$, retaining only baryons $\Omega_b = 0.049243$ and the local $L = 10$ km Yukawa field):

1. **Shift of the sound horizon and acoustic scale (numerically integrated background result):**

- Due to the reduced pre-recombination expansion rate $H(z)$, the sound horizon increases from $r_s(z_*) = 144.44$ Mpc to 195.11 **Mpc** (with full $N_{\rm eff} = 3.046$; or 209.14 Mpc in the semi-analytic scaling diagnostic).
- With flat Λ closure ($\Omega_\Lambda = 0.949257$), the comoving angular distance expands to $D_M(z_*) = \mathbf{27\,512.60}$ Mpc, yielding $100\,\theta_* = \mathbf{0.709167}$ and $\ell_A = \mathbf{442.9974}$ ($\ell_1^{\rm est} \approx 303.45$, vs. Planck $\ell_1 = 220.0$; or $\ell_1^{\rm est} \approx 152.02$ if evaluated at fixed ΛCDM $D_M = 13\,871.12$ Mpc). In an open dark-energy-free universe ($\Omega_K = +0.949257$), $D_M(z_*) = 118\,631.37$ Mpc, $100\,\theta_* = 0.164311$, and $\ell_A = 1911.97$.

2. **Gravitational potential wells and odd/even peak heights (literature benchmark comparison):**

- At recombination, the local Yukawa fifth-force modification is $\Delta G_{\rm eff}/G(z_*, \ell_1) = 2.50 \times 10^{-36}$, so the local $L = 10$ km field does not create baryon-free gravitational potential wells.
- Standard cosmological perturbation theory (Hu & Sugiyama 1996; Planck 2018 VI) shows that for $\Omega_c = 0$, radiation pressure erases gravitational potentials upon sound-horizon entry, suppressing the third acoustic peak ($H_3/H_2 \approx 0.415$ vs. the observed 0.985). In `cosmological_background_models.csv`, these peak-height ratios are recorded strictly as **fixed external literature benchmark inputs** (`boltzmann_spectrum_computed = False`).
- Extending the TAL macroscopic clock sector ($\rho_{\rm clock} = (c^2\Gamma_{\rm clock}/8\pi G)|\nabla\phi|^2$) to replace cosmological CDM would require deriving the input $\Omega_c/\Omega_b = 0.1200/0.02237 = 5.364327$ ratio and the evolution of acoustic potential wells from a coupled Einstein–Boltzmann system derived from a closed covariant action.

26.6 6. Cross-Step Physical Synthesis Table of Steps 22–25 (V26 — Reconciled Numerical Values)

In accordance with Review Point 4, we have corrected all three numerical discrepancies that remained in the earlier synthesis text so that every entry matches the machine-readable CSV and JSON outputs of Steps 22, 23, 24, and 25:

1. **Neutron stars (Step 22, SLy, $L = 10$ km, $\alpha_Y = 1$):** the maximum ADM mass from the exact 4-component curved exterior BVP is `M_ADM,max = 2.033214 M_sun` (`2.033213549 M_sun` at the turning point $\log_{10}\rho_{0c} = 15.316404$, with $M_{\rm surf} = 2.029106\ M_\odot$, $\Delta M_{\phi,\rm ext}^{\rm curved} = 0.004108\ M_\odot$, $M_{B,\max}^{\rm quad} = 2.423385\ M_\odot$, $R(M_{\rm ADM,max}) = 9.8853$ km, $R_{1.4}^{\rm ADM} = 11.5359$ km).

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2. **GW170817 sub-cutoff scalar amplitude suppression (Step 23, $D = 40$ Mpc, $f = 100$ Hz $\ll f_c = 4.7713$ kHz):** the base-10 logarithm of the evanescent suppression factor is `log10 |delta phi / delta phi_0| = -5.360e19` (`-5.360058e+19` at $f = 100$ Hz and `-5.360179e+19` in the static $f \to 0$ limit in `gw170817_dispersion_and_cT.csv`).
3. **Bullet Cluster synthetic 1D convergence test (Step 24):** the normalized synthetic 1D target MSE of the local $L = 10$ km Yukawa branch is `M_synth = 32.089` (`synth_mse_norm_vs_target = 32.088829` in `bullet_cluster_peak_offsets_summary.csv`) and is explicitly labeled as the synthetic 1D target MSE ($M_{\rm synth}$), not an observational $\chi^2_{\rm red}$.

Physical System / Test	Characteristic Scale r or k^{-1}	r/L or $(k_{\rm phys}L)^{-1}$ ($L = 10$ km)	Computed Result of the Local $L = 10$ km Yukawa Branch (`yukawa_action_el`)	Observational Constraint / Status (V26)	Is a Separate Macroscopic / Cosmological Branch Required?
Neutron Stars (TOV + Curved Exterior ADM + Stability, Step 22)	$R_* \approx$ 9.89–13.9 km	$r/L \sim 1.0$	SLy ($L =$ 10 km, $\alpha_Y = 1$): `M_ADM,max = 2.033214 M_sun` ($M_{\rm surf} =$ $2.029106\,M_\odot$, $\Delta M_{\phi,\rm ext} =$ $0.004108\,M_\odot$, $M_{B,\rm max}^{\rm quad} =$ $2.423385\,M_\odot$, $R(M_{\rm ADM,max}) =$ 9.8853 km, $R_{1.4}^{\rm ADM} =$ 11.5359 km); dynamical $\omega_0^2 = 0$ zero crossing within $+5.80 \times 10^{-6}$ dex of the ADM peak	Satisfies PSR J0740+6620 and NICER mass–radius bounds (global stability statement remains conditional pending independent free-surface variation closure)	**No** — the local $L = 10$ km Yukawa branch self-consistently describes compact-star structure
GW170817 / GRB 170817A Propagation and Polarization (Step 23)	$D = 40$ Mpc = 1.234×10^{21} km	$D/L =$ 1.234×10^{20}	On the purely conformal, universally coupled Maxwell branch: $c_T = c$, $\Delta c_T/c = 0$, $\Delta t_{g-\gamma} = 0$ s; $f_c = 4771.345$ Hz; at $f = 100$ Hz, evanescent scalar suppression: `log10|dphi/dphi0| = -5.360e19`; at $f = 5$ kHz, $\Delta t_\phi =$ $+1.4645 \times 10^{15}$ s	Satisfies the Abbott et al. (2017) $\lvert\Delta c_T/c\rvert \le$ 3×10^{-15} bound on the purely conformal branch	**No** (though $c_T = c$ must be proven separately on any disformal $D(\phi_L, X) \neq 0$ branch)

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Physical System / Test	Characteristic Scale r or k^{-1}	r/L or $(k_{\rm phys}L)^{-1}$ ($L = 10$ km)	Computed Result of the Local $L = 10$ km Yukawa Branch (`yukawa_action_el`)	Observational Constraint / Status (V26)	Is a Separate Macroscopic / Cosmological Branch Required?
Solar–Cassini PPN and Light Deflection ($1.000000\,M_\odot$, Step 23)	$b_{\rm Cassini} = 1.6\,R_\odot = 1.11312 \times 10^6$ km	$b/L = 1.11312 \times 10^5$	Dedicated $1.0\,M_\odot$ Solar model: $\hat{\alpha}_{\rm GR,\odot}(R_\odot) = 1.751190''$, $\hat{\alpha}_{\rm GR,\odot}(1.6\,R_\odot) = 1.094494''$; $\lvert\gamma_{\rm PPN,\odot}(1.6\,R_\odot) - 1\rvert = \mathbf{10^{-48\,343.00}}$ ($\beta_{\rm Yuk} = 1/(4\pi)$); conformal photon invariance yields exact null-geodesic deflection $\delta\hat{\alpha}_{\rm null} = 0$	$> 48\,338$ orders of magnitude below the Cassini $\lvert\gamma_{\rm PPN} - 1\rvert \le 2.3 \times 10^{-5}$ bound	**No** — the local Yukawa branch is naturally screened at $r \gg 10$ km
SPARC Galactic Rotation Curves (Steps 13, 17, 19)	$r \sim$ 1–50 kpc	$r/L \sim 3{\times}10^{15}$–$1.5{\times}10^{17}$	The local $L = 10$ km Yukawa force vanishes as $\exp(-r/L) \sim 10^{-10^{15}}$, reverting to pure baryonic Newtonian gravity	Reproducing SPARC RAR / BTFR requires macroscopic coupling	**Yes** — macroscopic clock-field closure ($\rho_{\rm clock} \propto \lvert\nabla\phi\rvert^2$) or effective a_0
Bullet Cluster (1E 0657-558) Weak Lensing (Step 24)	$R \sim$ 100–1000 kpc	$R/L \sim 3 \times 10^{17}$–$3 \times 10^{18}$	$\exp(-100\ \text{kpc}/10\ \text{km}) = 10^{-1.340\times10^{17}}$; κ peaks remain on the X-ray gas ($\Delta x = 0$ kpc); synthetic 1D target MSE: `M_synth = 32.089` (baryonic GR: `32.089`, gas-dominated $L_{\rm macro} = 300$ kpc: `20.723`, collisionless galaxy-sourced clock field: `1.637`)	The ~ 180–200 kpc lensing–X-ray offset cannot be produced by a local gas-sourced field	**Yes** — requires particle CDM or a galaxy-specific ($\eta_{\rm gal}/\eta_{\rm gas} \ge 25.665$) nonlocal clock branch

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Physical System / Test	Characteristic Scale r or k^{-1}	r/L or $(k_{\rm phys}L)^{-1}$ ($L = 10$ km)	Computed Result of the Local $L = 10$ km Yukawa Branch (`yukawa_action_el`)	Observational Constraint / Status (V26)	Is a Separate Macroscopic / Cosmological Branch Required?
Cosmological FLRW Background and CMB Acoustic Scale (Step 25)	$k^{-1} \sim 5$–10^4 Mpc	$(k_{\rm phys}L)^{-1} \sim 10^{16}$–$10^{20}$	$\Omega_{\phi,\rm can}(z_*) = 9.85 \times 10^{-38} \geq 0$; $\Delta G_{\rm eff}/G(z_*, \ell_1 = 220) =$ $\mathbf{2.50 \times 10^{-36}}$, $\Delta G_{\rm eff}/G(z_*, \ell = 3000) =$ $\mathbf{4.65 \times 10^{-34}}$; local Yukawa sector leaves input ΛCDM background unchanged ($100\,\theta_* = 1.041320$, $\ell_A = 301.6932$)	Local Yukawa modification is > 31–33 orders below Planck 2018 $G_{\rm eff}$ bounds; $\Omega_c/\Omega_b = 5.364327$ is an input Planck ratio	**Yes**, if CDM (Ω_c) or dark energy (Ω_Λ) is to be derived from the clock sector (requires coupled Boltzmann calculation)

26.7 7. Checks and Reproducibility

All calculations in Step 25 can be reproduced with a single command:

```
python3 cmb_and_cosmology_audit.py
```

Running the script generates and verifies the following files (in the root directory and synced to `csv/` and `data/v23/`):

- `cosmological_background_models.csv`: numerically integrated FLRW background quantities ($E^2(z=0)$, $E(z_*)$, q_0, $a_{\rm max}$, $r_s(z_*)$, $\chi(z_*)$, $D_M(z_*)$, $D_A(z_*)$, $100\,\theta_*$, ℓ_A) for all 8 models, together with dedicated columns for fixed phase-shift estimates (`l1_analytic_phase_est` – `l3_analytic_phase_est`) and fixed literature peak-height benchmark inputs (`H2_H1_lit_benchmark_input`, `H3_H2_lit_benchmark_input`, `boltzmann_spectrum_computed = False`).
- `cmb_geff_and_acoustic_scales.csv`: $\Delta G_{\rm eff}(a,k)/G$, Weyl slip, and ISW modifications across 13 cosmological and CMB wavenumbers ($\ell = 2$ through $k = 10\,h$ Mpc^{-1}).
- `cmb_and_cosmology_summary.json`: machine-readable audit protocol recording the V26 theoretical disclosures (`computed_vs_external_inputs_disclosure`, `conditional_separate_branch_hypothesis`), background integrals, and reconciled synthesis numbers across Steps 22–25 (`v26_synthesis_table_alignment`).

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27 Reproducibility Manifest and SHA-256 Checksums (MANIFEST_en.txt)

Source file: MANIFEST_en.txt

```
# MANIFEST --- Scalar--Tensor and Yukawa/Disformal Action Mapping (Steps 1--25 Derivation and Audit Reproduction Package,
↪  V26: V25 Steps 1--22 + Extended Modules 23--25)
# Total files tracked in this manifest: 388 (including the 25 Hungarian chapters, 25 English translation chapters,
↪  README.md, and README_en.md)
# Publication status: The earlier Zenodo DOI (10.5281/zenodo.23680715) is cited strictly as a reference to the prior V24
↪  release; the two sign fixes and three diagnostics belong to V25 (DOI: 10.5281/zenodo.23699685, with a conditional
↪  stability closing statement), and modules step23--step25 provide the extended observational tests at fixed L = 10 km,
↪  alpha_Y = 1.
# Format: <SHA-256>  <size_in_bytes>  <relative_path>

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82d55c6e8f2351451afe7dcd9941f5414e21bbfaea6ebd4d5a7fa0b800b3d82b       97492  README_en.md
febb1bfe434b22cc54a4f0bdfafb62efa24ad361b96cda05cf9290eafd80ec7c        1989  SHA256SUMS_step23_25.txt
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8c28105dda15834a0bac4a05d17436969a13187fe07d04800d15a346785e06d4       77525  bullet_cluster_convergence_profile.csv
8caca2431b9f2cda1847e9d5851ccf4b1b9511327d665ba8101a0b8d58f98914       31459  bullet_cluster_lensing_audit.py
1c55be84757dc68f7d4aeecb48430bda1bf07443371fc896783302335f7b758e        1096  bullet_cluster_peak_offsets_summary.csv
ad054b61c758080335f2d9d5827ab5a2fe2dcff4183b6e788119fd1f15cd19ab       14002  closure_tov_comparison.csv
34f98435de79e59d962f34e77fcf52ed91a2e42a57c70ae02fc55b214e04ceb3       42828  cmb_and_cosmology_audit.py
8aeb25850578a519b94eedcd0983be1360eefce7acc71f3f8a012101b8cb7c06       40762  cmb_and_cosmology_summary.json
0f0d86f1139a982f867132d141211d71ee3ee1db78134eaf058580bd55c8c9e2        3898  cmb_geff_and_acoustic_scales.csv
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5c6453d1fd17c4933839adaf343a40a4475698d471a05d0fe8e740aee77a1841       10416  code/bullet_cluster_audit_summary.json
8c28105dda15834a0bac4a05d17436969a13187fe07d04800d15a346785e06d4       77525  code/bullet_cluster_convergence_profile.csv
8caca2431b9f2cda1847e9d5851ccf4b1b9511327d665ba8101a0b8d58f98914       31459  code/bullet_cluster_lensing_audit.py
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↪  code/bullet_cluster_peak_offsets_summary.csv
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0f0d86f1139a982f867132d141211d71ee3ee1db78134eaf058580bd55c8c9e2      3898  data/v23/cmb_geff_and_acoustic_scales.csv
059ac76b1b8fc4dcbf1537a2b3fa0e95dcca0e88415e51a400f4f1f88360a226    115783  data/v23/conservation_Residuals.json
0ec03152b731c02b3f6a51f279a5d8a5920e63b48fbc4a95eacc2d60d4836cb9      4355  data/v23/cosmological_background_models.csv
95fa3eab530daa8f159f02b6433b7a726f47c0d698214886c69d7dae3d1bcdee     18245  data/v23/coupled_stability_results.csv
acf334f7e5594b9af1ec1f2f9ecb6c39eba34e9abbeca864fae5d62a3227526c      6133  data/v23/curved_yukawa_adm_sequences.csv
a97669f24dade5a83a26e476946313c161cf692db7f845b368e5d1e51aebbaed     97385  data/v23/curved_yukawa_audit_summary.json
f4fe6d056841cfa9c0a072388624a5ffe51a732fd8da8655e1fbacb3c6b8747e      7033
↪  data/v23/curved_yukawa_coupled_stability.csv
95dda8d26c9c6d8bd8d8c1ca1d15abe7408be41f0f1ea6f1d1e2d7a6efe600d7     81582  data/v23/gw170817_and_lensing_summary.json
e03014bc93cb6de1d85d82398d737cb31b1da176c90d8ea84b5c3c274901591c      3945  data/v23/gw170817_dispersion_and_cT.csv
```

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daa67c9a98e57704ea185a1c4682b3a9df47e808494610e739f25e54717e9f46       10382
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18435bb1f964c72c08ad333716db5ee7d227a52a668888c6158436ac6c3e5ddd       70431  gw170817_and_lensing_audit.py
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2b4022fef37a705aa9c6a963ba80528b733667102d43ad0c8c352a0d243ec79d      423790
↪  hetedik_modell_kimenet/points_tal_fL_FITTED_forward.csv
daa67c9a98e57704ea185a1c4682b3a9df47e808494610e739f25e54717e9f46       10382  lensing_deflection_and_ppn_slip.csv
93ca2b3955333869c5f29eadc381d26c1932b5a4560a01db1a3025b2aaed9533        5405  make_revision3_checks.py
f55e556dd081ba5b813ae0aa210081d7e59c002b384ae87b2783d94871730c26       39044  manuscripts/en_part1.tex
b60816ea4b993915a1a2794f2c124fe775c0c65e0dce763c0917079d2111f38e       30422  manuscripts/en_part2.tex
dd8a8b54113bd594e9a900ff29d4a4df58f0f6df12342bfc6ef787434617288a       33377  manuscripts/hu_p1.tex
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ab8bd81310bd098556ec496a40cbf46bb2ed0b8fbaaf602ae33c9d23563f332b       23913  manuscripts/hu_p3.tex
d4cbd4b6c2a7670c86842f7683d016ba9ba6cc0e715c6559a807e71661034051       39119
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a97537d7835440117997dd5a8deb520d0c6165853c3e8dcbb2a4221a867652ea       27450
↪  manuscripts/tal_covariant_closure_50aa2db0_EN_2026-10-24.tex
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a7469307e1ccfe2da873225b288e09a29dcfc347af7ca7f1deea3829c09061b7      223977  run_output/points_vbar.csv
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eb40036842162dc9a42216bd7a4d35c7f2ce8addcf7e3ed8ffd3626427b22313        6259  sparc_full_runner.py
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637f892bb25468fc0c058fd1ecdd60ca53d51059675b94e61b885398ad162bdb         876  sparkto_b_raw.py
cb93c90fb261009c987ebc98a516fc15d66c7f8c34ae8098b5617fd7ddd21906       27472  stellar_tal_v21.py
3cfe4aaa90e66f3f2dba00611c31d86344c408b997e38c02f0e3e38f2e52df0b       13420
↪  step10_bilateral_kernel_reciprocity_eft_synthesis.md
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↪  step10_bilateral_kernel_reciprocity_eft_synthesis_en.md
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↪  step11_chameleon_screening_multiscale_eft_synthesis.md
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↪  step11_chameleon_screening_multiscale_eft_synthesis_en.md
35a16b76cb419e0adb23fea36130527c971ba8de5bb558288e4729eb9864b76e       19550
↪  step12_second_variation_coupled_pulsation_hessian.md
798f21c32075f3a77aaff6c66e0a7a3fcd0d114ca5903c4729209525bbd5eac2       19010
↪  step12_second_variation_coupled_pulsation_hessian_en.md
261d16eadbf89cf4b3691c12f00defbd3f051820e8ec067f6b1e4c986c05d13d       11595
↪  step13_curved_window_metric_variation_synge_world_function.md
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↪  step13_curved_window_metric_variation_synge_world_function_en.md
d377321aba8ddfffdc16403e974c366ba2b464fb9a4894489227c8d417fd06da       16324
↪  step14_gravitational_lensing_null_geodesics_cosmological_background.md
43051f7ff852c02546ae7b084d61716bf7ae53826e91fb493e49617232233407       15515
↪  step14_gravitational_lensing_null_geodesics_cosmological_background_en.md
75332aacb1edc9f3be3995955728194b47e86b7c19158b1157ca314ec825bff4       13340
↪  step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20.md
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↪  step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md
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↪  step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress.md
f3886101a664595b5dd2c3a4a922f8fa349e13411780ce069001b66f02786dba       13530
↪  step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md
fce1edf9f49c8d622e830d6001ccb9a6e7159f5c99c1884cd1f4b3add46db6a3       14383
↪  step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness.md
672dc1adfa28d46f8b4115afe314f8d8c5b129ee93c02af3256eb66abadbf8dd       13765
↪  step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md
bac44dce1493f220e1e85311b9f313ef965eca1bab6319ecb9d8dae78dde8c6c       21179
↪  step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance.md
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↪  step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md
bf6eea32f1a730c04c7437057ad4cbeccd320d2759800e95c965b569c9d054b8       18882
↪  step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy.md
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↪  step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md
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83a2e8e88808373947d5b078ee3f20f616a9c23c865f3a75d384697bf11e05bf       10890  step1_dilaton_form_en.md
56eb853ee017a0d35b8fed73461c203d197ba173fbc86f3ebadeb4bae4e32175       17366
↪  step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature.md
800ff0bb696640791b0baedf6a95ae5349b560dc3d4bd036097a3fd3037d151b       16820
↪  step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md
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↪  step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse.md
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↪  step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md
dc84080449d2fcba870df8d7273c9495cafe153624e27f8a8867f89df020686c       36896
↪  step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md
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↪  step23_25_gw170817_lensing_bullet_cmb_en.pdf
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↪  step23_25_gw170817_lensing_bullet_cmb_hu.pdf
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7a3761a4adcca6fd38443be0666b49577b49ca18582ffe395fce4038623e2a3d       18402  step2_conformal_disformal.md
abd9db8d104e3bcf157194a5796e75d9695c20d7ef0b47403fb5fc8a1028bfc9       18014  step2_conformal_disformal_en.md
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52b1cab653da99d60994267b39b1799c91a9401bc37a21a08b5012e17b879442       12356  step4_covariant_conservation_hydrostatic.md
1deaba07182e7cfa7c65db0462577f37ea3652269982afc0673906015da67baa       11986
↪  step4_covariant_conservation_hydrostatic_en.md
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8fbdd2157d0a4aa9fbd78dca87444c4689f7ecb35f660424bfa8a252a1b32594       14615  step6_sparc_galactic_ngc3198_consistency.md
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↪  step6_sparc_galactic_ngc3198_consistency_en.md
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↪  step8_microscopic_clock_coupling_siriusb_lab_v20.md
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↪  step8_microscopic_clock_coupling_siriusb_lab_v20_en.md
af5b8d0ab66ff02c831ac4774b84446dc589a7d43e4c485d2819410a427af197       12164
↪  step9_vhe_photon_propagation_hess_dispersion.md
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70cad9a97511e3b0d37a15a58eb22b656e323615d1cce5df1ab2c778250fb3a2       18181  ta_pipeline.py
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3637d2f5bda6b2b97e645c5fba5cf39d310a1826bee388286c2c74b9171aa283        1039  v17_package/github/data/blind_123.txt
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